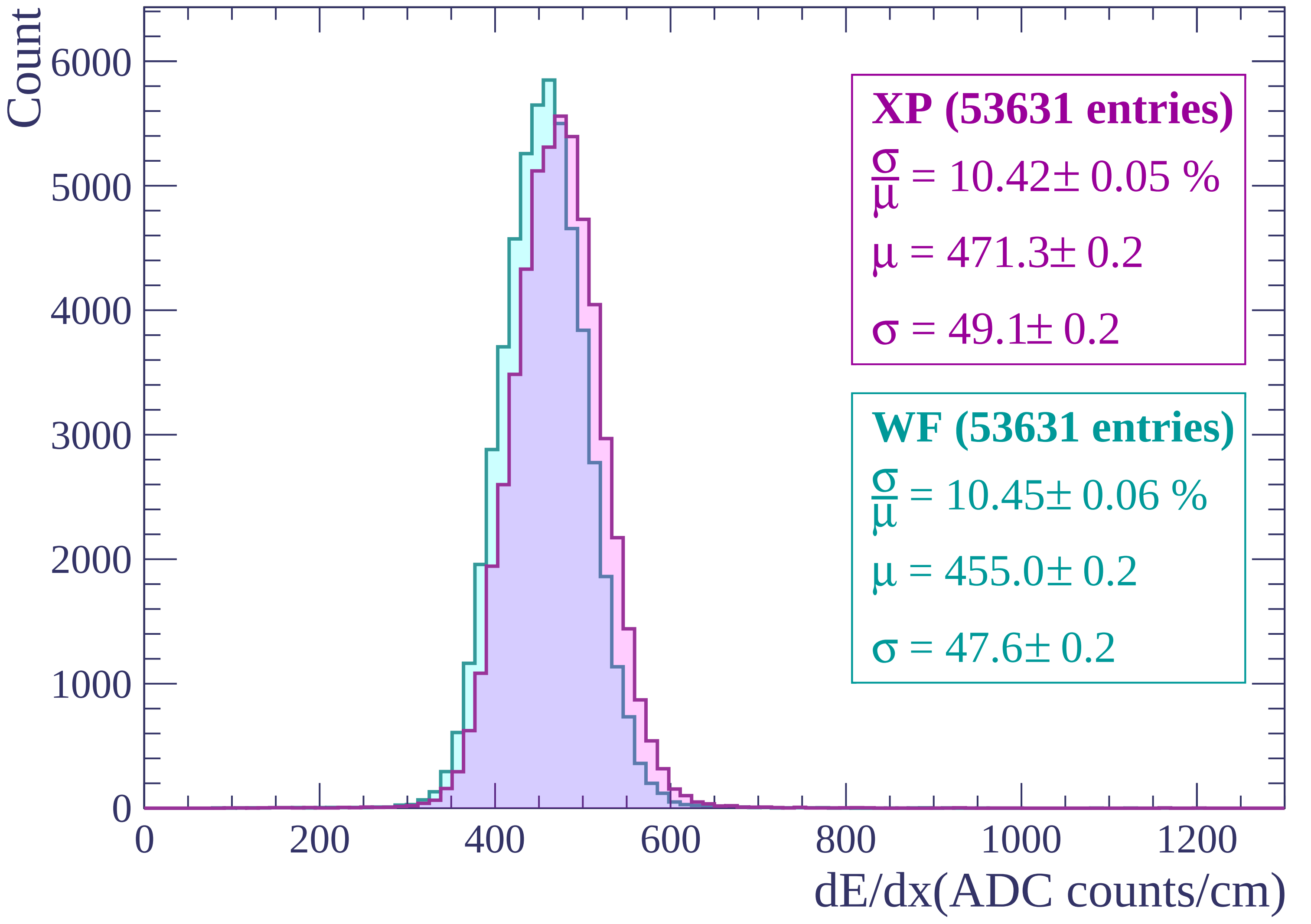
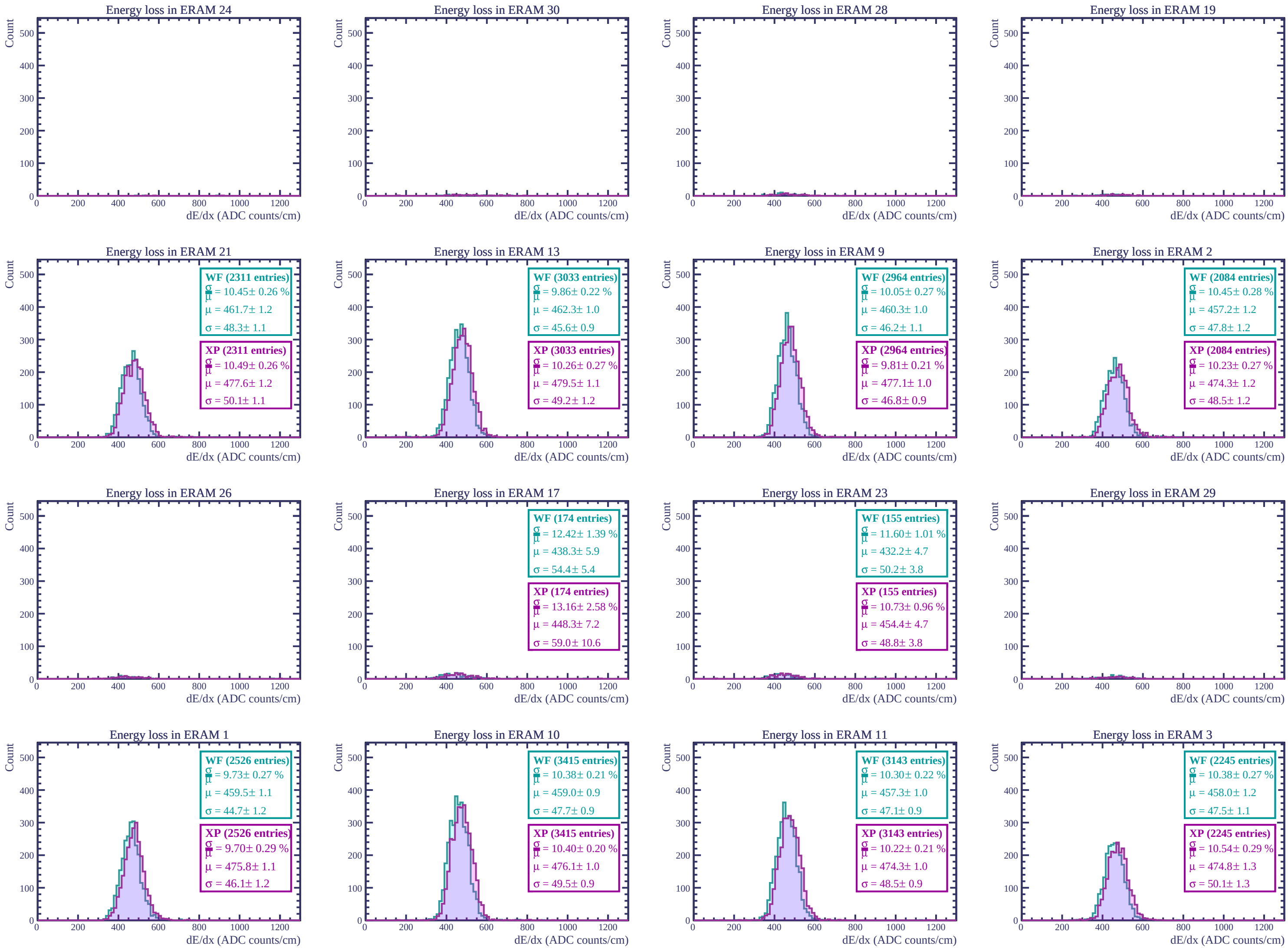
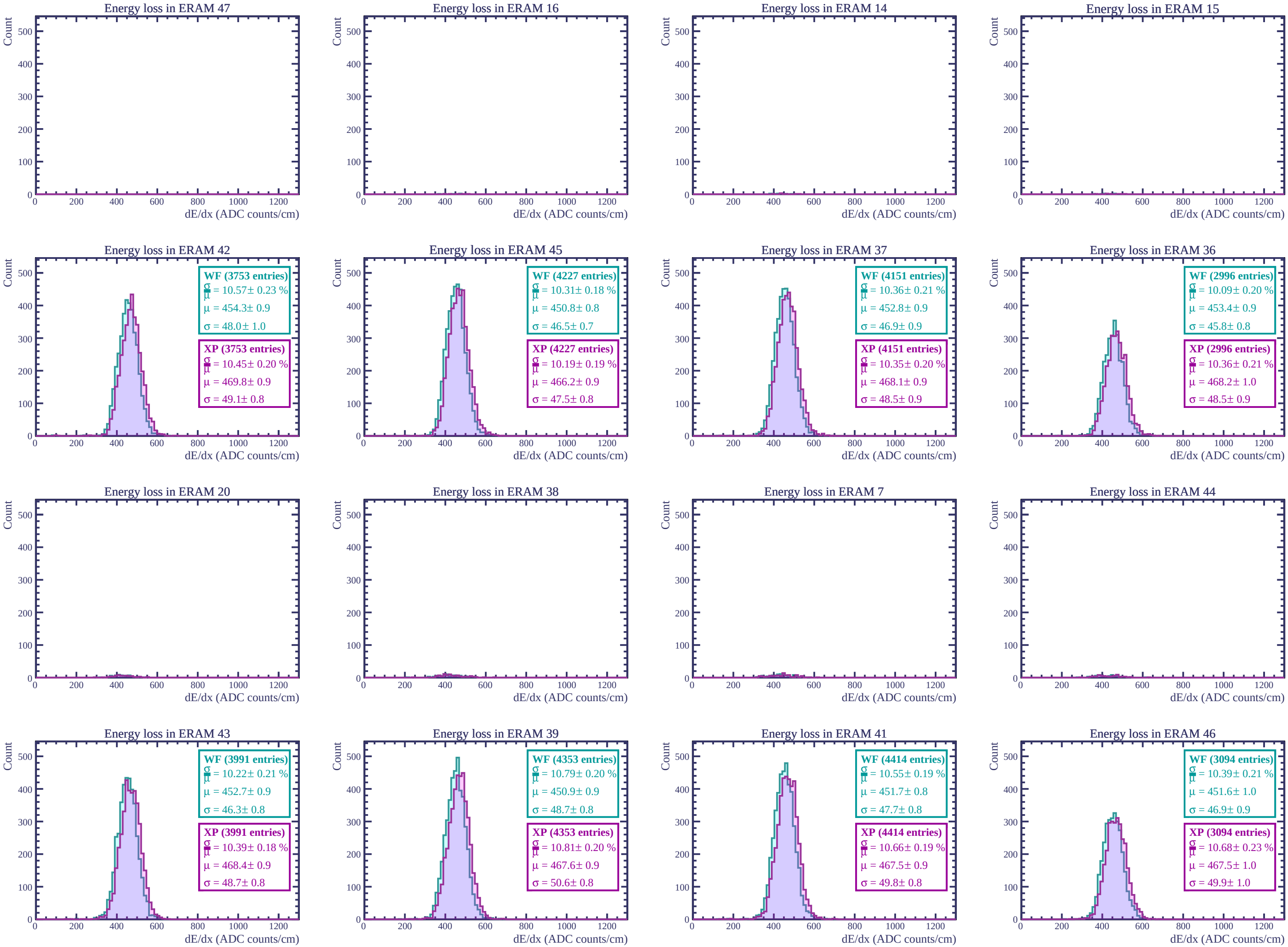


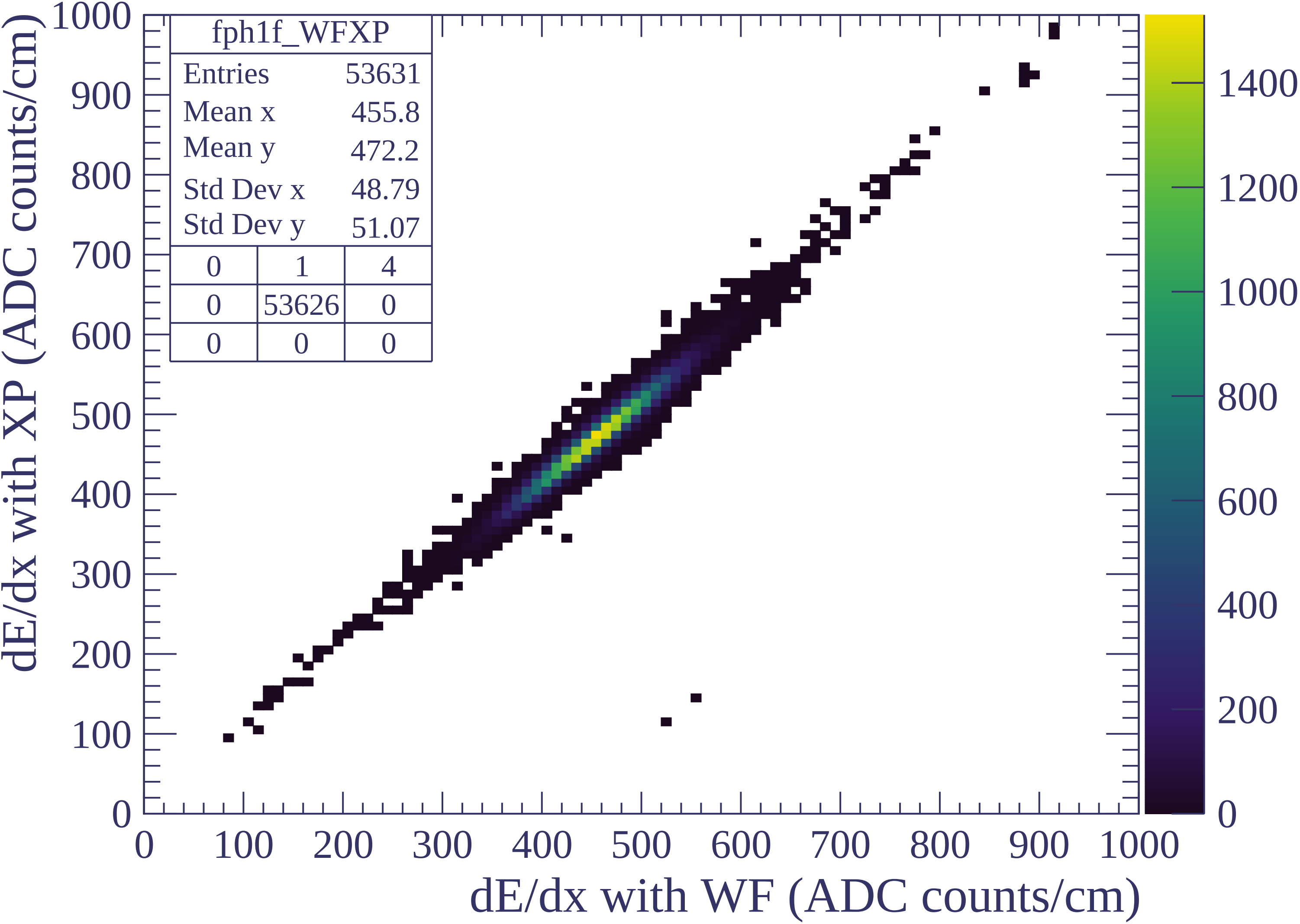
Count
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (53631 entries)
σ/μ = 10.42± 0.05 %
μ = 471.3± 0.2
σ = 49.1± 0.2
WF (53631 entries)
σ/μ = 10.45± 0.06 %
μ = 455.0± 0.2
σ = 47.6± 0.2



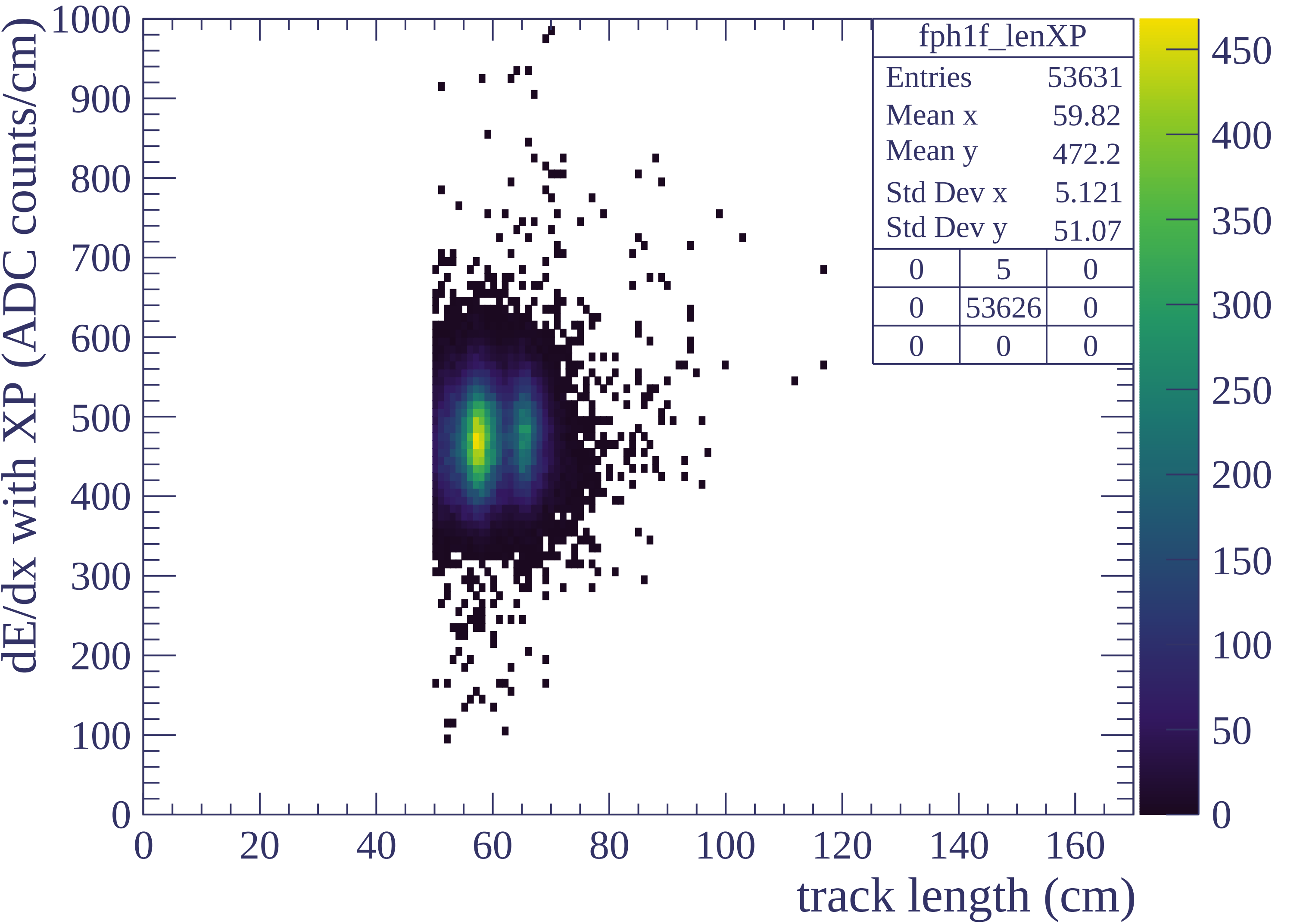
Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
WF (2311 entries)
σ/μ = 10.45± 0.26 %
μ = 461.7± 1.2
σ = 48.3± 1.1
XP (2311 entries)
σ/μ = 10.49± 0.26 %
μ = 477.6± 1.2
σ = 50.1± 1.1
Energy loss in ERAM 13
WF (3033 entries)
σ/μ = 9.86± 0.22 %
μ = 462.3± 1.0
σ = 45.6± 0.9
XP (3033 entries)
σ/μ = 10.26± 0.27 %
μ = 479.5± 1.1
σ = 49.2± 1.2
Energy loss in ERAM 9
WF (2964 entries)
σ/μ = 10.05± 0.27 %
μ = 460.3± 1.0
σ = 46.2± 1.1
XP (2964 entries)
σ/μ = 9.81± 0.21 %
μ = 477.1± 1.0
σ = 46.8± 0.9
Energy loss in ERAM 2
WF (2084 entries)
σ/μ = 10.45± 0.28 %
μ = 457.2± 1.2
σ = 47.8± 1.2
XP (2084 entries)
σ/μ = 10.23± 0.27 %
μ = 474.3± 1.2
σ = 48.5± 1.2
Energy loss in ERAM 26
Energy loss in ERAM 17
WF (174 entries)
σ/μ = 12.42± 1.39 %
μ = 438.3± 5.9
σ = 54.4± 5.4
XP (174 entries)
σ/μ = 13.16± 2.58 %
μ = 448.3± 7.2
σ = 59.0± 10.6
Energy loss in ERAM 23
WF (155 entries)
σ/μ = 11.60± 1.01 %
μ = 432.2± 4.7
σ = 50.2± 3.8
XP (155 entries)
σ/μ = 10.73± 0.96 %
μ = 454.4± 4.7
σ = 48.8± 3.8
Energy loss in ERAM 29
Energy loss in ERAM 1
WF (2526 entries)
σ/μ = 9.73± 0.27 %
μ = 459.5± 1.1
σ = 44.7± 1.2
XP (2526 entries)
σ/μ = 9.70± 0.29 %
μ = 475.8± 1.1
σ = 46.1± 1.2
Energy loss in ERAM 10
WF (3415 entries)
σ/μ = 10.38± 0.21 %
μ = 459.0± 0.9
σ = 47.7± 0.9
XP (3415 entries)
σ/μ = 10.40± 0.20 %
μ = 476.1± 1.0
σ = 49.5± 0.9
Energy loss in ERAM 11
WF (3143 entries)
σ/μ = 10.30± 0.22 %
μ = 457.3± 1.0
σ = 47.1± 0.9
XP (3143 entries)
σ/μ = 10.22± 0.21 %
μ = 474.3± 1.0
σ = 48.5± 0.9
Energy loss in ERAM 3
WF (2245 entries)
σ/μ = 10.38± 0.27 %
μ = 458.0± 1.2
σ = 47.5± 1.1
XP (2245 entries)
σ/μ = 10.54± 0.29 %
μ = 474.8± 1.3
σ = 50.1± 1.3
Count
dE/dx (ADC counts/cm)



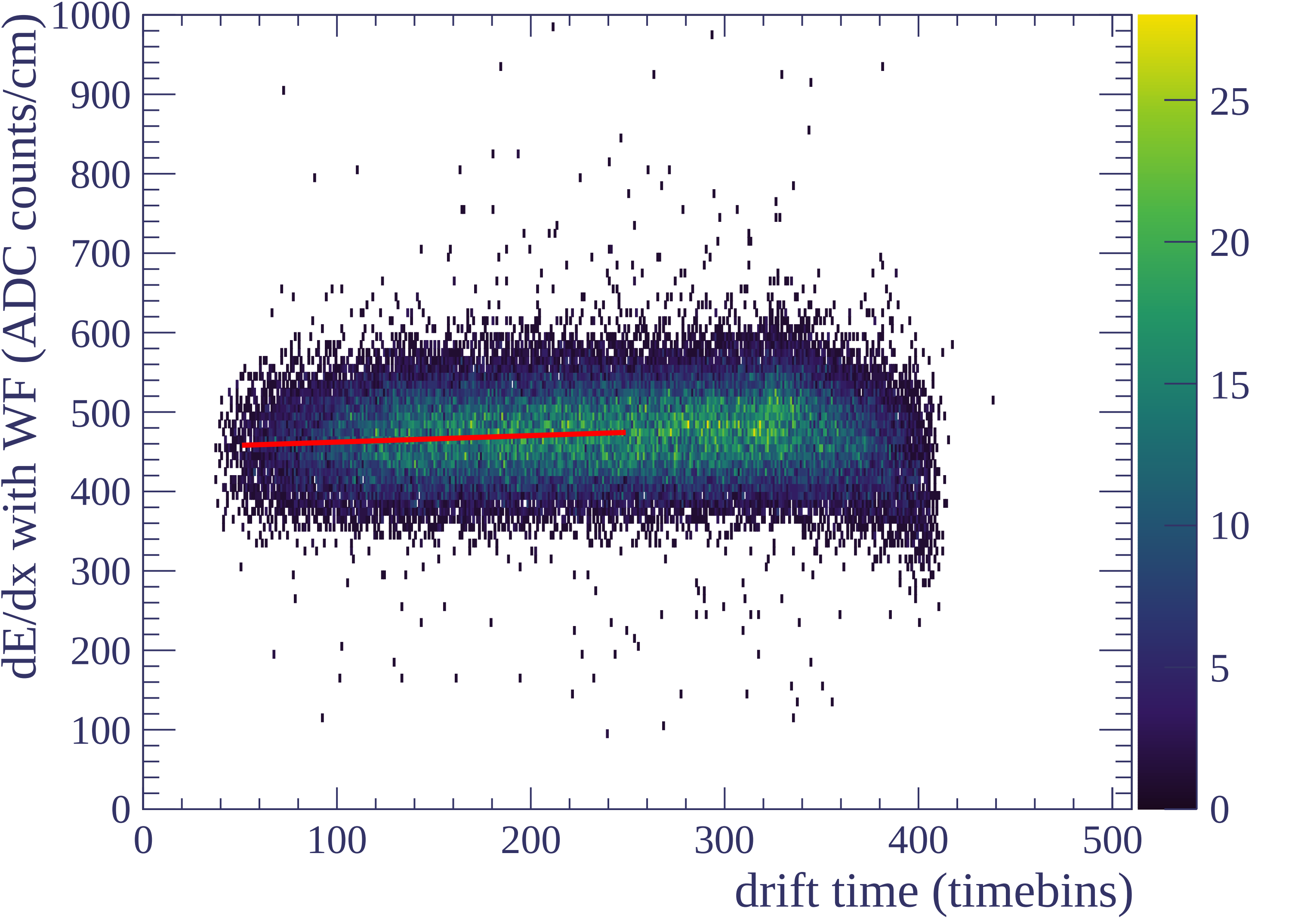
Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
Energy loss in ERAM 42
WF (3753 entries)
σ/μ = 10.57± 0.23 %
μ = 454.3± 0.9
σ = 48.0± 1.0
XP (3753 entries)
σ/μ = 10.45± 0.20 %
μ = 469.8± 0.9
σ = 49.1± 0.8
Energy loss in ERAM 45
WF (4227 entries)
σ/μ = 10.31± 0.18 %
μ = 450.8± 0.8
σ = 46.5± 0.7
XP (4227 entries)
σ/μ = 10.19± 0.19 %
μ = 466.2± 0.9
σ = 47.5± 0.8
Energy loss in ERAM 37
WF (4151 entries)
σ/μ = 10.36± 0.21 %
μ = 452.8± 0.9
σ = 46.9± 0.9
XP (4151 entries)
σ/μ = 10.35± 0.20 %
μ = 468.1± 0.9
σ = 48.5± 0.9
Energy loss in ERAM 36
WF (2996 entries)
σ/μ = 10.09± 0.20 %
μ = 453.4± 0.9
σ = 45.8± 0.8
XP (2996 entries)
σ/μ = 10.36± 0.21 %
μ = 468.2± 1.0
σ = 48.5± 0.9
Energy loss in ERAM 20
Energy loss in ERAM 38
Energy loss in ERAM 7
Energy loss in ERAM 44
Energy loss in ERAM 43
WF (3991 entries)
σ/μ = 10.22± 0.21 %
μ = 452.7± 0.9
σ = 46.3± 0.8
XP (3991 entries)
σ/μ = 10.39± 0.18 %
μ = 468.4± 0.9
σ = 48.7± 0.8
Energy loss in ERAM 39
WF (4353 entries)
σ/μ = 10.79± 0.20 %
μ = 450.9± 0.9
σ = 48.7± 0.8
XP (4353 entries)
σ/μ = 10.81± 0.20 %
μ = 467.6± 0.9
σ = 50.6± 0.8
Energy loss in ERAM 41
WF (4414 entries)
σ/μ = 10.55± 0.19 %
μ = 451.7± 0.8
σ = 47.7± 0.8
XP (4414 entries)
σ/μ = 10.66± 0.19 %
μ = 467.5± 0.9
σ = 49.8± 0.8
Energy loss in ERAM 46
WF (3094 entries)
σ/μ = 10.39± 0.21 %
μ = 451.6± 1.0
σ = 46.9± 0.9
XP (3094 entries)
σ/μ = 10.68± 0.23 %
μ = 467.5± 1.0
σ = 49.9± 1.0
Count
dE/dx (ADC counts/cm)



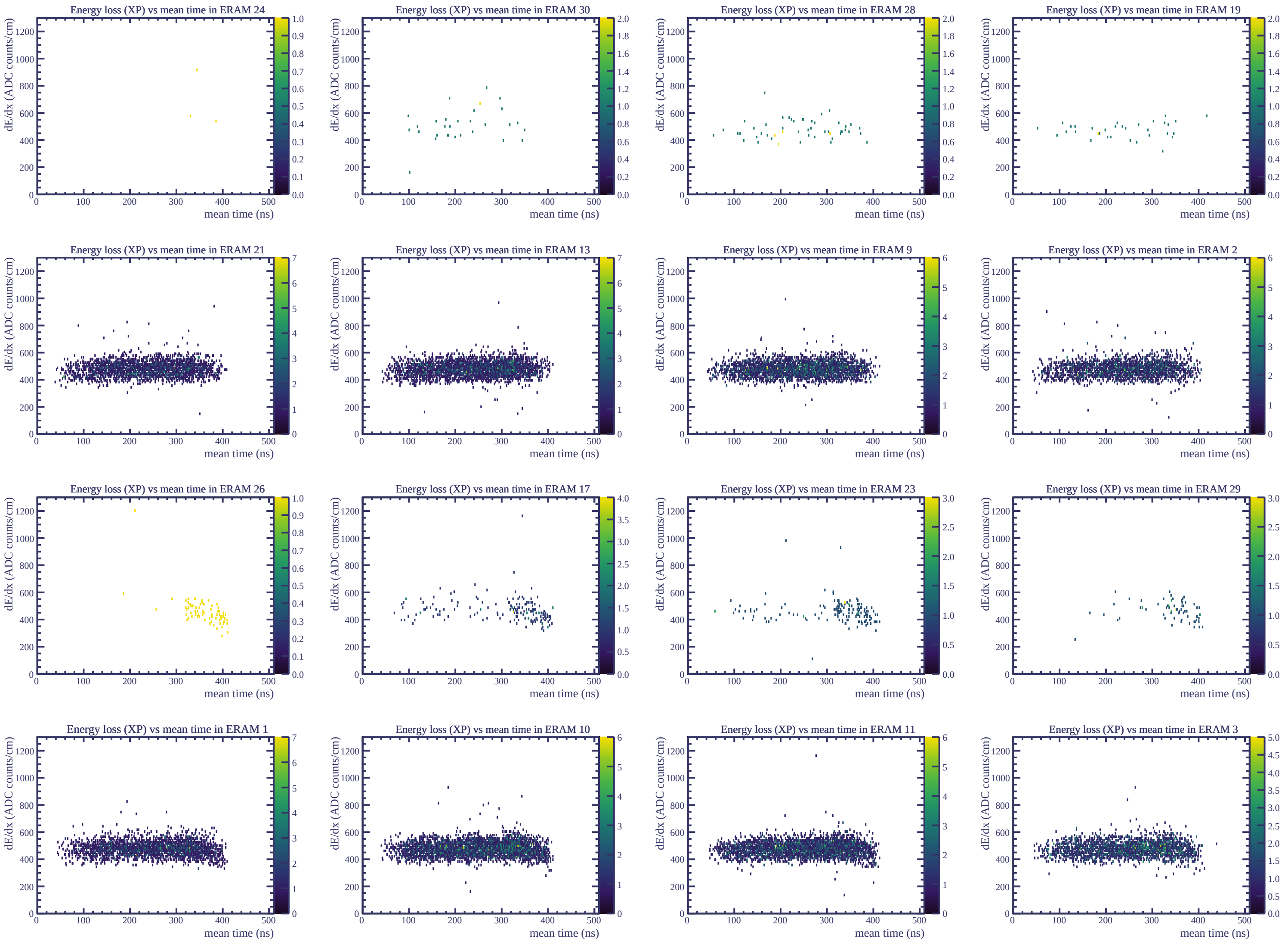
fph1f_WFXP
Entries 53631
Mean x 455.8
Mean y 472.2
Std Dev x 48.79
Std Dev y 51.07
0 1 4
0 53626 0
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0 100 200 300 400 500 600 700 800 900 1000
0 200 400 600 800 1000 1200 1400



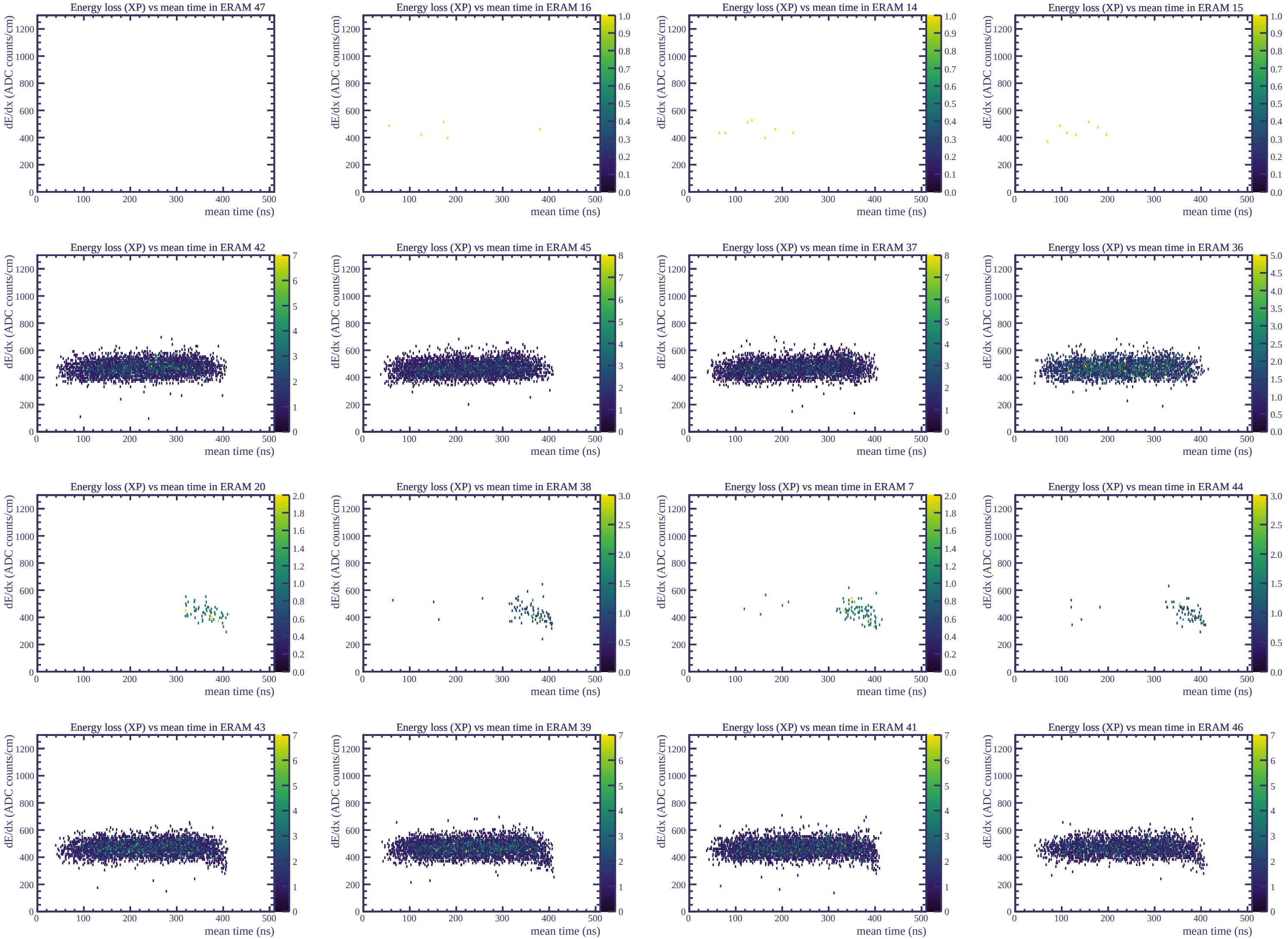
fph1f_lenXP
Entries 53631
Mean x 59.82
Mean y 472.2
Std Dev x 5.121
Std Dev y 51.07
0 5 0
0 53626 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
50
100
150
200
250
300
350
400
450



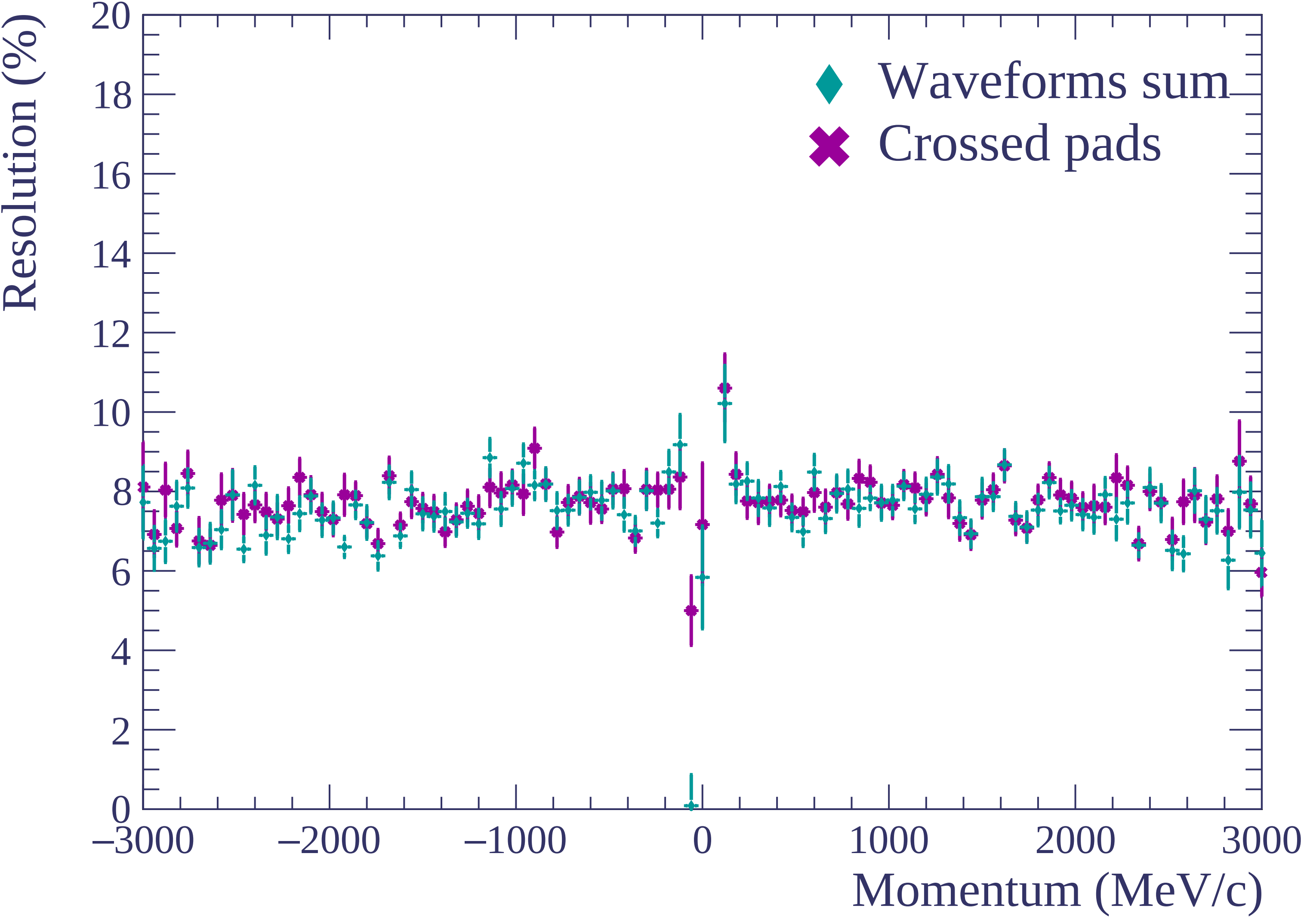
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
100
200
300
400
500
drift time (timebins)
25
20
15
10
5
0



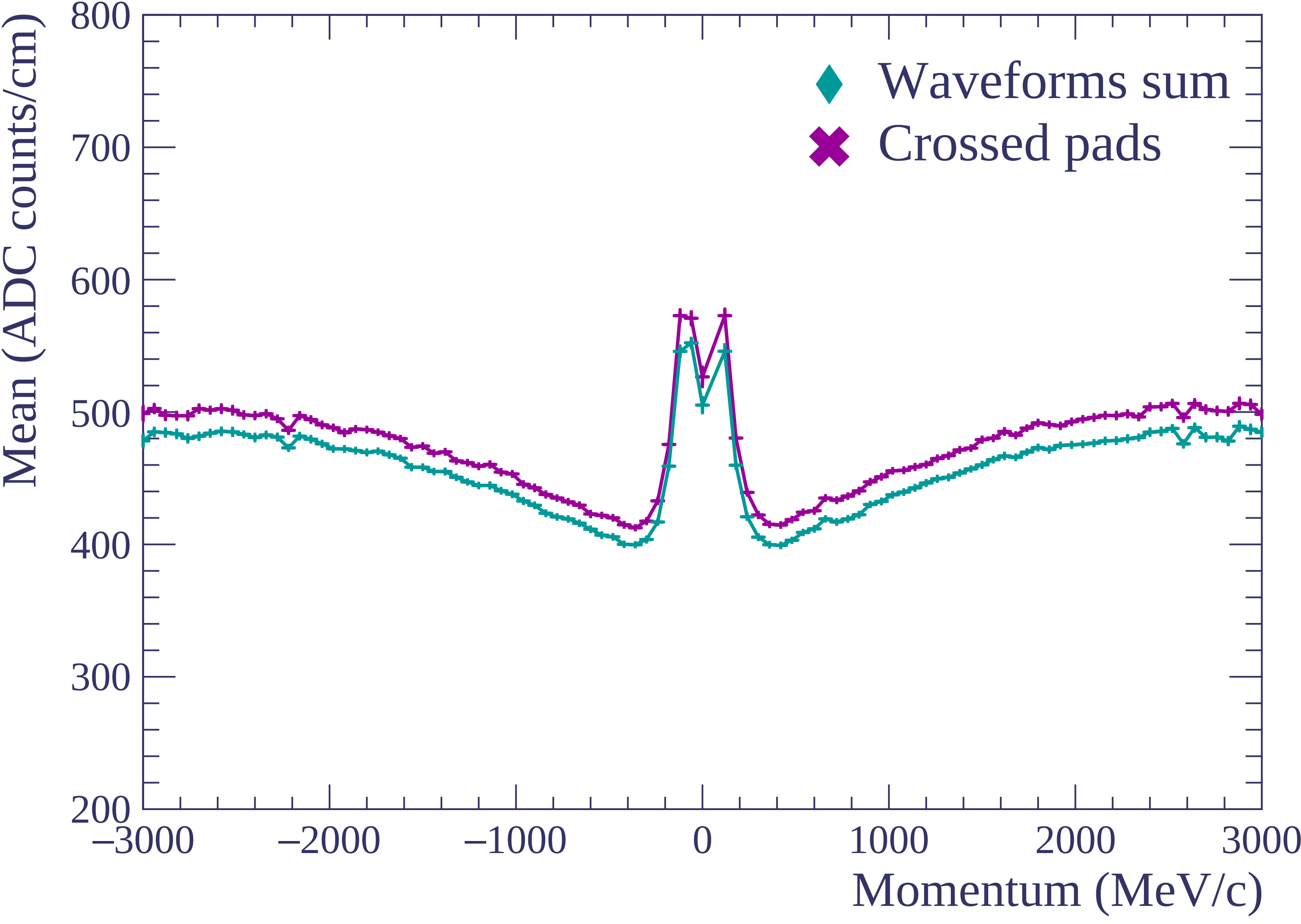
Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)



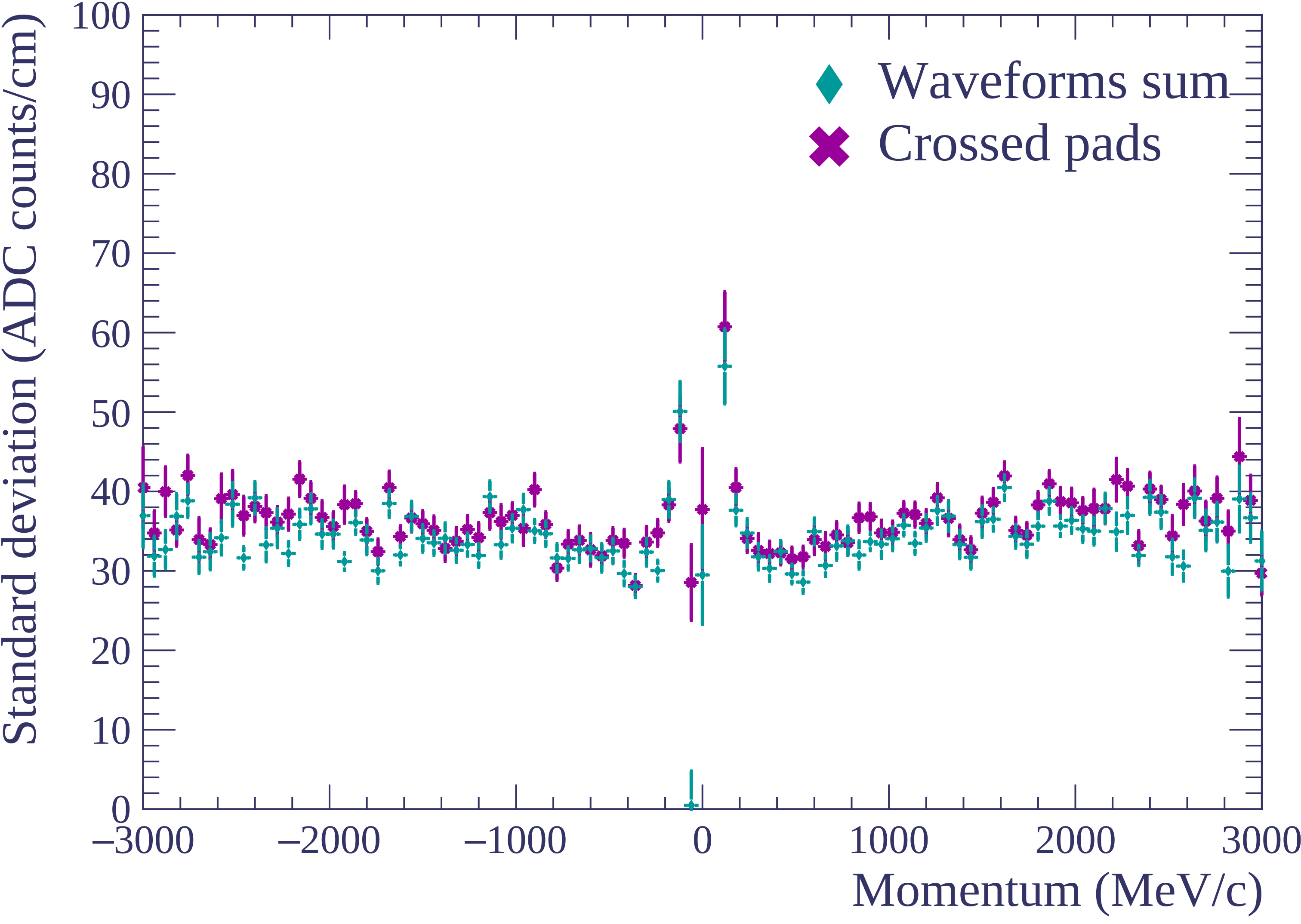
Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)



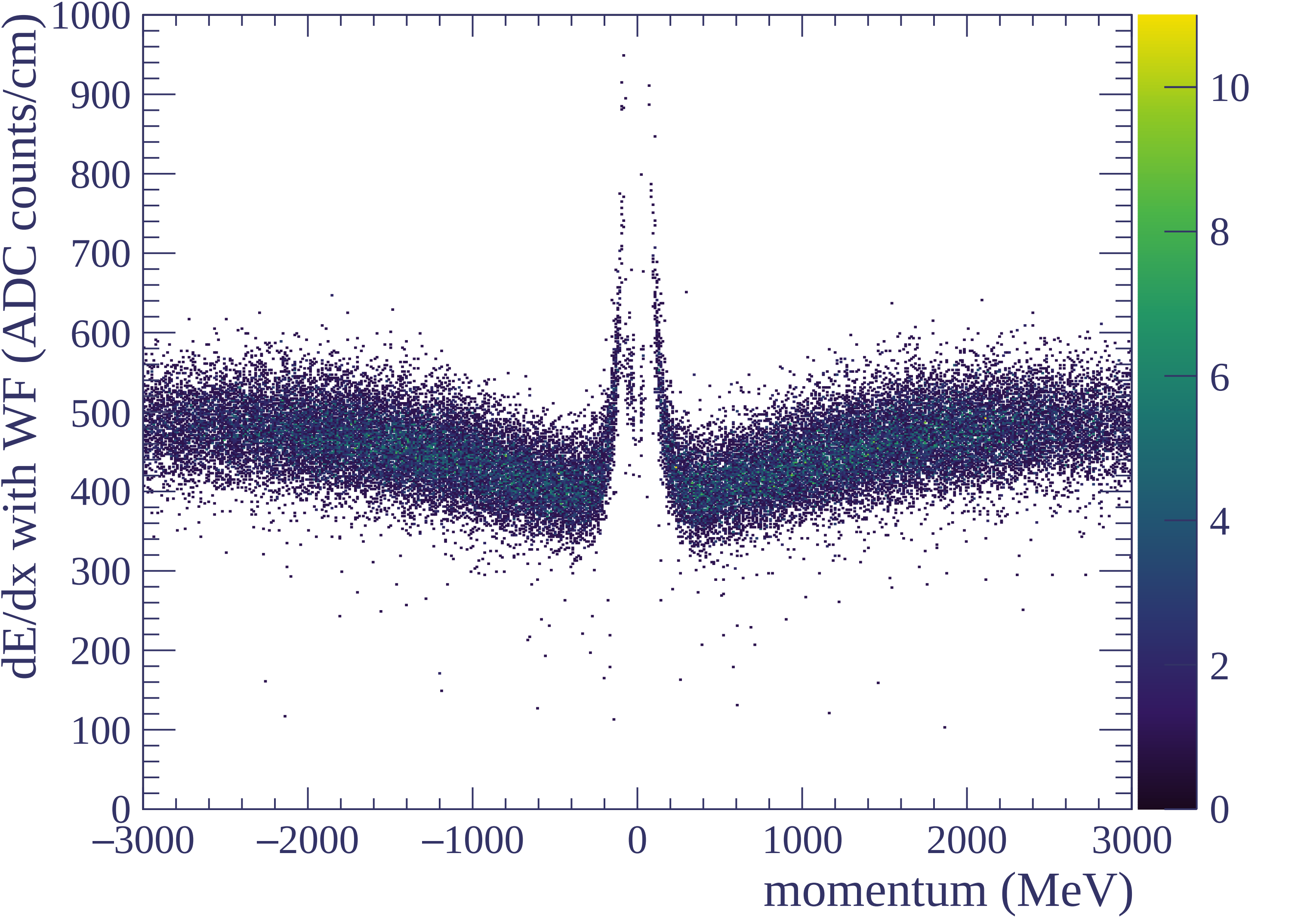
Resolution (%)
Waveforms sum
Crossed pads
Momentum (MeV/c)
0
2
4
6
8
10
12
14
16
18
20
−3000
−2000
−1000
0
1000
2000
3000



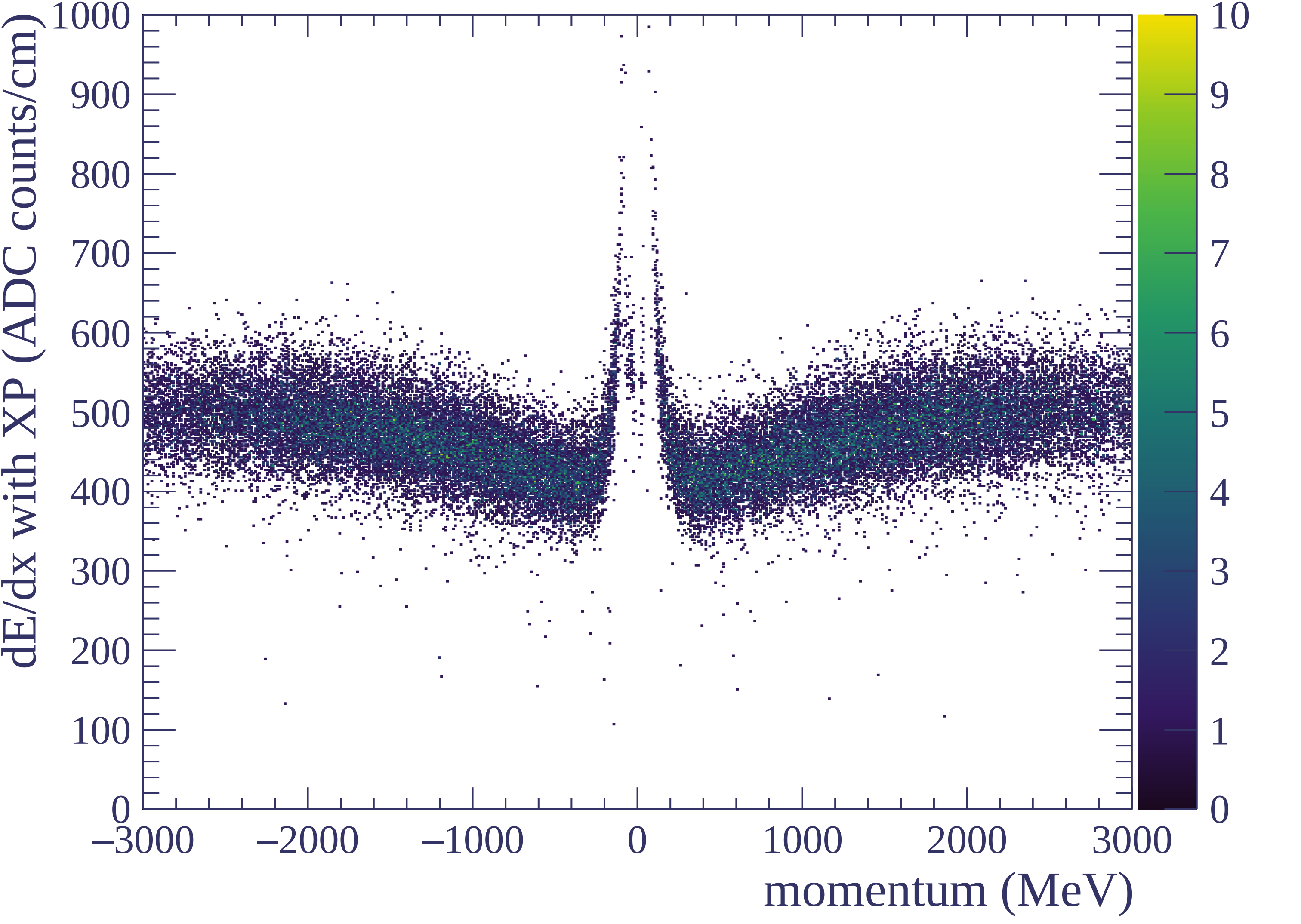
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads



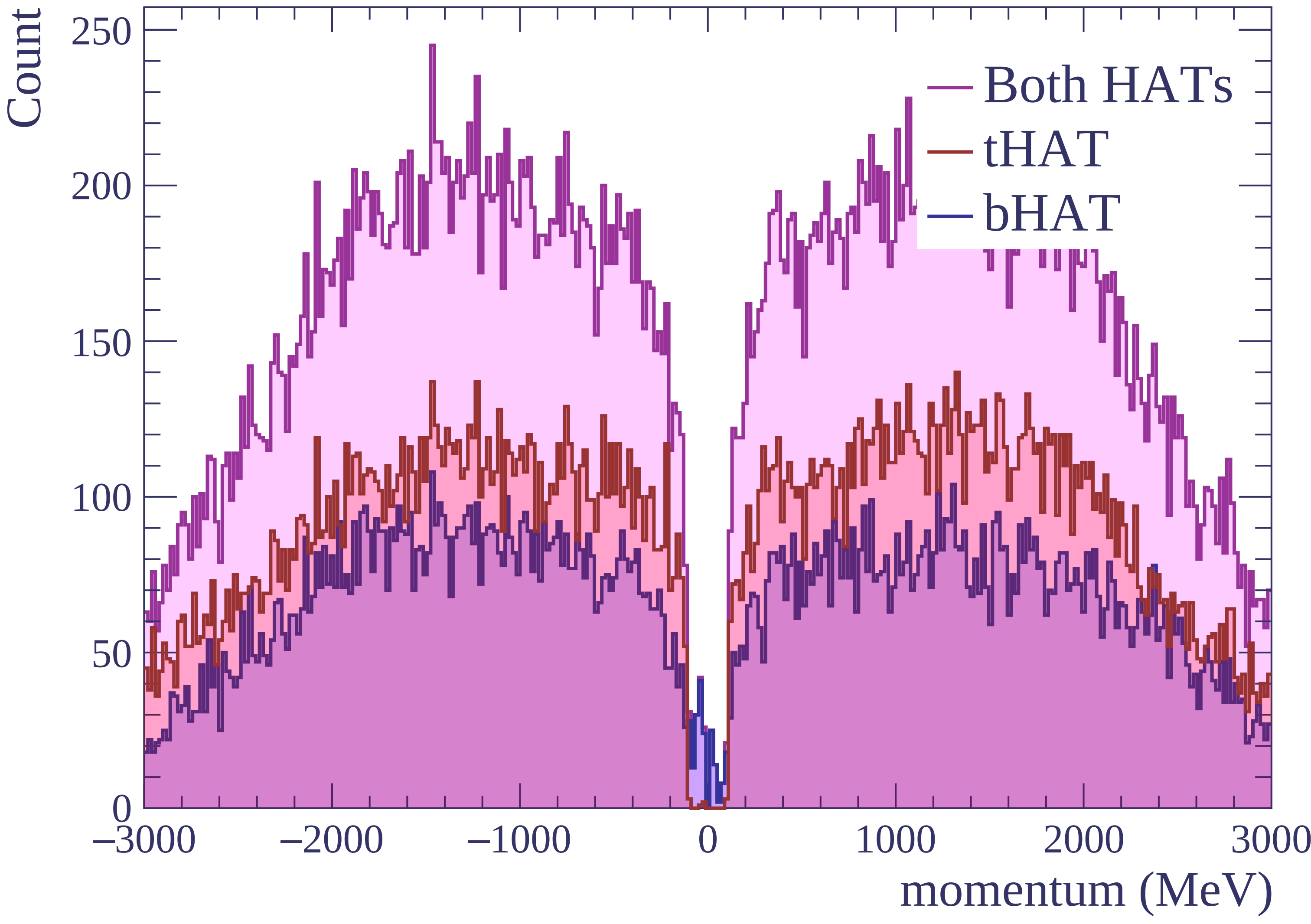
Waveforms sum
Crossed pads
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
100
90
80
70
60
50
40
30
20
10
0
−3000
−2000
−1000
0
1000
2000
3000



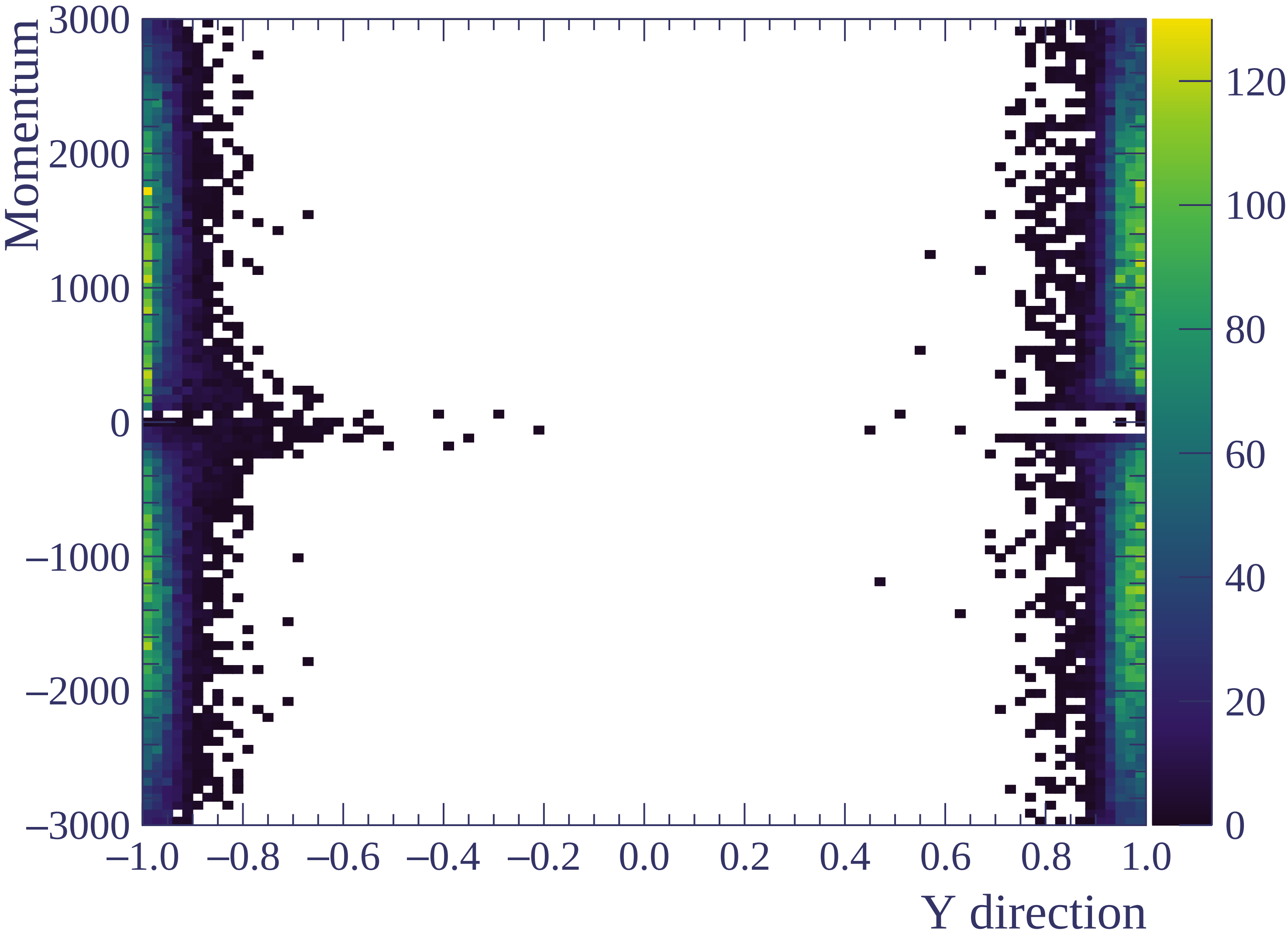
dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
10
8
6
4
2
0



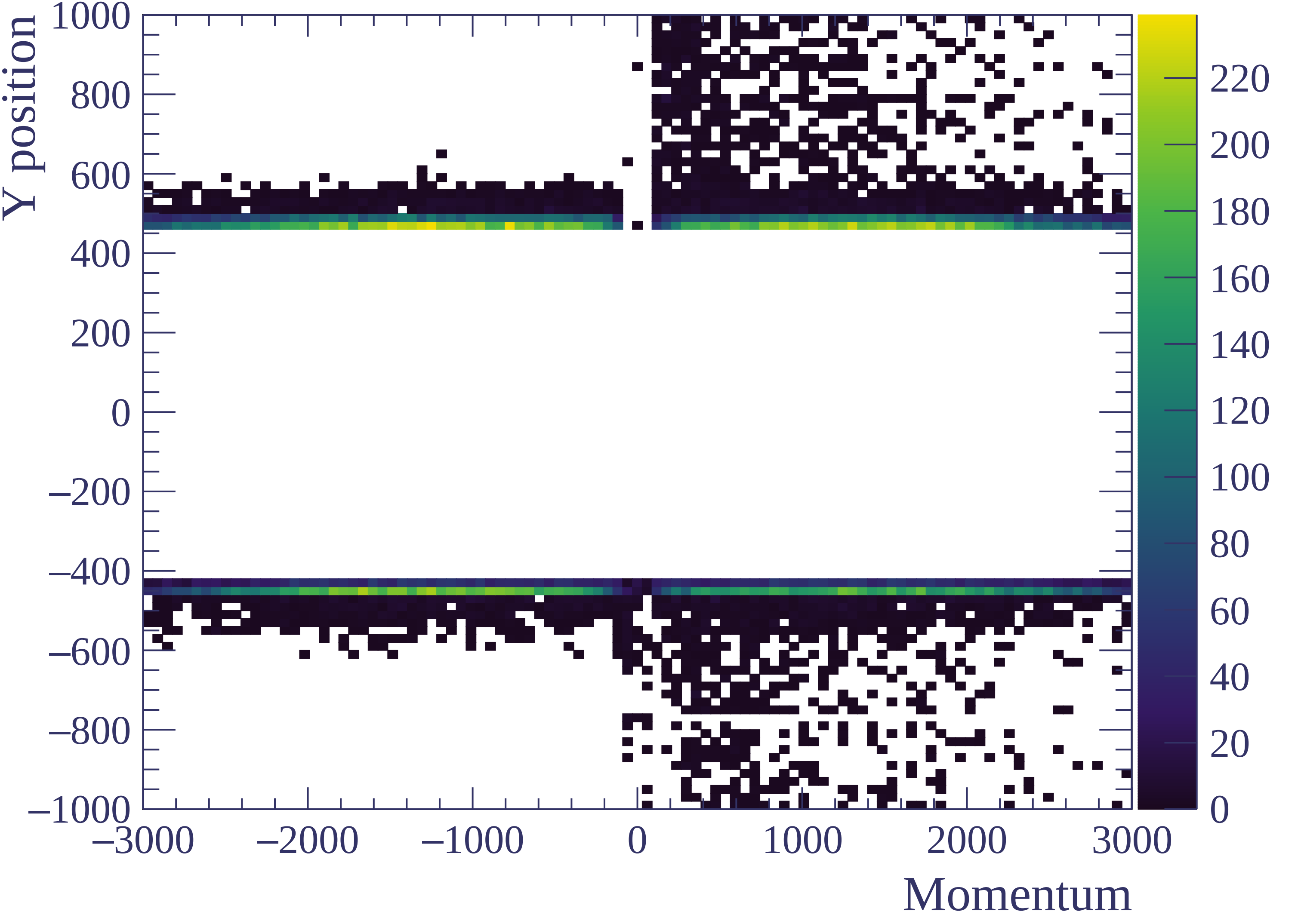
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
10
9
8
7
6
5
4
3
2
1
0

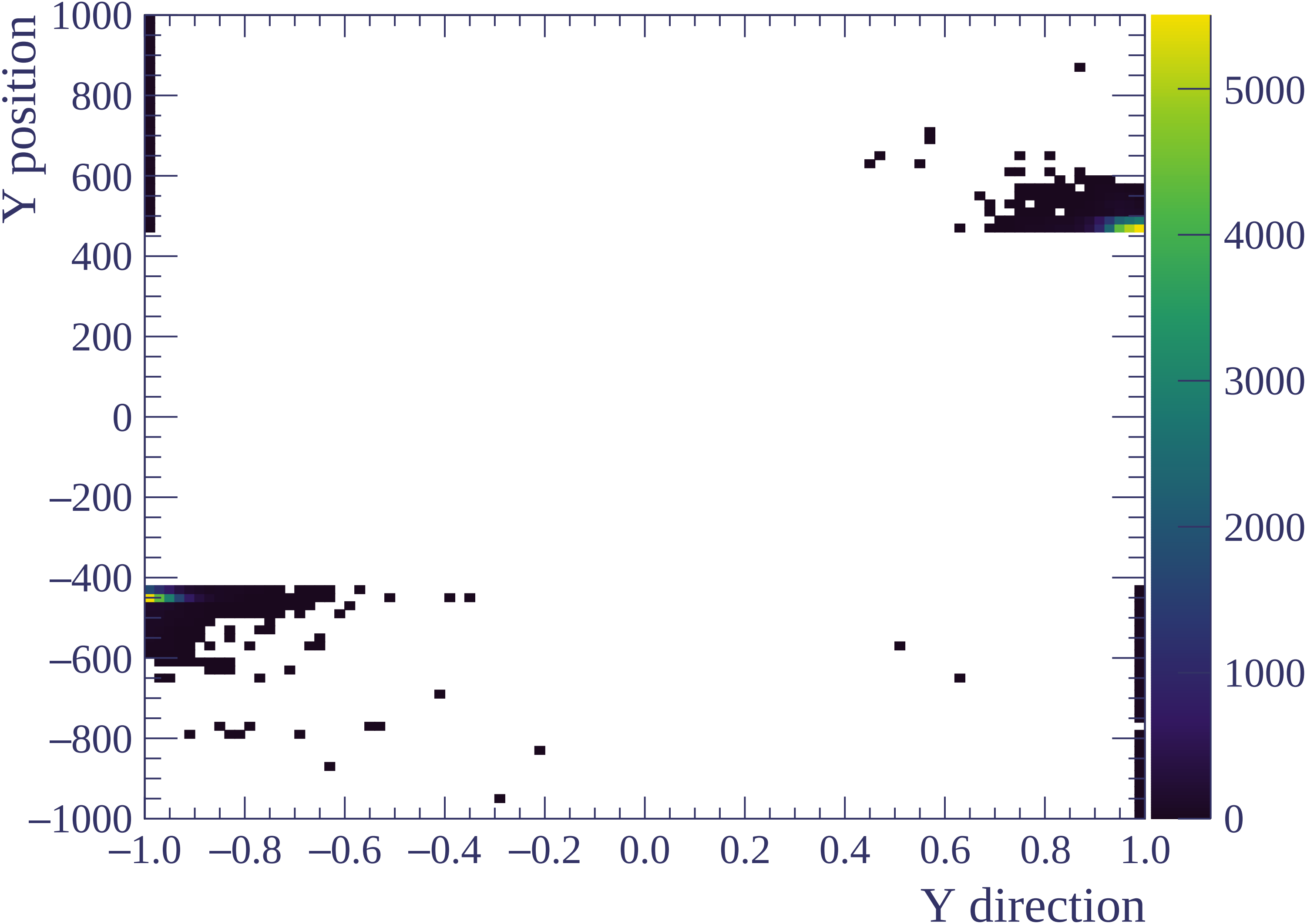


Count
250
200
150
100
50
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT

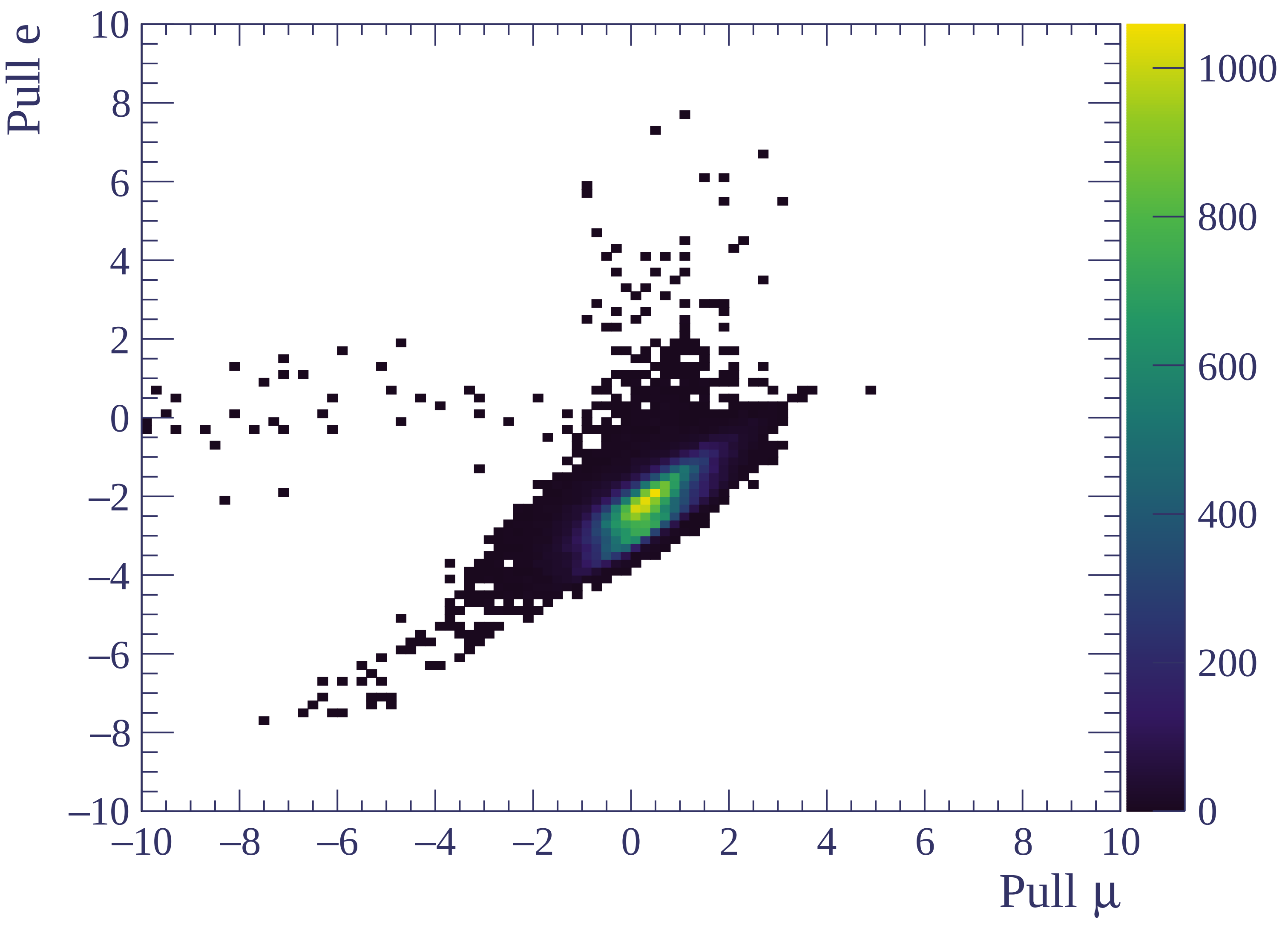


Momentum
3000
2000
1000
0
−1000
−2000
−3000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
120
100
80
60
40
20
0

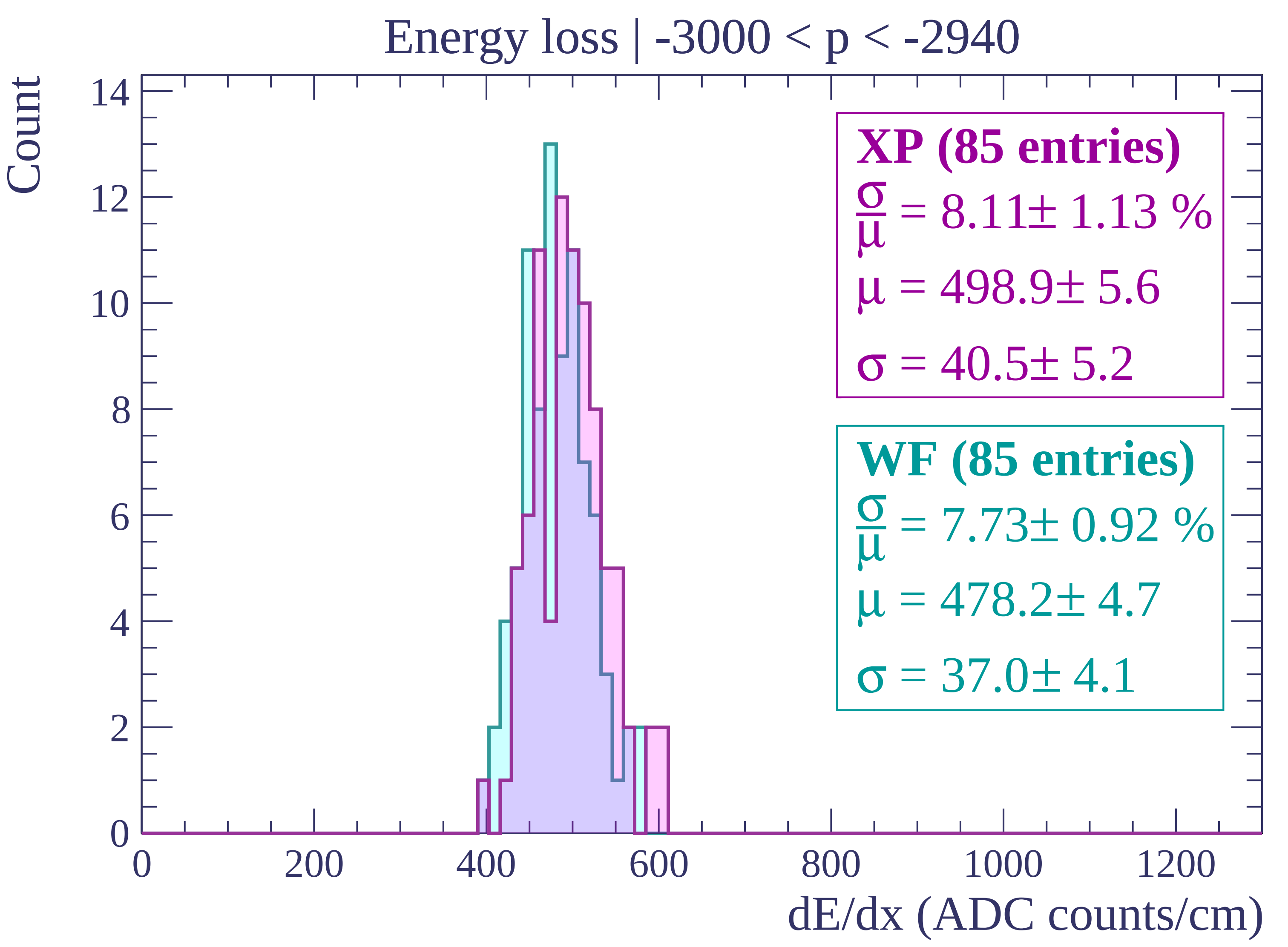

Y position
Momentum
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−3000
−2000
−1000
0
1000
2000
3000
220
200
180
160
140
120
100
80
60
40
20
0



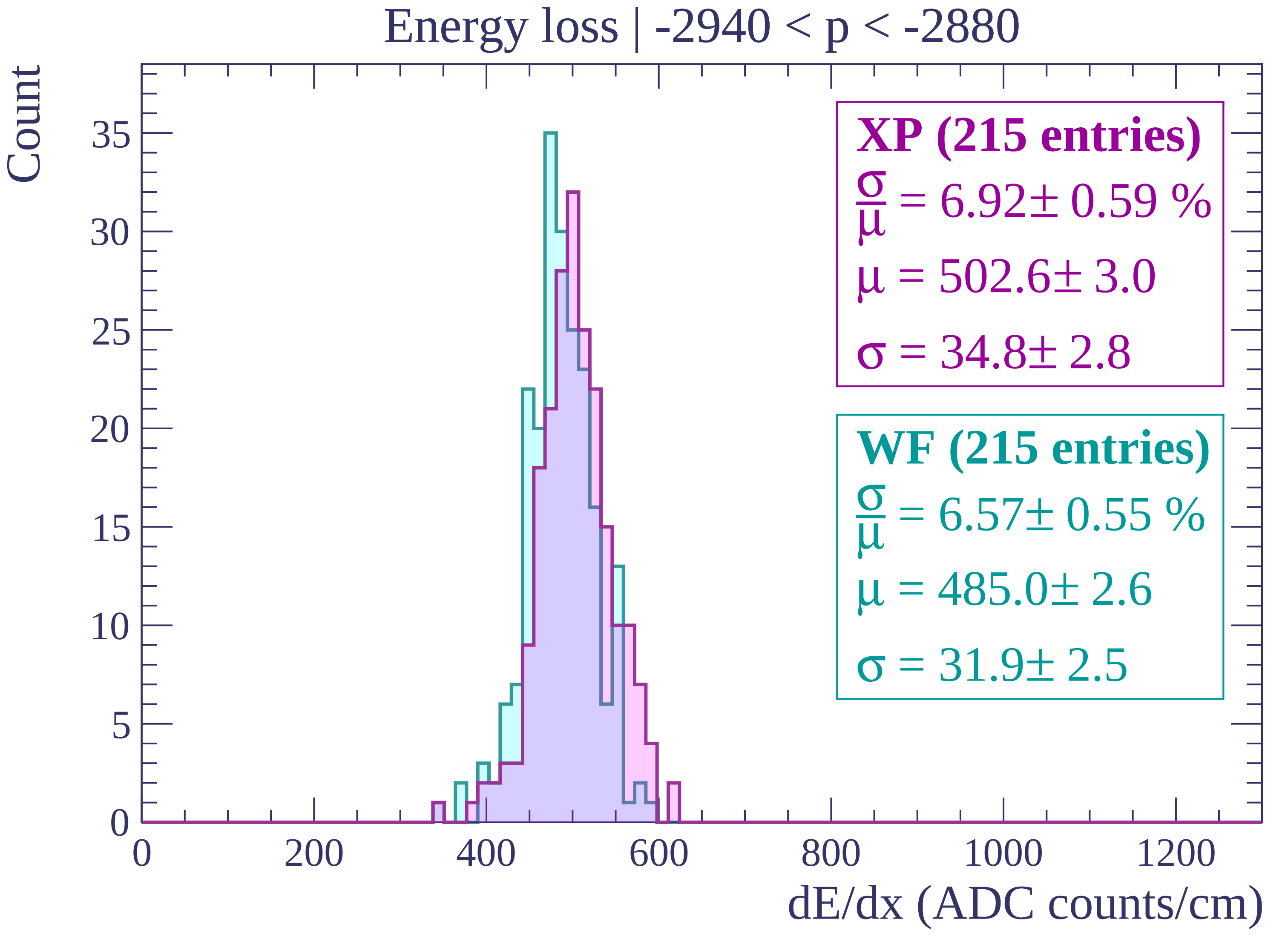
Y position
1000
800
600
400
200
0
−200
−400
−600
−800
−1000
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction
5000
4000
3000
2000
1000
0



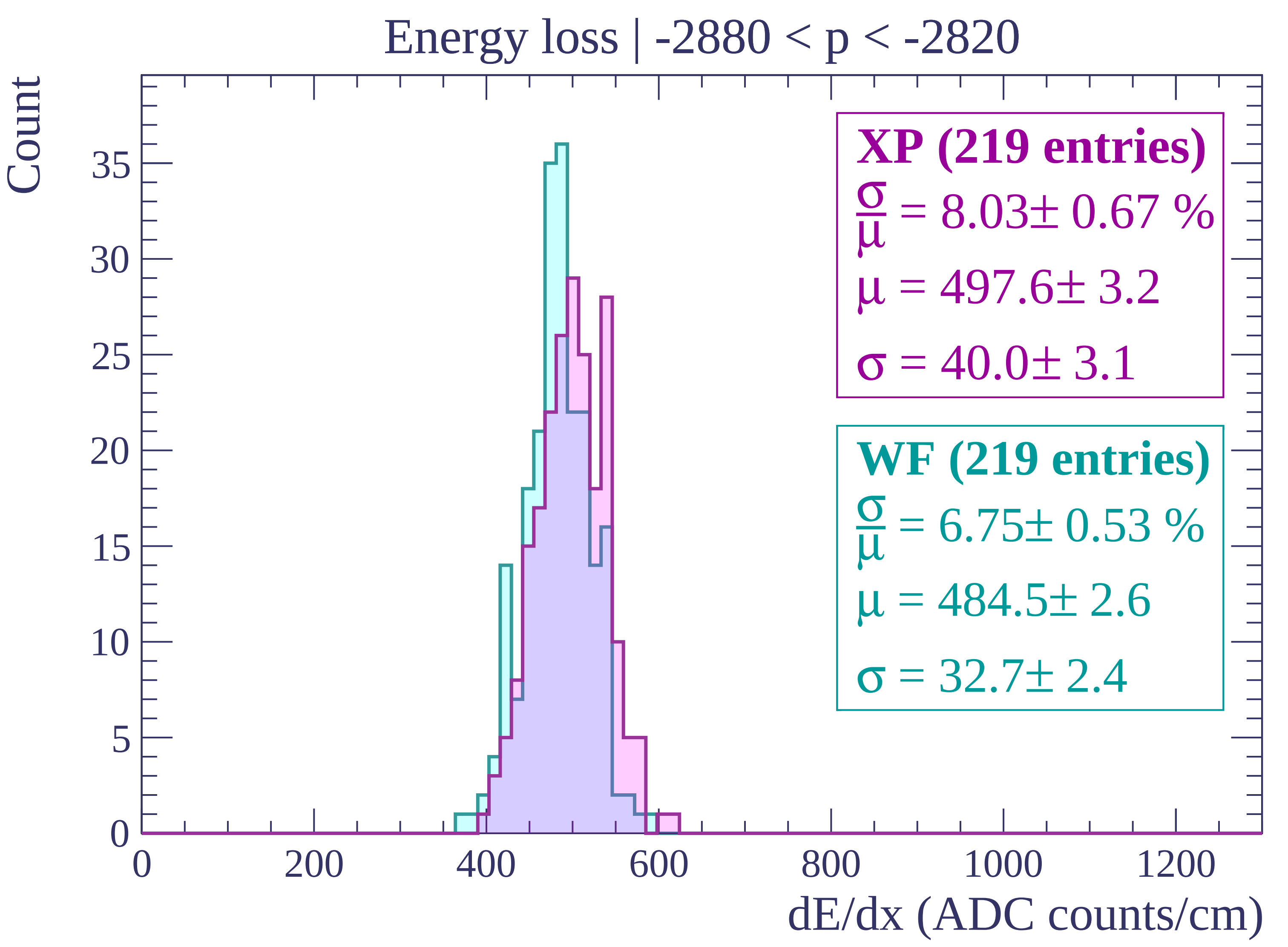
Pull e
Pull μ
10
8
6
4
2
0
−2
−4
−6
−8
−10
1000
800
600
400
200



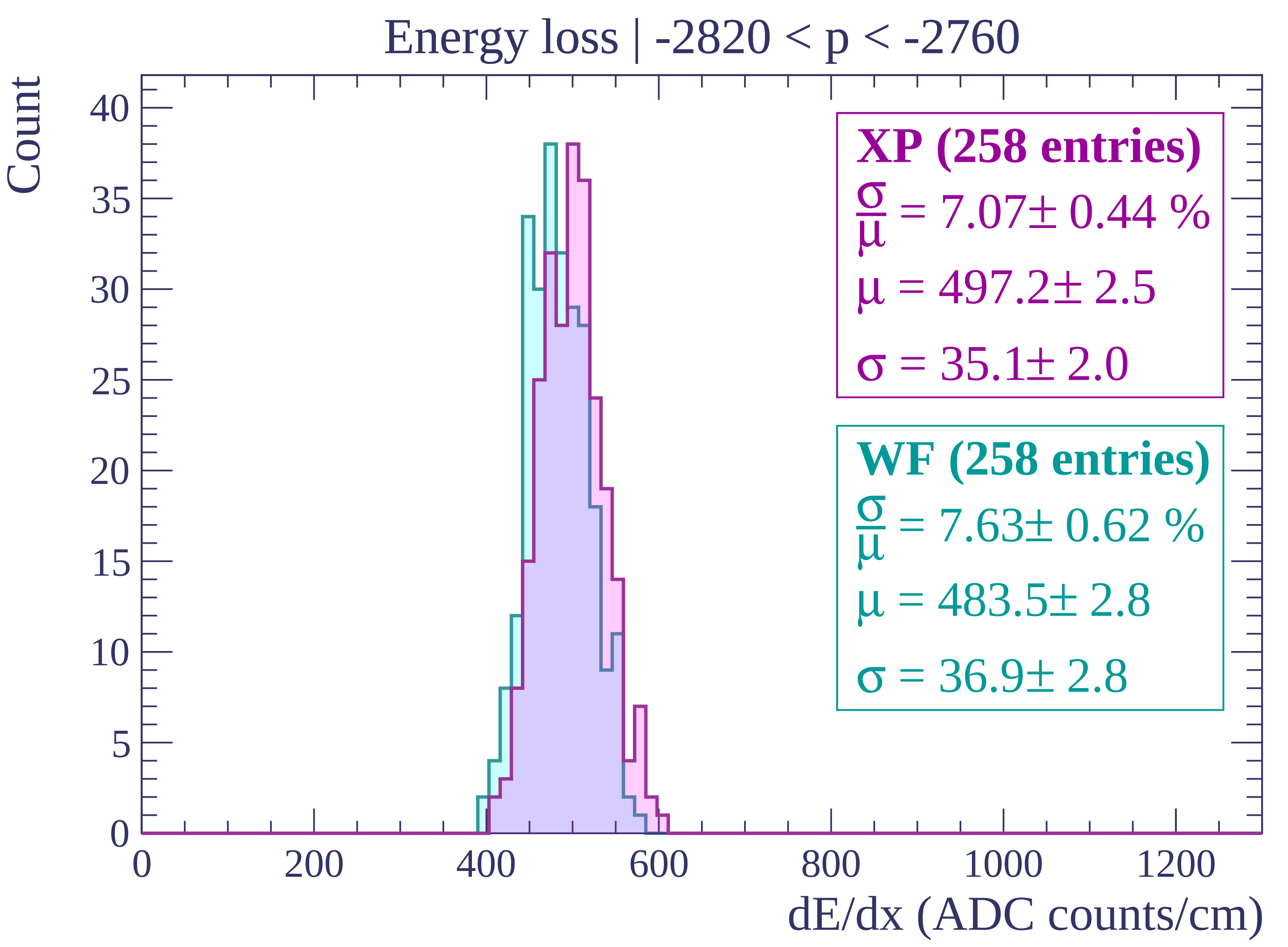
Energy loss | -3000 < p < -2940
Count
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (85 entries)
σ/μ = 8.11± 1.13 %
μ = 498.9± 5.6
σ = 40.5± 5.2
WF (85 entries)
σ/μ = 7.73± 0.92 %
μ = 478.2± 4.7
σ = 37.0± 4.1



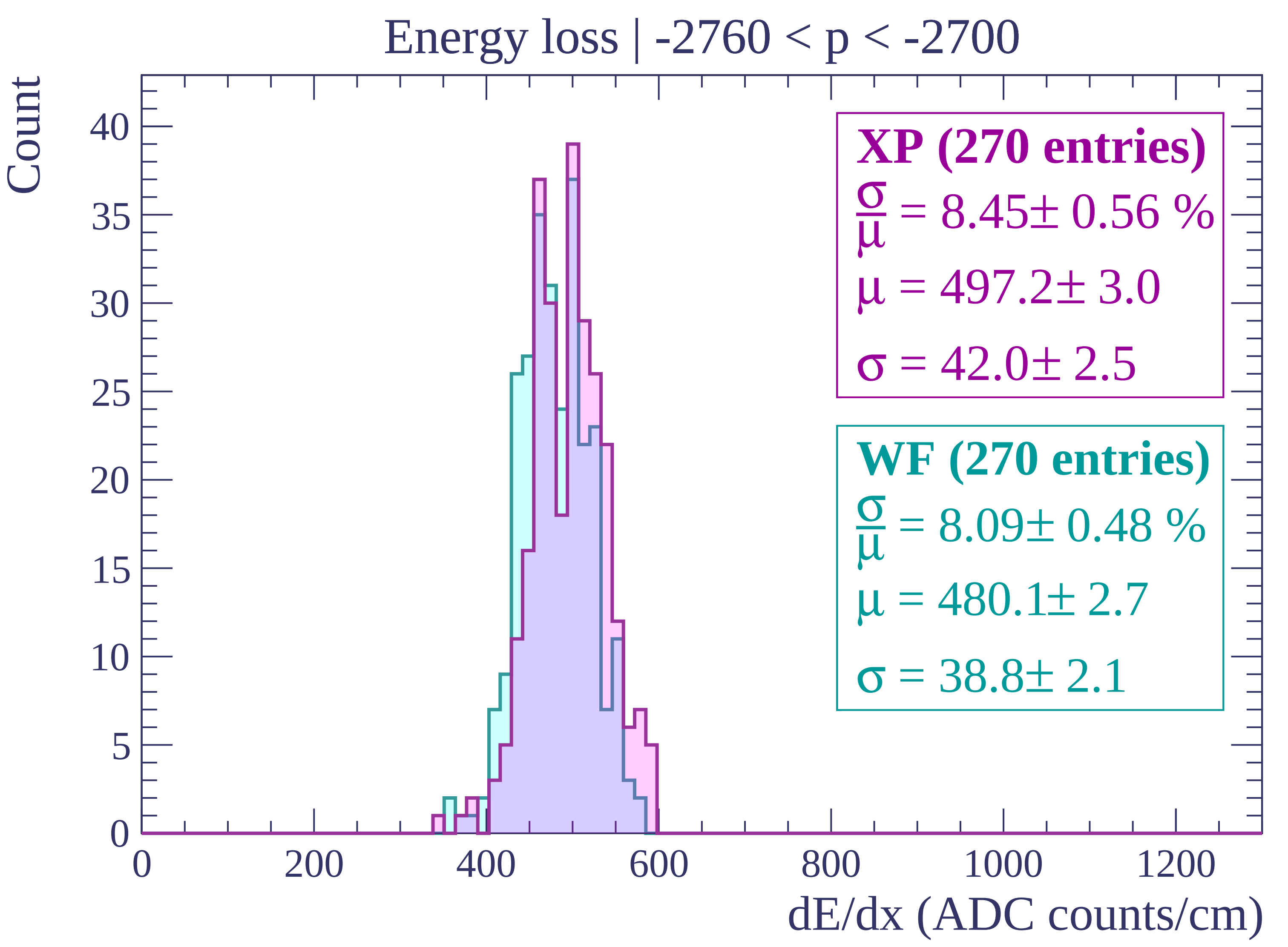
Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP (215 entries)
σ/μ = 6.92± 0.59 %
μ = 502.6± 3.0
σ = 34.8± 2.8
WF (215 entries)
σ/μ = 6.57± 0.55 %
μ = 485.0± 2.6
σ = 31.9± 2.5

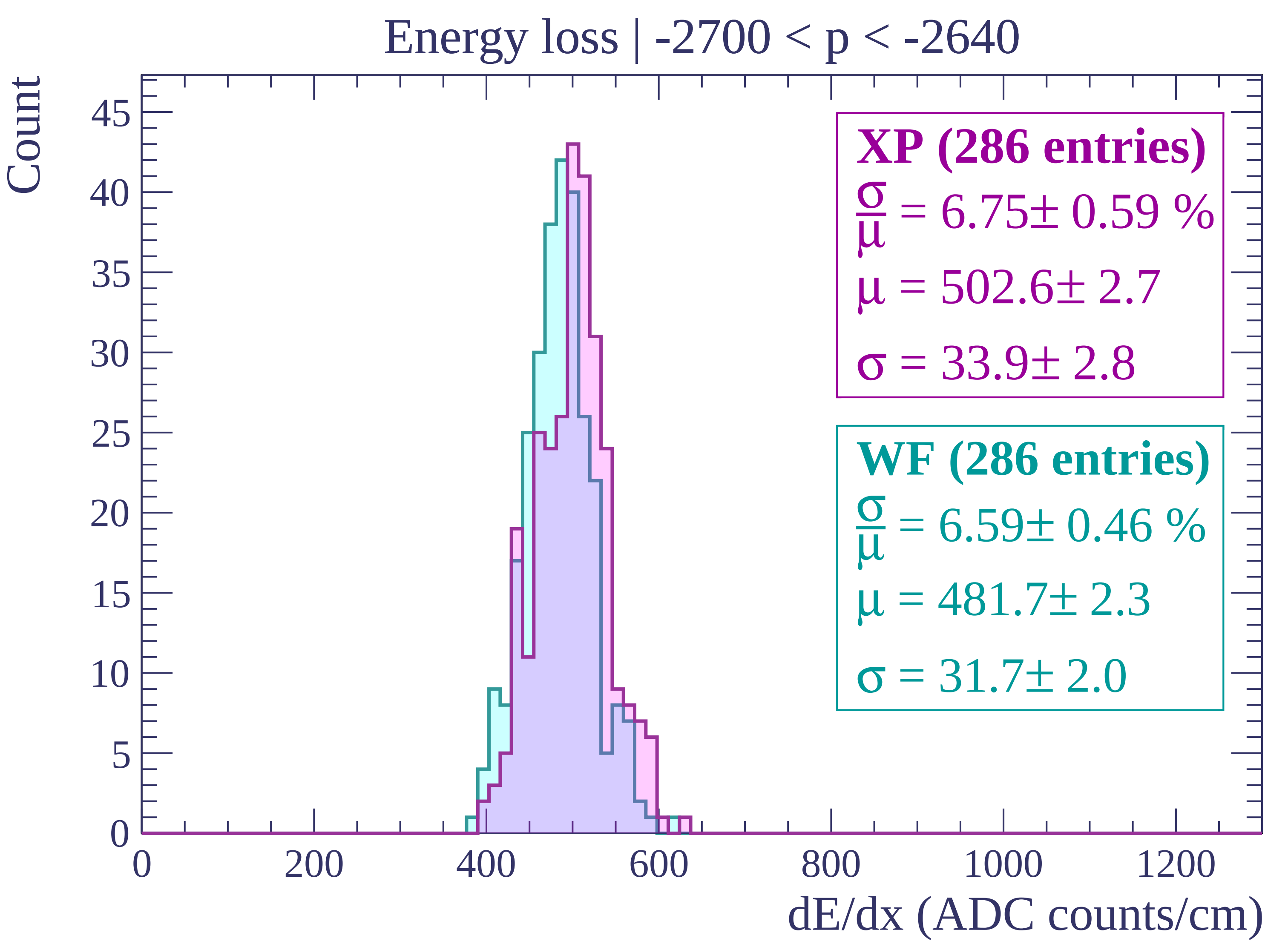


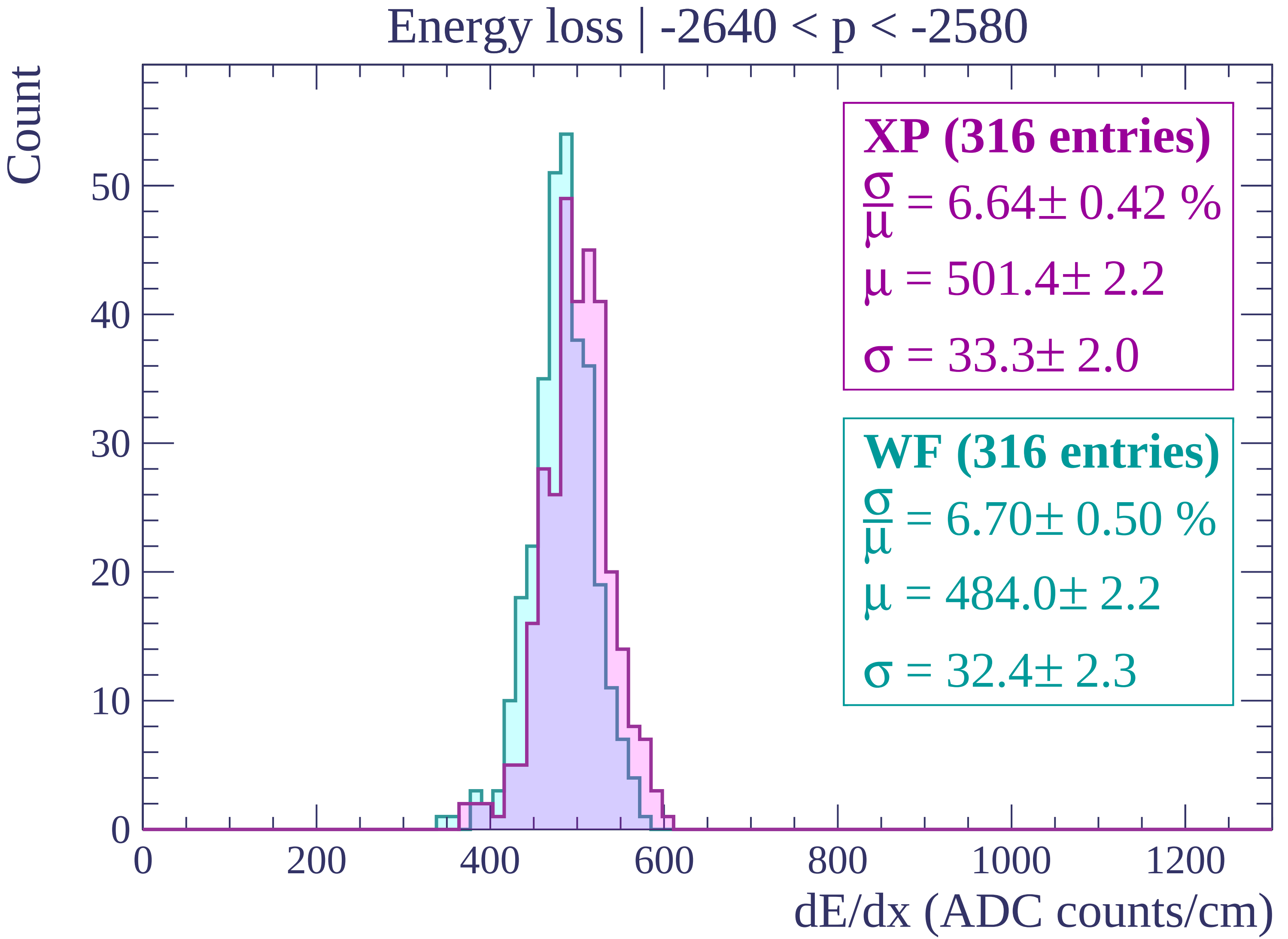
Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP (219 entries)
$\frac{\sigma}{\mu}$ = 8.03± 0.67 %
μ = 497.6± 3.2
σ = 40.0± 3.1
WF (219 entries)
$\frac{\sigma}{\mu}$ = 6.75± 0.53 %
μ = 484.5± 2.6
σ = 32.7± 2.4



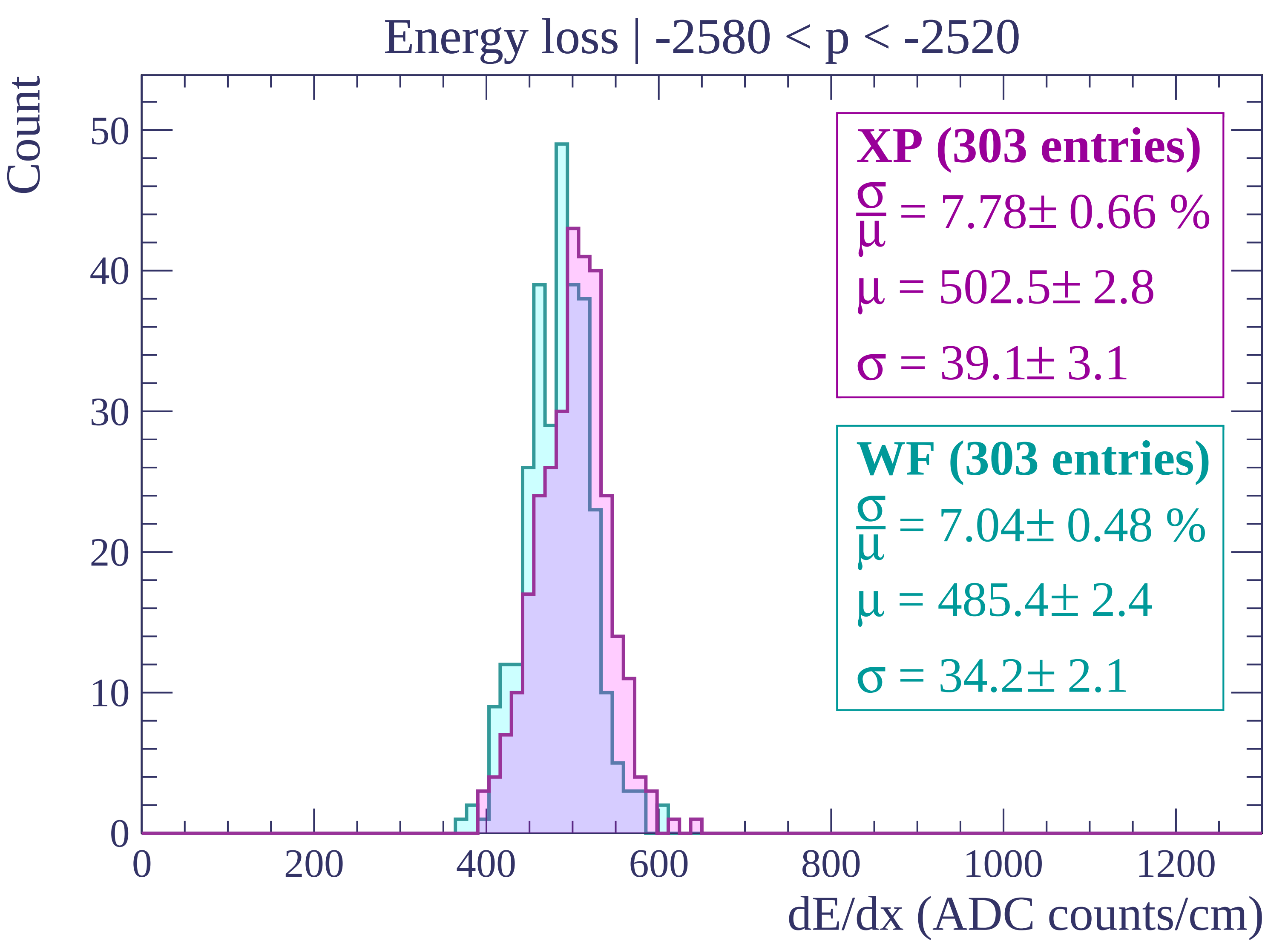
Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP (258 entries)
σ/μ = 7.07± 0.44 %
μ = 497.2± 2.5
σ = 35.1± 2.0
WF (258 entries)
σ/μ = 7.63± 0.62 %
μ = 483.5± 2.8
σ = 36.9± 2.8


Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP (270 entries)
σ/μ = 8.45± 0.56 %
μ = 497.2± 3.0
σ = 42.0± 2.5
WF (270 entries)
σ/μ = 8.09± 0.48 %
μ = 480.1± 2.7
σ = 38.8± 2.1

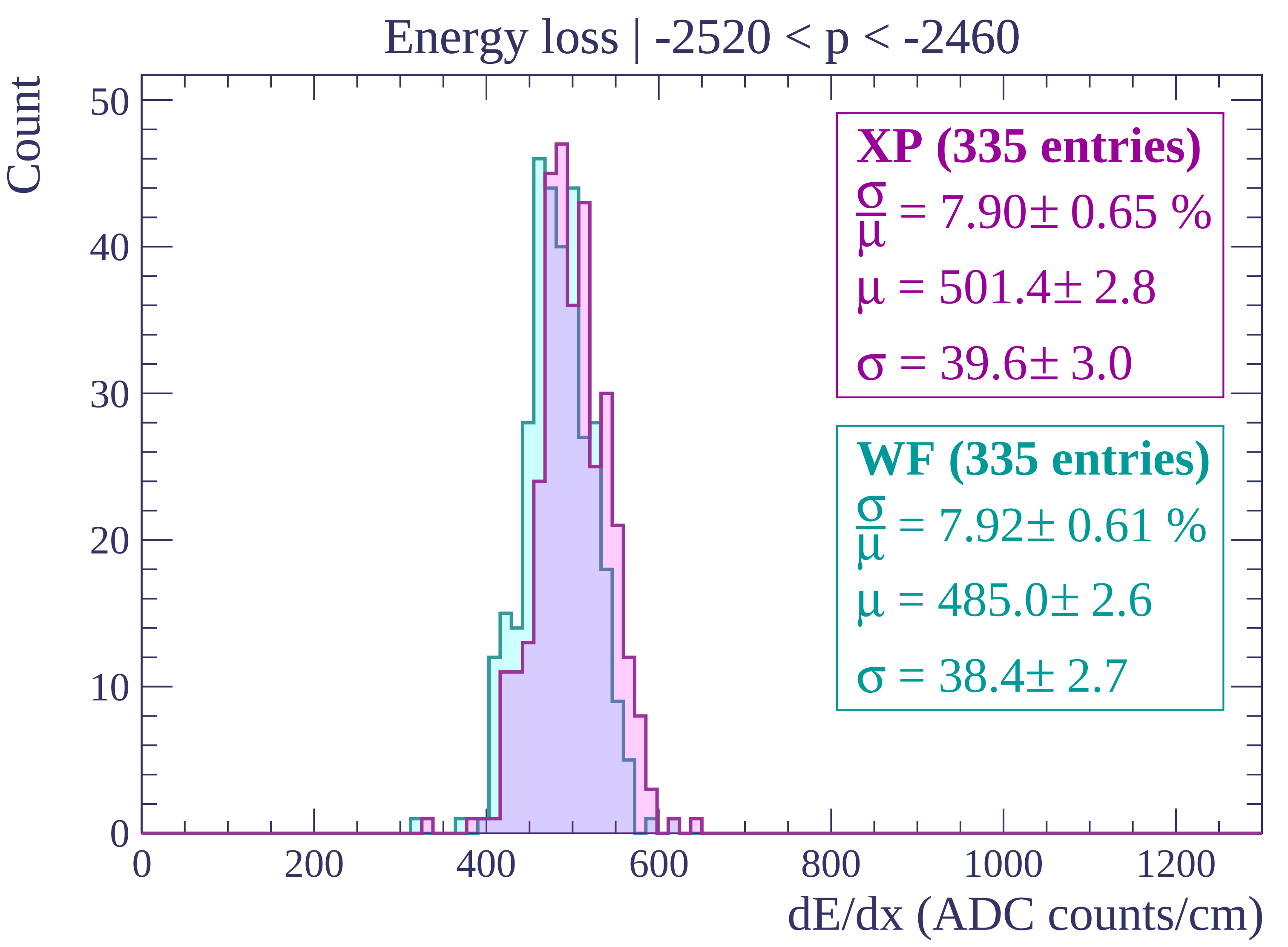

Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45
XP (286 entries)
σ/μ = 6.75± 0.59 %
μ = 502.6± 2.7
σ = 33.9± 2.8
WF (286 entries)
σ/μ = 6.59± 0.46 %
μ = 481.7± 2.3
σ = 31.7± 2.0



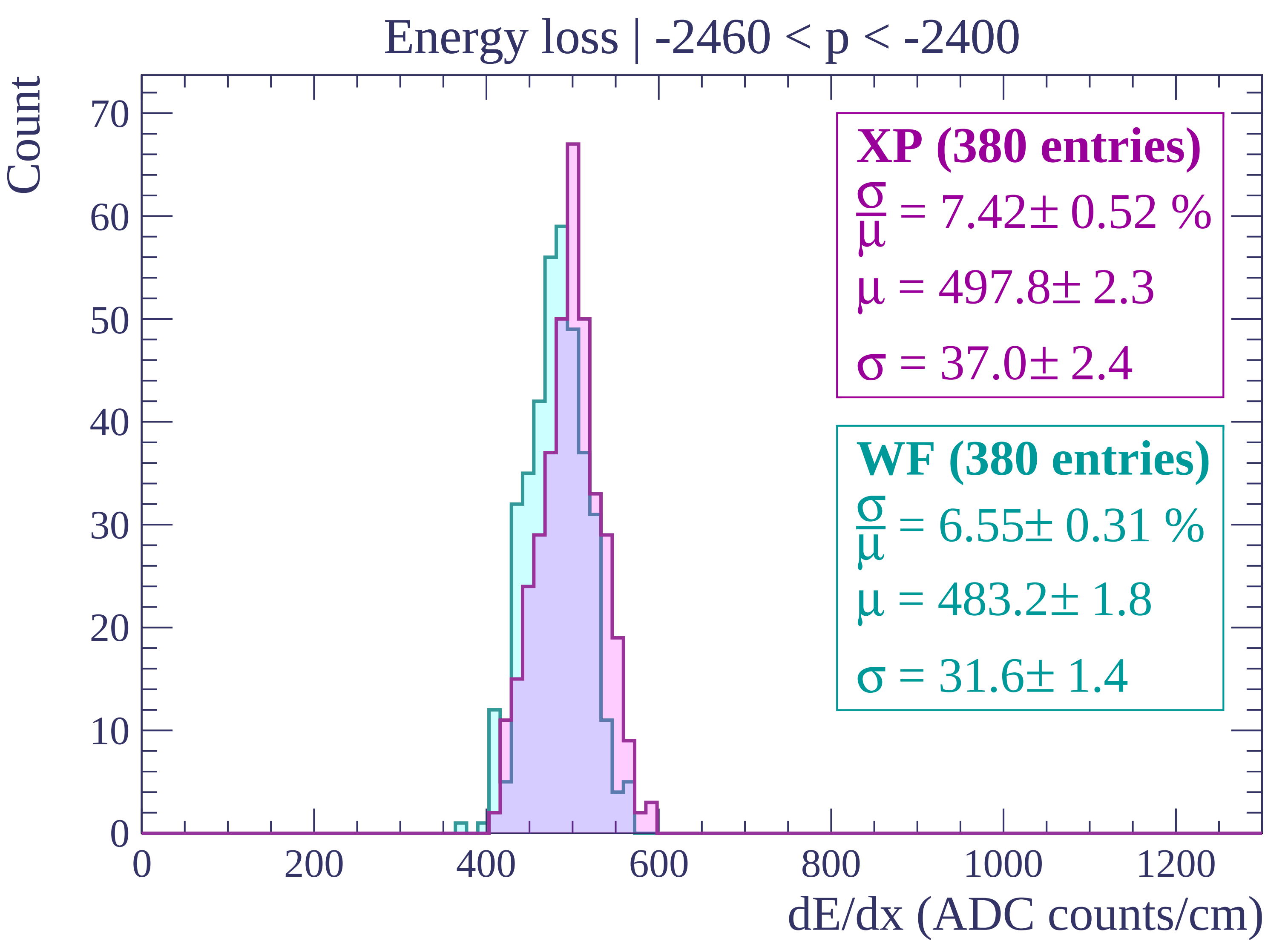
Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP (316 entries)
σ/μ = 6.64± 0.42 %
μ = 501.4± 2.2
σ = 33.3± 2.0
WF (316 entries)
σ/μ = 6.70± 0.50 %
μ = 484.0± 2.2
σ = 32.4± 2.3

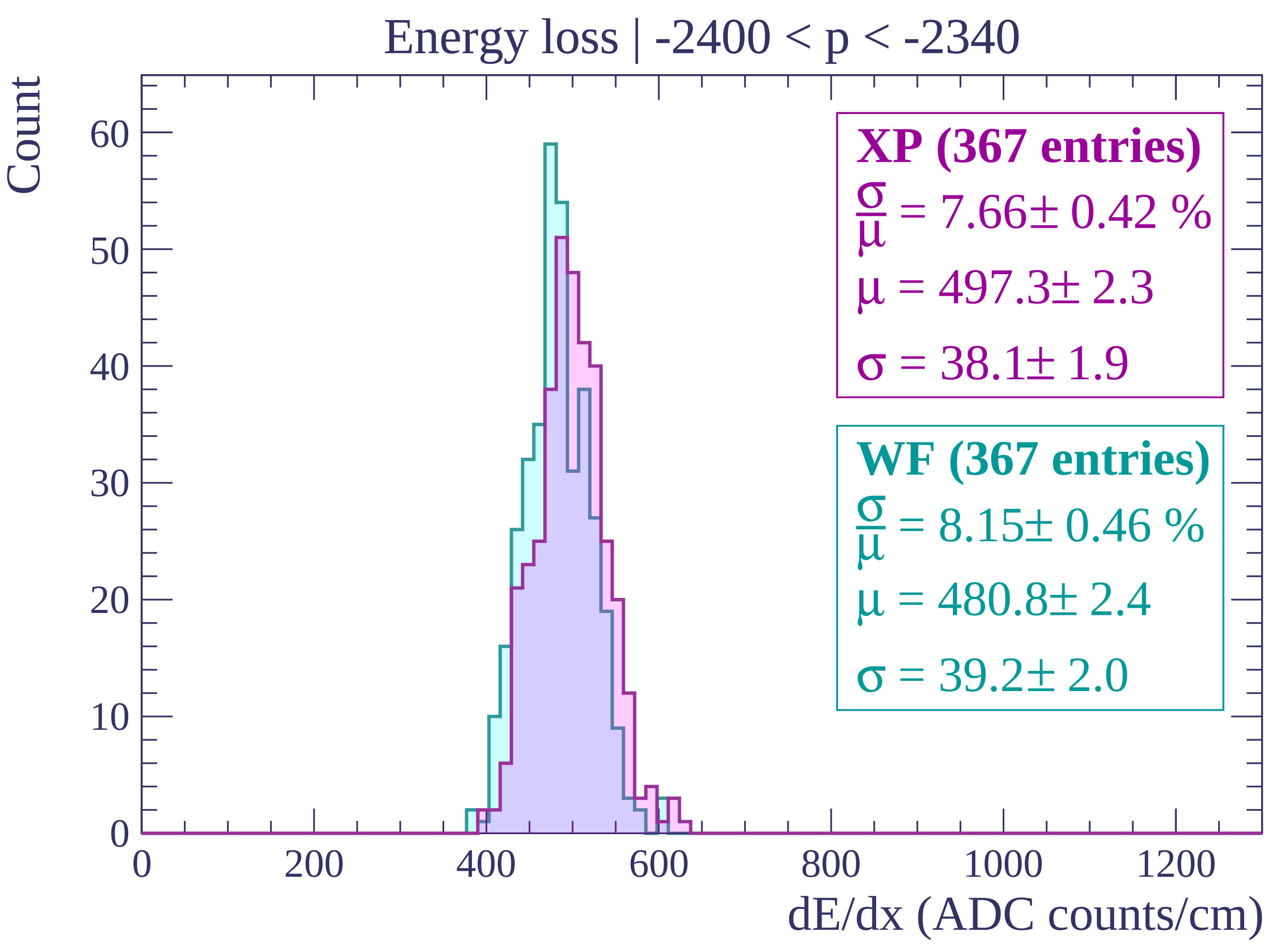


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP (303 entries)
σ/μ = 7.78± 0.66 %
μ = 502.5± 2.8
σ = 39.1± 3.1
WF (303 entries)
σ/μ = 7.04± 0.48 %
μ = 485.4± 2.4
σ = 34.2± 2.1

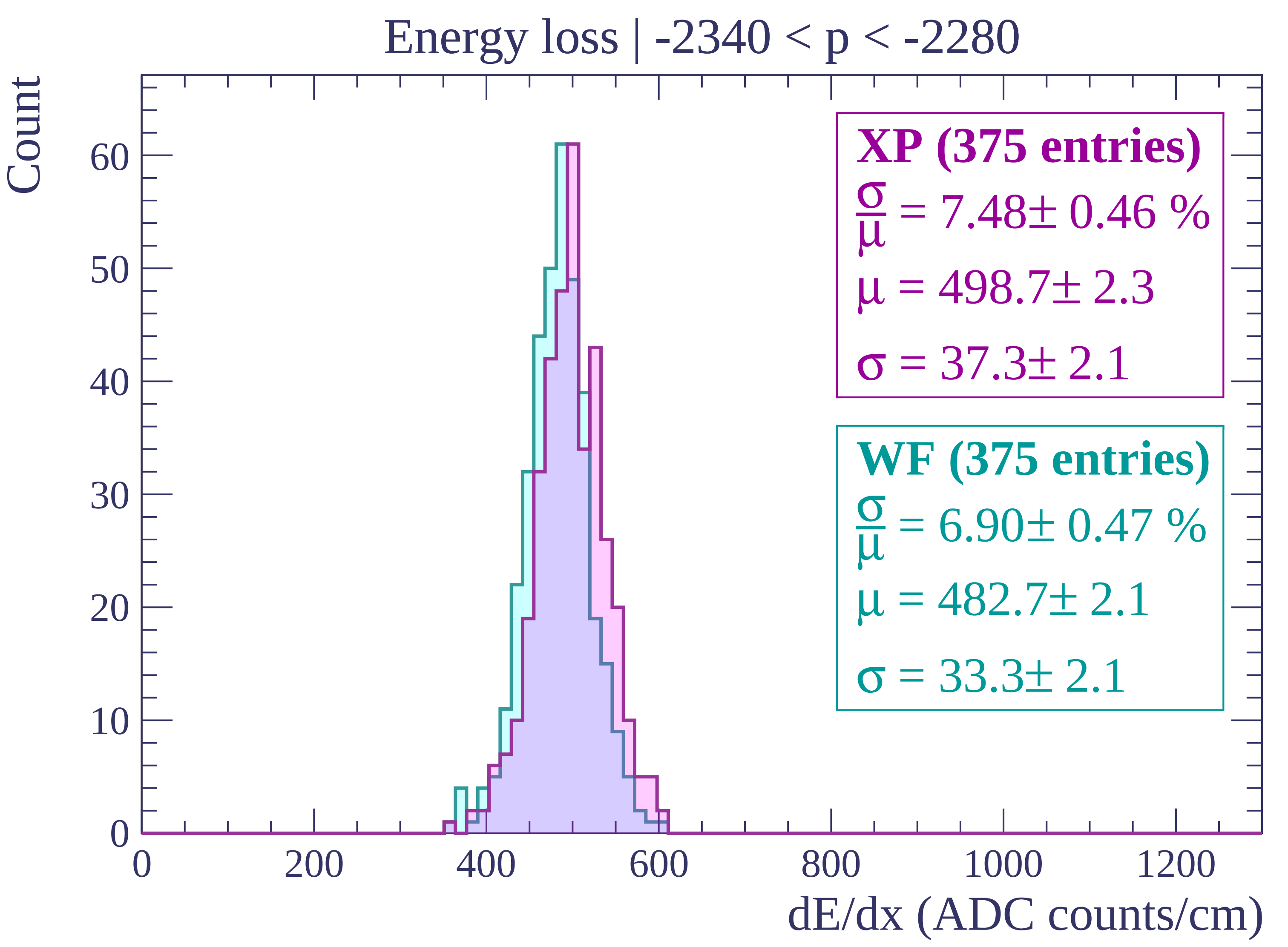


Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (335 entries)
$\frac{\sigma}{\mu}$ = 7.90± 0.65 %
μ = 501.4± 2.8
σ = 39.6± 3.0
WF (335 entries)
$\frac{\sigma}{\mu}$ = 7.92± 0.61 %
μ = 485.0± 2.6
σ = 38.4± 2.7

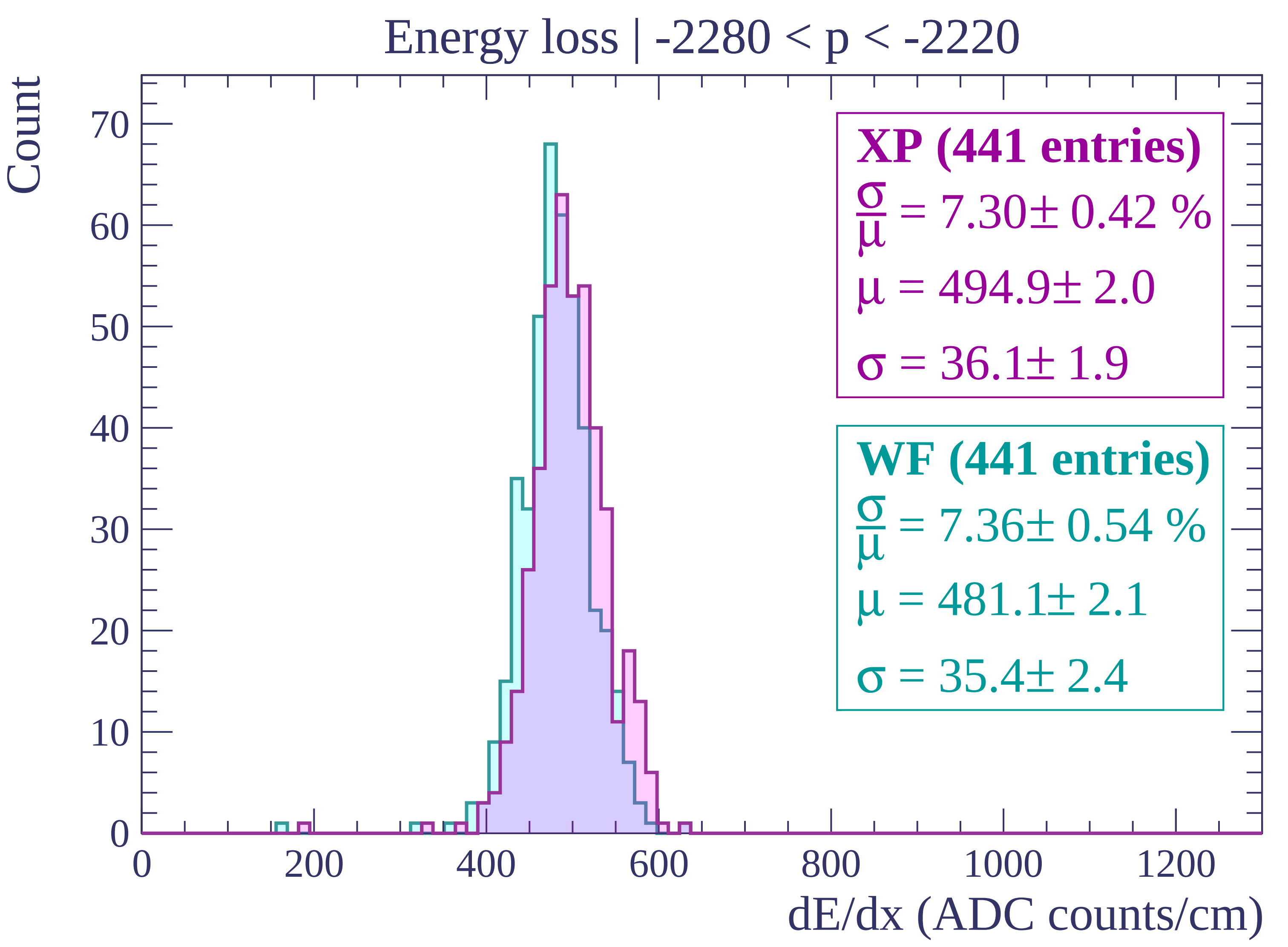

Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
XP (380 entries)
$\frac{\sigma}{\mu}$ = 7.42± 0.52 %
μ = 497.8± 2.3
σ = 37.0± 2.4
WF (380 entries)
$\frac{\sigma}{\mu}$ = 6.55± 0.31 %
μ = 483.2± 1.8
σ = 31.6± 1.4
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200

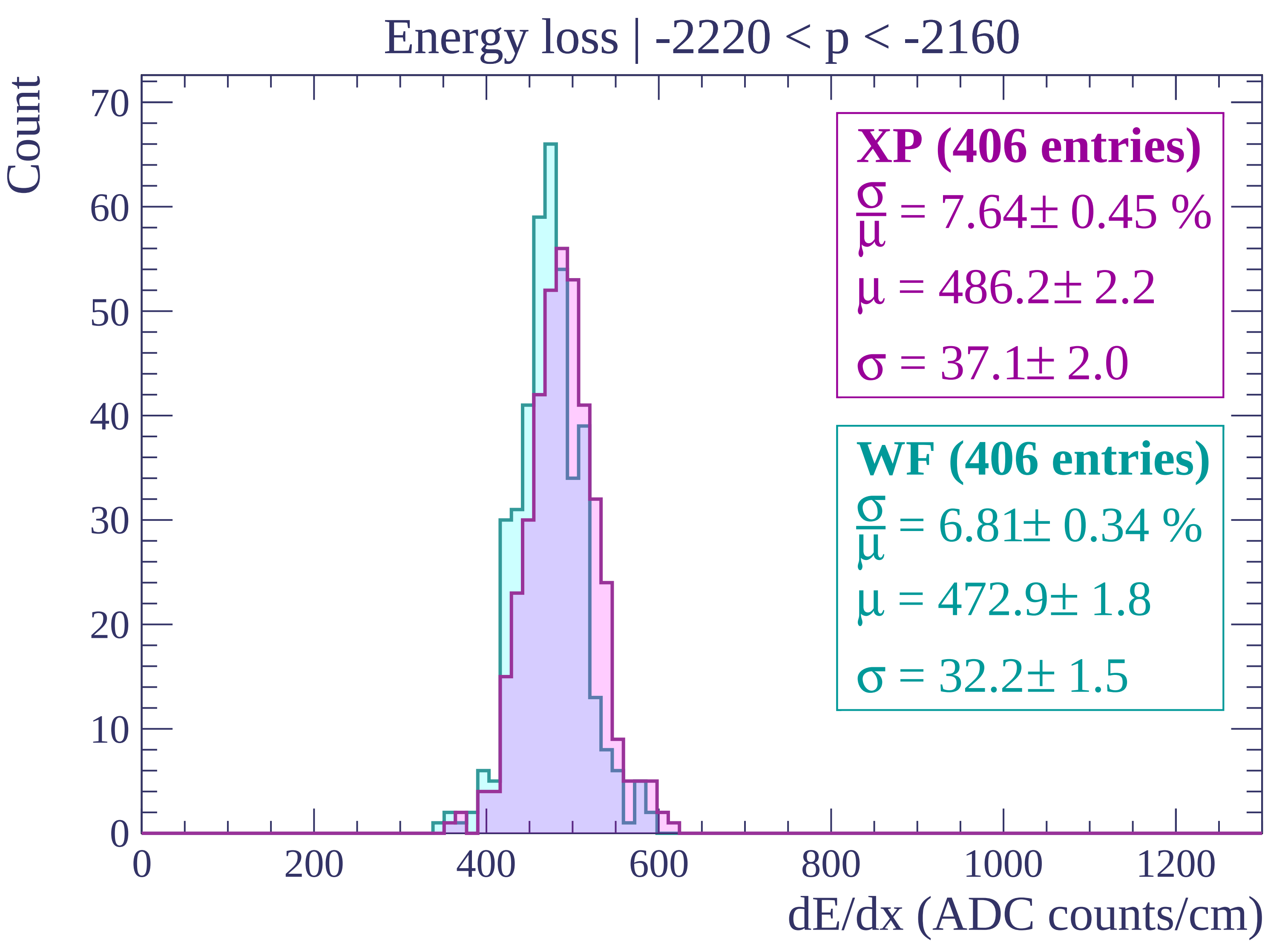


Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP (367 entries)
σ/μ = 7.66± 0.42 %
μ = 497.3± 2.3
σ = 38.1± 1.9
WF (367 entries)
σ/μ = 8.15± 0.46 %
μ = 480.8± 2.4
σ = 39.2± 2.0

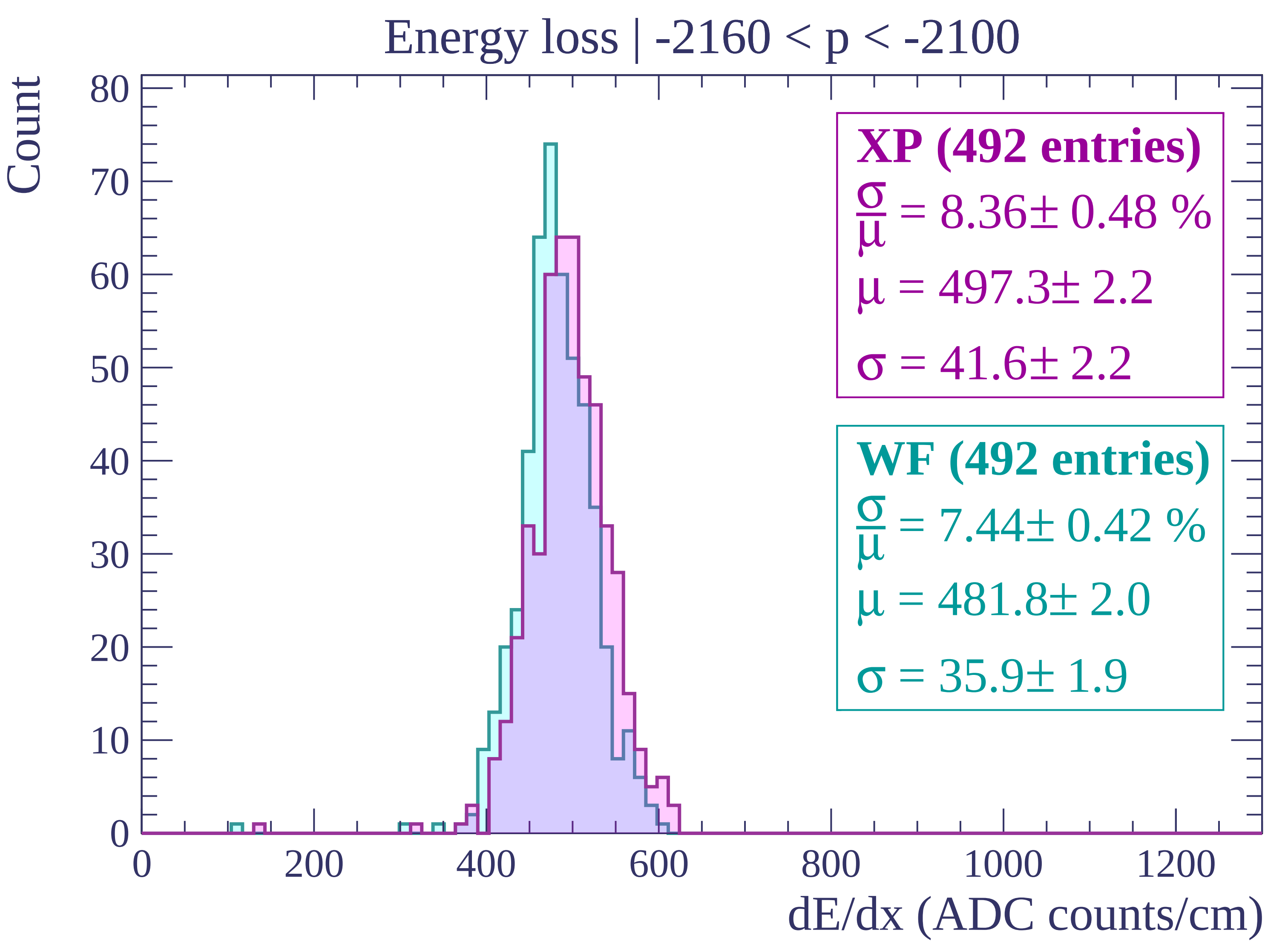


Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (375 entries)
σ/μ = 7.48± 0.46 %
μ = 498.7± 2.3
σ = 37.3± 2.1
WF (375 entries)
σ/μ = 6.90± 0.47 %
μ = 482.7± 2.1
σ = 33.3± 2.1

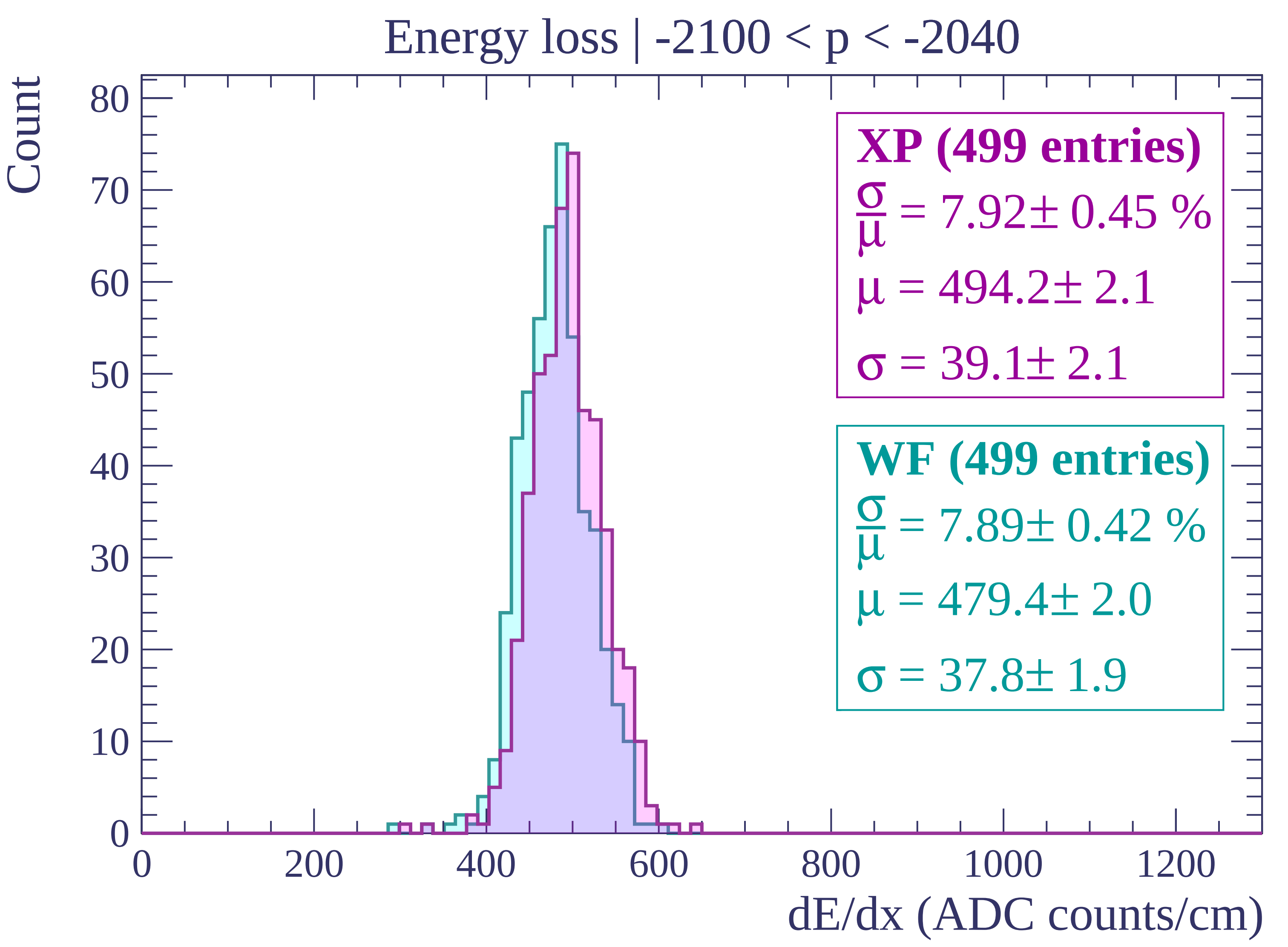

Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP (441 entries)
σ/μ = 7.30± 0.42 %
μ = 494.9± 2.0
σ = 36.1± 1.9
WF (441 entries)
σ/μ = 7.36± 0.54 %
μ = 481.1± 2.1
σ = 35.4± 2.4



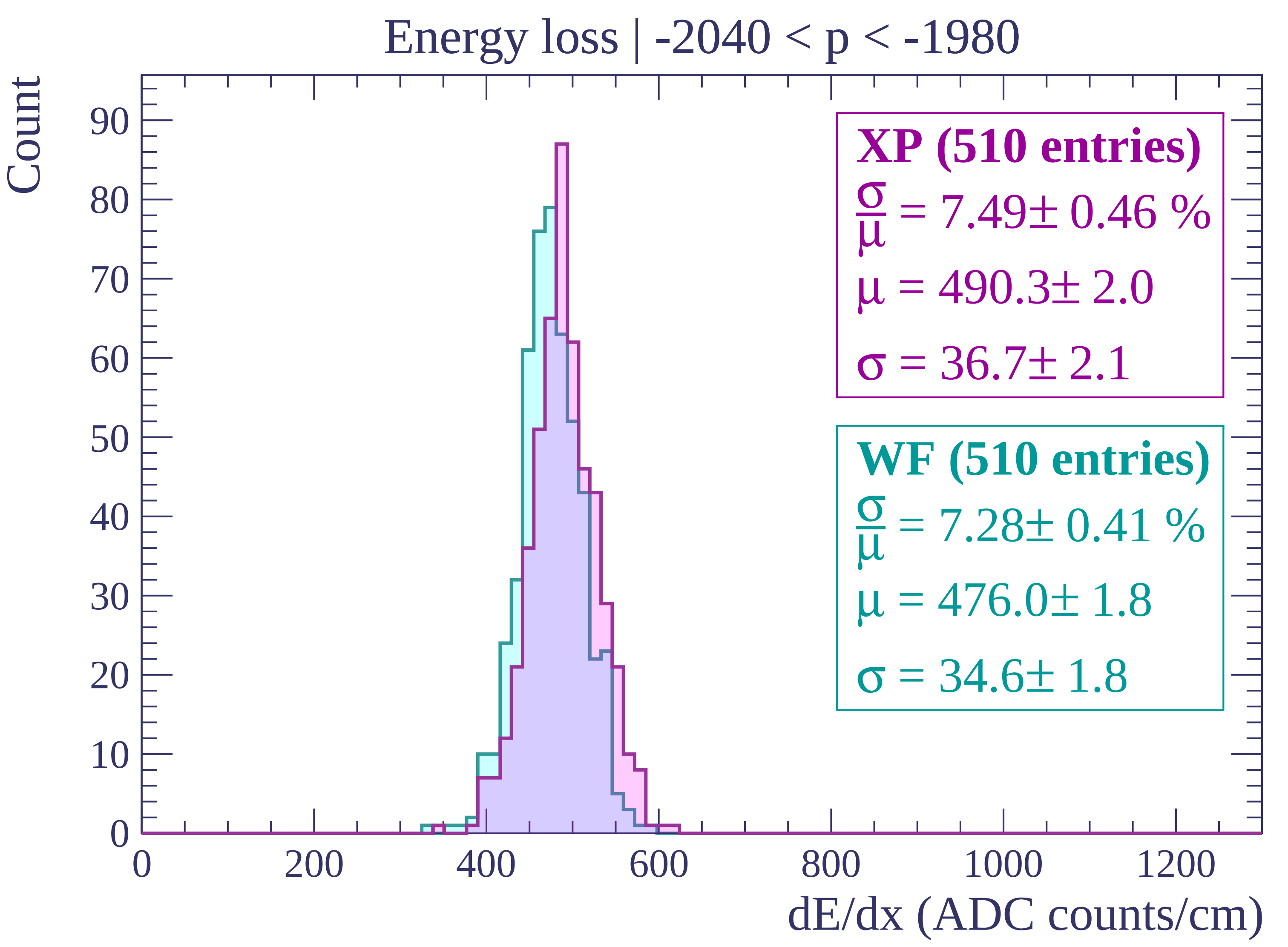
Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP (406 entries)
σ/μ = 7.64± 0.45 %
μ = 486.2± 2.2
σ = 37.1± 2.0
WF (406 entries)
σ/μ = 6.81± 0.34 %
μ = 472.9± 1.8
σ = 32.2± 1.5



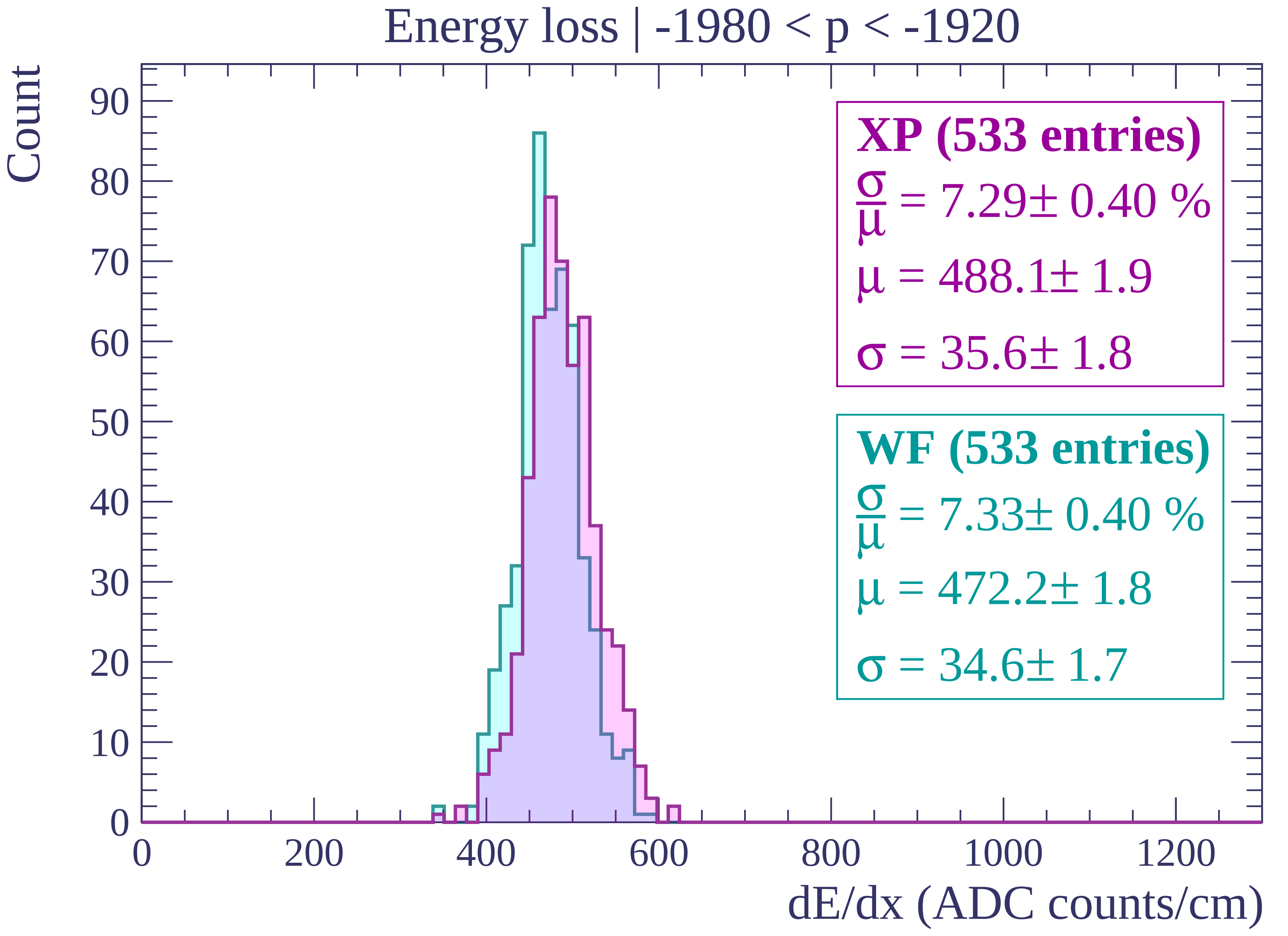
Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
XP (492 entries)
σ/μ = 8.36± 0.48 %
μ = 497.3± 2.2
σ = 41.6± 2.2
WF (492 entries)
σ/μ = 7.44± 0.42 %
μ = 481.8± 2.0
σ = 35.9± 1.9
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200



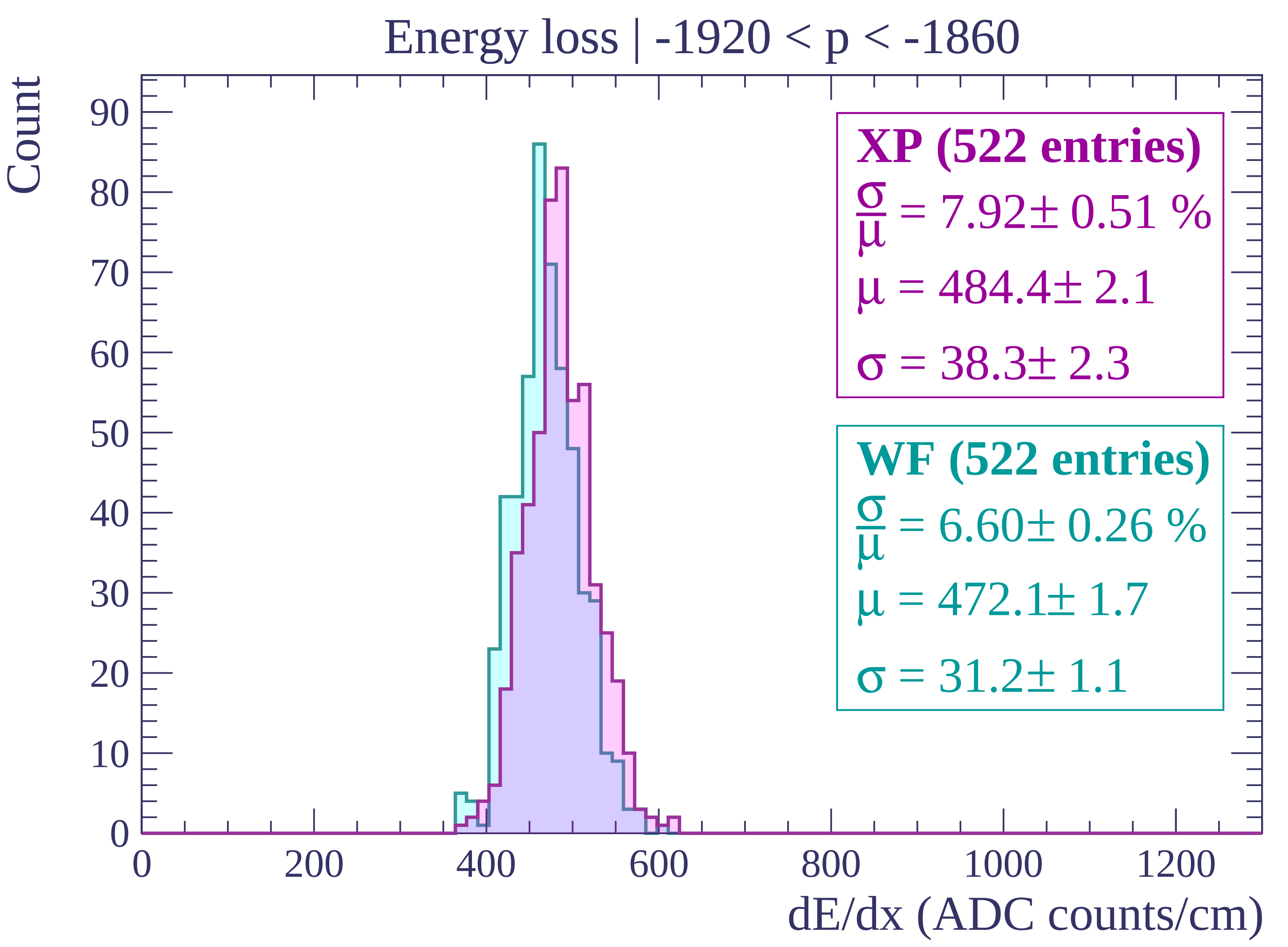
Energy loss | -2100 < p < -2040
Count
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (499 entries)
σ/μ = 7.92± 0.45 %
μ = 494.2± 2.1
σ = 39.1± 2.1
WF (499 entries)
σ/μ = 7.89± 0.42 %
μ = 479.4± 2.0
σ = 37.8± 1.9



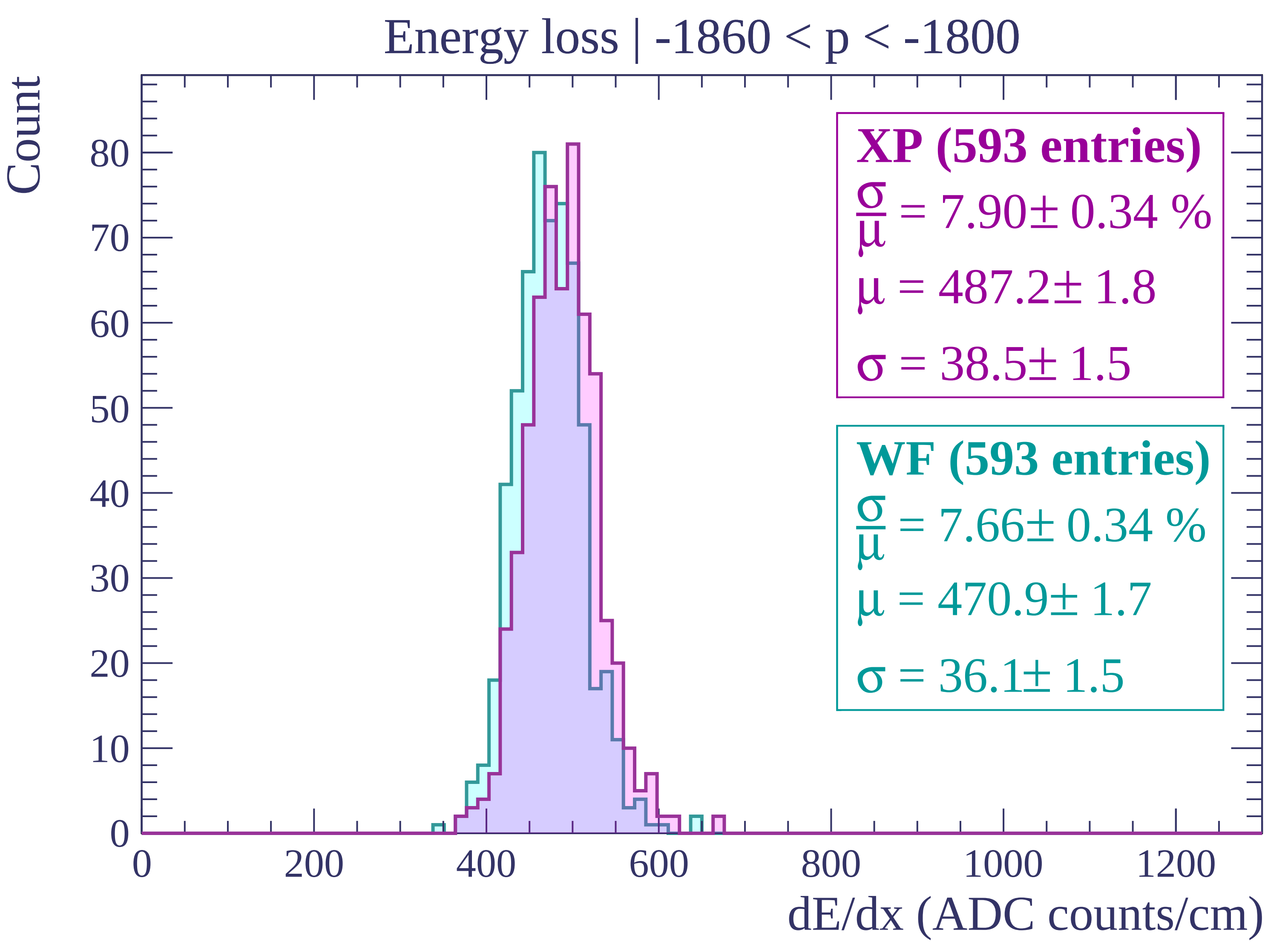
Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (510 entries)
σ/μ = 7.49± 0.46 %
μ = 490.3± 2.0
σ = 36.7± 2.1
WF (510 entries)
σ/μ = 7.28± 0.41 %
μ = 476.0± 1.8
σ = 34.6± 1.8



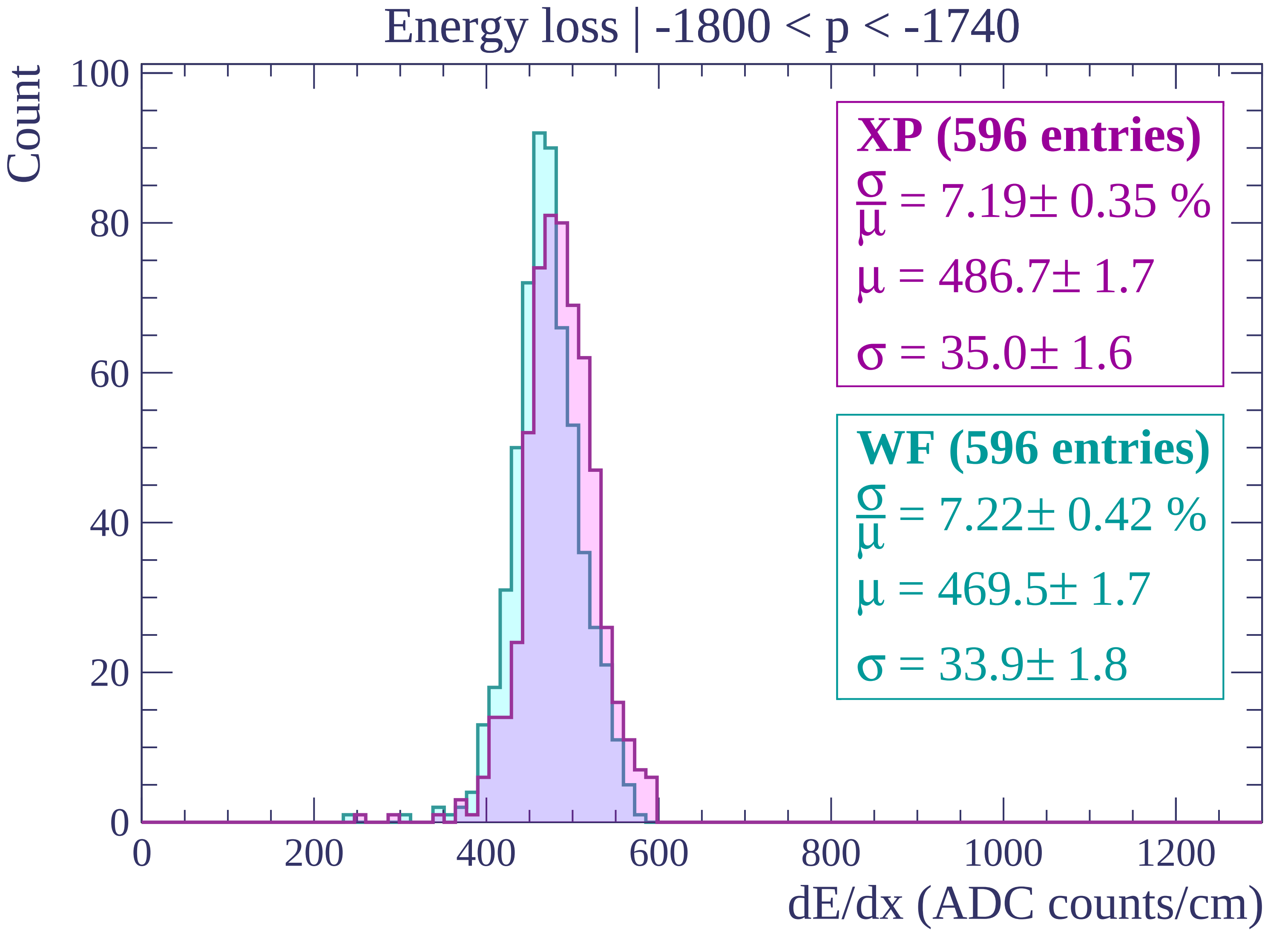
Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (533 entries)
$\frac{\sigma}{\mu}$ = 7.29± 0.40 %
μ = 488.1± 1.9
σ = 35.6± 1.8
WF (533 entries)
$\frac{\sigma}{\mu}$ = 7.33± 0.40 %
μ = 472.2± 1.8
σ = 34.6± 1.7

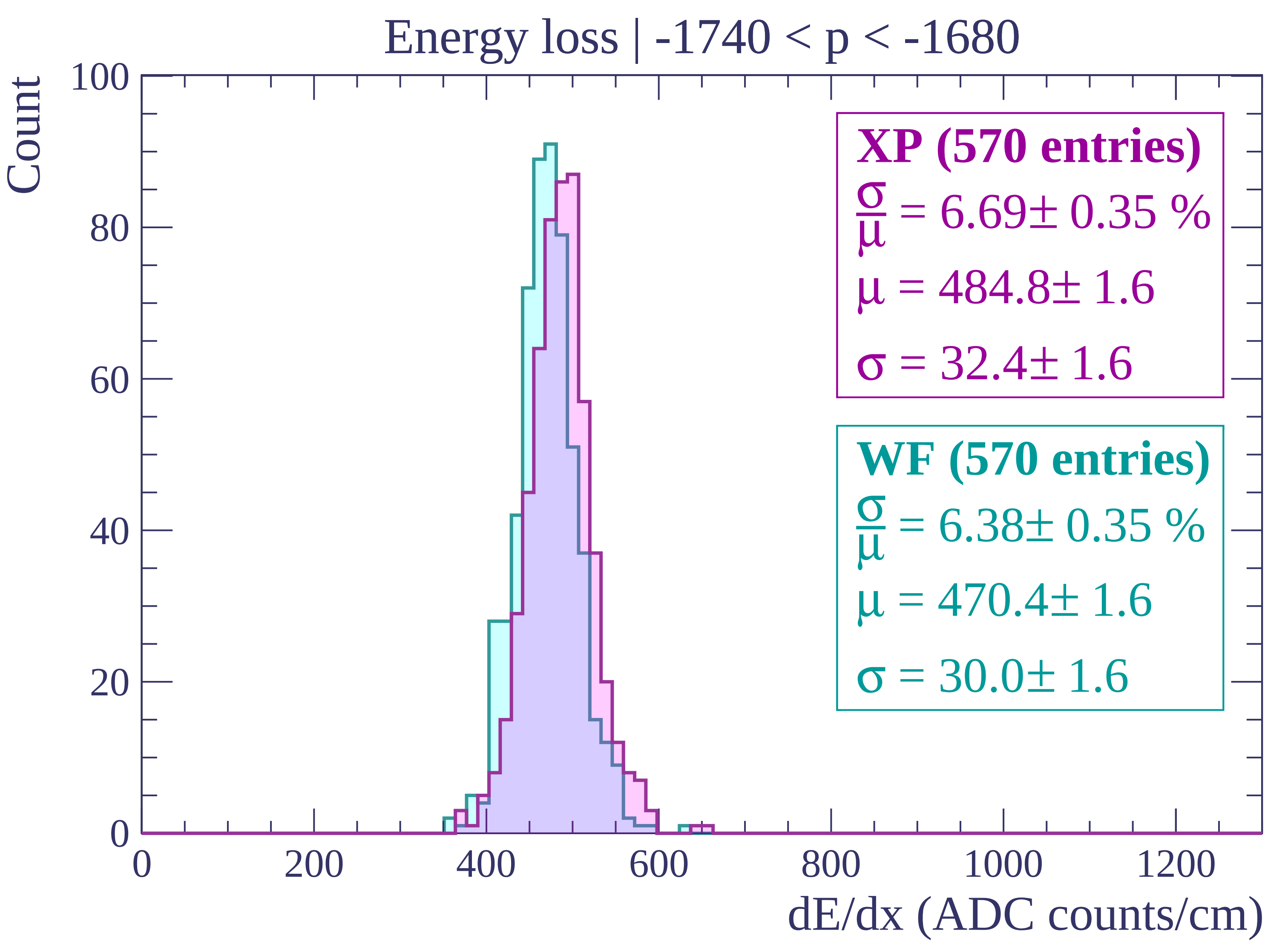


Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (522 entries)
$\frac{\sigma}{\mu}$ = 7.92± 0.51 %
μ = 484.4± 2.1
σ = 38.3± 2.3
WF (522 entries)
$\frac{\sigma}{\mu}$ = 6.60± 0.26 %
μ = 472.1± 1.7
σ = 31.2± 1.1

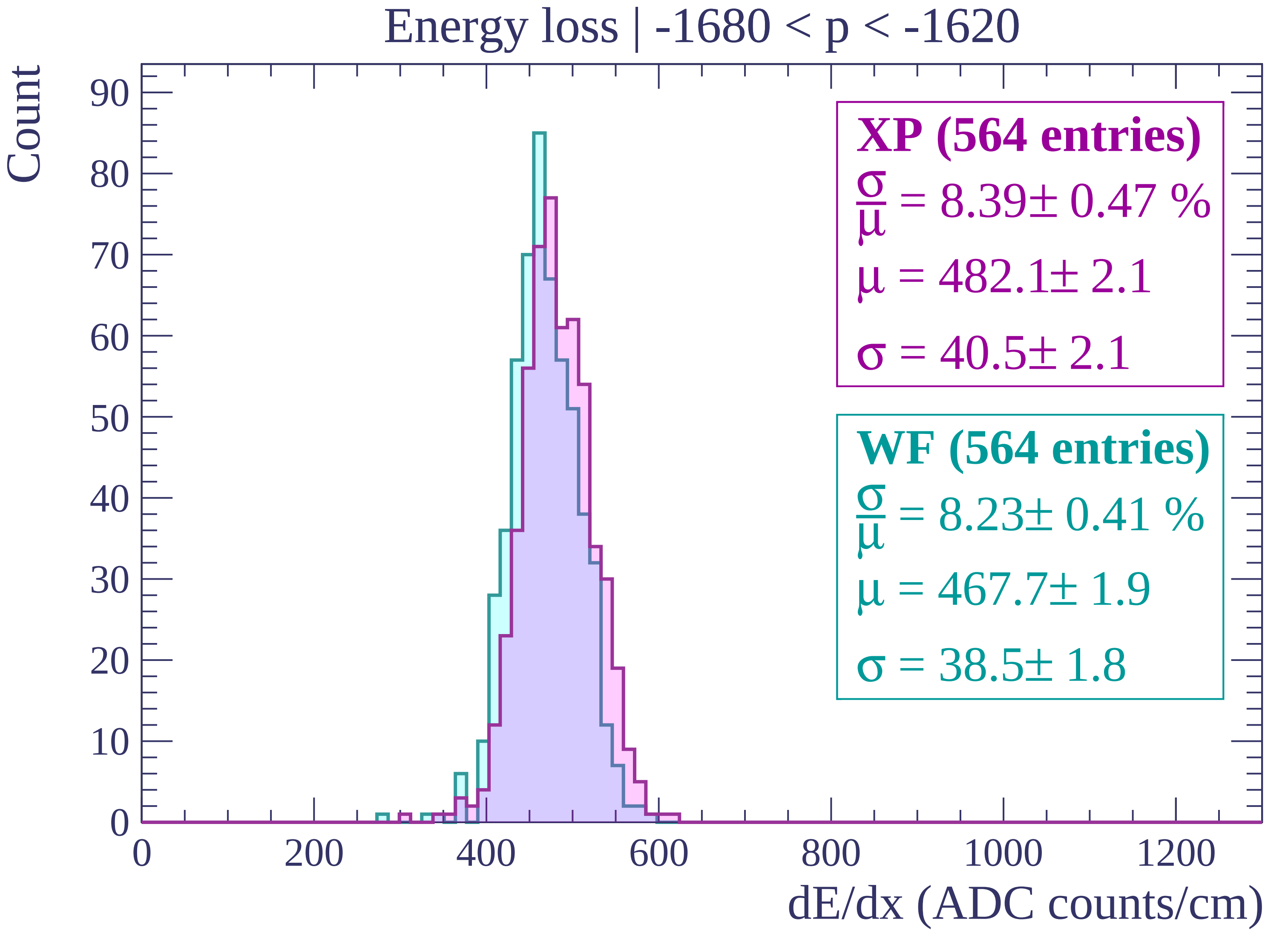


Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (593 entries)
σ/μ = 7.90± 0.34 %
μ = 487.2± 1.8
σ = 38.5± 1.5
WF (593 entries)
σ/μ = 7.66± 0.34 %
μ = 470.9± 1.7
σ = 36.1± 1.5

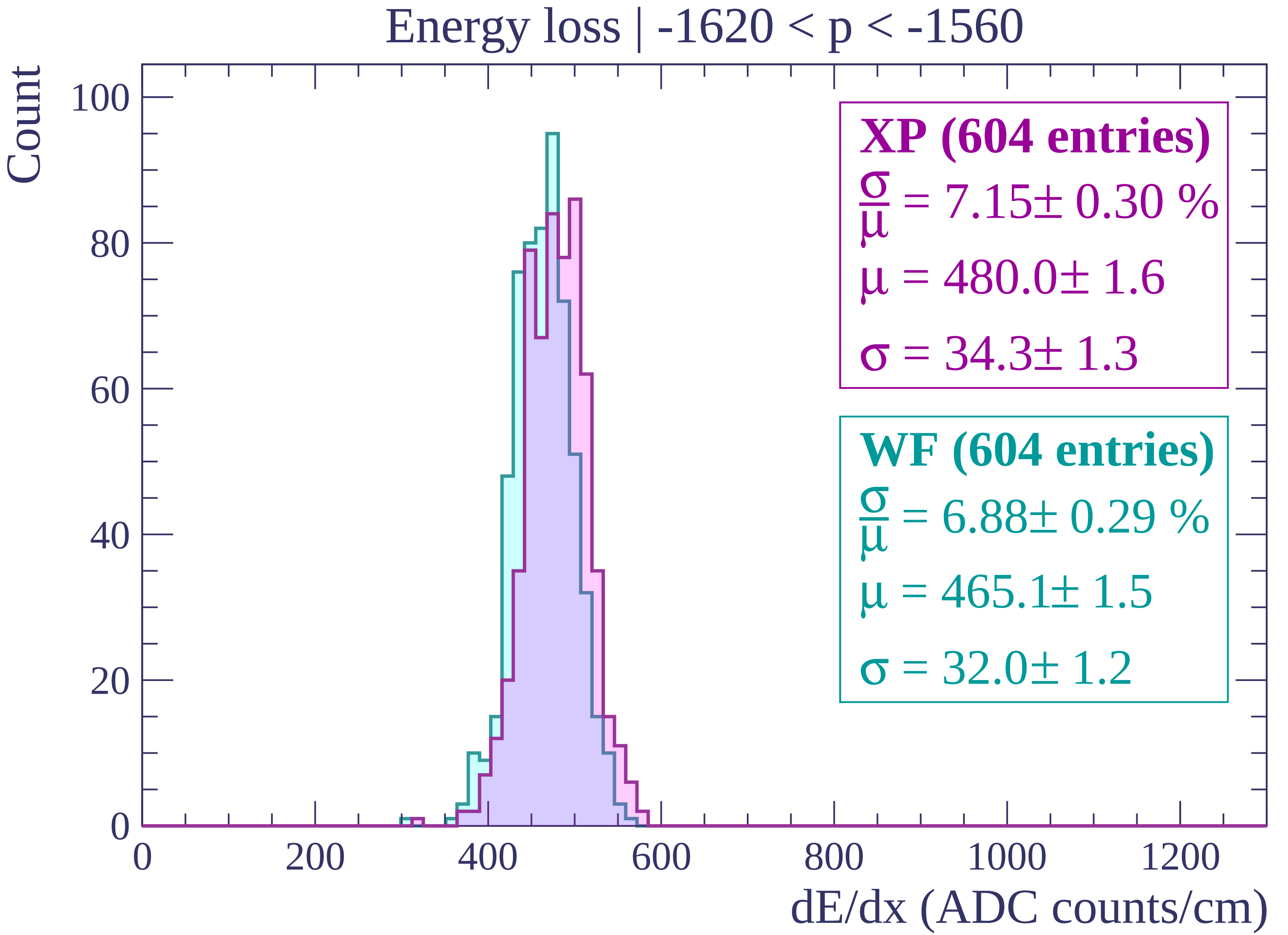

Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (596 entries)
σ/μ = 7.19± 0.35 %
μ = 486.7± 1.7
σ = 35.0± 1.6
WF (596 entries)
σ/μ = 7.22± 0.42 %
μ = 469.5± 1.7
σ = 33.9± 1.8

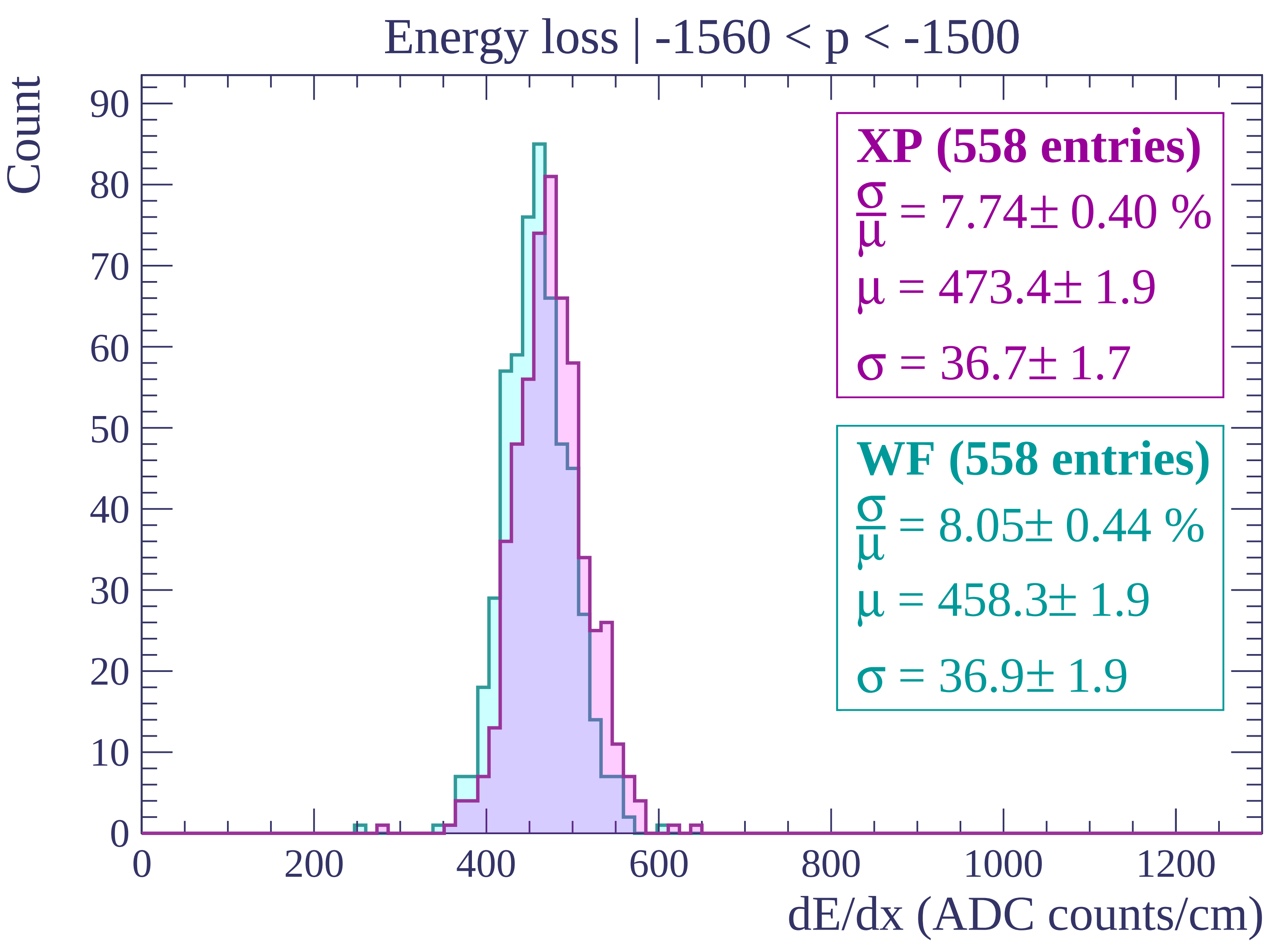


Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (570 entries)
σ/μ = 6.69± 0.35 %
μ = 484.8± 1.6
σ = 32.4± 1.6
WF (570 entries)
σ/μ = 6.38± 0.35 %
μ = 470.4± 1.6
σ = 30.0± 1.6

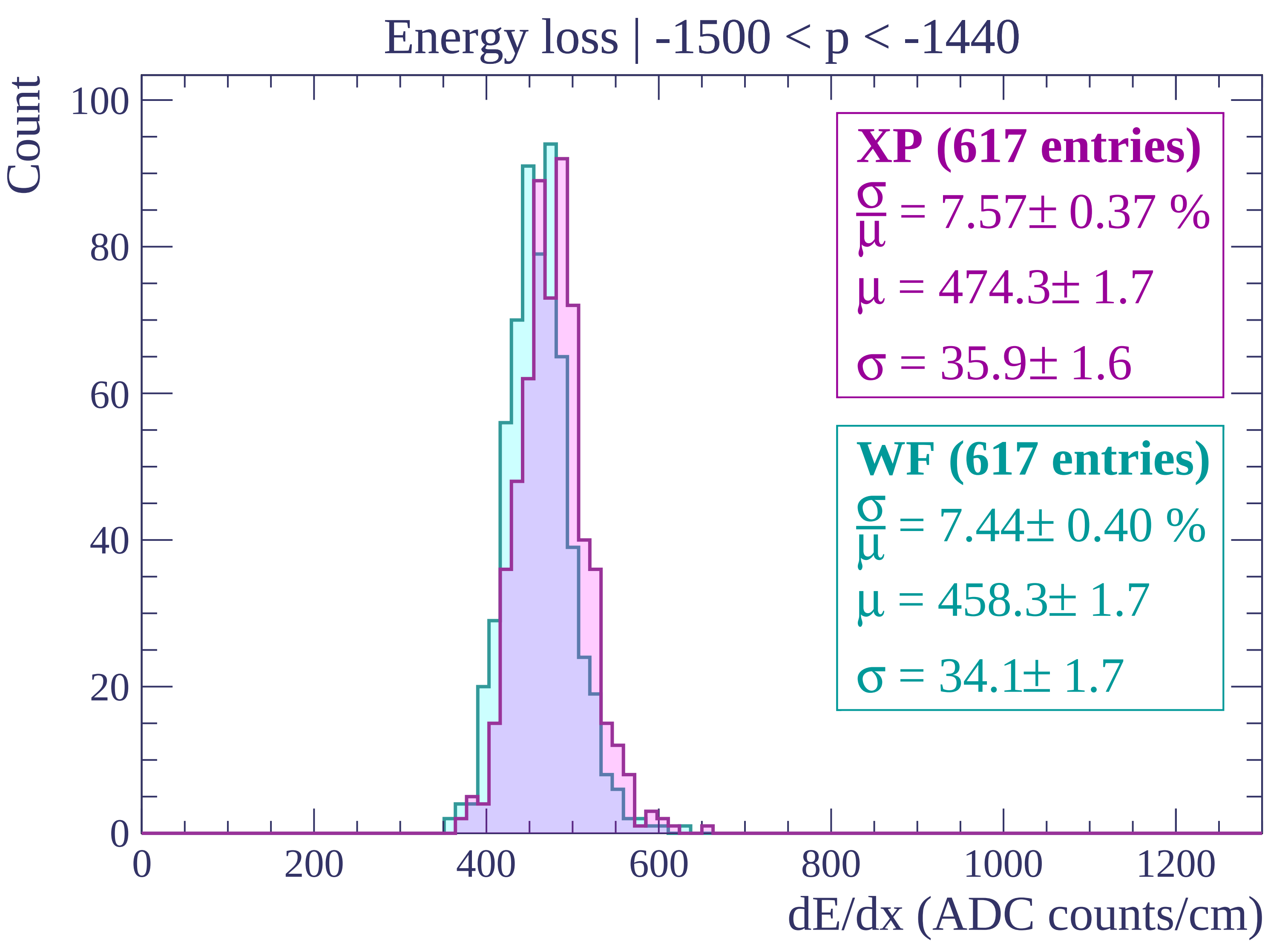


Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
XP (564 entries)
σ/μ = 8.39± 0.47 %
μ = 482.1± 2.1
σ = 40.5± 2.1
WF (564 entries)
σ/μ = 8.23± 0.41 %
μ = 467.7± 1.9
σ = 38.5± 1.8

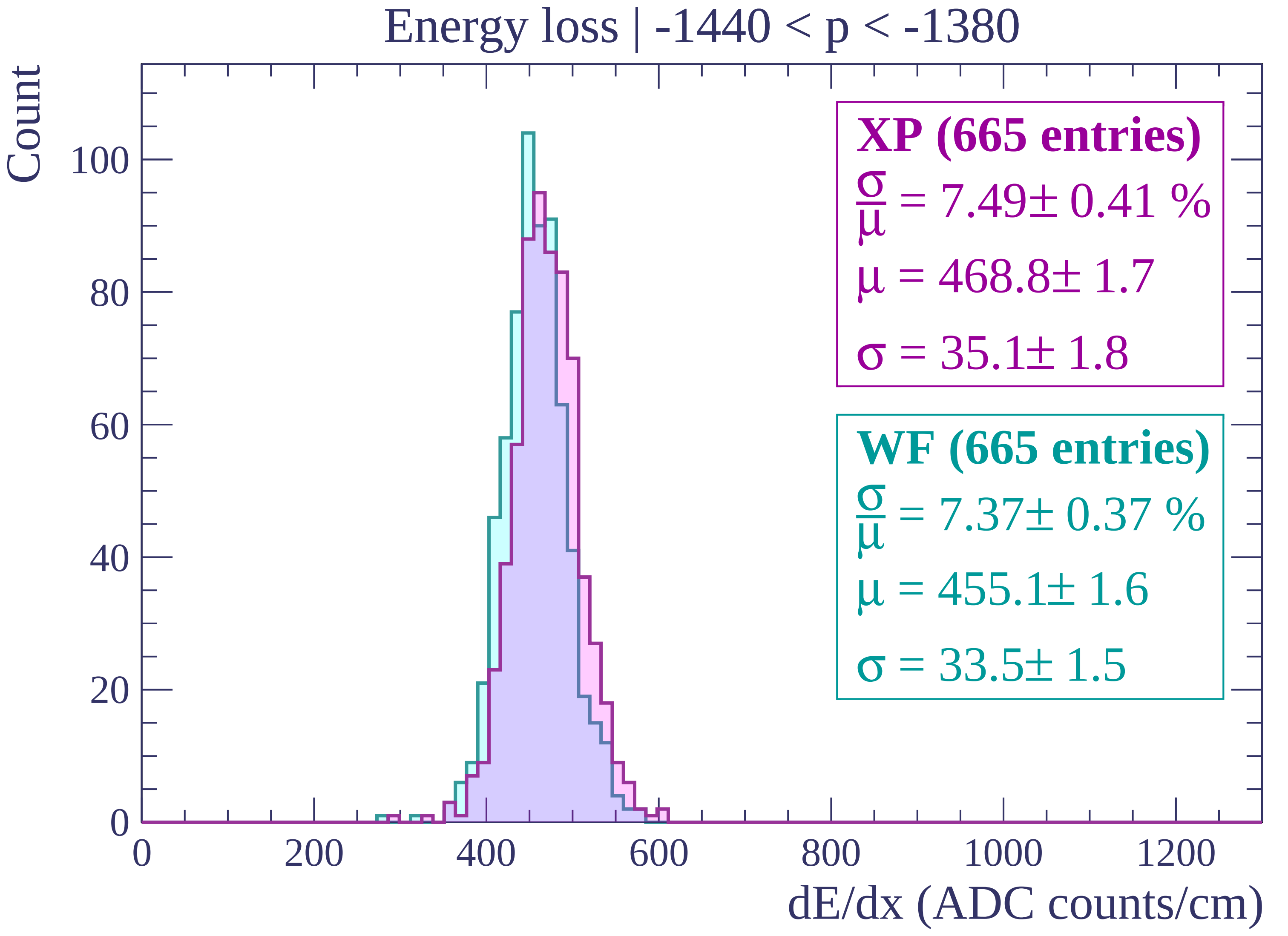

Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (604 entries)
σ/μ = 7.15± 0.30 %
μ = 480.0± 1.6
σ = 34.3± 1.3
WF (604 entries)
σ/μ = 6.88± 0.29 %
μ = 465.1± 1.5
σ = 32.0± 1.2



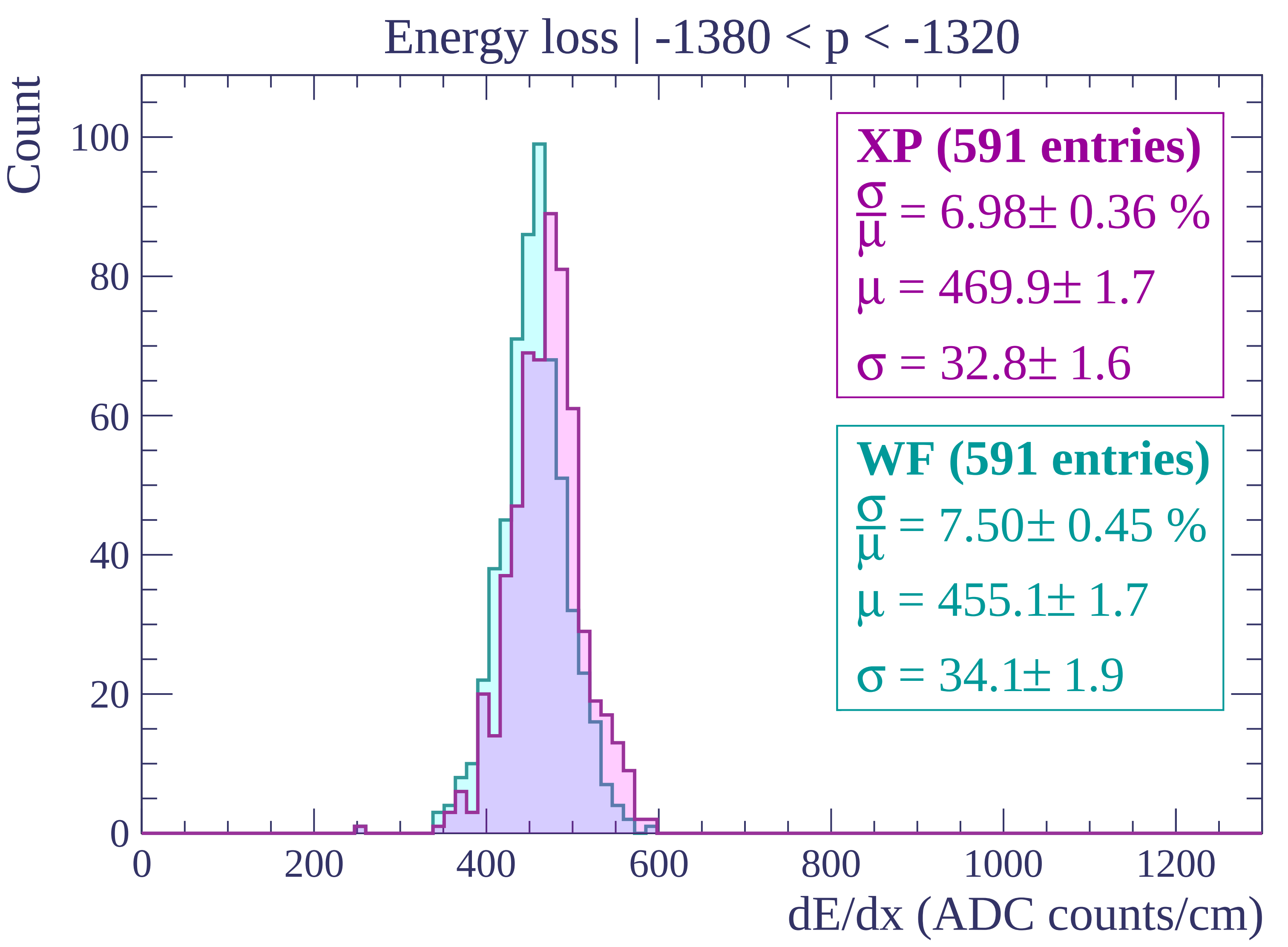
Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (558 entries)
σ/μ = 7.74± 0.40 %
μ = 473.4± 1.9
σ = 36.7± 1.7
WF (558 entries)
σ/μ = 8.05± 0.44 %
μ = 458.3± 1.9
σ = 36.9± 1.9



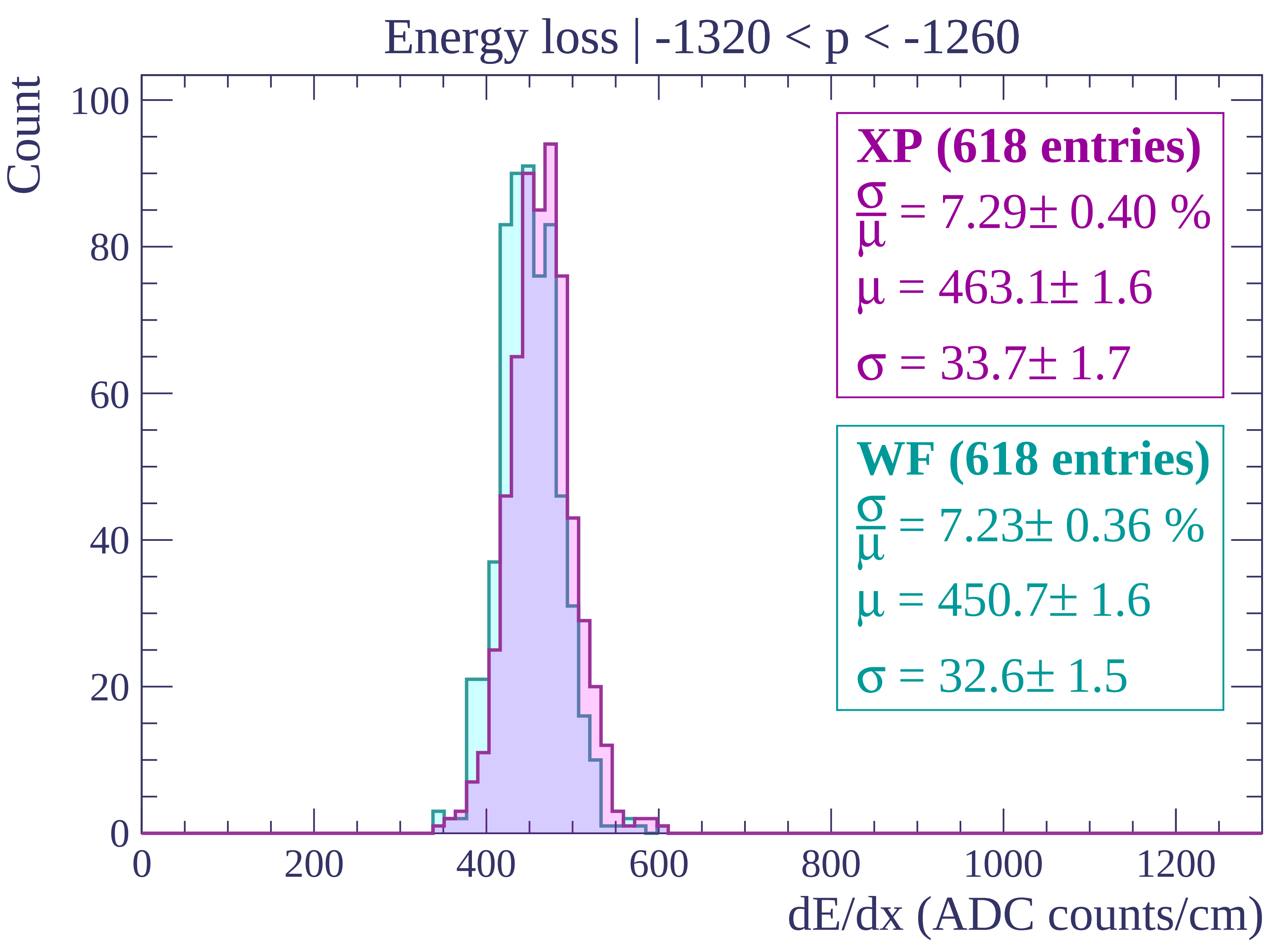
Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (617 entries)
σ/μ = 7.57± 0.37 %
μ = 474.3± 1.7
σ = 35.9± 1.6
WF (617 entries)
σ/μ = 7.44± 0.40 %
μ = 458.3± 1.7
σ = 34.1± 1.7

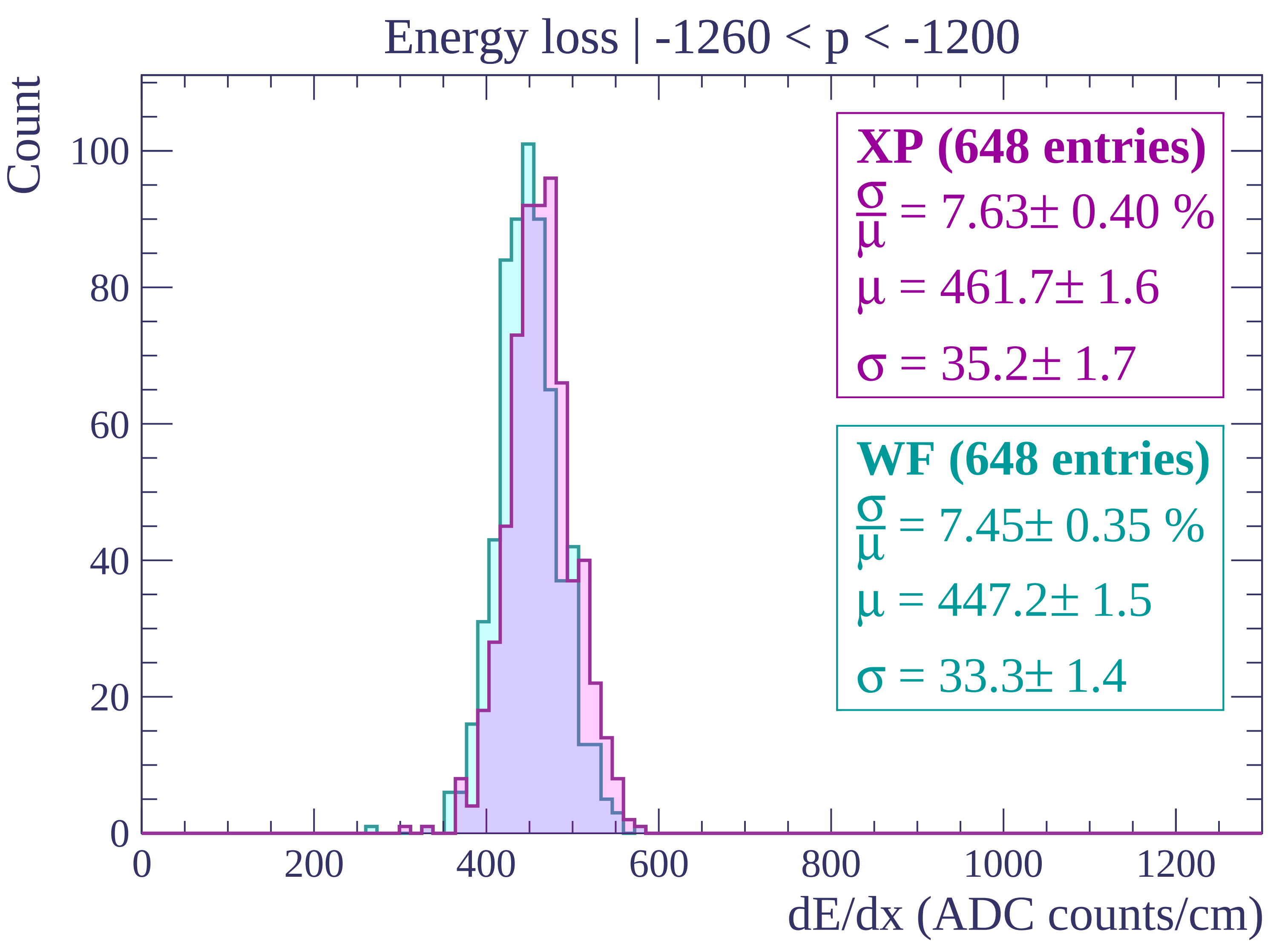


Energy loss | -1440 < p < -1380
Count
XP (665 entries)
σ/μ = 7.49± 0.41 %
μ = 468.8± 1.7
σ = 35.1± 1.8
WF (665 entries)
σ/μ = 7.37± 0.37 %
μ = 455.1± 1.6
σ = 33.5± 1.5
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

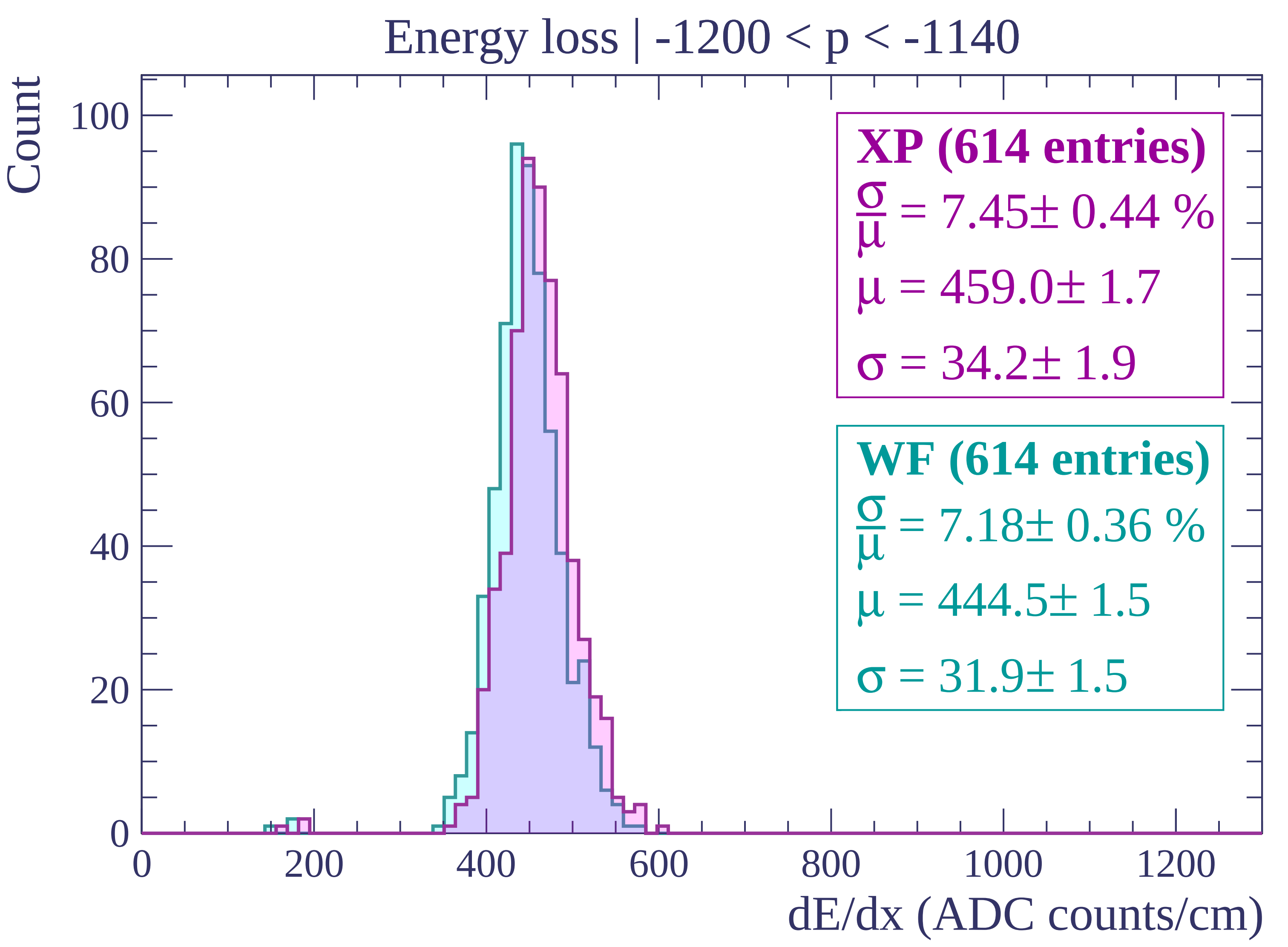


Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (591 entries)
σ/μ = 6.98± 0.36 %
μ = 469.9± 1.7
σ = 32.8± 1.6
WF (591 entries)
σ/μ = 7.50± 0.45 %
μ = 455.1± 1.7
σ = 34.1± 1.9

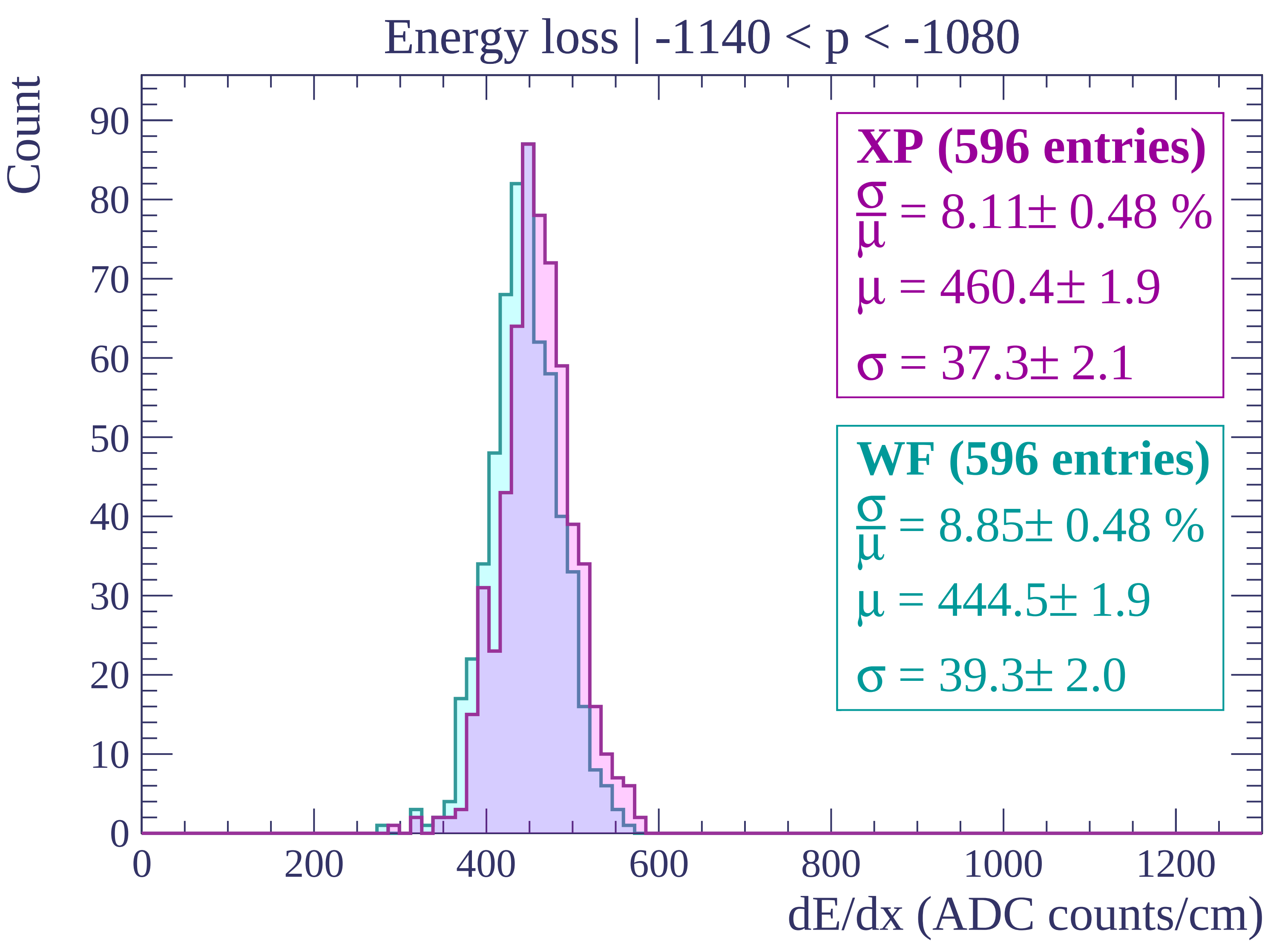

Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (618 entries)
σ/μ = 7.29± 0.40 %
μ = 463.1± 1.6
σ = 33.7± 1.7
WF (618 entries)
σ/μ = 7.23± 0.36 %
μ = 450.7± 1.6
σ = 32.6± 1.5



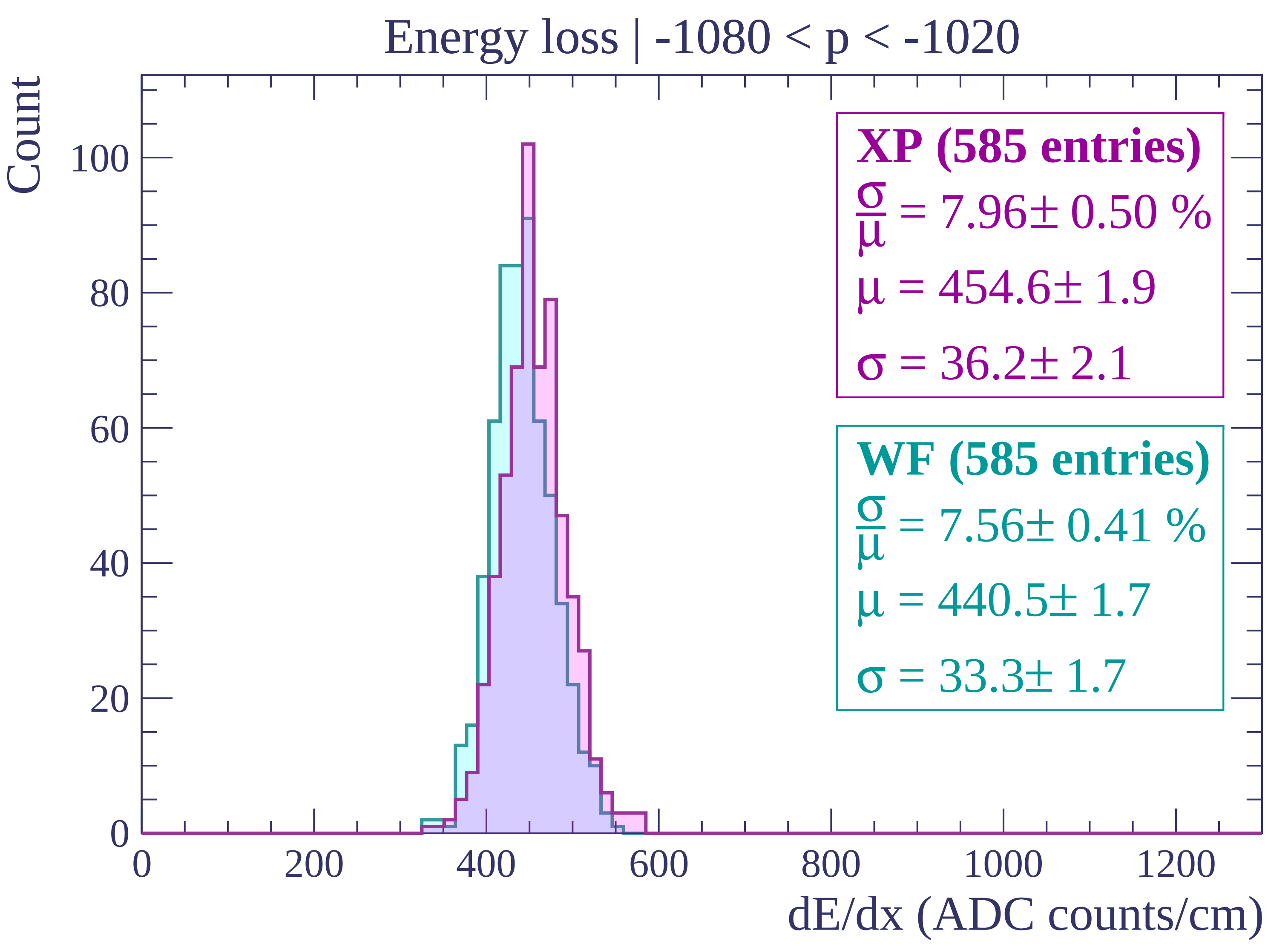
Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (648 entries)
σ/μ = 7.63± 0.40 %
μ = 461.7± 1.6
σ = 35.2± 1.7
WF (648 entries)
σ/μ = 7.45± 0.35 %
μ = 447.2± 1.5
σ = 33.3± 1.4



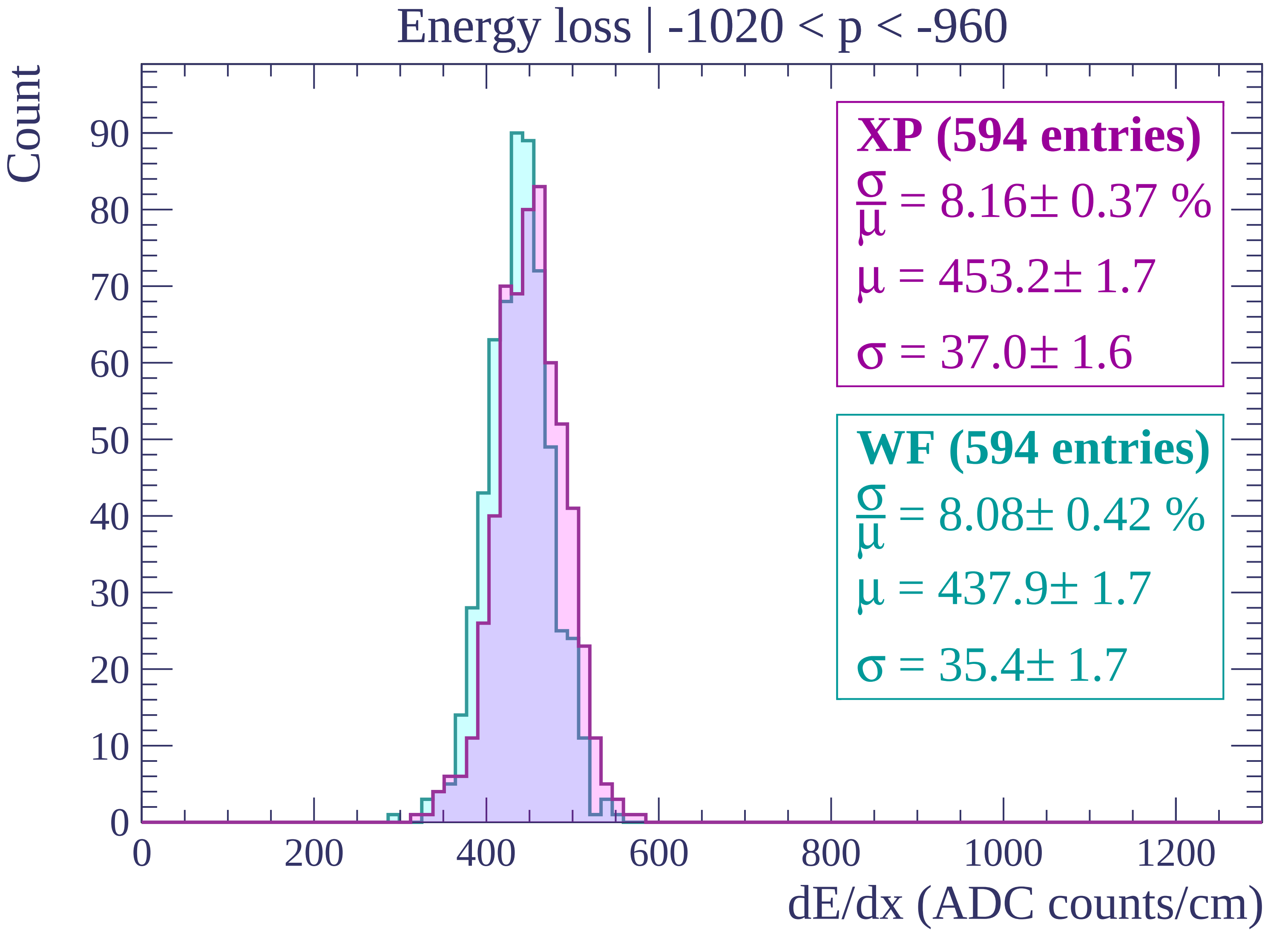
Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (614 entries)
σ/μ = 7.45± 0.44 %
μ = 459.0± 1.7
σ = 34.2± 1.9
WF (614 entries)
σ/μ = 7.18± 0.36 %
μ = 444.5± 1.5
σ = 31.9± 1.5



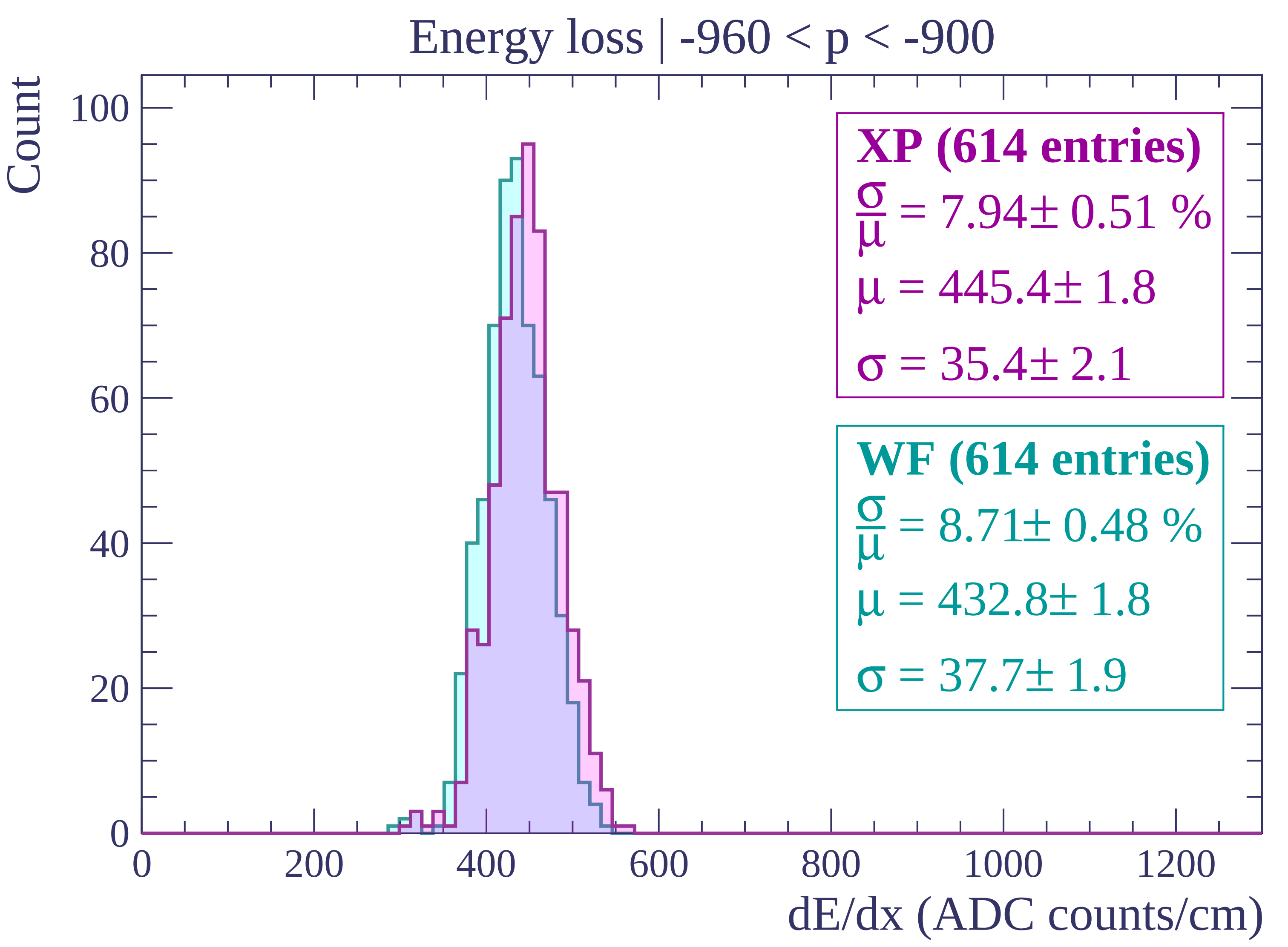
Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (596 entries)
σ/μ = 8.11± 0.48 %
μ = 460.4± 1.9
σ = 37.3± 2.1
WF (596 entries)
σ/μ = 8.85± 0.48 %
μ = 444.5± 1.9
σ = 39.3± 2.0



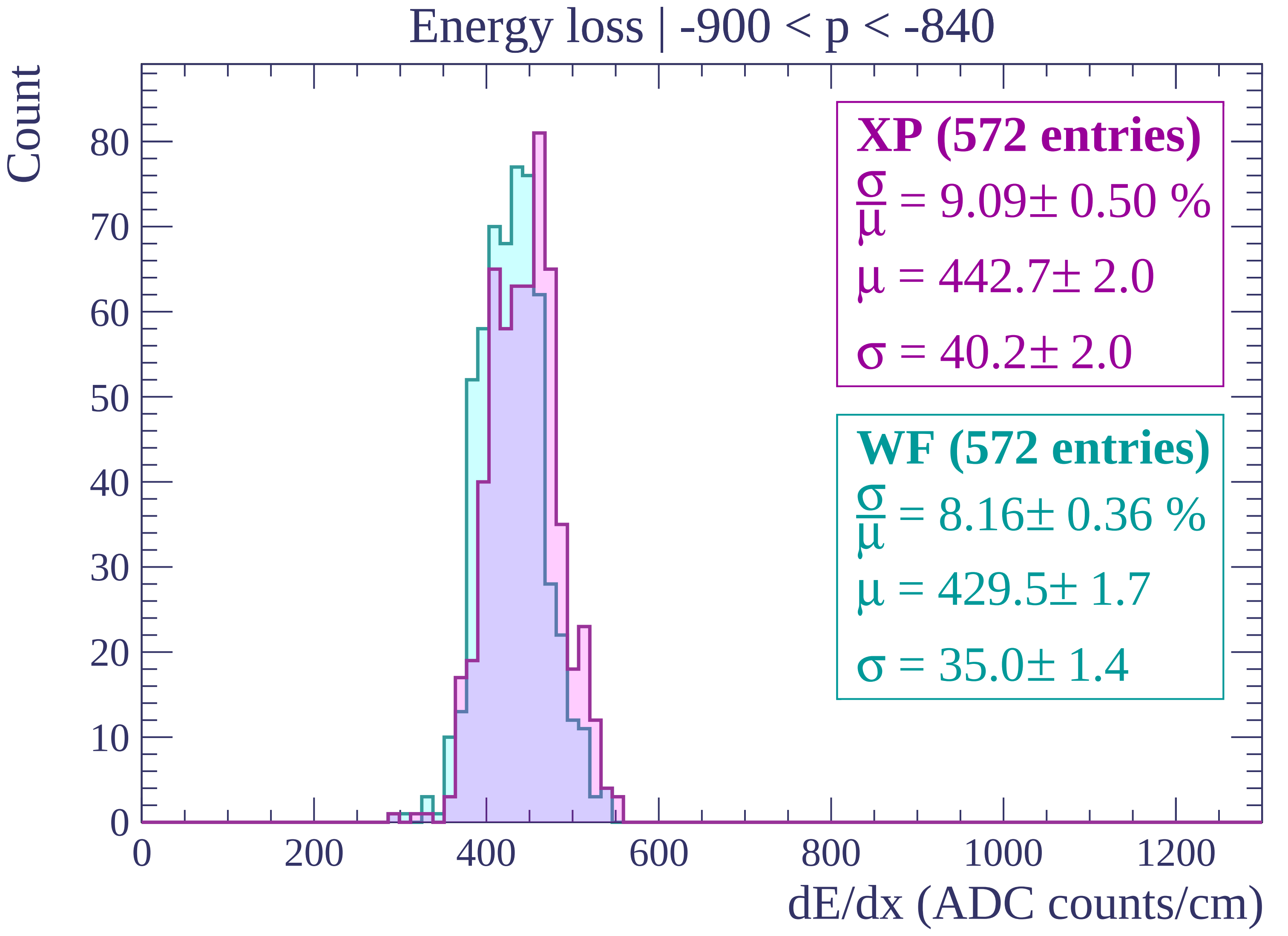
Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (585 entries)
σ/μ = 7.96± 0.50 %
μ = 454.6± 1.9
σ = 36.2± 2.1
WF (585 entries)
σ/μ = 7.56± 0.41 %
μ = 440.5± 1.7
σ = 33.3± 1.7

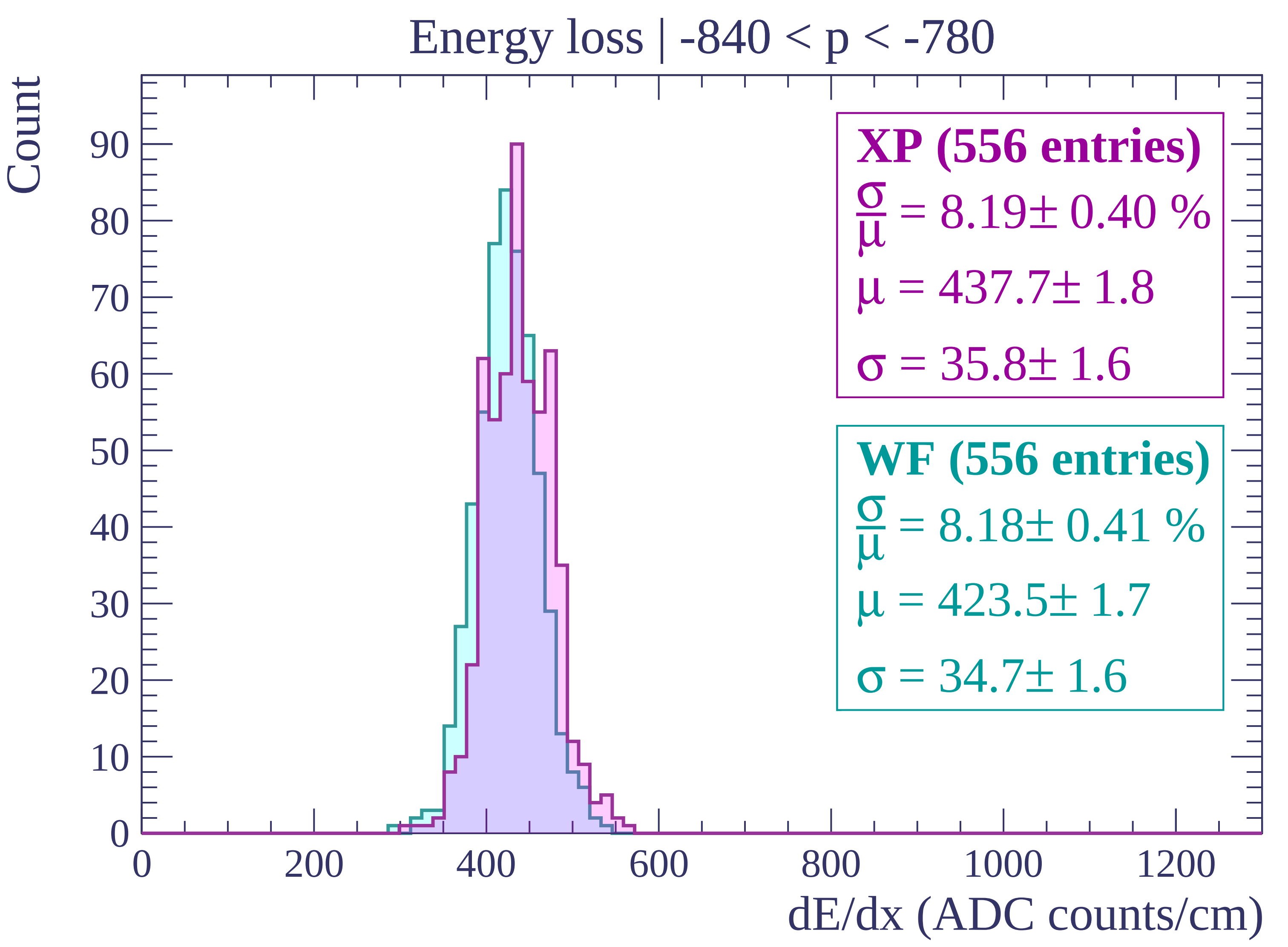


Energy loss | -1020 < p < -960
Count
XP (594 entries)
σ/μ = 8.16± 0.37 %
μ = 453.2± 1.7
σ = 37.0± 1.6
WF (594 entries)
σ/μ = 8.08± 0.42 %
μ = 437.9± 1.7
σ = 35.4± 1.7
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200

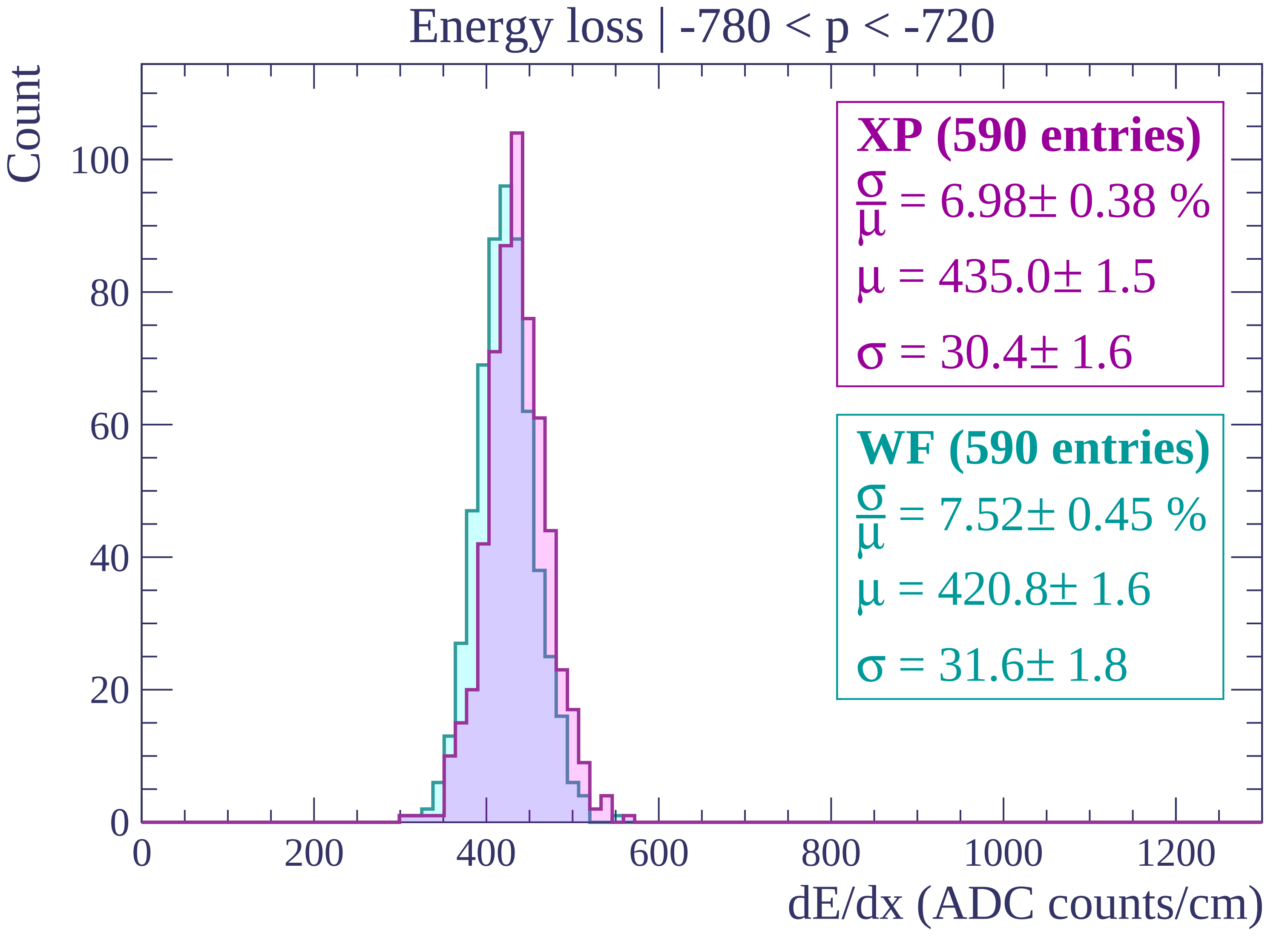


Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (614 entries)
σ/μ = 7.94± 0.51 %
μ = 445.4± 1.8
σ = 35.4± 2.1
WF (614 entries)
σ/μ = 8.71± 0.48 %
μ = 432.8± 1.8
σ = 37.7± 1.9

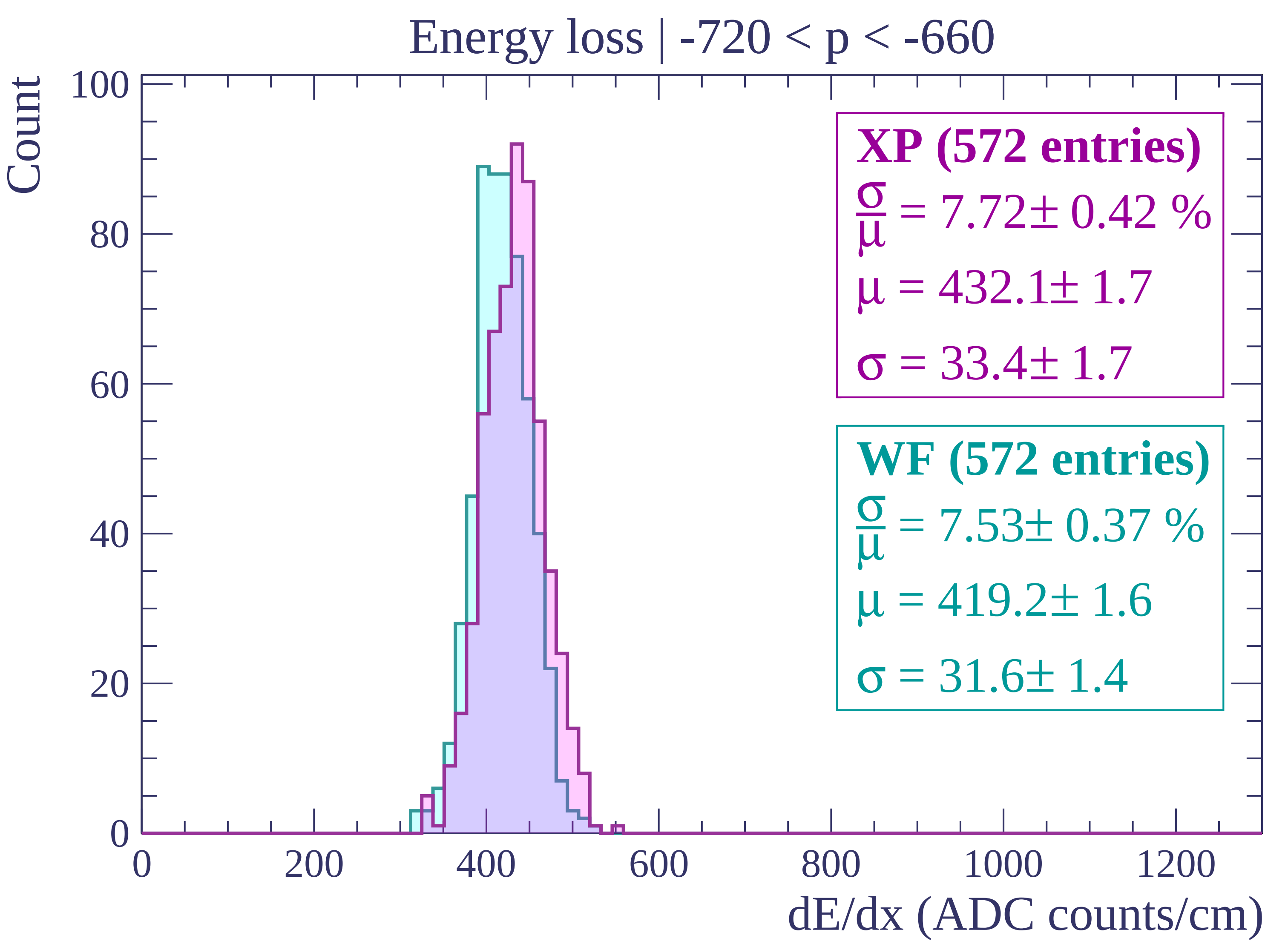

Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
XP (572 entries)
σ/μ = 9.09± 0.50 %
μ = 442.7± 2.0
σ = 40.2± 2.0
WF (572 entries)
σ/μ = 8.16± 0.36 %
μ = 429.5± 1.7
σ = 35.0± 1.4
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200



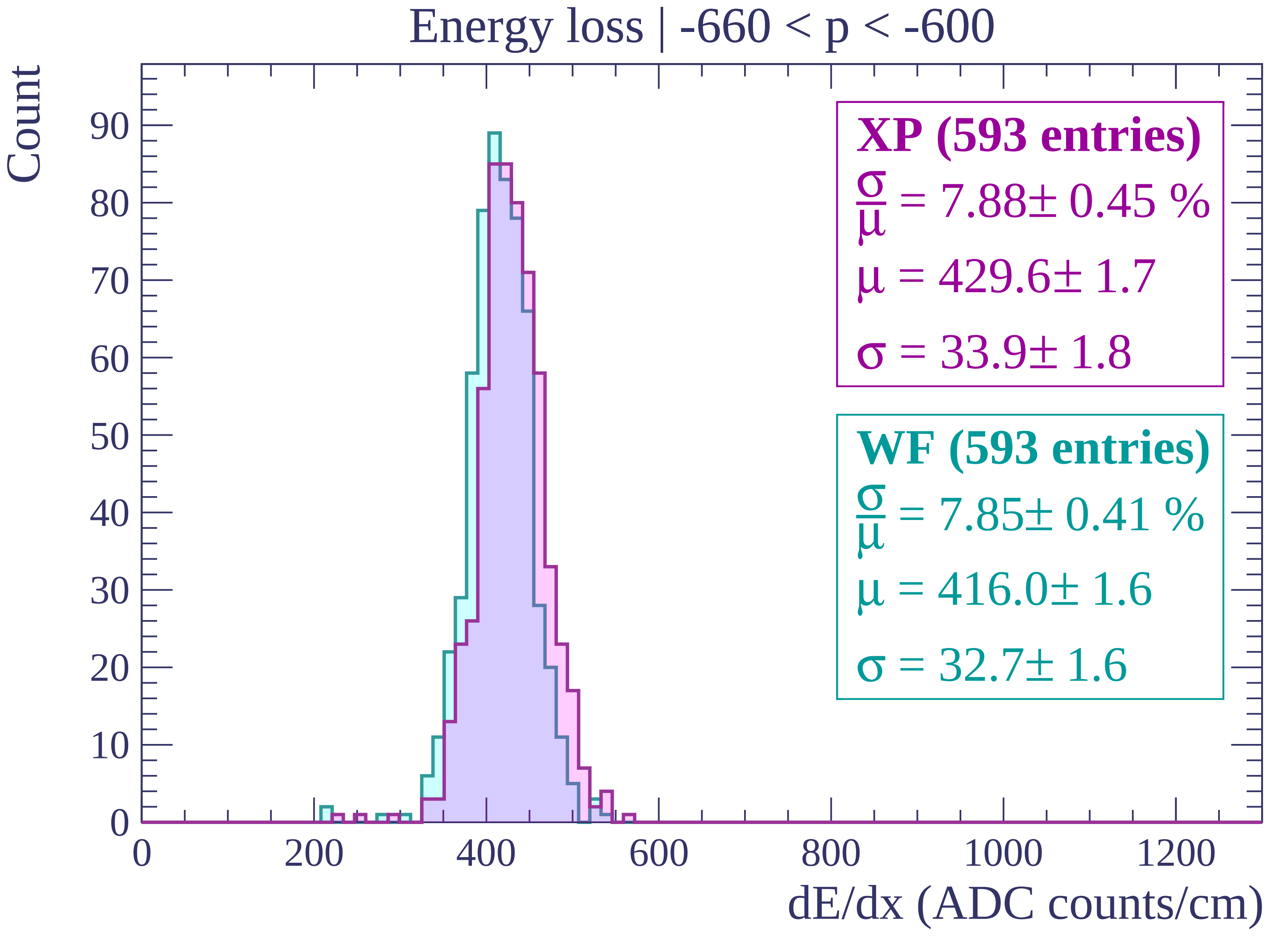
Energy loss | -840 < p < -780
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (556 entries)
σ/μ = 8.19± 0.40 %
μ = 437.7± 1.8
σ = 35.8± 1.6
WF (556 entries)
σ/μ = 8.18± 0.41 %
μ = 423.5± 1.7
σ = 34.7± 1.6



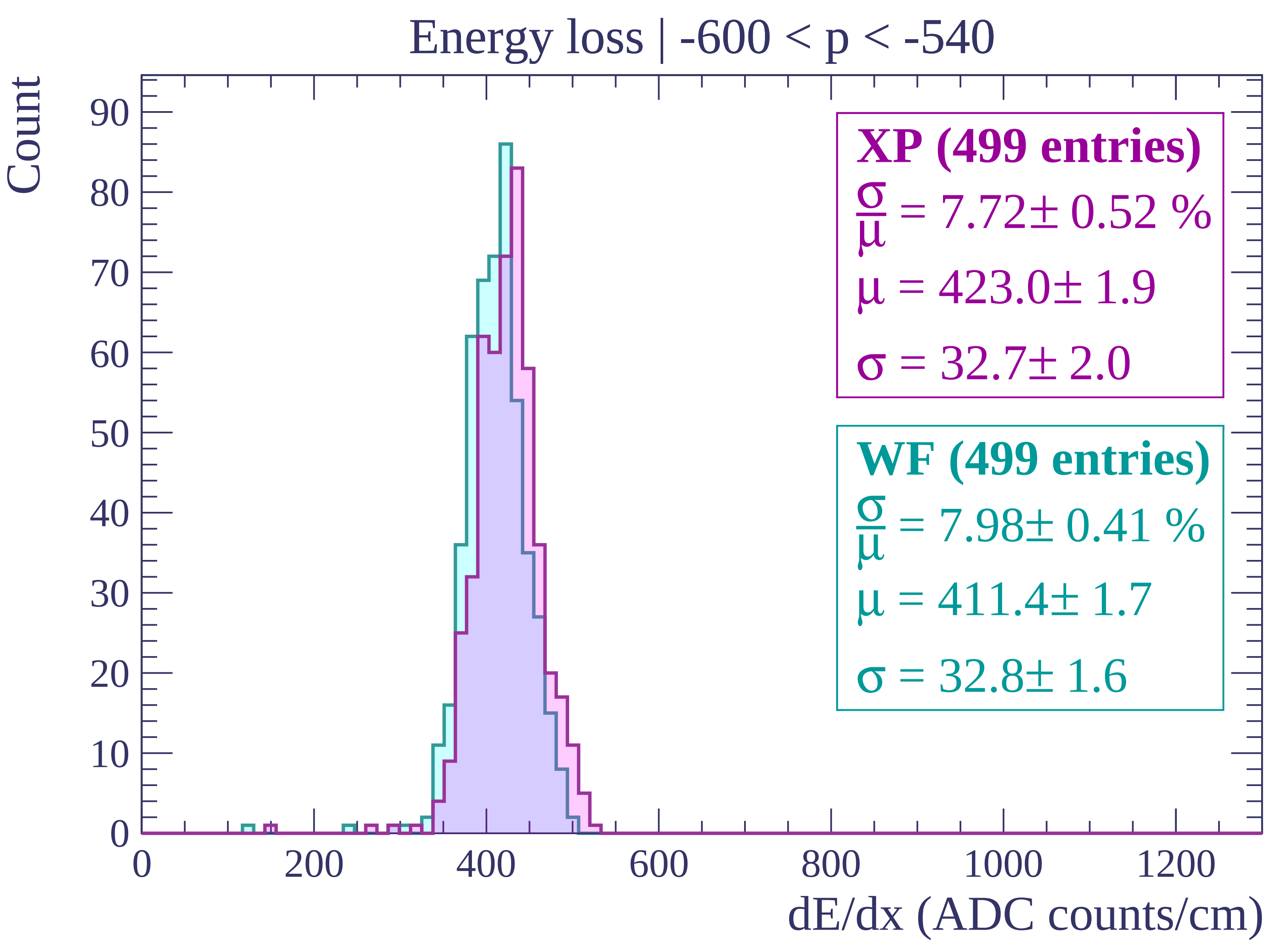
Energy loss | -780 < p < -720
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (590 entries)
σ/μ = 6.98± 0.38 %
μ = 435.0± 1.5
σ = 30.4± 1.6
WF (590 entries)
σ/μ = 7.52± 0.45 %
μ = 420.8± 1.6
σ = 31.6± 1.8



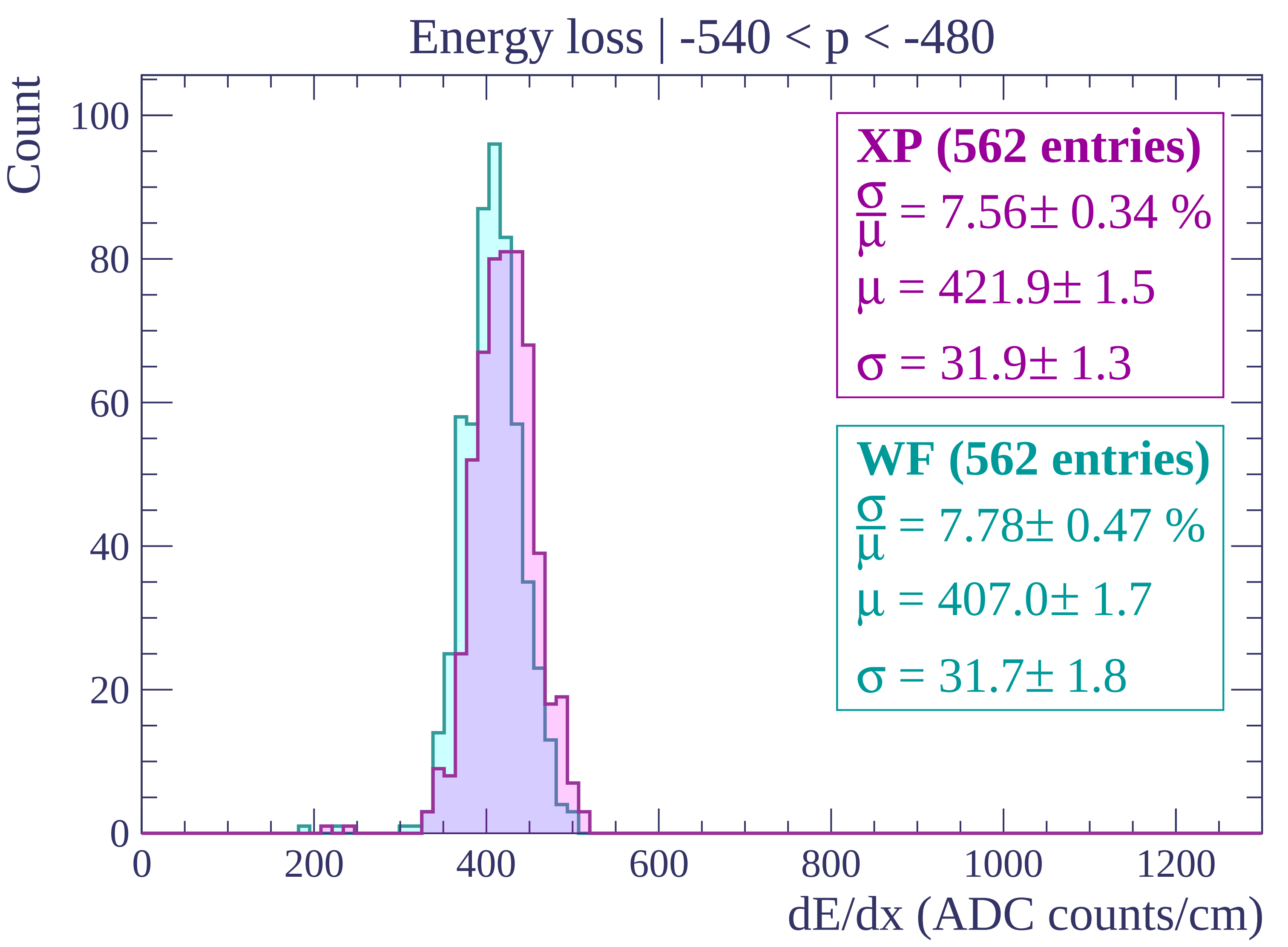
Energy loss | -720 < p < -660
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (572 entries)
σ/μ = 7.72± 0.42 %
μ = 432.1± 1.7
σ = 33.4± 1.7
WF (572 entries)
σ/μ = 7.53± 0.37 %
μ = 419.2± 1.6
σ = 31.6± 1.4



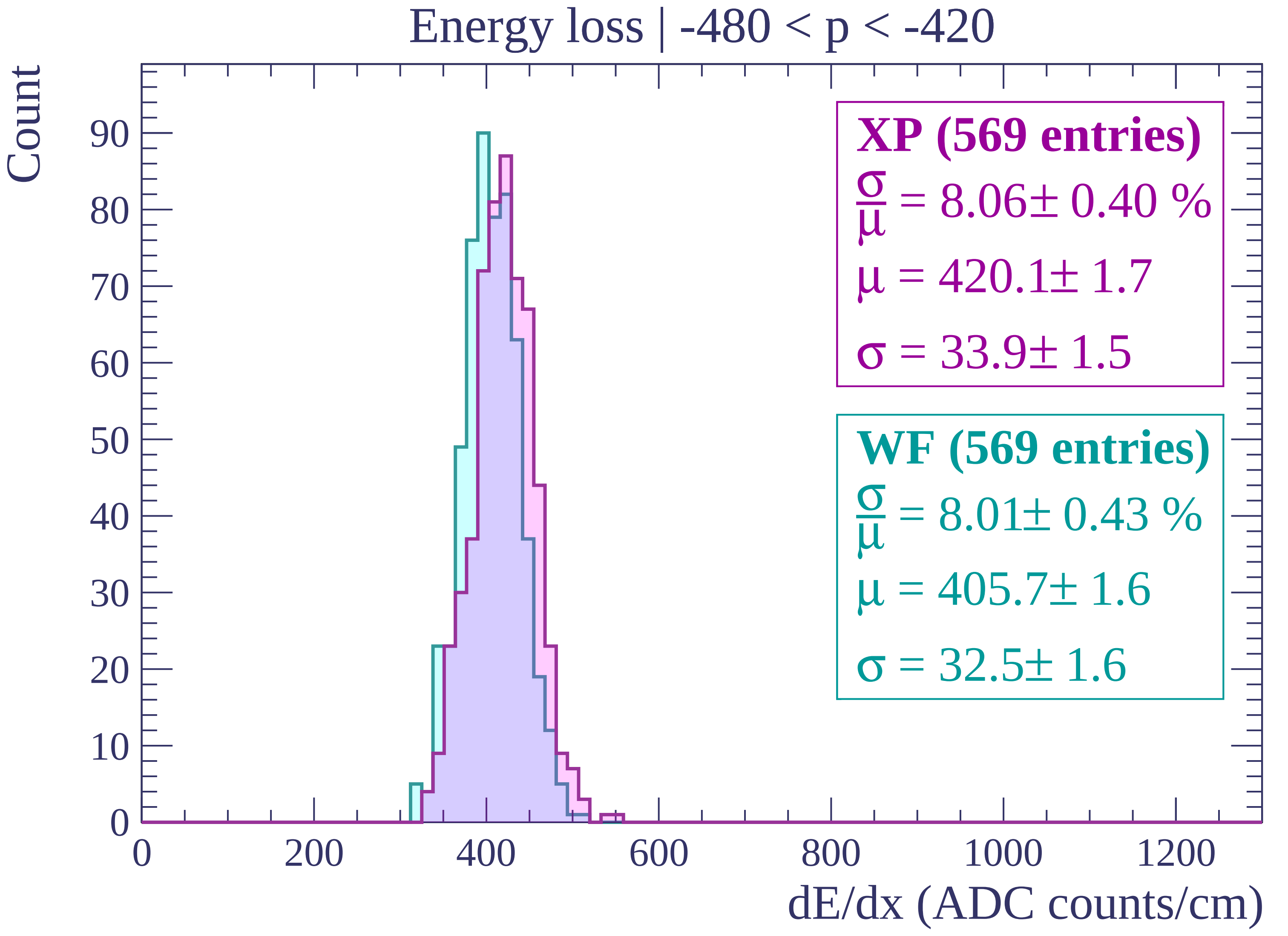
Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (593 entries)
$\frac{\sigma}{\mu}$ = 7.88± 0.45 %
μ = 429.6± 1.7
σ = 33.9± 1.8
WF (593 entries)
$\frac{\sigma}{\mu}$ = 7.85± 0.41 %
μ = 416.0± 1.6
σ = 32.7± 1.6

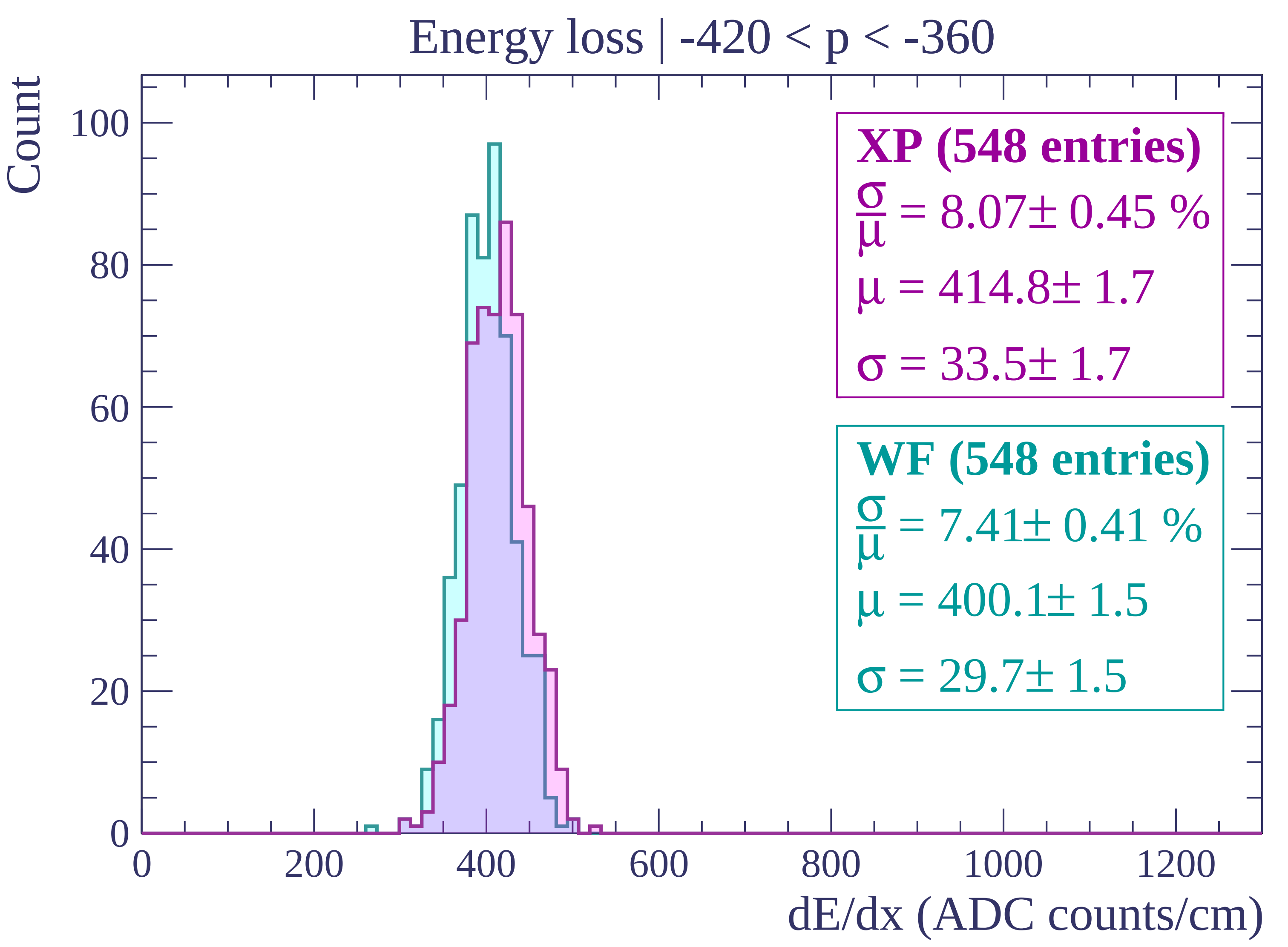


Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (499 entries)
σ/μ = 7.72± 0.52 %
μ = 423.0± 1.9
σ = 32.7± 2.0
WF (499 entries)
σ/μ = 7.98± 0.41 %
μ = 411.4± 1.7
σ = 32.8± 1.6

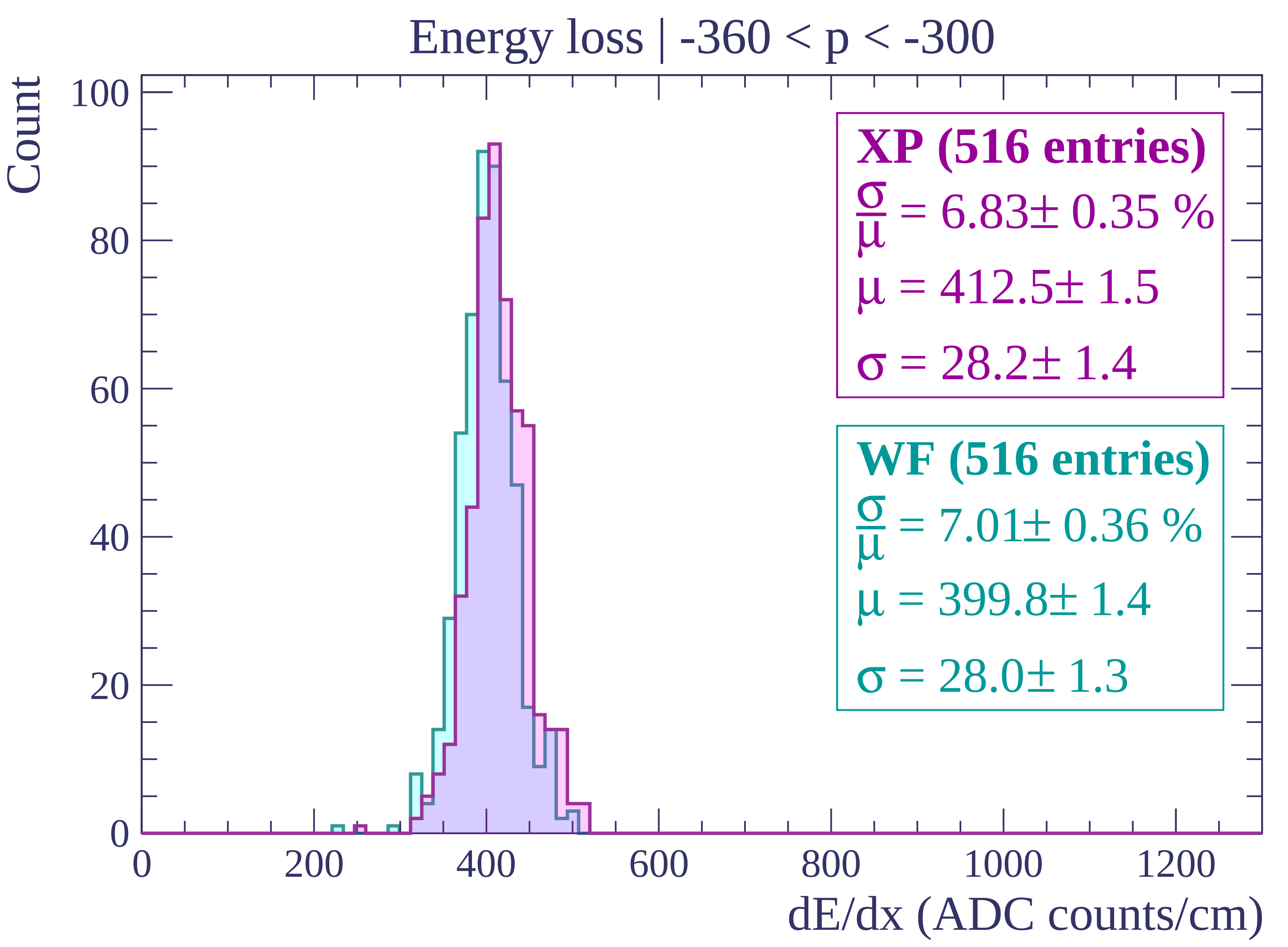


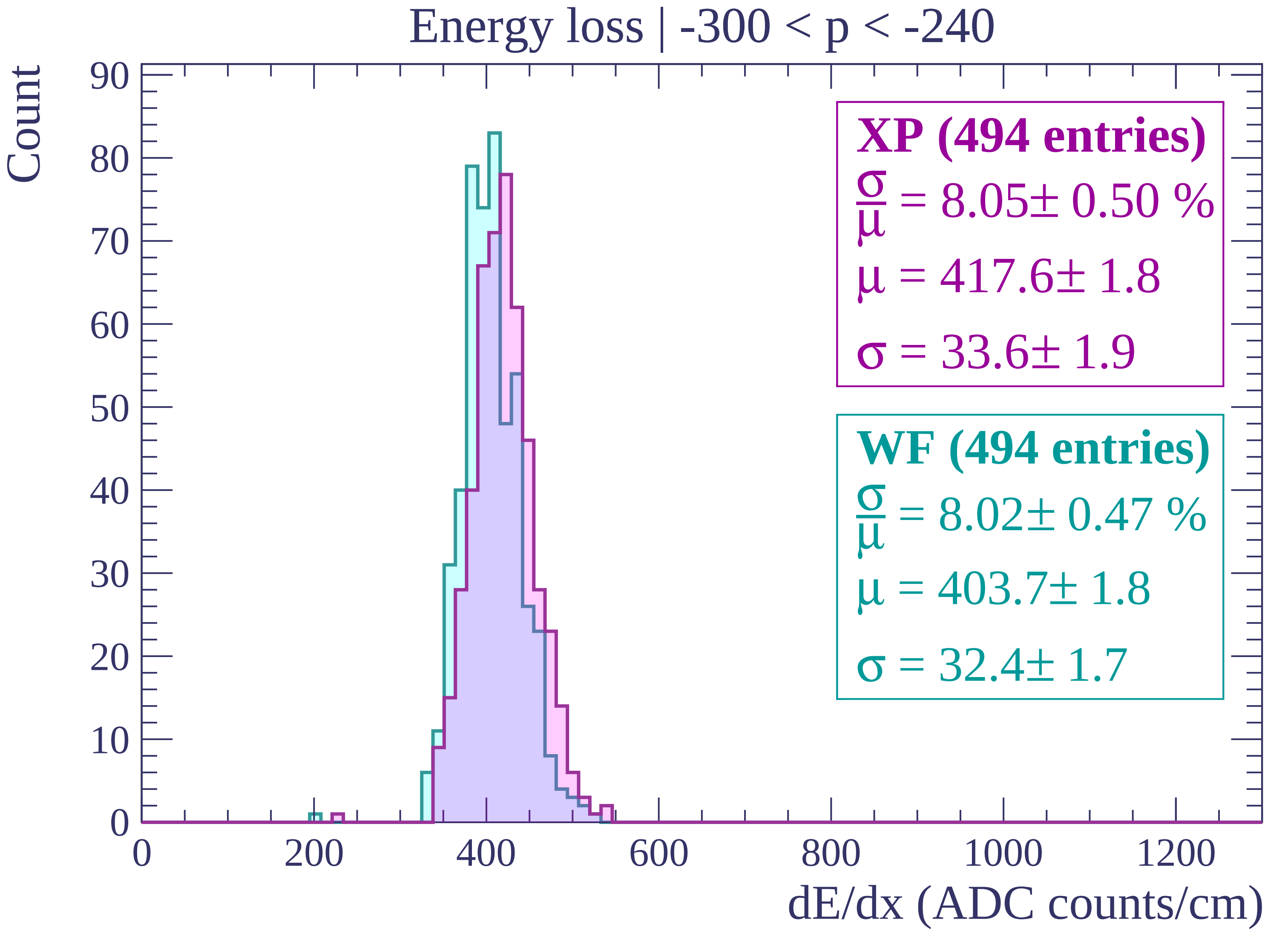
Energy loss | -540 < p < -480
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (562 entries)
σ/μ = 7.56± 0.34 %
μ = 421.9± 1.5
σ = 31.9± 1.3
WF (562 entries)
σ/μ = 7.78± 0.47 %
μ = 407.0± 1.7
σ = 31.7± 1.8


Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (569 entries)
σ/μ = 8.06± 0.40 %
μ = 420.1± 1.7
σ = 33.9± 1.5
WF (569 entries)
σ/μ = 8.01± 0.43 %
μ = 405.7± 1.6
σ = 32.5± 1.6

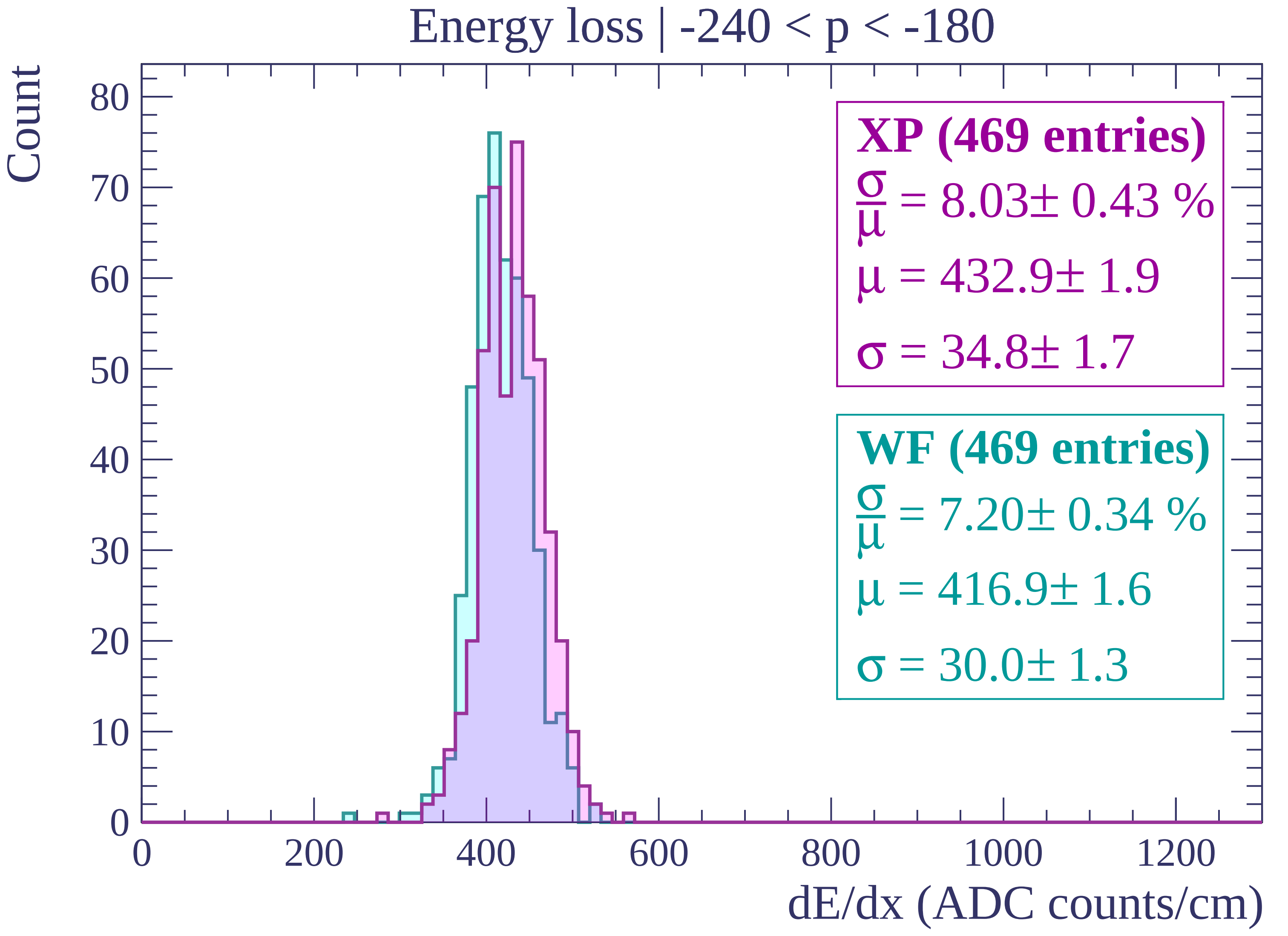


Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (548 entries)
σ/μ = 8.07± 0.45 %
μ = 414.8± 1.7
σ = 33.5± 1.7
WF (548 entries)
σ/μ = 7.41± 0.41 %
μ = 400.1± 1.5
σ = 29.7± 1.5

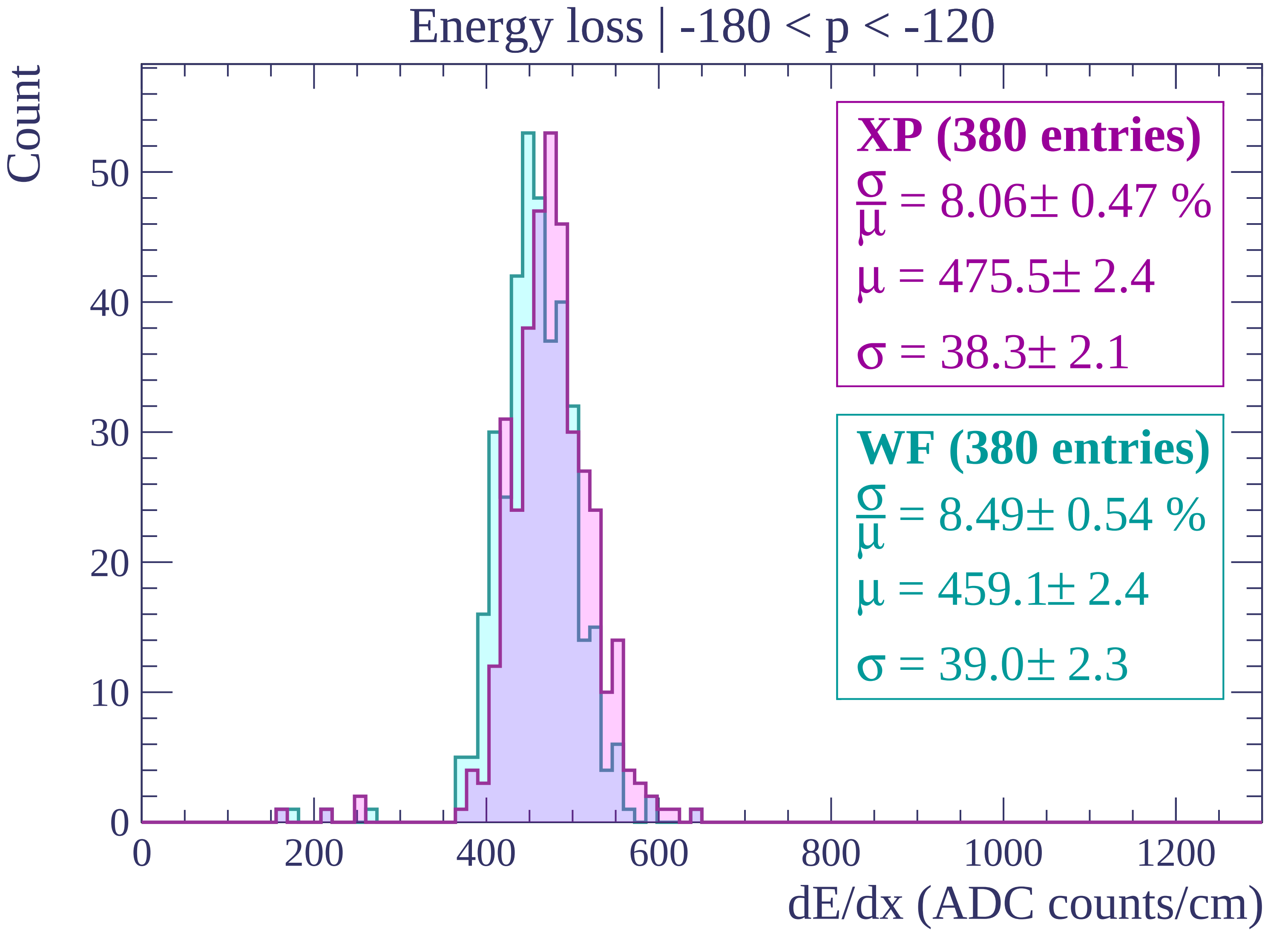

Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (516 entries)
σ/μ = 6.83± 0.35 %
μ = 412.5± 1.5
σ = 28.2± 1.4
WF (516 entries)
σ/μ = 7.01± 0.36 %
μ = 399.8± 1.4
σ = 28.0± 1.3



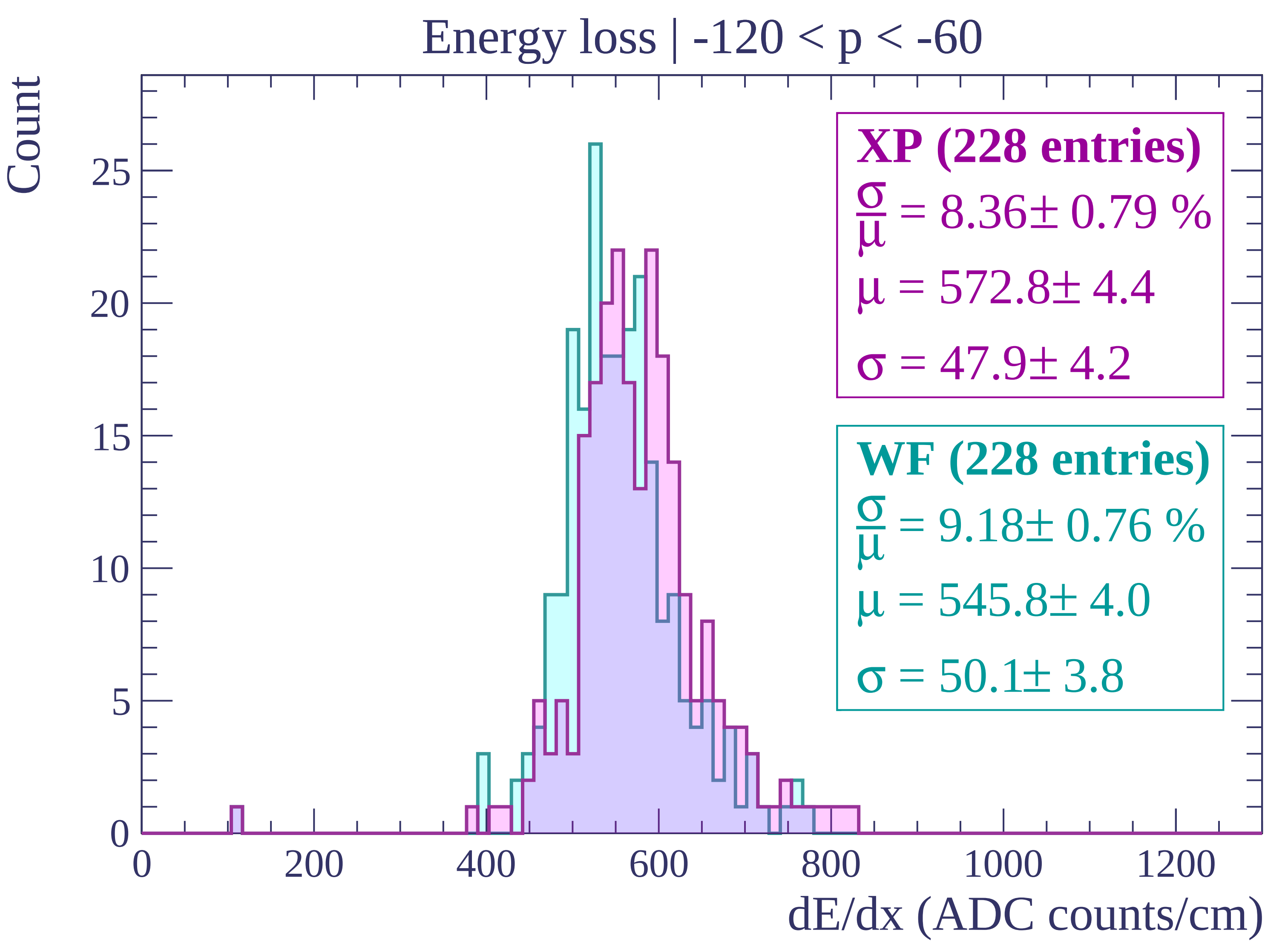
Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
XP (494 entries)
σ/μ = 8.05± 0.50 %
μ = 417.6± 1.8
σ = 33.6± 1.9
WF (494 entries)
σ/μ = 8.02± 0.47 %
μ = 403.7± 1.8
σ = 32.4± 1.7
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200



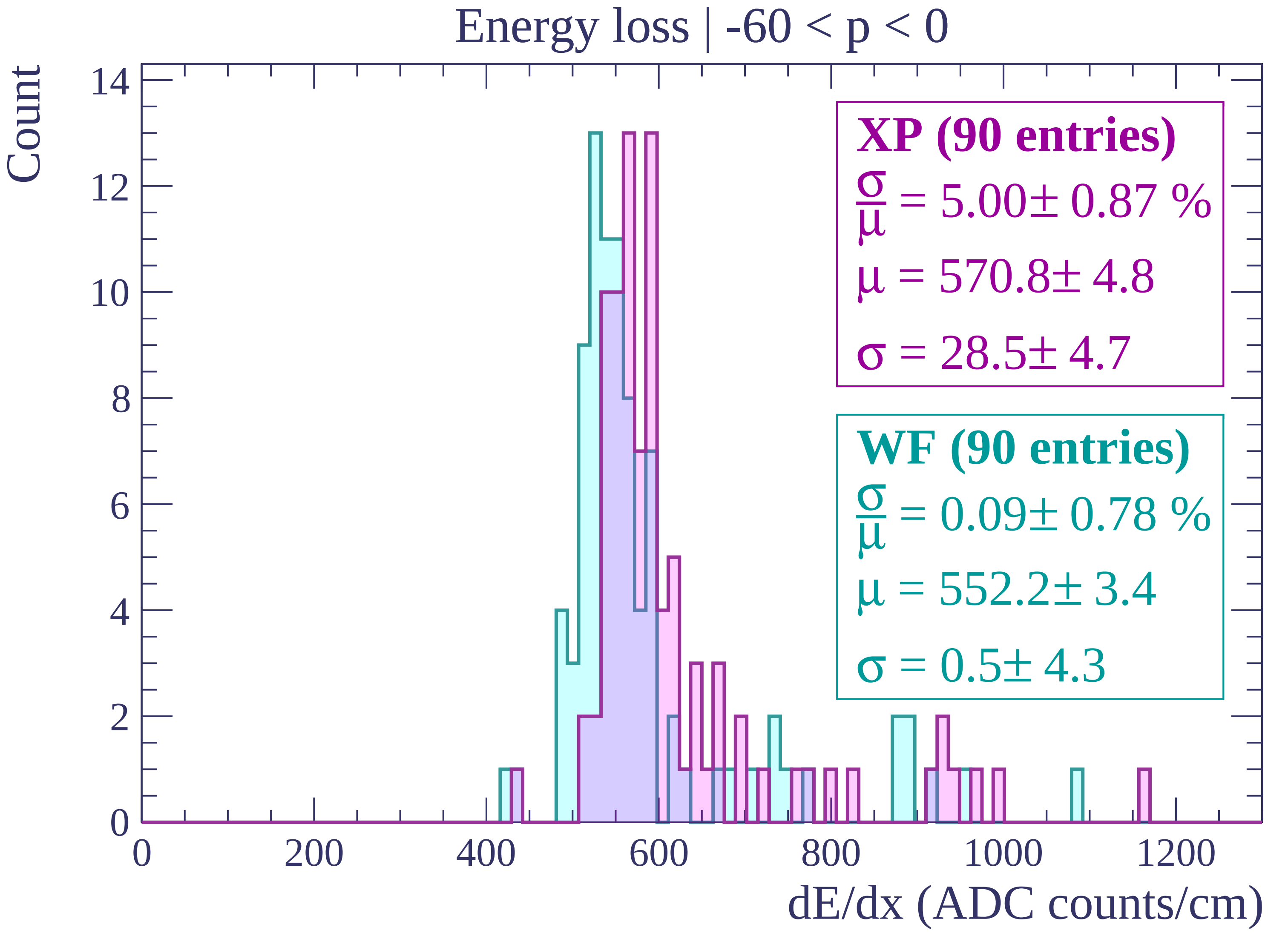
Energy loss | -240 < p < -180
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (469 entries)
σ/μ = 8.03± 0.43 %
μ = 432.9± 1.9
σ = 34.8± 1.7
WF (469 entries)
σ/μ = 7.20± 0.34 %
μ = 416.9± 1.6
σ = 30.0± 1.3

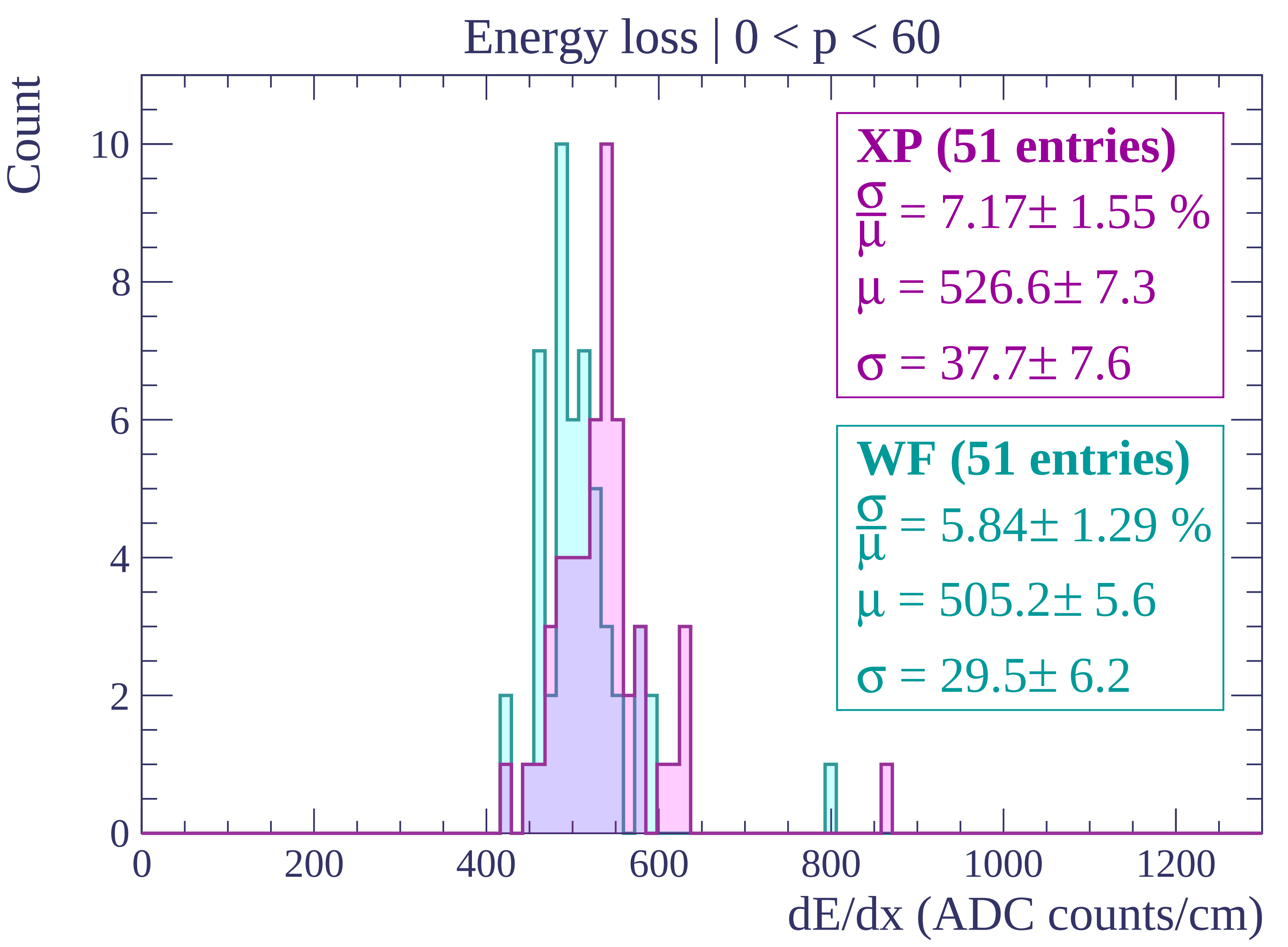


Energy loss | -180 < p < -120
Count
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (380 entries)
σ/μ = 8.06± 0.47 %
μ = 475.5± 2.4
σ = 38.3± 2.1
WF (380 entries)
σ/μ = 8.49± 0.54 %
μ = 459.1± 2.4
σ = 39.0± 2.3

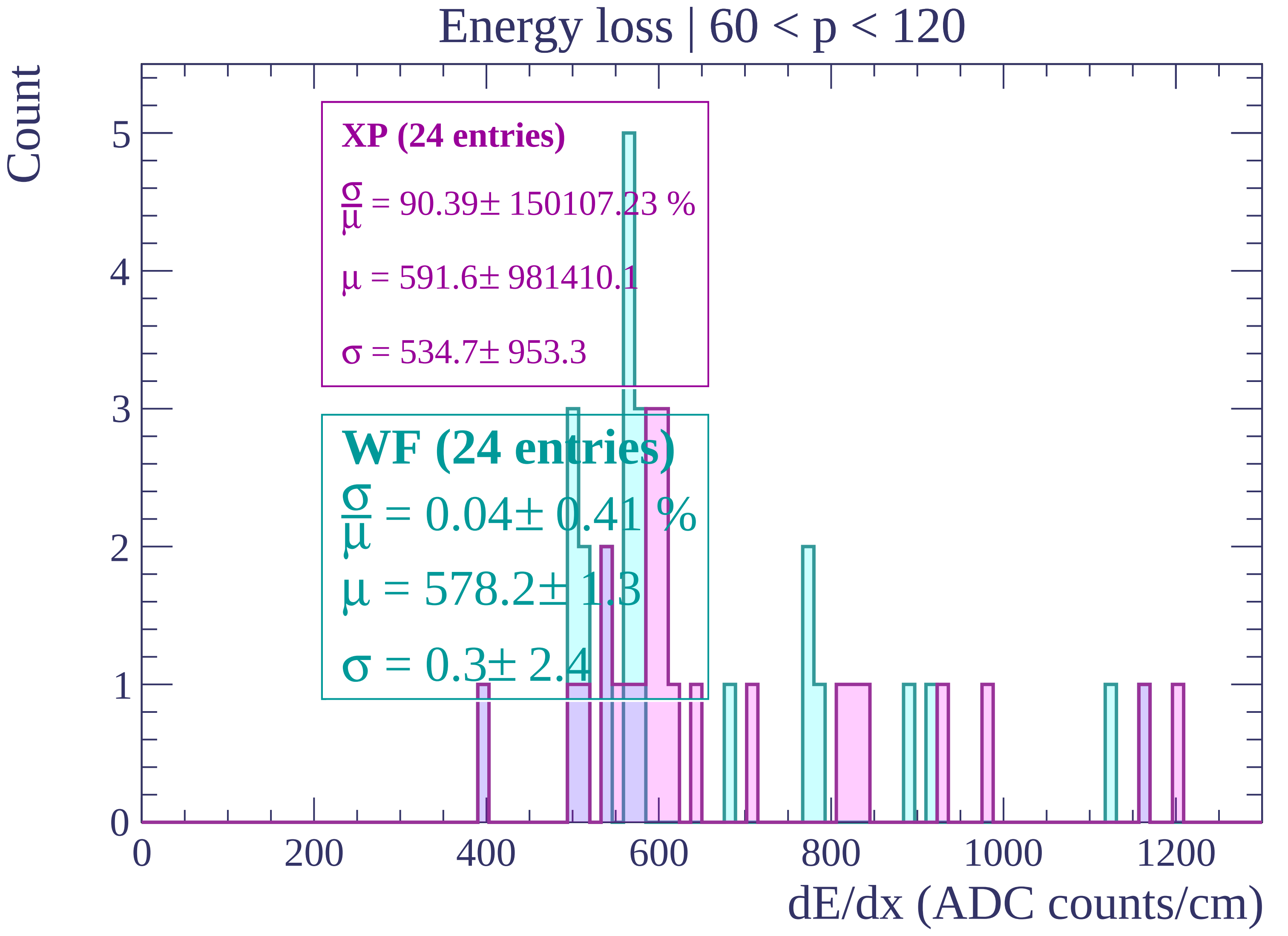


Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP (228 entries)
σ/μ = 8.36± 0.79 %
μ = 572.8± 4.4
σ = 47.9± 4.2
WF (228 entries)
σ/μ = 9.18± 0.76 %
μ = 545.8± 4.0
σ = 50.1± 3.8

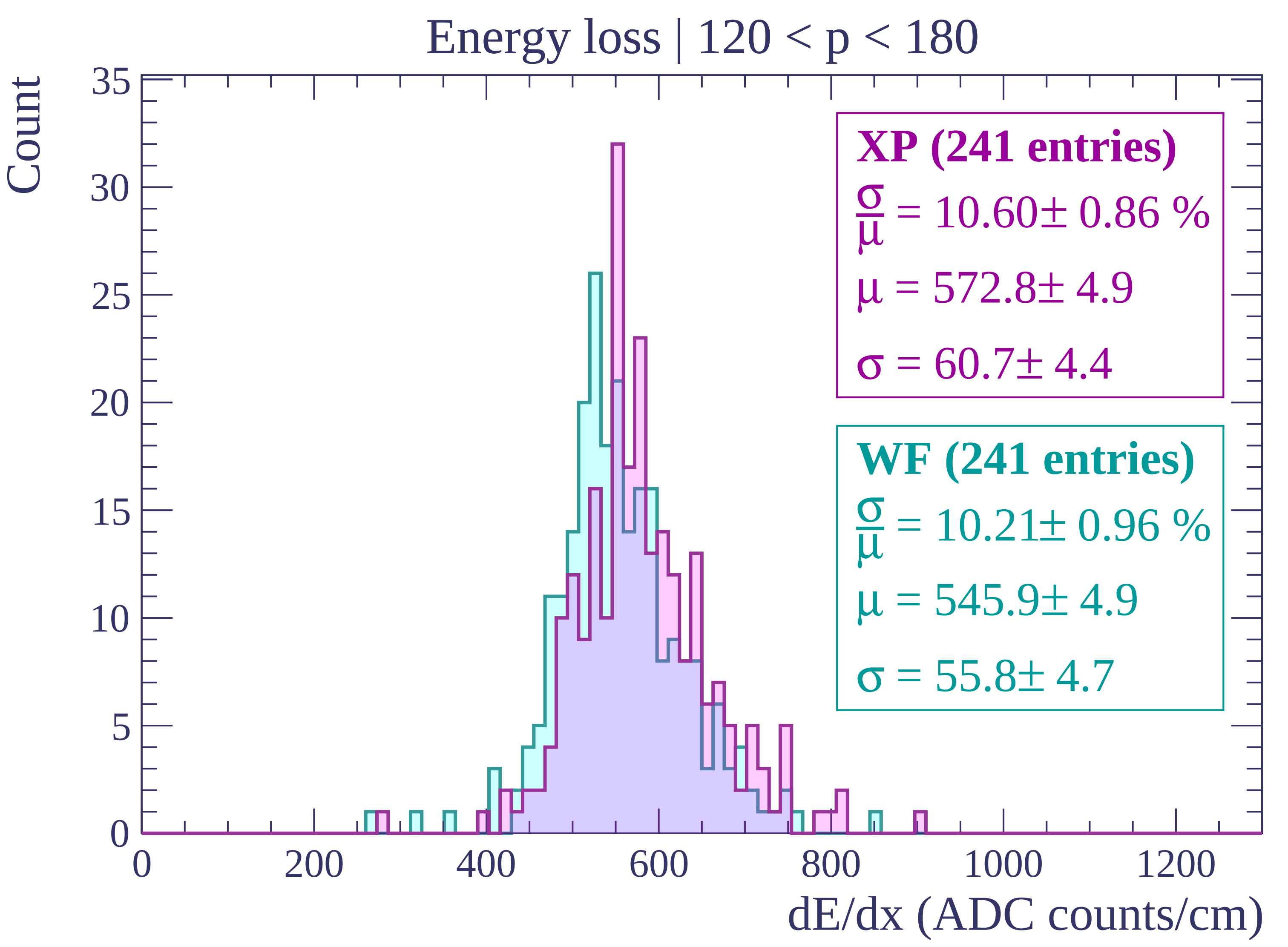

Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
XP (90 entries)
σ/μ = 5.00± 0.87 %
μ = 570.8± 4.8
σ = 28.5± 4.7
WF (90 entries)
σ/μ = 0.09± 0.78 %
μ = 552.2± 3.4
σ = 0.5± 4.3

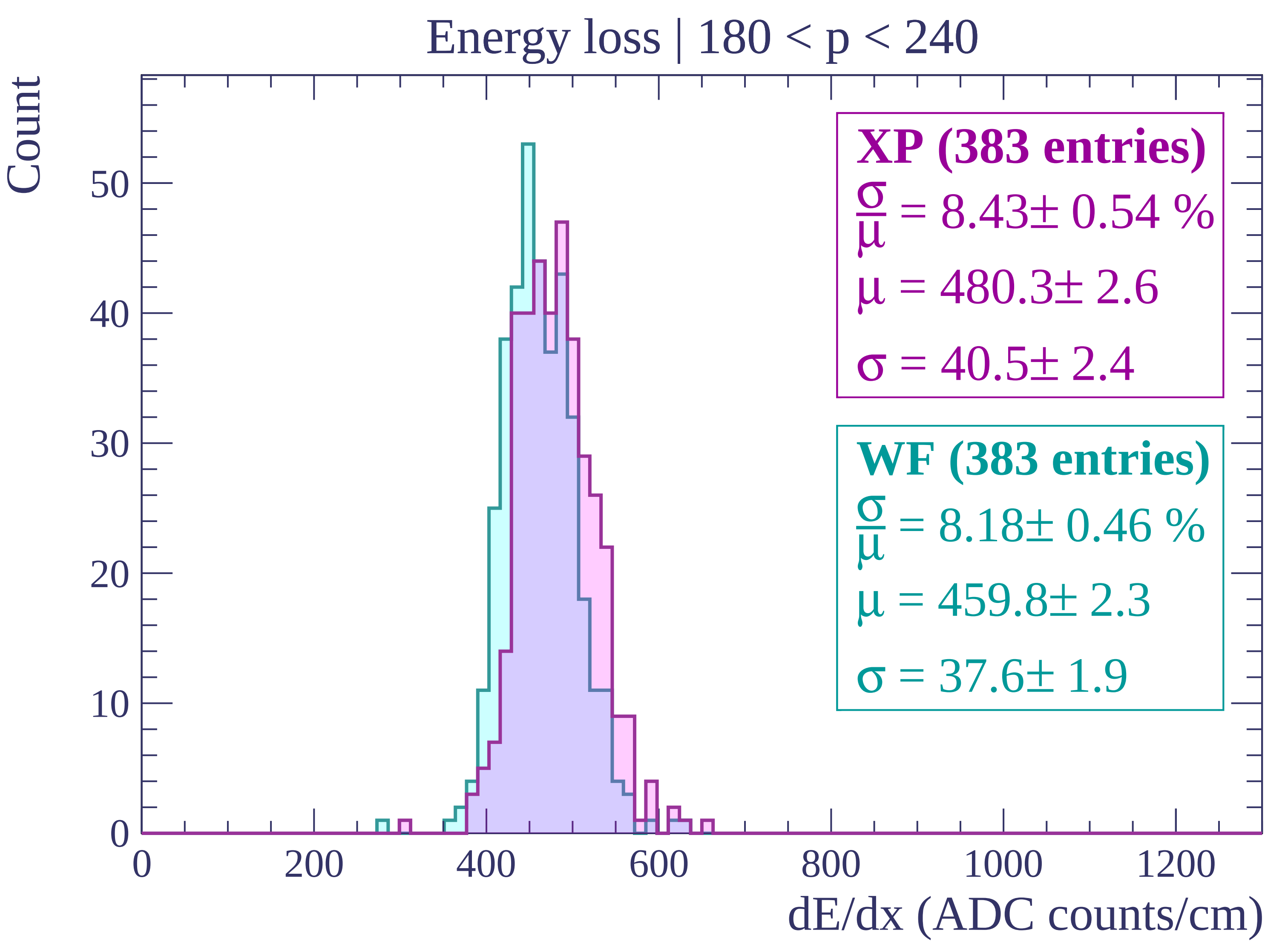


Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
XP (51 entries)
σ/μ = 7.17± 1.55 %
μ = 526.6± 7.3
σ = 37.7± 7.6
WF (51 entries)
σ/μ = 5.84± 1.29 %
μ = 505.2± 5.6
σ = 29.5± 6.2

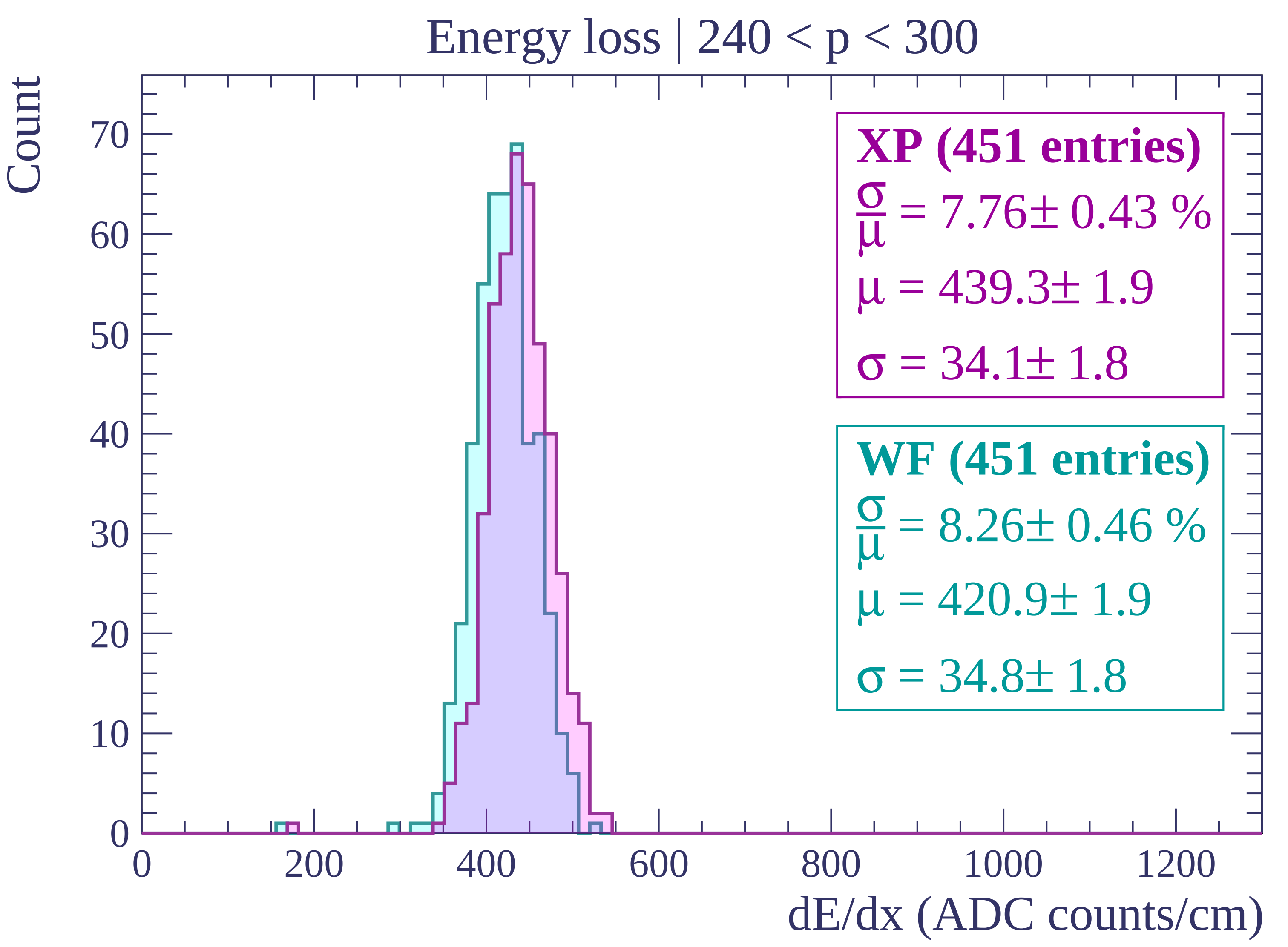


Energy loss | 60 < p < 120
Count
XP (24 entries)
σ/μ = 90.39± 150107.23 %
μ = 591.6± 981410.1
σ = 534.7± 953.3
WF (24 entries)
σ/μ = 0.04± 0.41 %
μ = 578.2± 1.3
σ = 0.3± 2.4
0
1
2
3
4
5
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

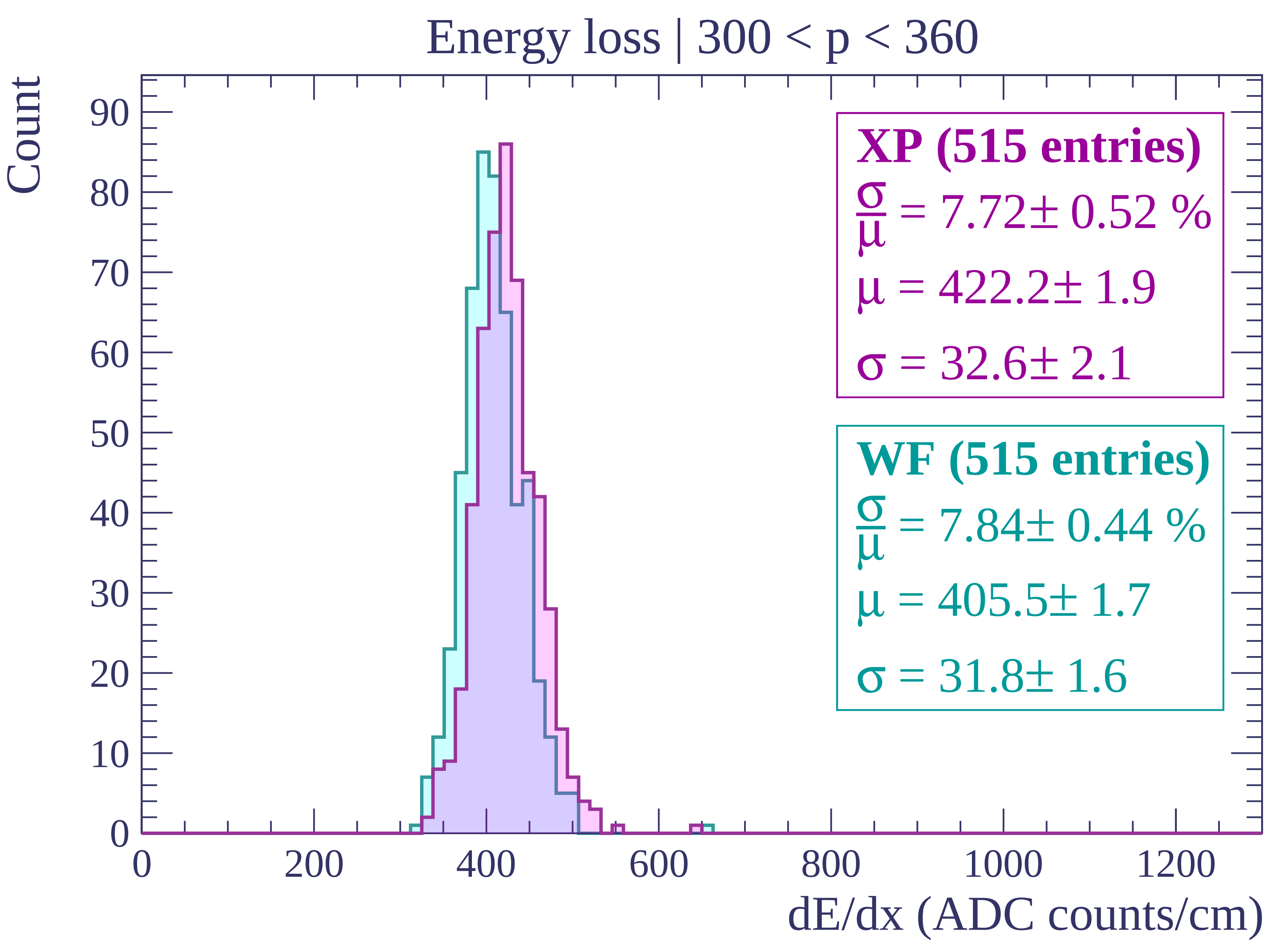

Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP (241 entries)
σ/μ = 10.60± 0.86 %
μ = 572.8± 4.9
σ = 60.7± 4.4
WF (241 entries)
σ/μ = 10.21± 0.96 %
μ = 545.9± 4.9
σ = 55.8± 4.7



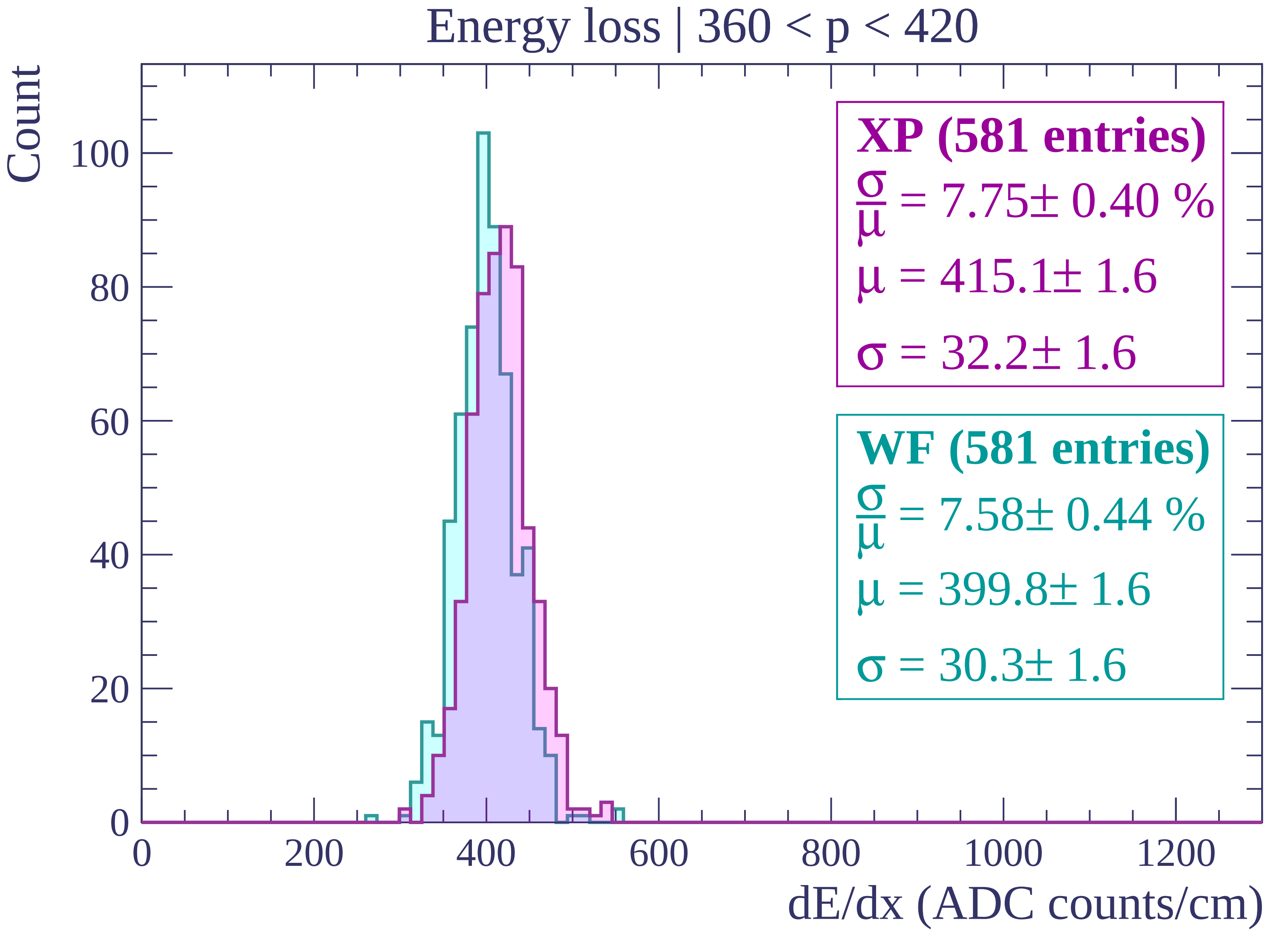
Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP (383 entries)
σ/μ = 8.43± 0.54 %
μ = 480.3± 2.6
σ = 40.5± 2.4
WF (383 entries)
σ/μ = 8.18± 0.46 %
μ = 459.8± 2.3
σ = 37.6± 1.9



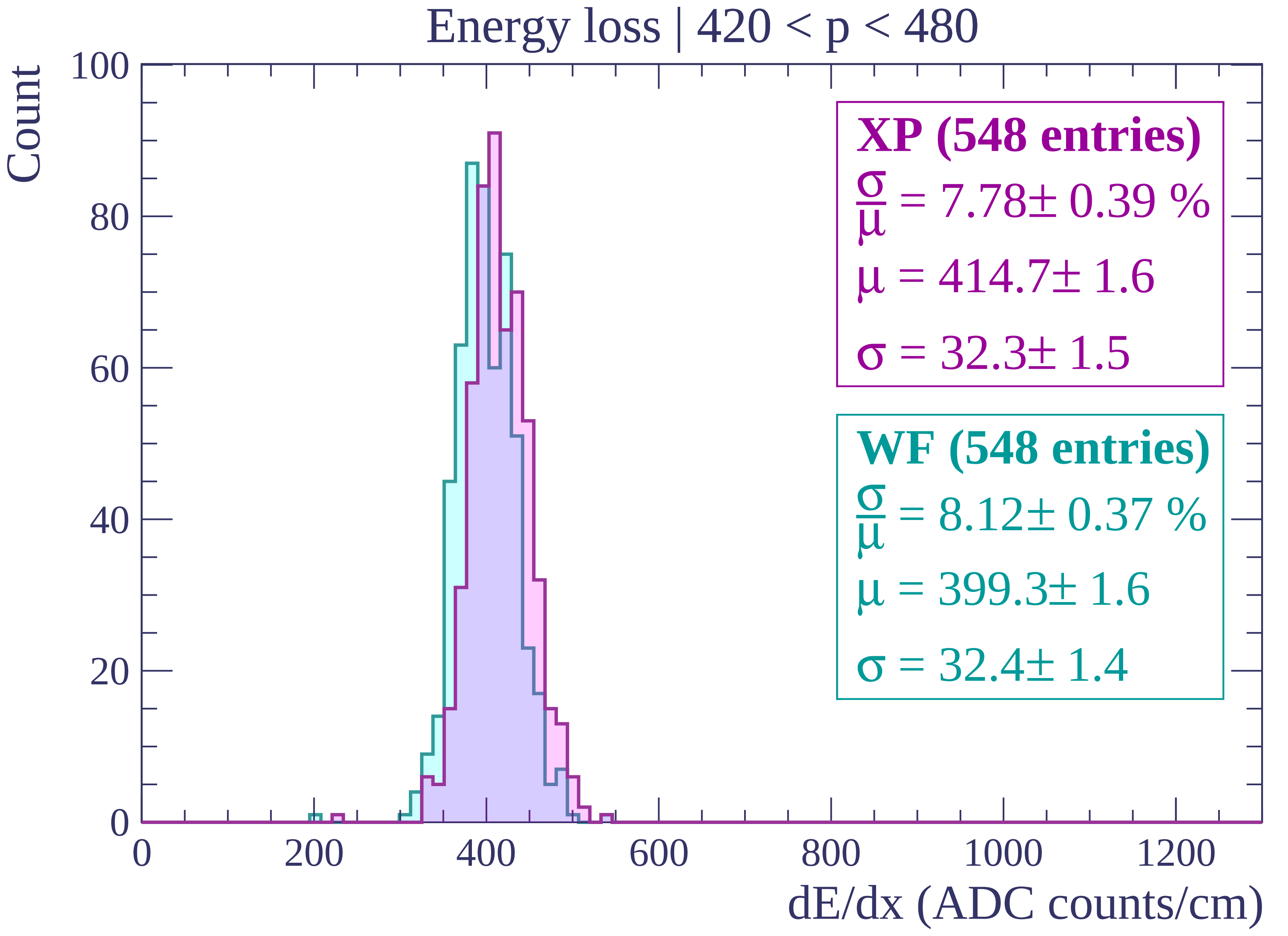
Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
XP (451 entries)
$\frac{\sigma}{\mu}$ = 7.76± 0.43 %
μ = 439.3± 1.9
σ = 34.1± 1.8
WF (451 entries)
$\frac{\sigma}{\mu}$ = 8.26± 0.46 %
μ = 420.9± 1.9
σ = 34.8± 1.8
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200

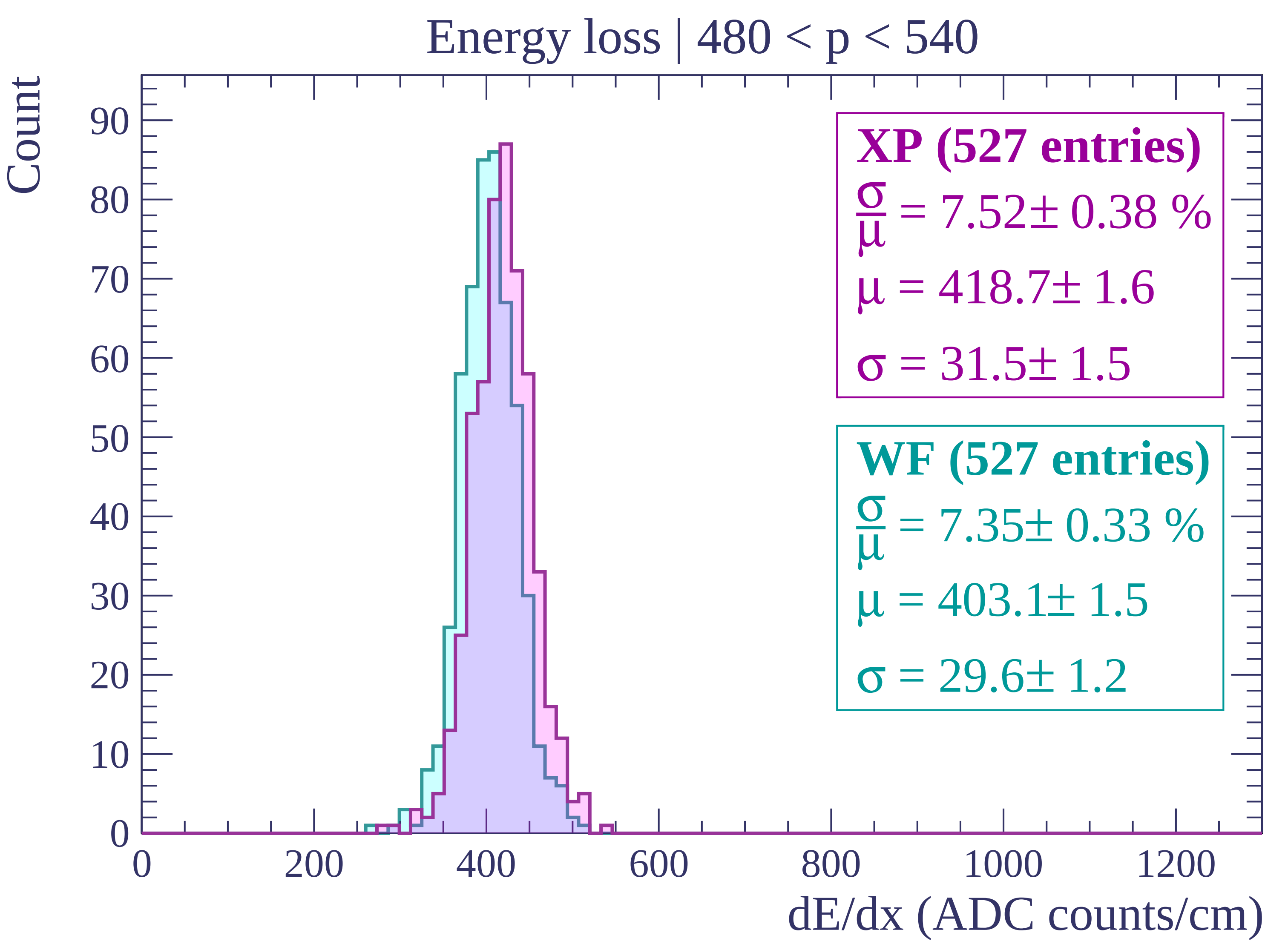


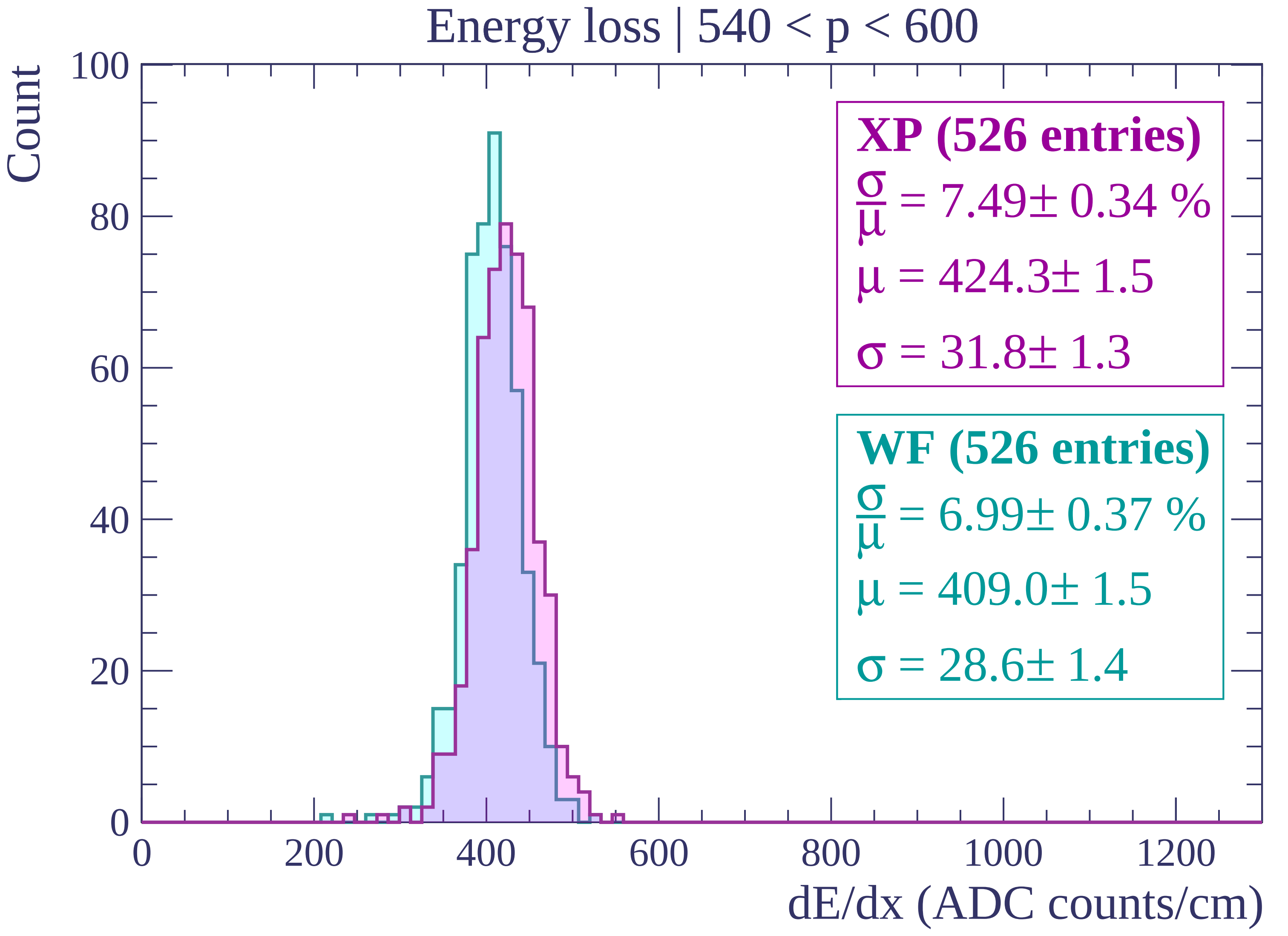
Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (515 entries)
$\frac{\sigma}{\mu}$ = 7.72± 0.52 %
μ = 422.2± 1.9
σ = 32.6± 2.1
WF (515 entries)
$\frac{\sigma}{\mu}$ = 7.84± 0.44 %
μ = 405.5± 1.7
σ = 31.8± 1.6



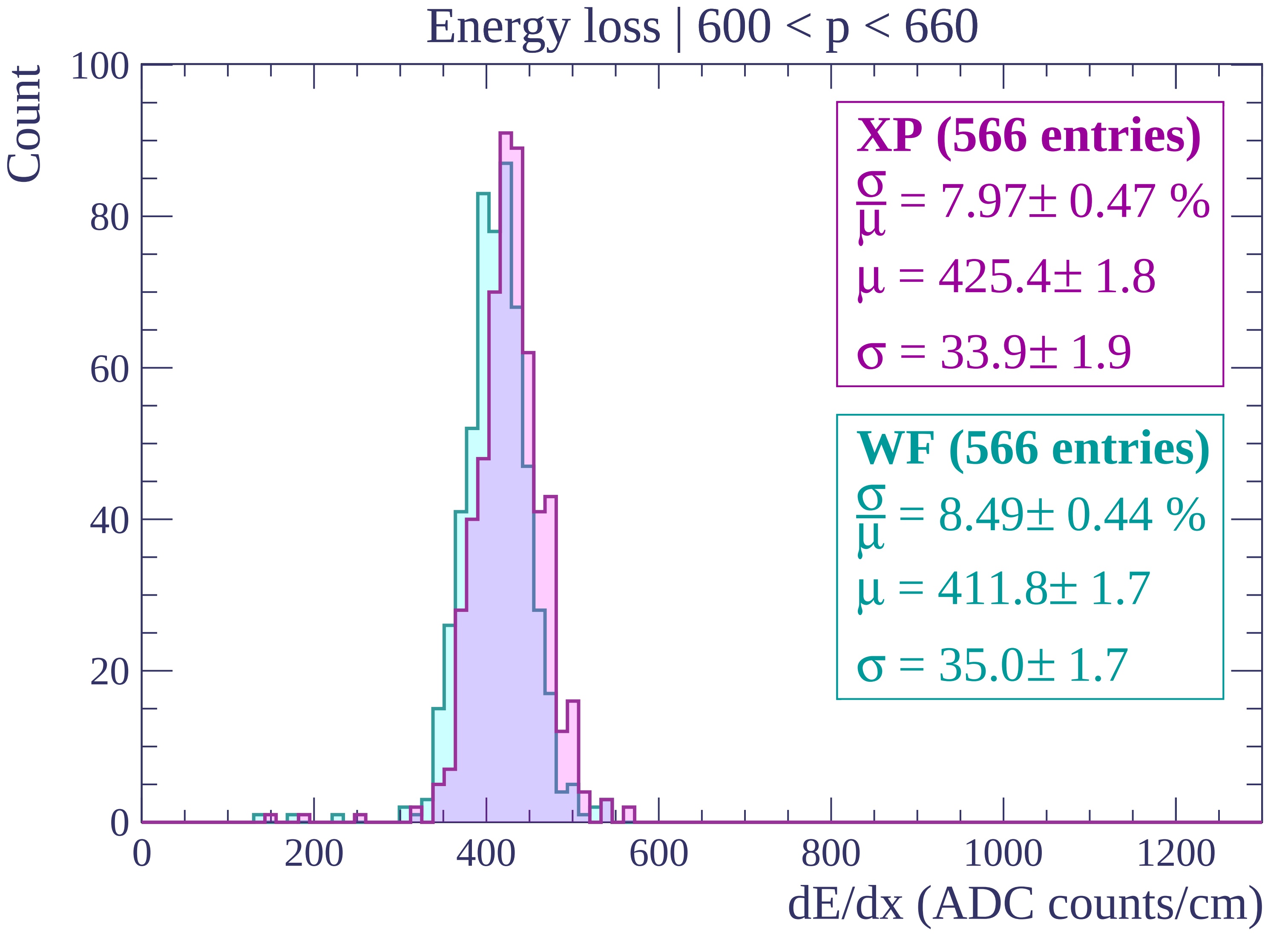
Energy loss | 360 < p < 420
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (581 entries)
σ/μ = 7.75± 0.40 %
μ = 415.1± 1.6
σ = 32.2± 1.6
WF (581 entries)
σ/μ = 7.58± 0.44 %
μ = 399.8± 1.6
σ = 30.3± 1.6


Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
XP (548 entries)
$\frac{\sigma}{\mu}$ = 7.78± 0.39 %
μ = 414.7± 1.6
σ = 32.3± 1.5
WF (548 entries)
$\frac{\sigma}{\mu}$ = 8.12± 0.37 %
μ = 399.3± 1.6
σ = 32.4± 1.4
0
20
40
60
80
100
0
200
400
600
800
1000
1200

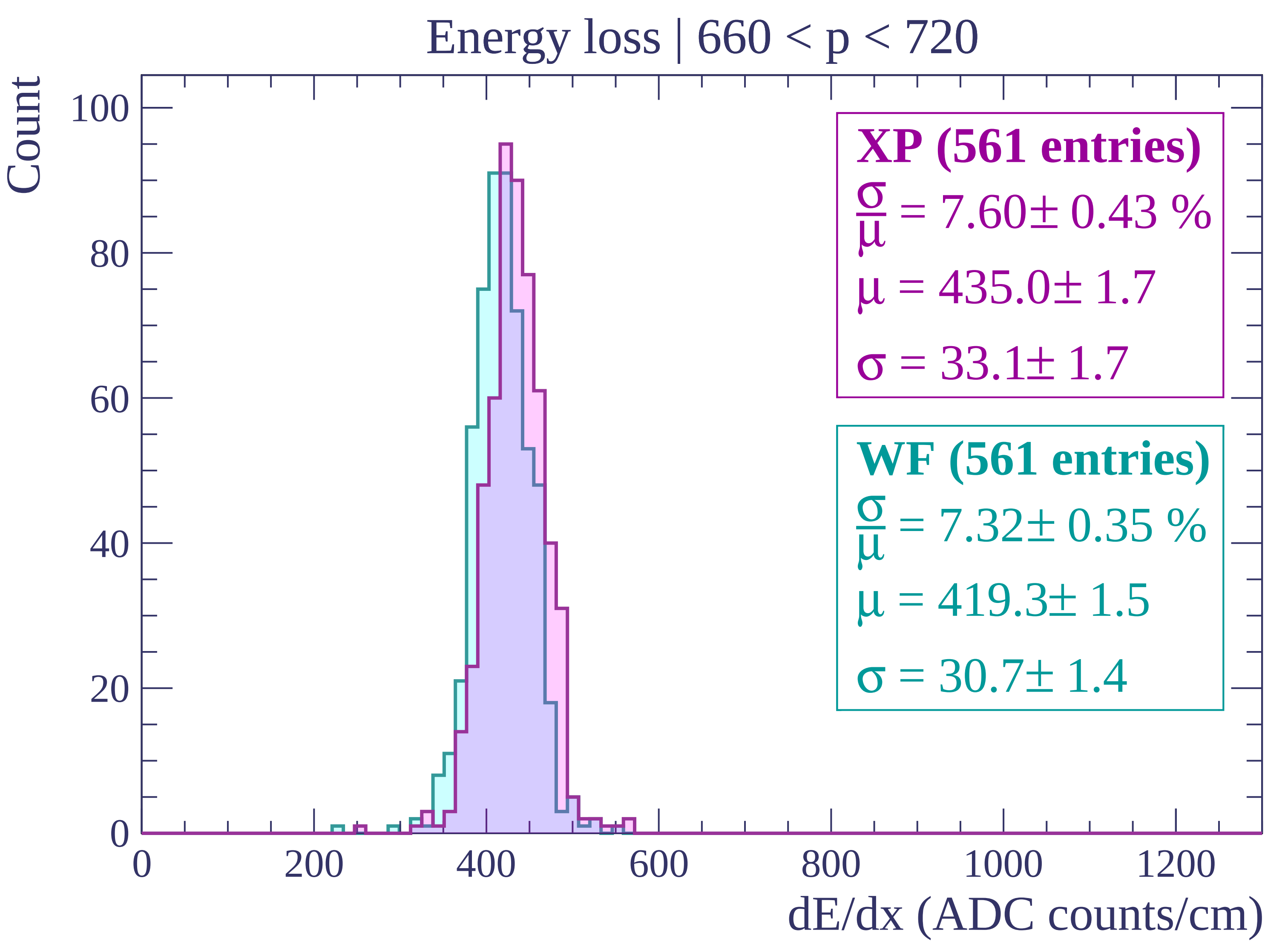

Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (527 entries)
σ/μ = 7.52± 0.38 %
μ = 418.7± 1.6
σ = 31.5± 1.5
WF (527 entries)
σ/μ = 7.35± 0.33 %
μ = 403.1± 1.5
σ = 29.6± 1.2



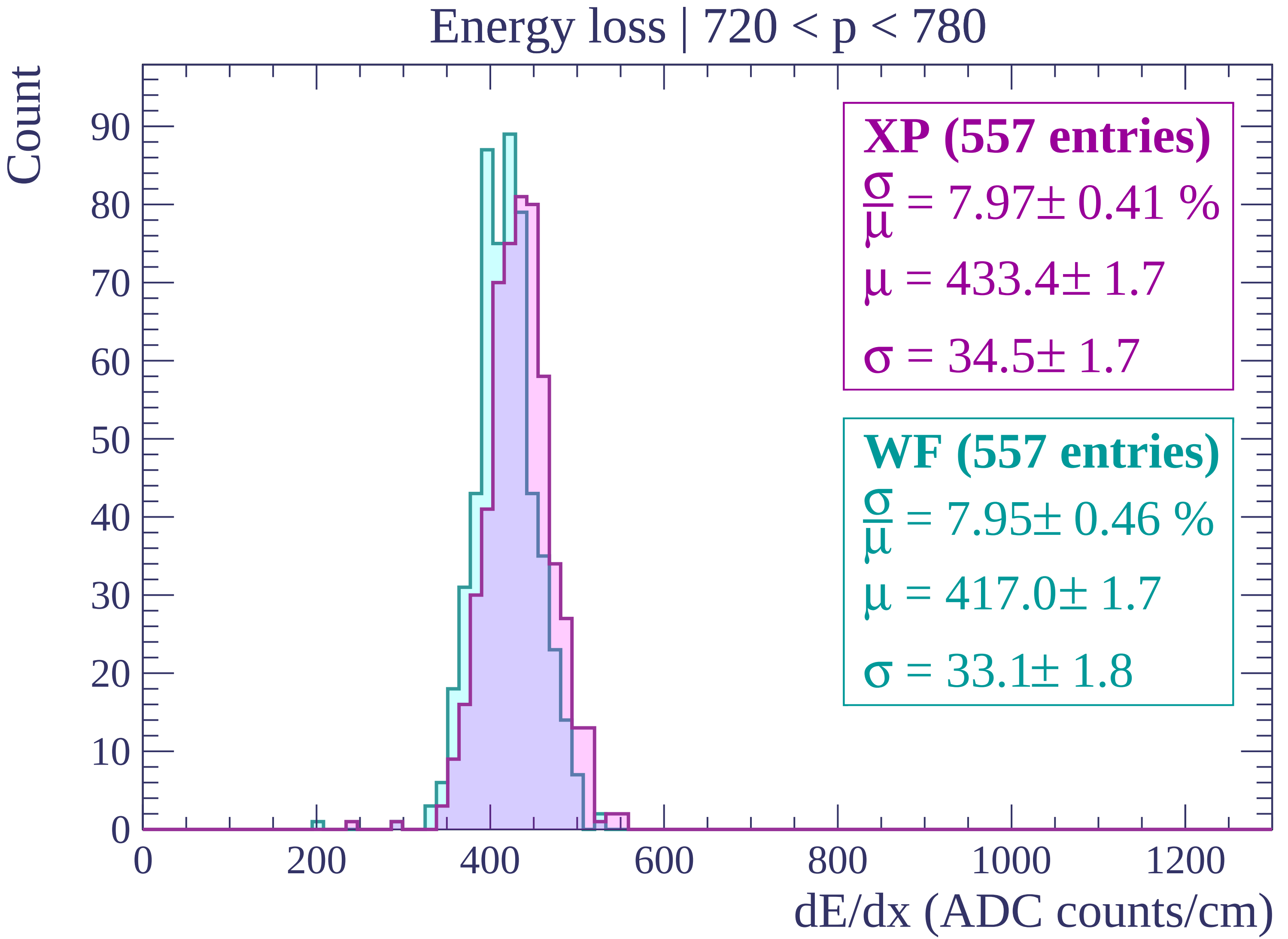
Energy loss | 540 < p < 600
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (526 entries)
σ/μ = 7.49± 0.34 %
μ = 424.3± 1.5
σ = 31.8± 1.3
WF (526 entries)
σ/μ = 6.99± 0.37 %
μ = 409.0± 1.5
σ = 28.6± 1.4



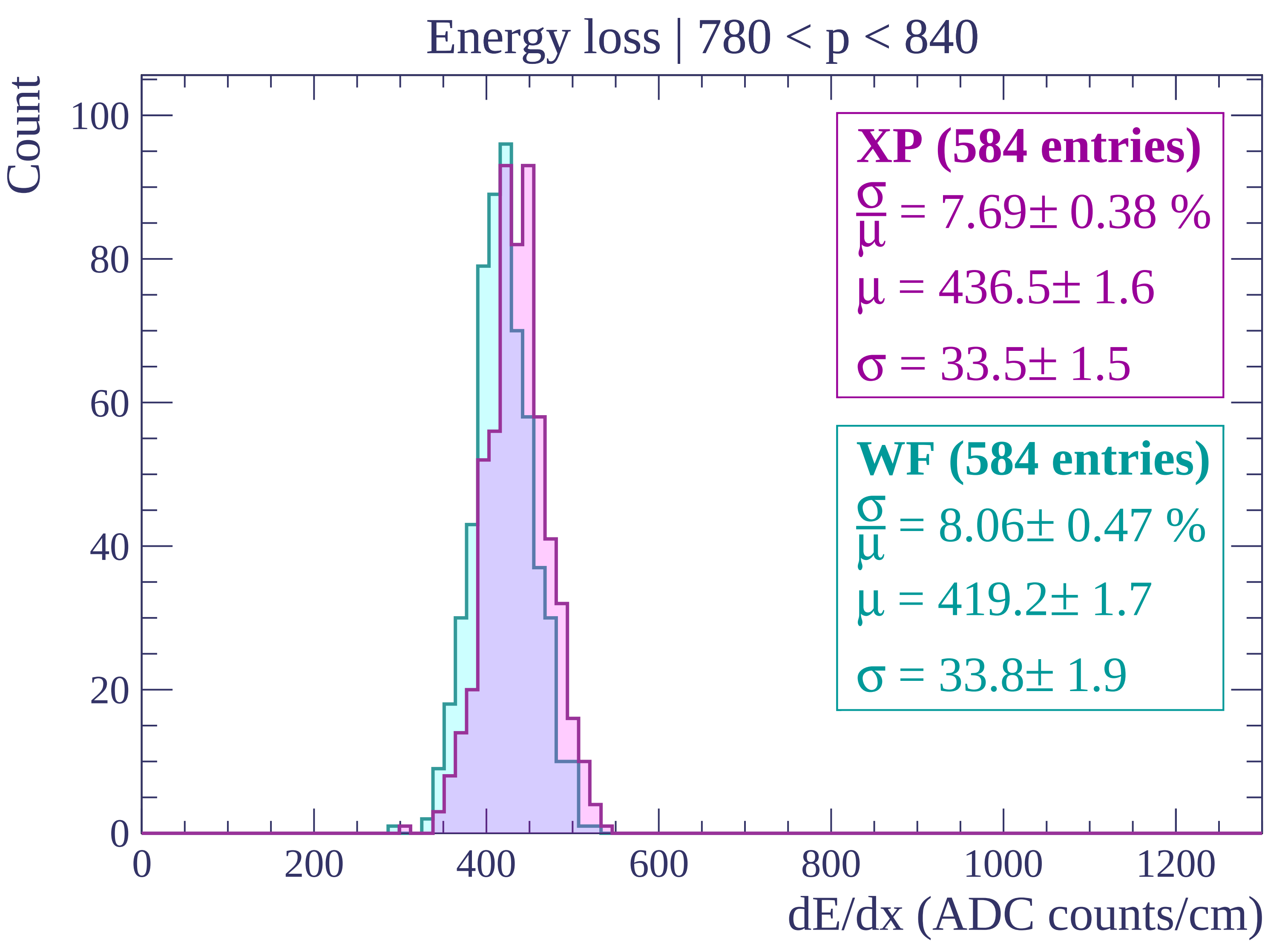
Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (566 entries)
σ/μ = 7.97± 0.47 %
μ = 425.4± 1.8
σ = 33.9± 1.9
WF (566 entries)
σ/μ = 8.49± 0.44 %
μ = 411.8± 1.7
σ = 35.0± 1.7

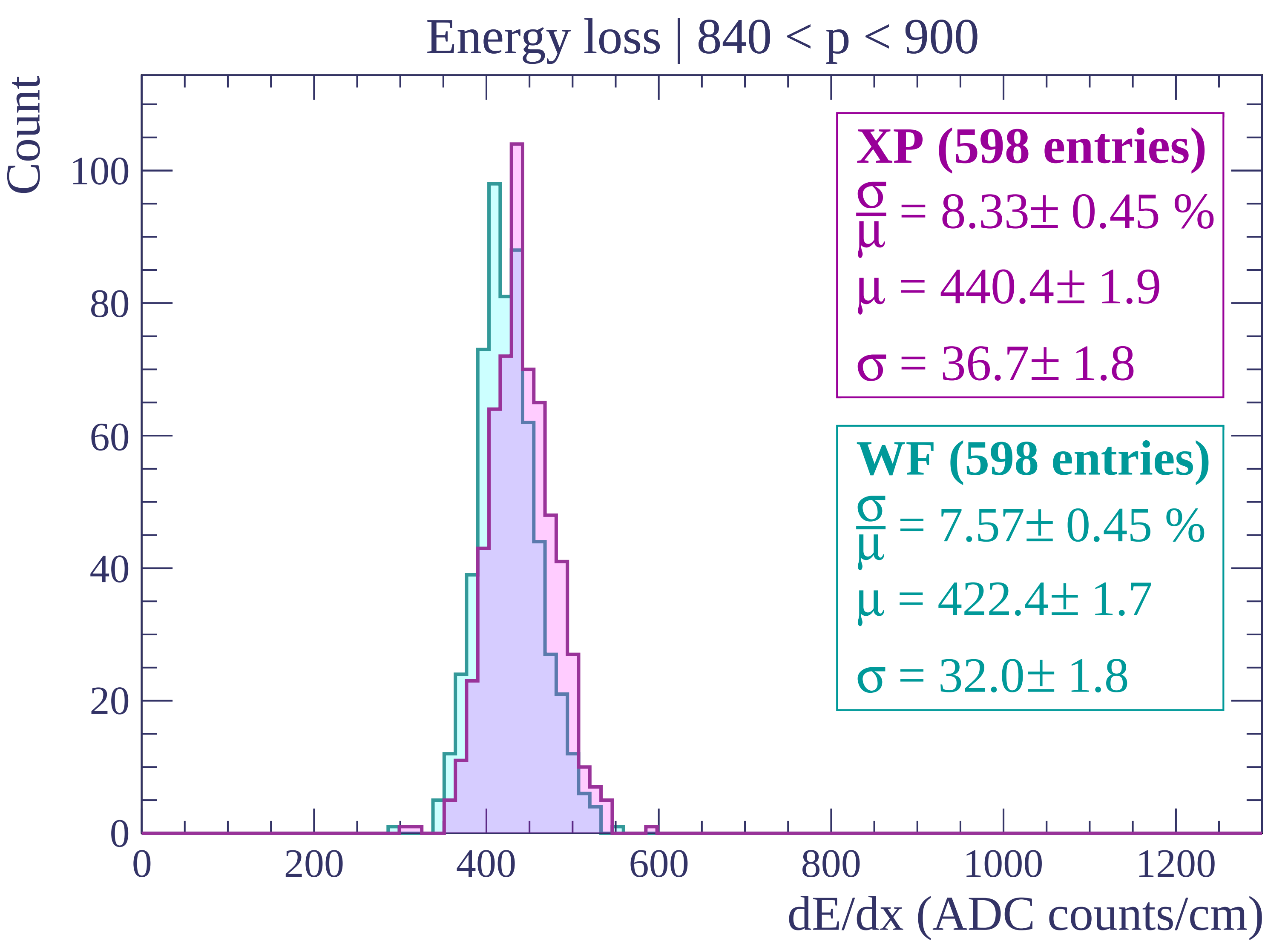


Energy loss | 660 < p < 720
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (561 entries)
σ/μ = 7.60± 0.43 %
μ = 435.0± 1.7
σ = 33.1± 1.7
WF (561 entries)
σ/μ = 7.32± 0.35 %
μ = 419.3± 1.5
σ = 30.7± 1.4

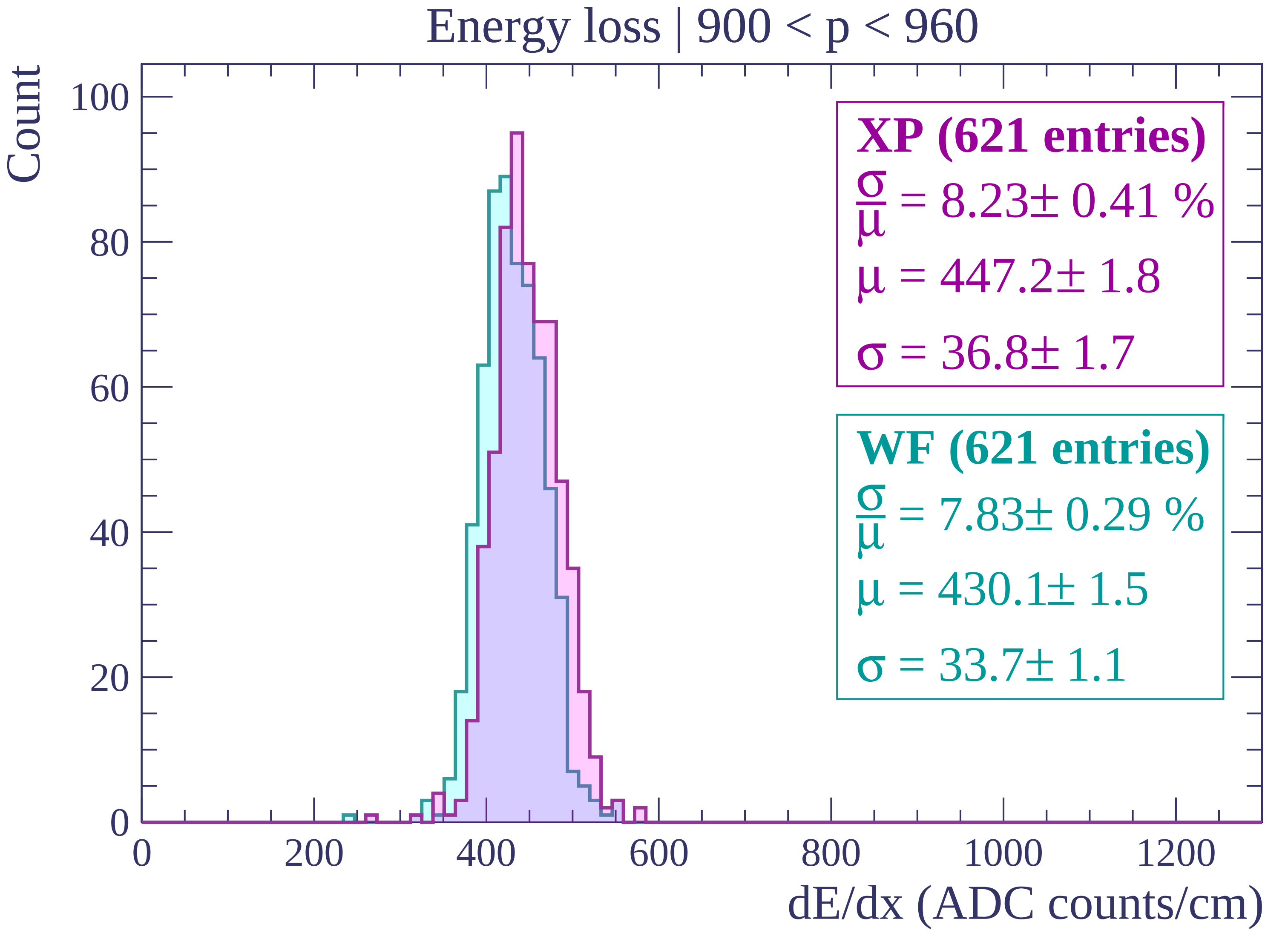


Energy loss | 720 < p < 780
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
90
XP (557 entries)
σ/μ = 7.97± 0.41 %
μ = 433.4± 1.7
σ = 34.5± 1.7
WF (557 entries)
σ/μ = 7.95± 0.46 %
μ = 417.0± 1.7
σ = 33.1± 1.8

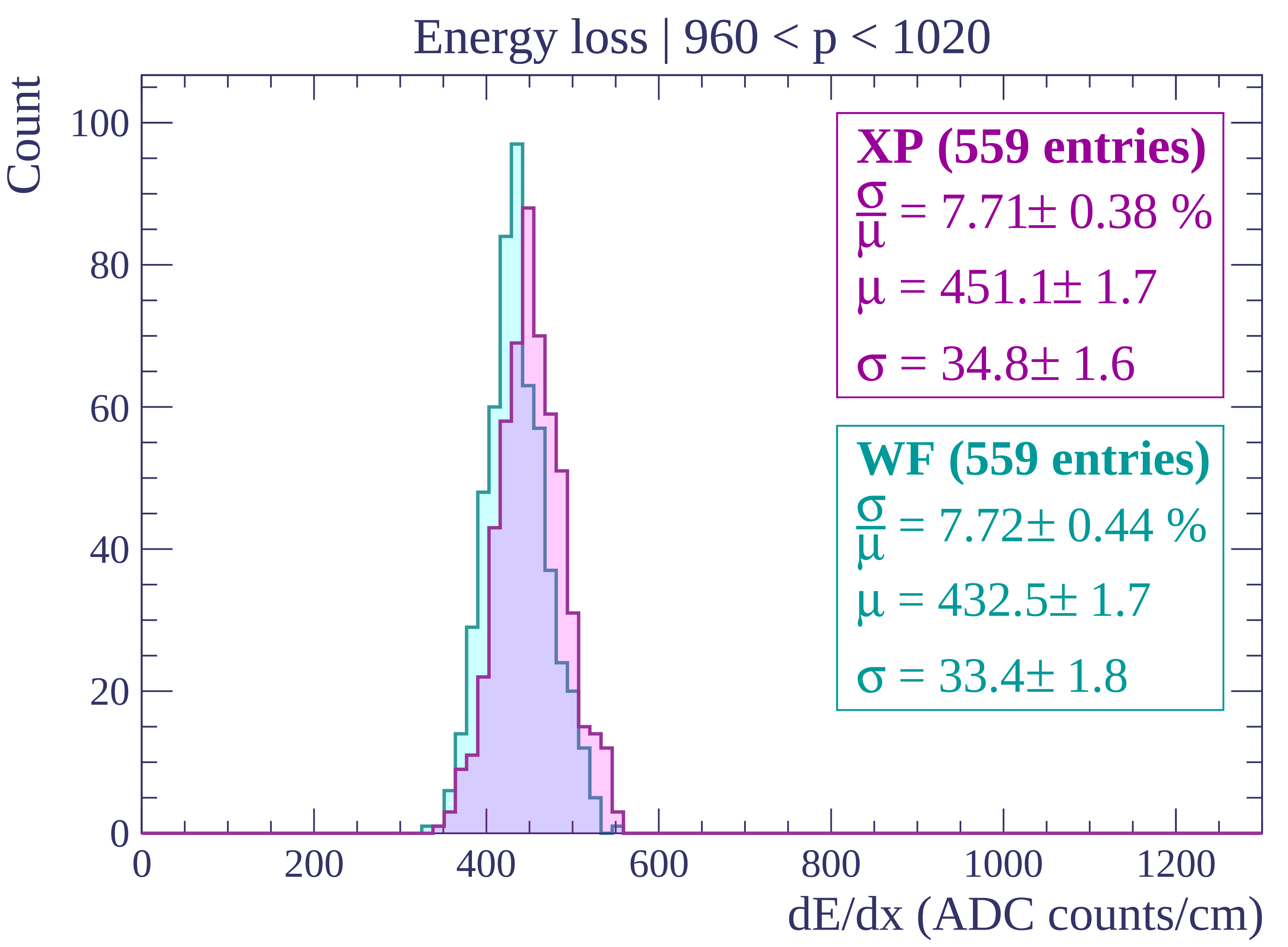

Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
XP (584 entries)
σ/μ = 7.69± 0.38 %
μ = 436.5± 1.6
σ = 33.5± 1.5
WF (584 entries)
σ/μ = 8.06± 0.47 %
μ = 419.2± 1.7
σ = 33.8± 1.9
0
20
40
60
80
100
0
200
400
600
800
1000
1200



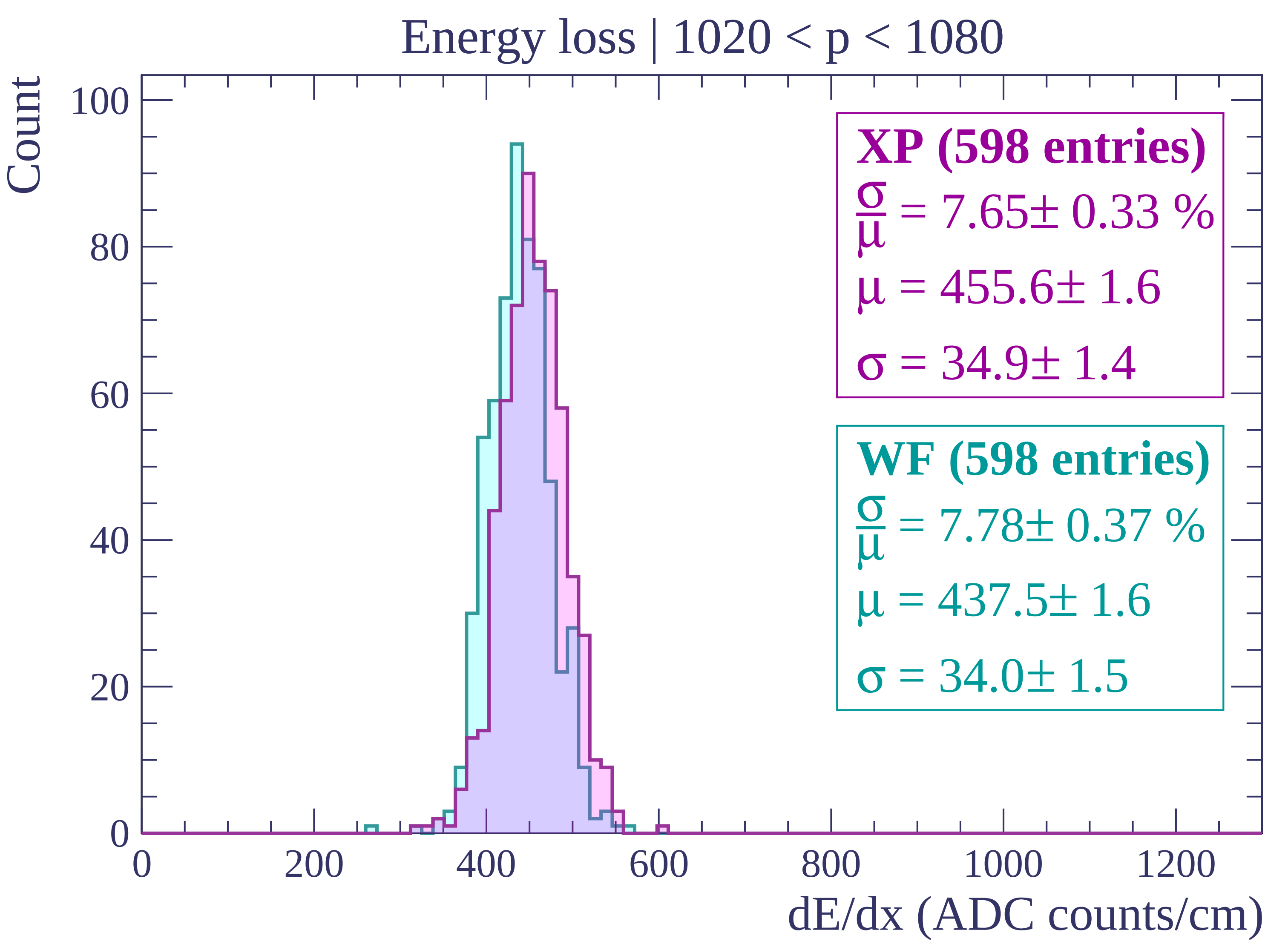
Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (598 entries)
$\frac{\sigma}{\mu}$ = 8.33± 0.45 %
μ = 440.4± 1.9
σ = 36.7± 1.8
WF (598 entries)
$\frac{\sigma}{\mu}$ = 7.57± 0.45 %
μ = 422.4± 1.7
σ = 32.0± 1.8



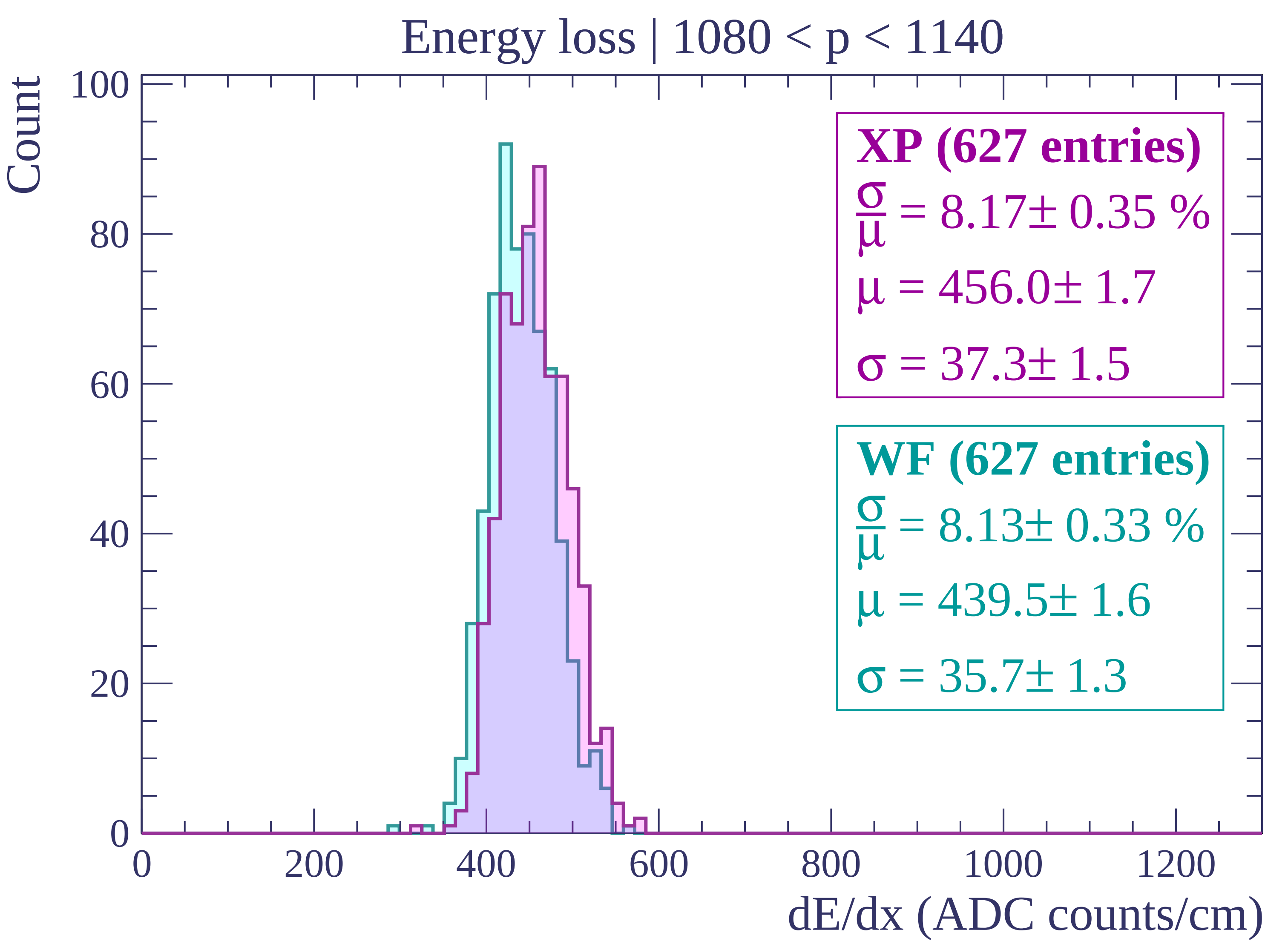
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP (621 entries)
σ/μ = 8.23± 0.41 %
μ = 447.2± 1.8
σ = 36.8± 1.7
WF (621 entries)
σ/μ = 7.83± 0.29 %
μ = 430.1± 1.5
σ = 33.7± 1.1



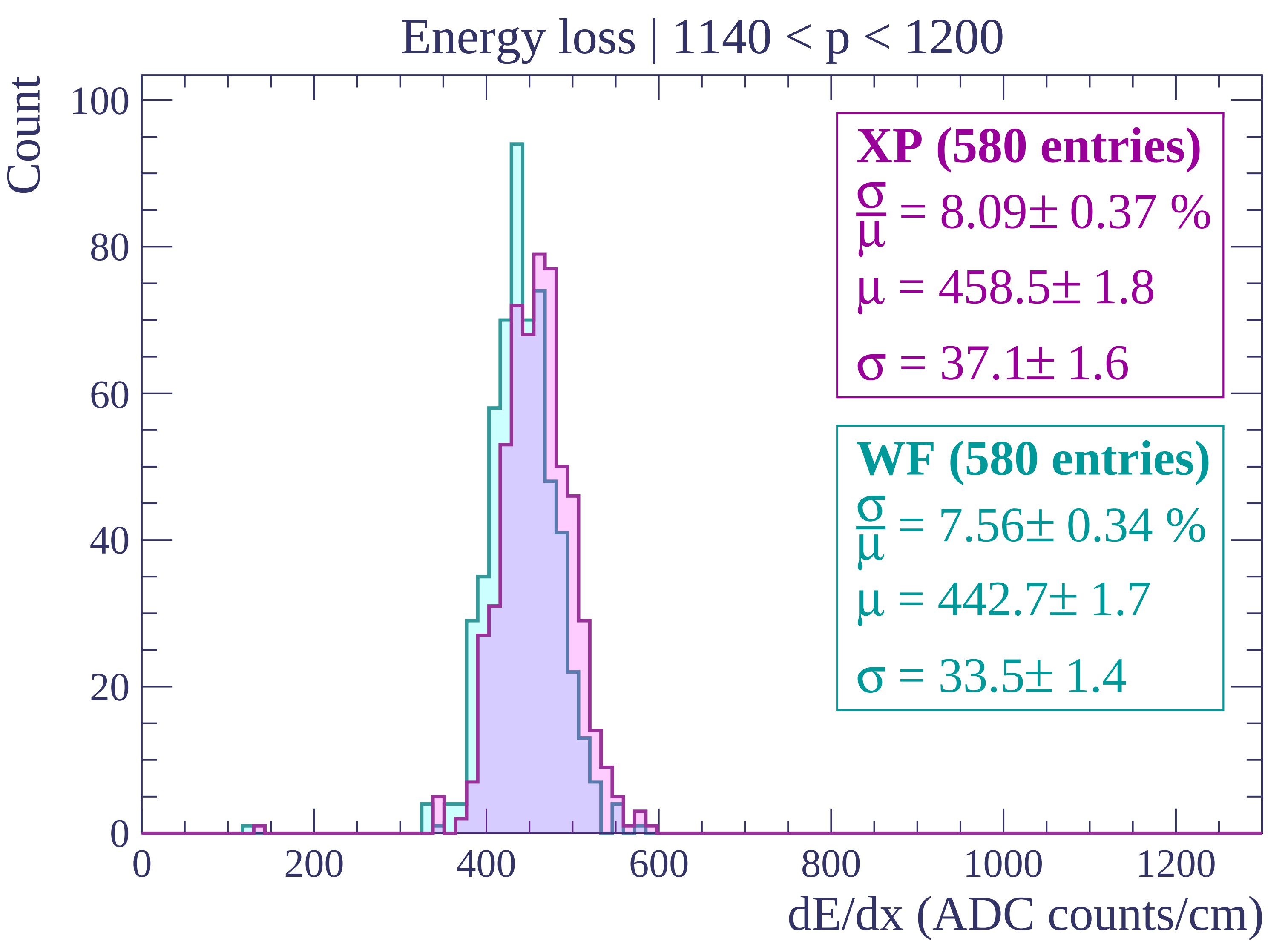
Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (559 entries)
σ/μ = 7.71± 0.38 %
μ = 451.1± 1.7
σ = 34.8± 1.6
WF (559 entries)
σ/μ = 7.72± 0.44 %
μ = 432.5± 1.7
σ = 33.4± 1.8



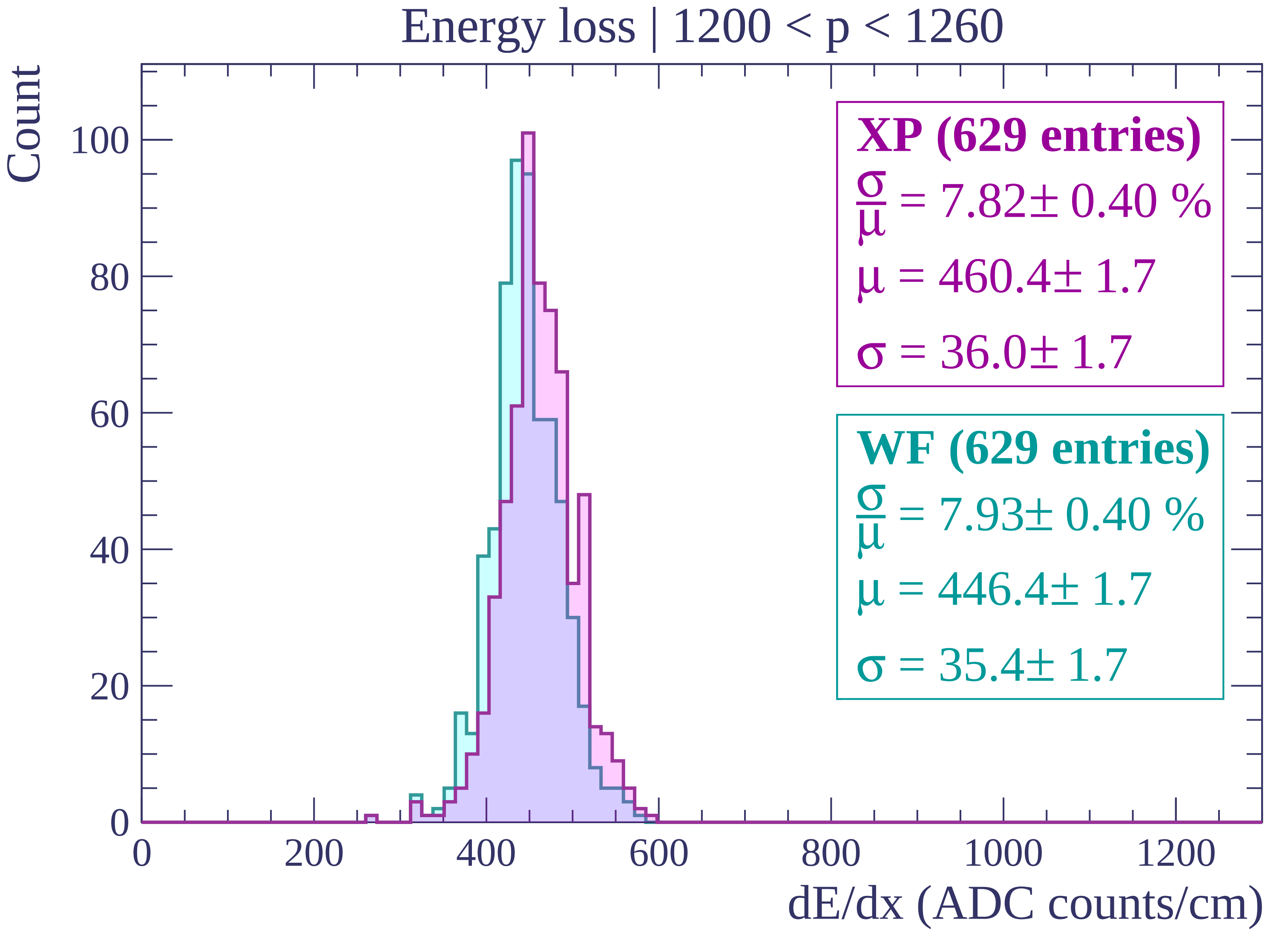
Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (598 entries)
$\frac{\sigma}{\mu}$ = 7.65± 0.33 %
μ = 455.6± 1.6
σ = 34.9± 1.4
WF (598 entries)
$\frac{\sigma}{\mu}$ = 7.78± 0.37 %
μ = 437.5± 1.6
σ = 34.0± 1.5



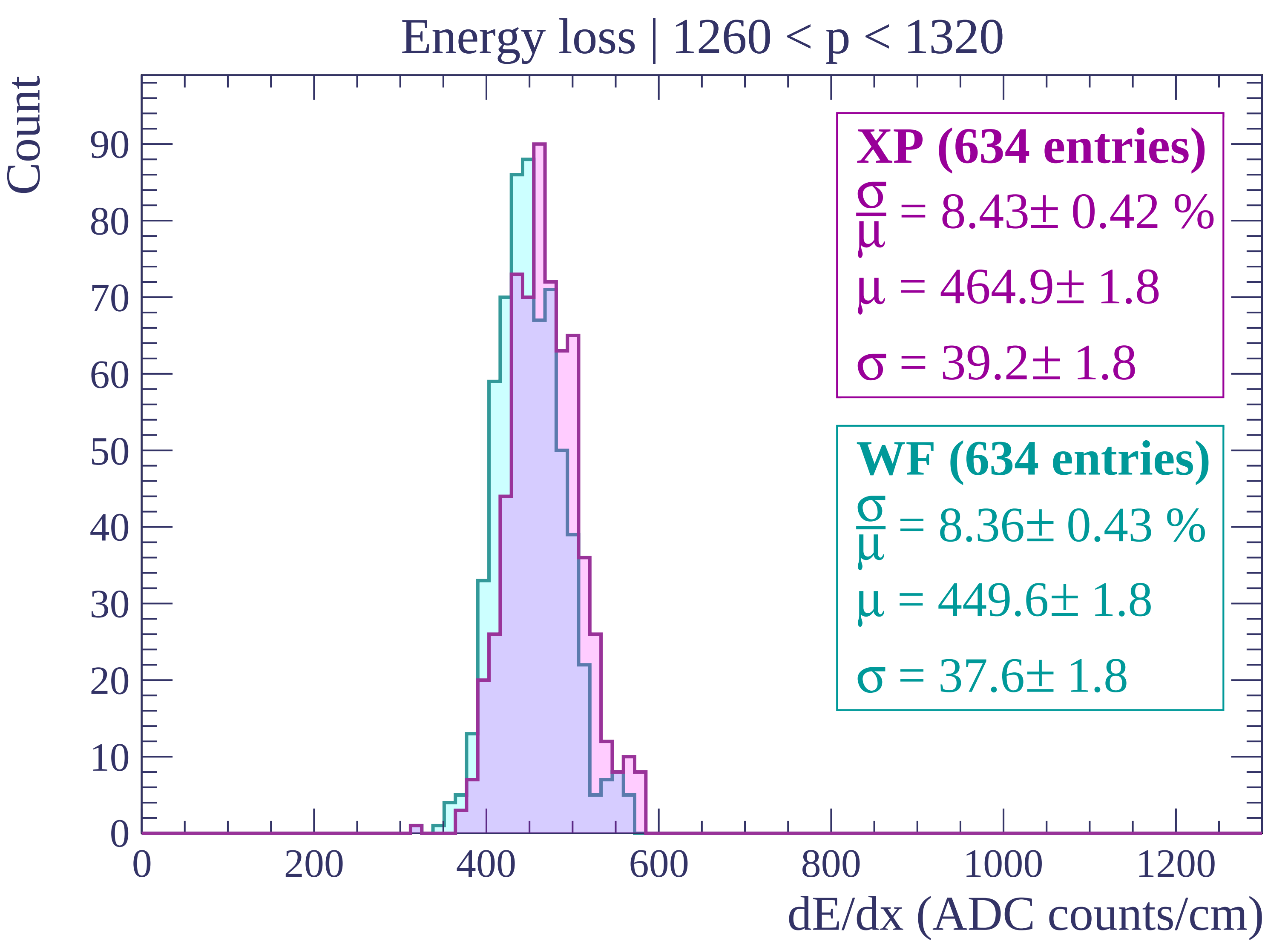
Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (627 entries)
σ/μ = 8.17± 0.35 %
μ = 456.0± 1.7
σ = 37.3± 1.5
WF (627 entries)
σ/μ = 8.13± 0.33 %
μ = 439.5± 1.6
σ = 35.7± 1.3



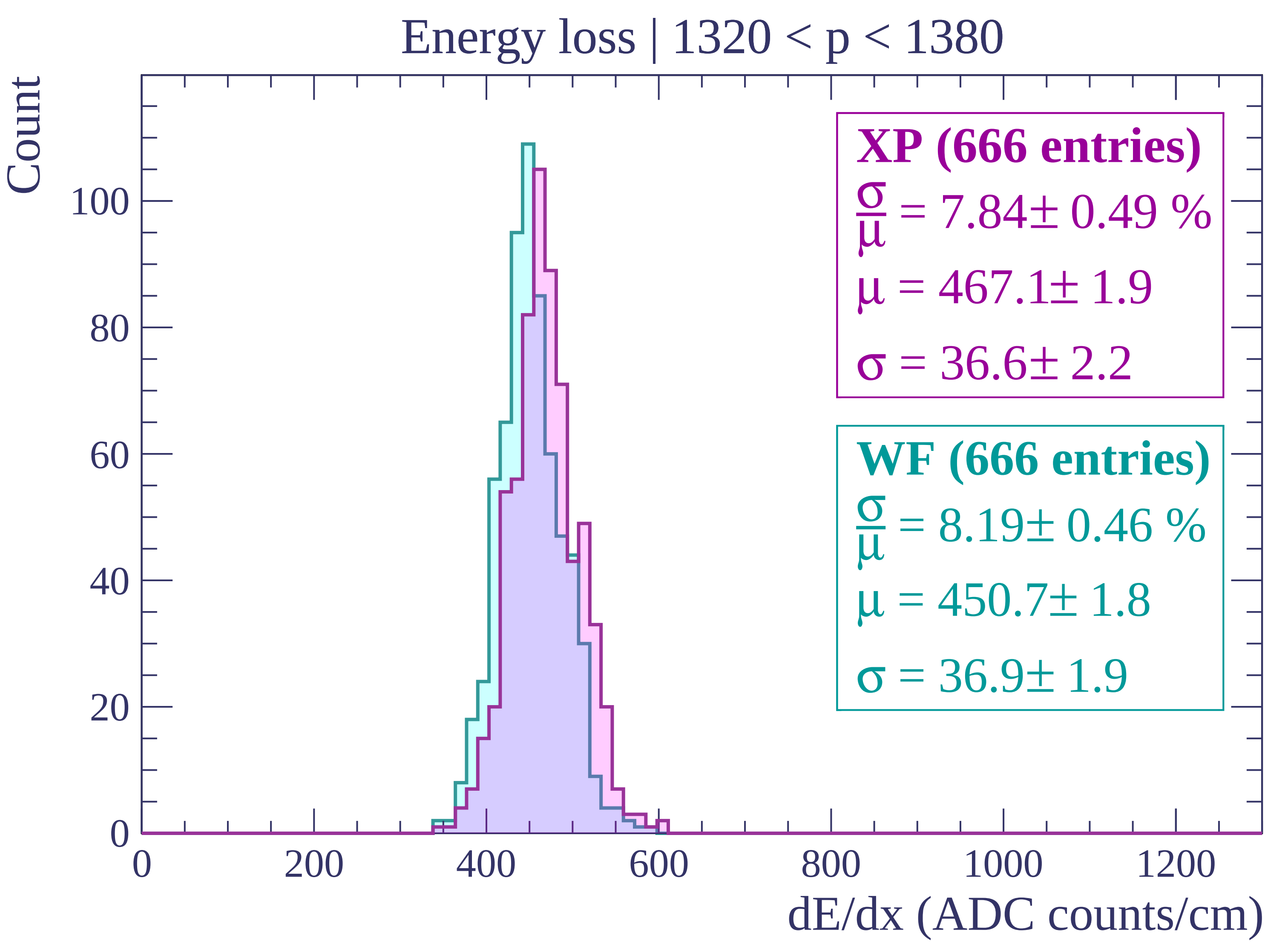
Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (580 entries)
$\frac{\sigma}{\mu}$ = 8.09± 0.37 %
μ = 458.5± 1.8
σ = 37.1± 1.6
WF (580 entries)
$\frac{\sigma}{\mu}$ = 7.56± 0.34 %
μ = 442.7± 1.7
σ = 33.5± 1.4



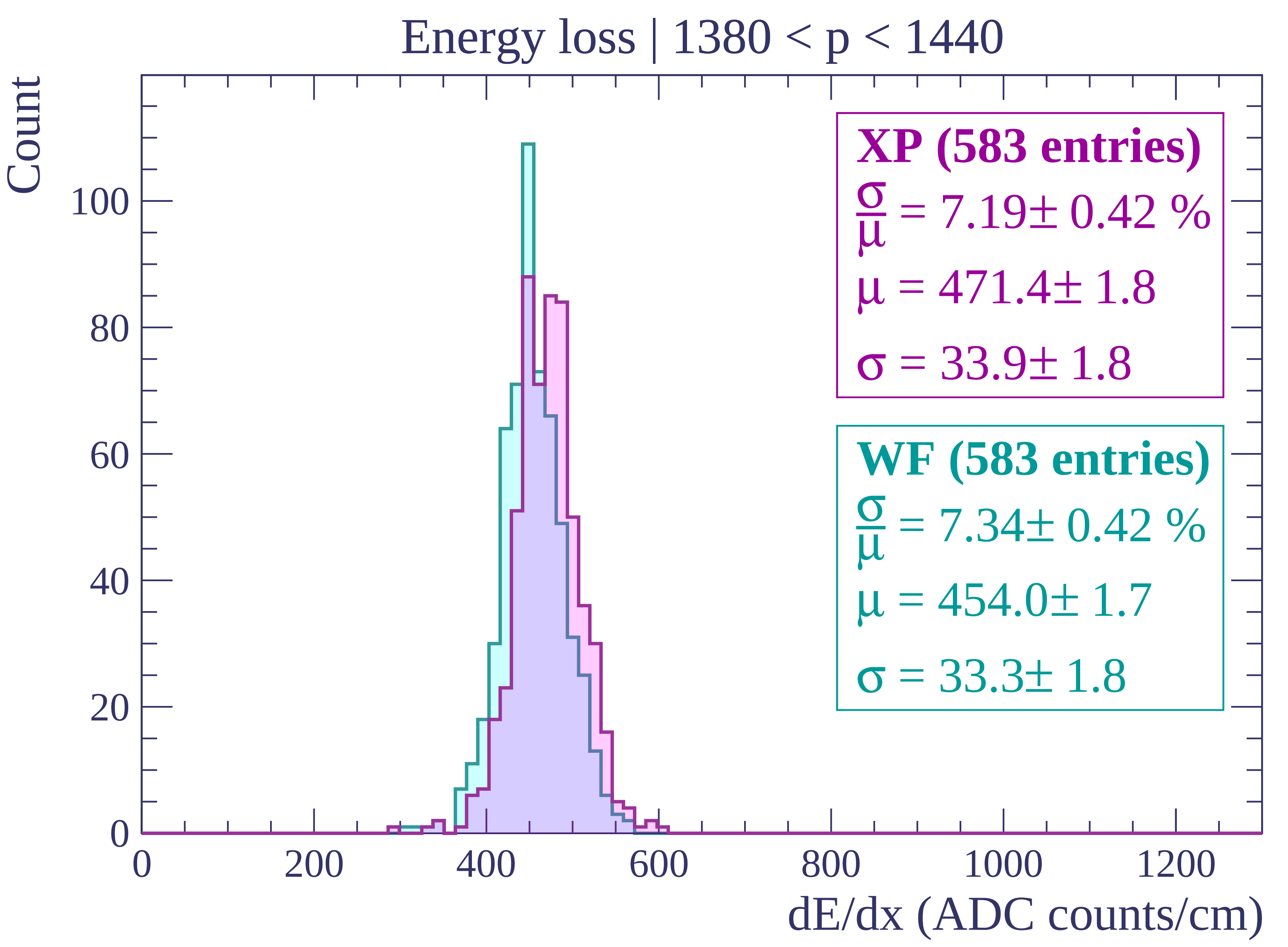
Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (629 entries)
σ/μ = 7.82± 0.40 %
μ = 460.4± 1.7
σ = 36.0± 1.7
WF (629 entries)
σ/μ = 7.93± 0.40 %
μ = 446.4± 1.7
σ = 35.4± 1.7



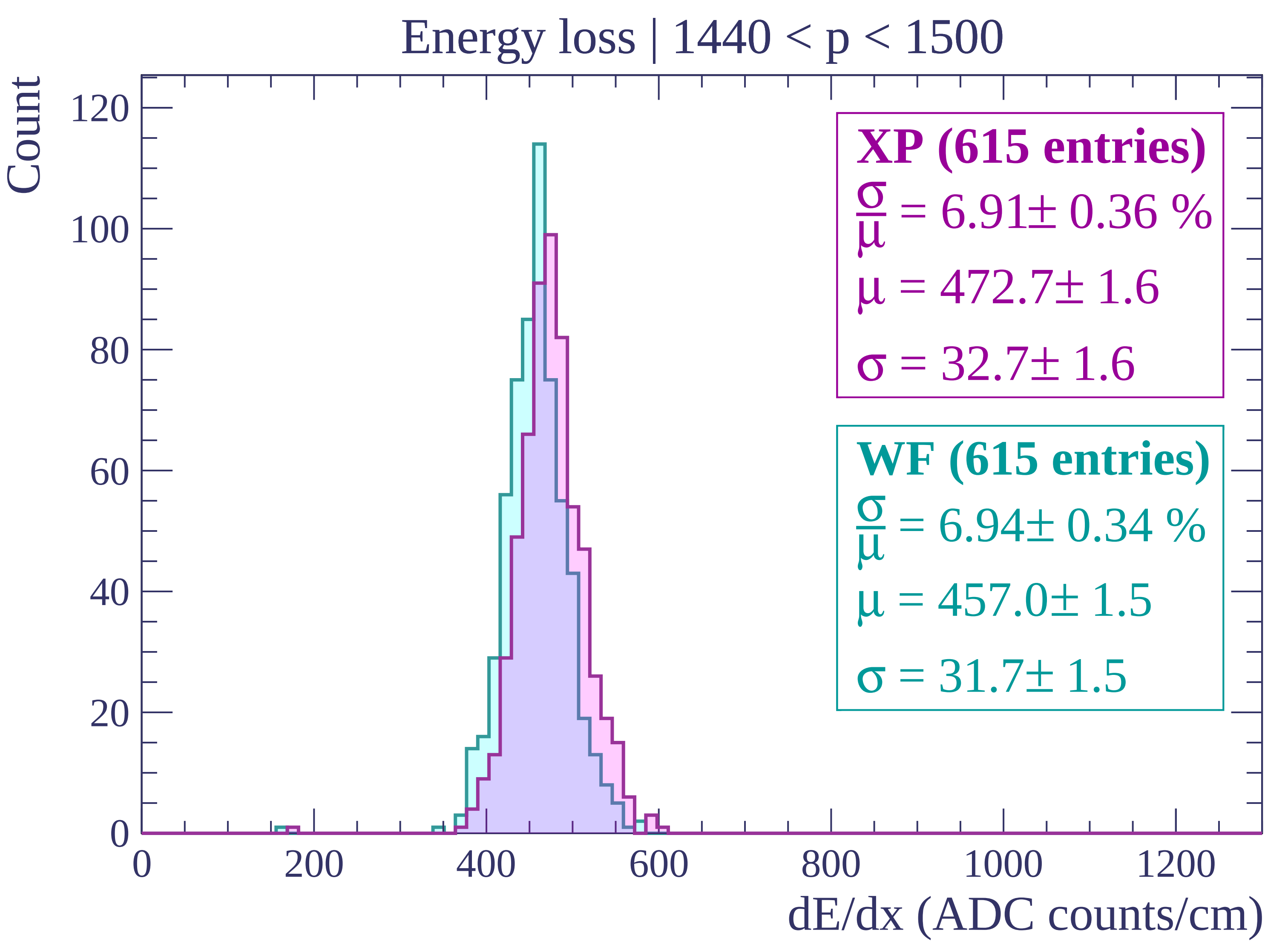
Energy loss | 1260 < p < 1320
Count
XP (634 entries)
σ/μ = 8.43± 0.42 %
μ = 464.9± 1.8
σ = 39.2± 1.8
WF (634 entries)
σ/μ = 8.36± 0.43 %
μ = 449.6± 1.8
σ = 37.6± 1.8
dE/dx (ADC counts/cm)



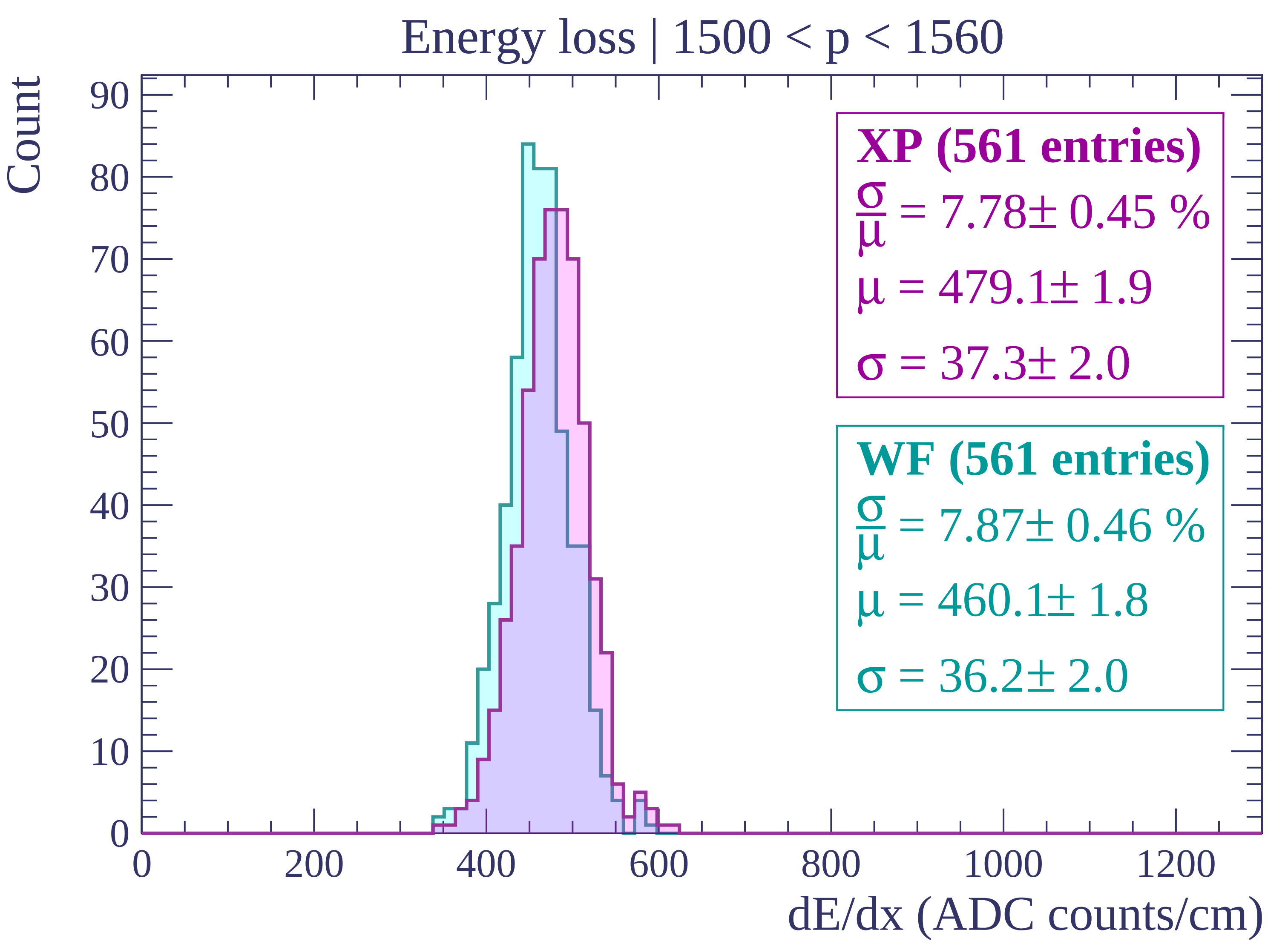
Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (666 entries)
σ/μ = 7.84± 0.49 %
μ = 467.1± 1.9
σ = 36.6± 2.2
WF (666 entries)
σ/μ = 8.19± 0.46 %
μ = 450.7± 1.8
σ = 36.9± 1.9



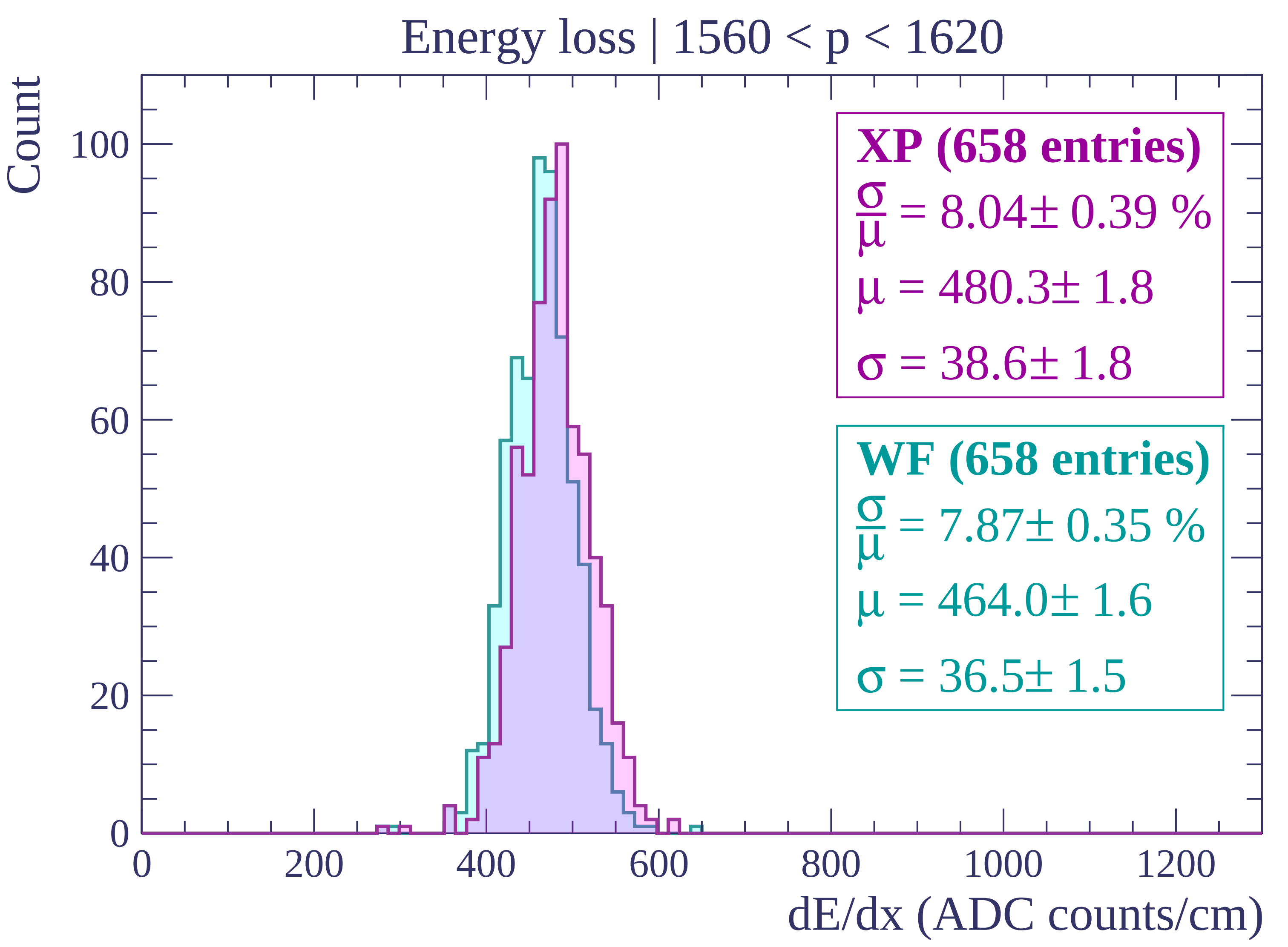
Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
XP (583 entries)
σ/μ = 7.19± 0.42 %
μ = 471.4± 1.8
σ = 33.9± 1.8
WF (583 entries)
σ/μ = 7.34± 0.42 %
μ = 454.0± 1.7
σ = 33.3± 1.8
0
20
40
60
80
100
0
200
400
600
800
1000
1200



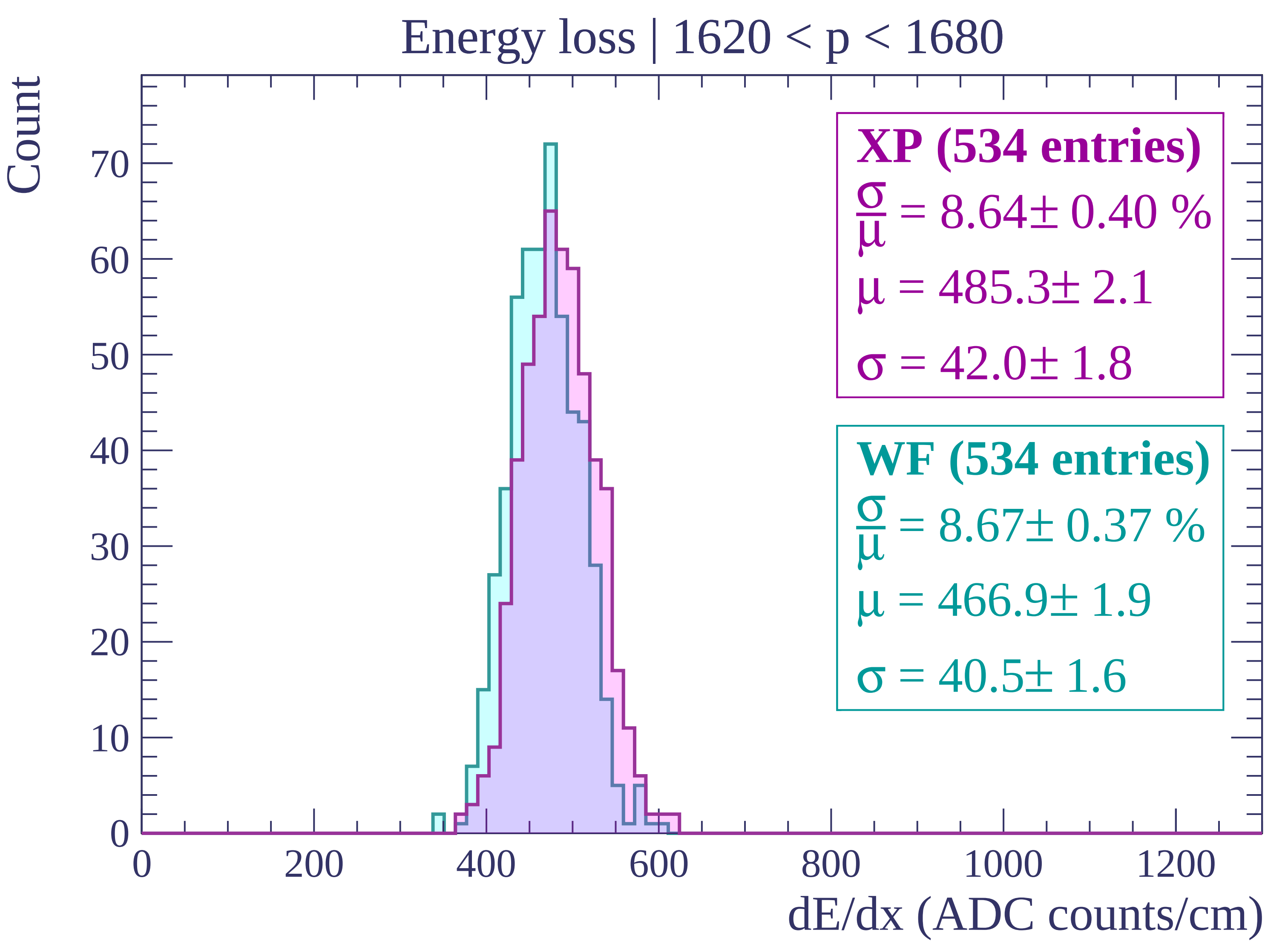
Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
XP (615 entries)
σ/μ = 6.91± 0.36 %
μ = 472.7± 1.6
σ = 32.7± 1.6
WF (615 entries)
σ/μ = 6.94± 0.34 %
μ = 457.0± 1.5
σ = 31.7± 1.5
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200



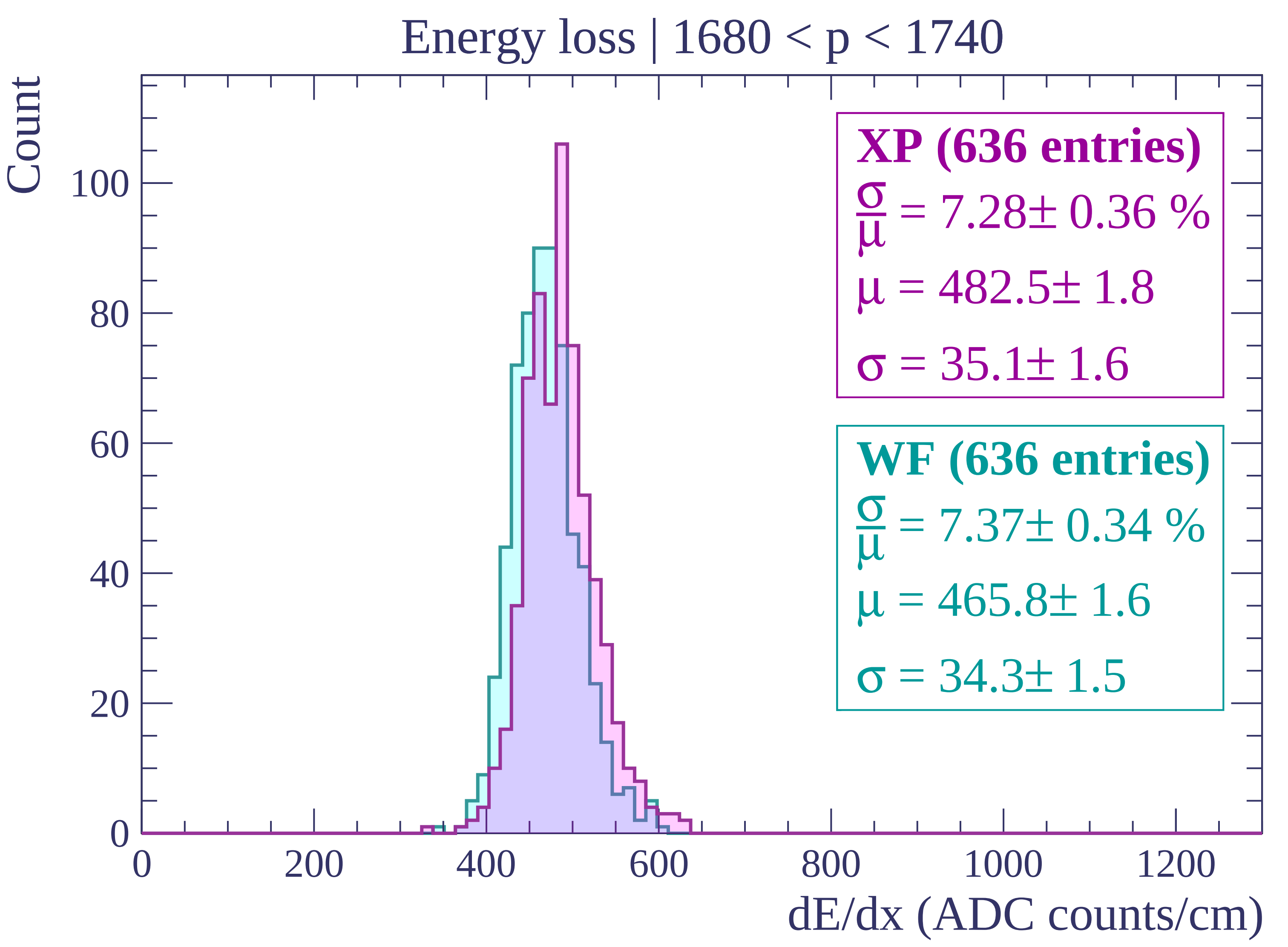
Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (561 entries)
σ/μ = 7.78± 0.45 %
μ = 479.1± 1.9
σ = 37.3± 2.0
WF (561 entries)
σ/μ = 7.87± 0.46 %
μ = 460.1± 1.8
σ = 36.2± 2.0



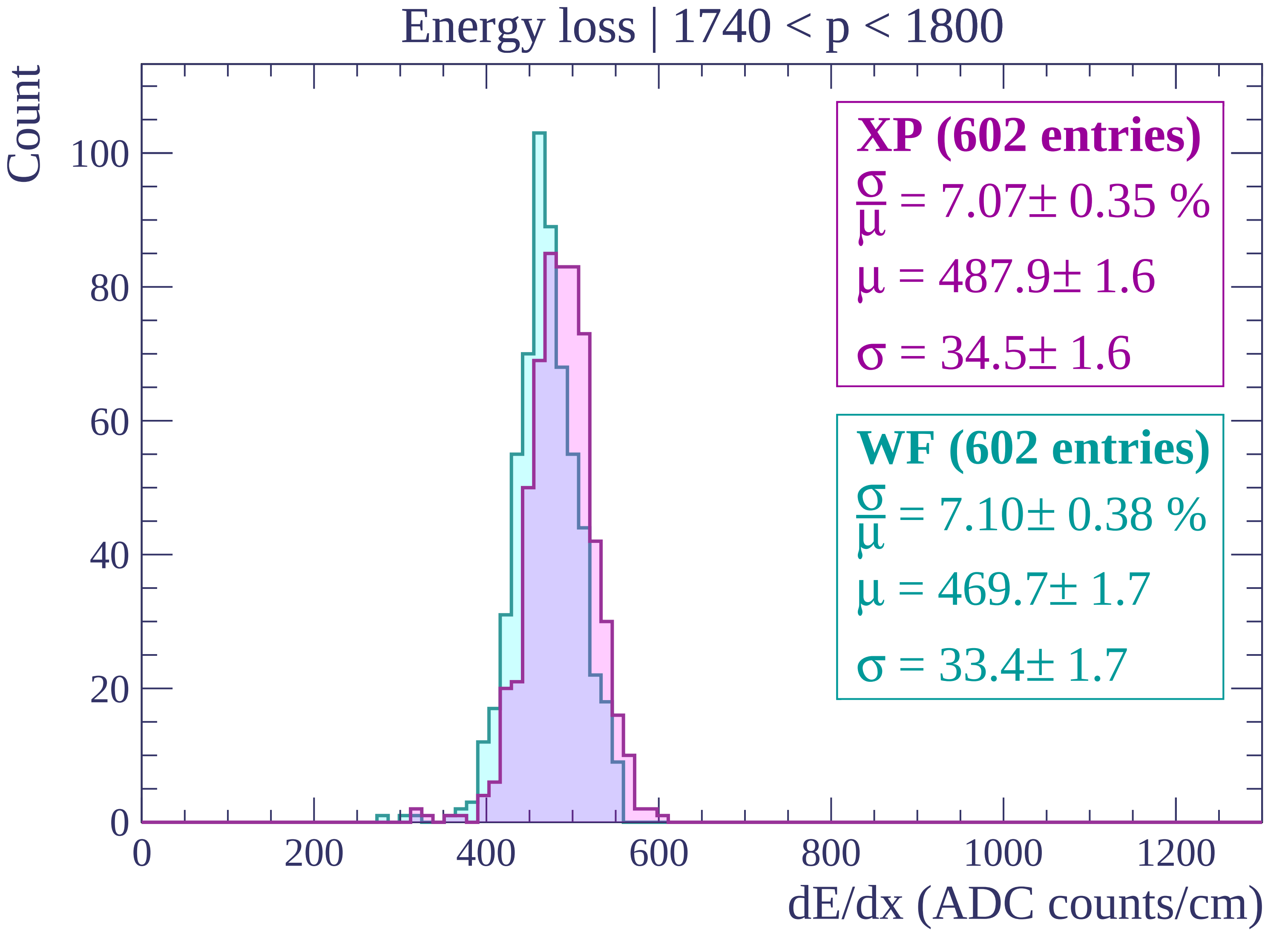
Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
XP (658 entries)
σ/μ = 8.04± 0.39 %
μ = 480.3± 1.8
σ = 38.6± 1.8
WF (658 entries)
σ/μ = 7.87± 0.35 %
μ = 464.0± 1.6
σ = 36.5± 1.5
0
20
40
60
80
100
0
200
400
600
800
1000
1200



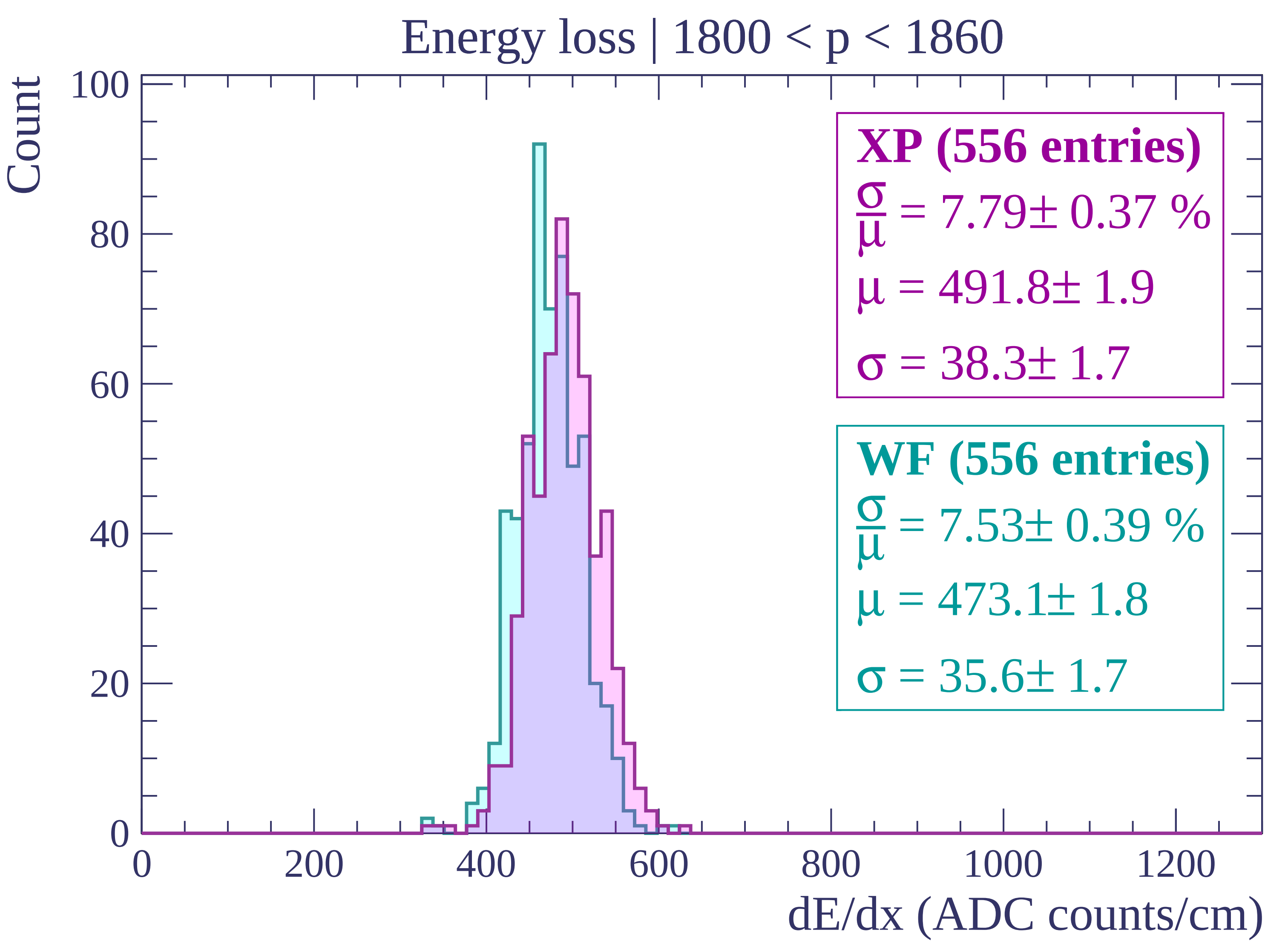
Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
XP (534 entries)
σ/μ = 8.64± 0.40 %
μ = 485.3± 2.1
σ = 42.0± 1.8
WF (534 entries)
σ/μ = 8.67± 0.37 %
μ = 466.9± 1.9
σ = 40.5± 1.6



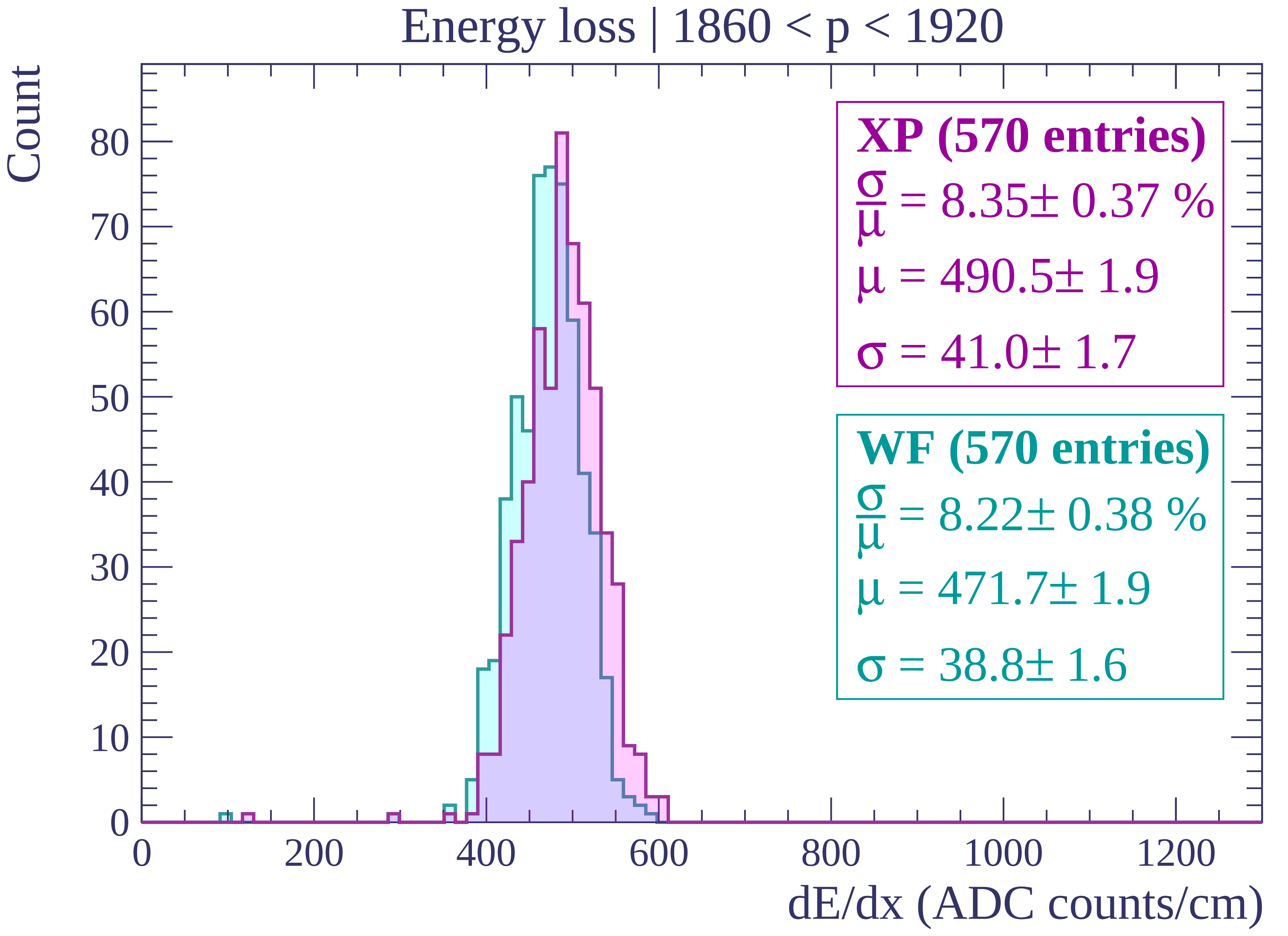
Energy loss | 1680 < p < 1740
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (636 entries)
σ/μ = 7.28± 0.36 %
μ = 482.5± 1.8
σ = 35.1± 1.6
WF (636 entries)
σ/μ = 7.37± 0.34 %
μ = 465.8± 1.6
σ = 34.3± 1.5



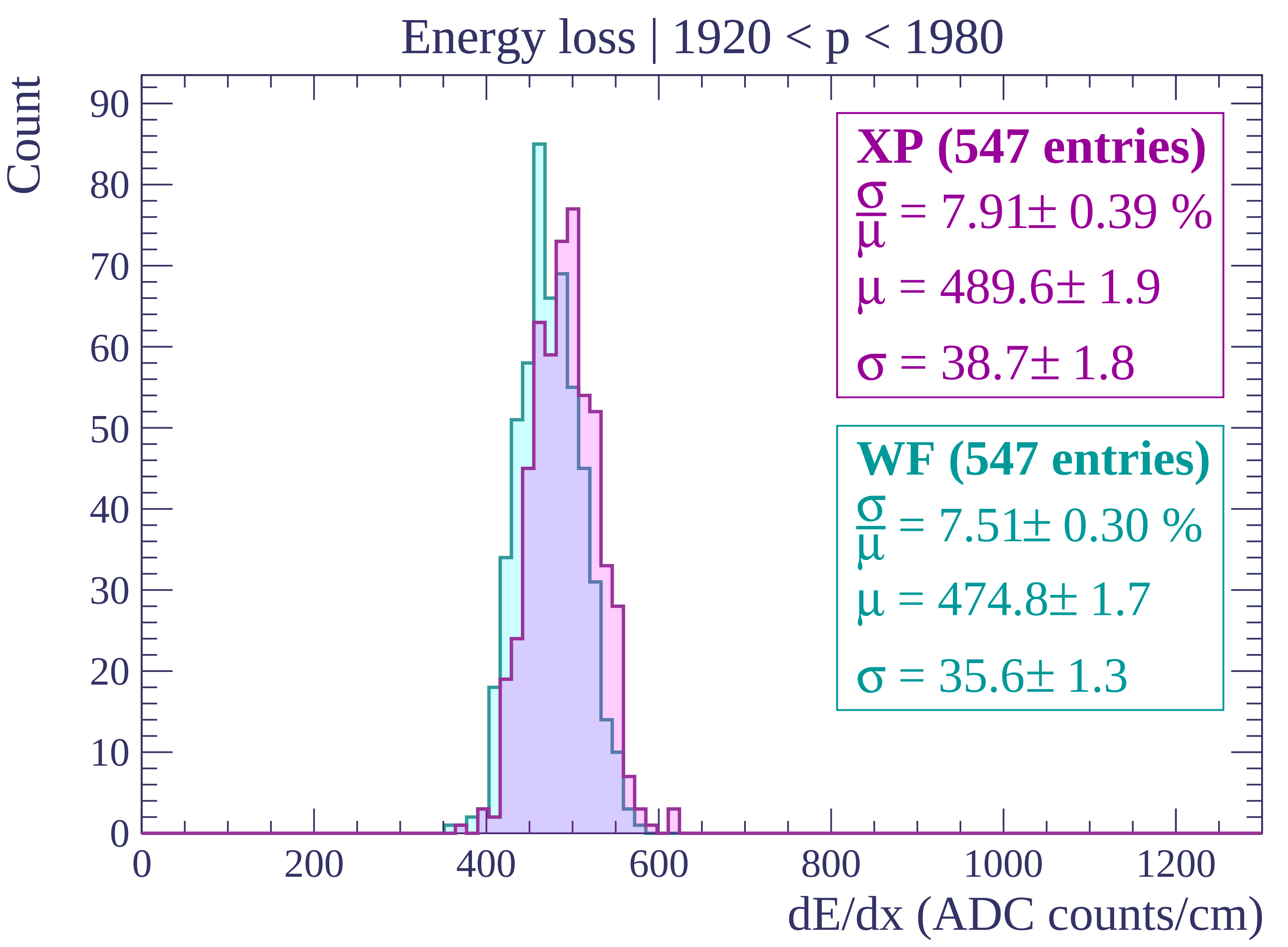
Energy loss | 1740 < p < 1800
Count
0
20
40
60
80
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (602 entries)
σ/μ = 7.07± 0.35 %
μ = 487.9± 1.6
σ = 34.5± 1.6
WF (602 entries)
σ/μ = 7.10± 0.38 %
μ = 469.7± 1.7
σ = 33.4± 1.7

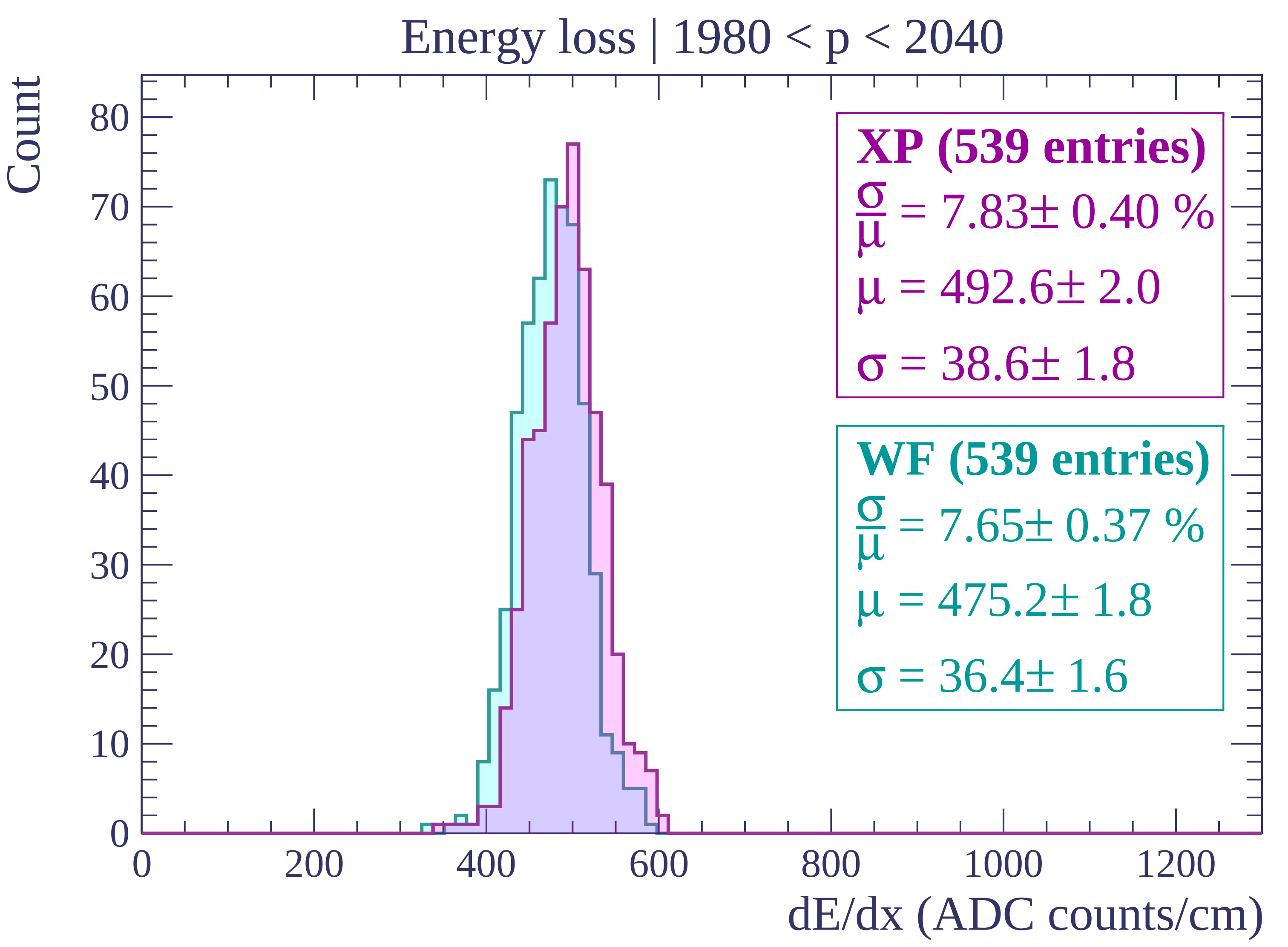


Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
XP (556 entries)
σ/μ = 7.79± 0.37 %
μ = 491.8± 1.9
σ = 38.3± 1.7
WF (556 entries)
σ/μ = 7.53± 0.39 %
μ = 473.1± 1.8
σ = 35.6± 1.7

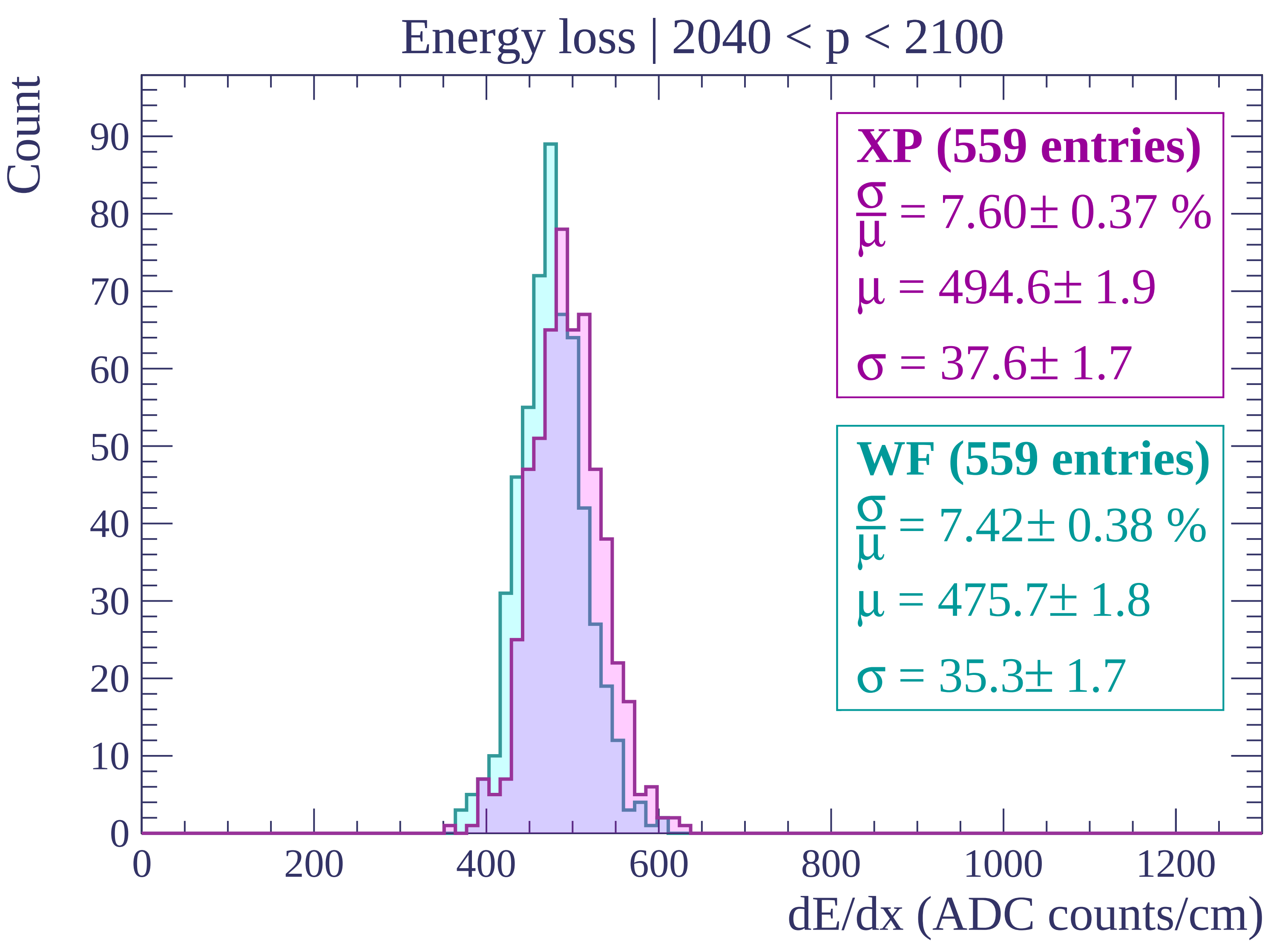


Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
XP (570 entries)
σ/μ = 8.35± 0.37 %
μ = 490.5± 1.9
σ = 41.0± 1.7
WF (570 entries)
σ/μ = 8.22± 0.38 %
μ = 471.7± 1.9
σ = 38.8± 1.6

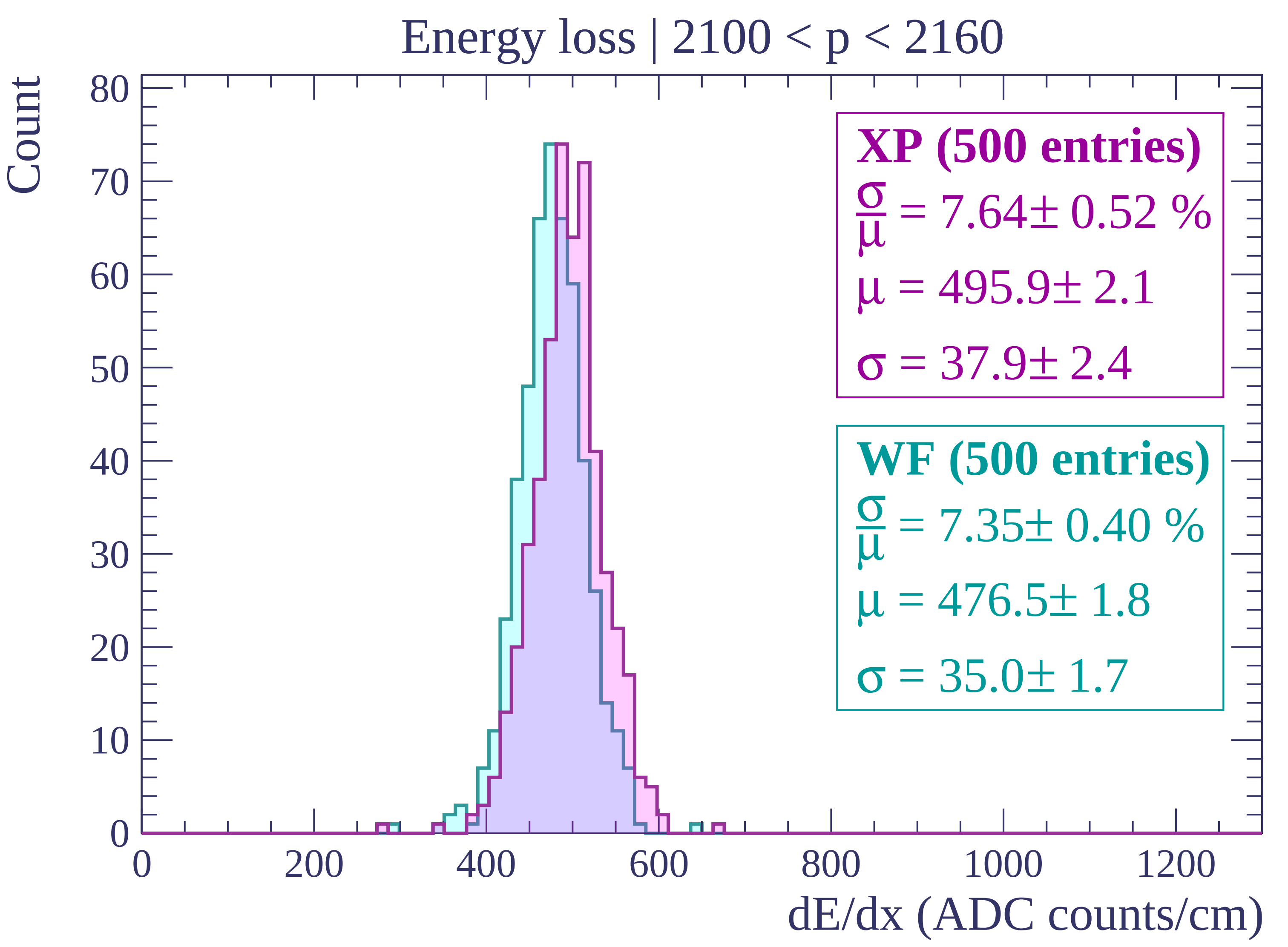

Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
XP (547 entries)
$\frac{\sigma}{\mu}$ = 7.91± 0.39 %
μ = 489.6± 1.9
σ = 38.7± 1.8
WF (547 entries)
$\frac{\sigma}{\mu}$ = 7.51± 0.30 %
μ = 474.8± 1.7
σ = 35.6± 1.3

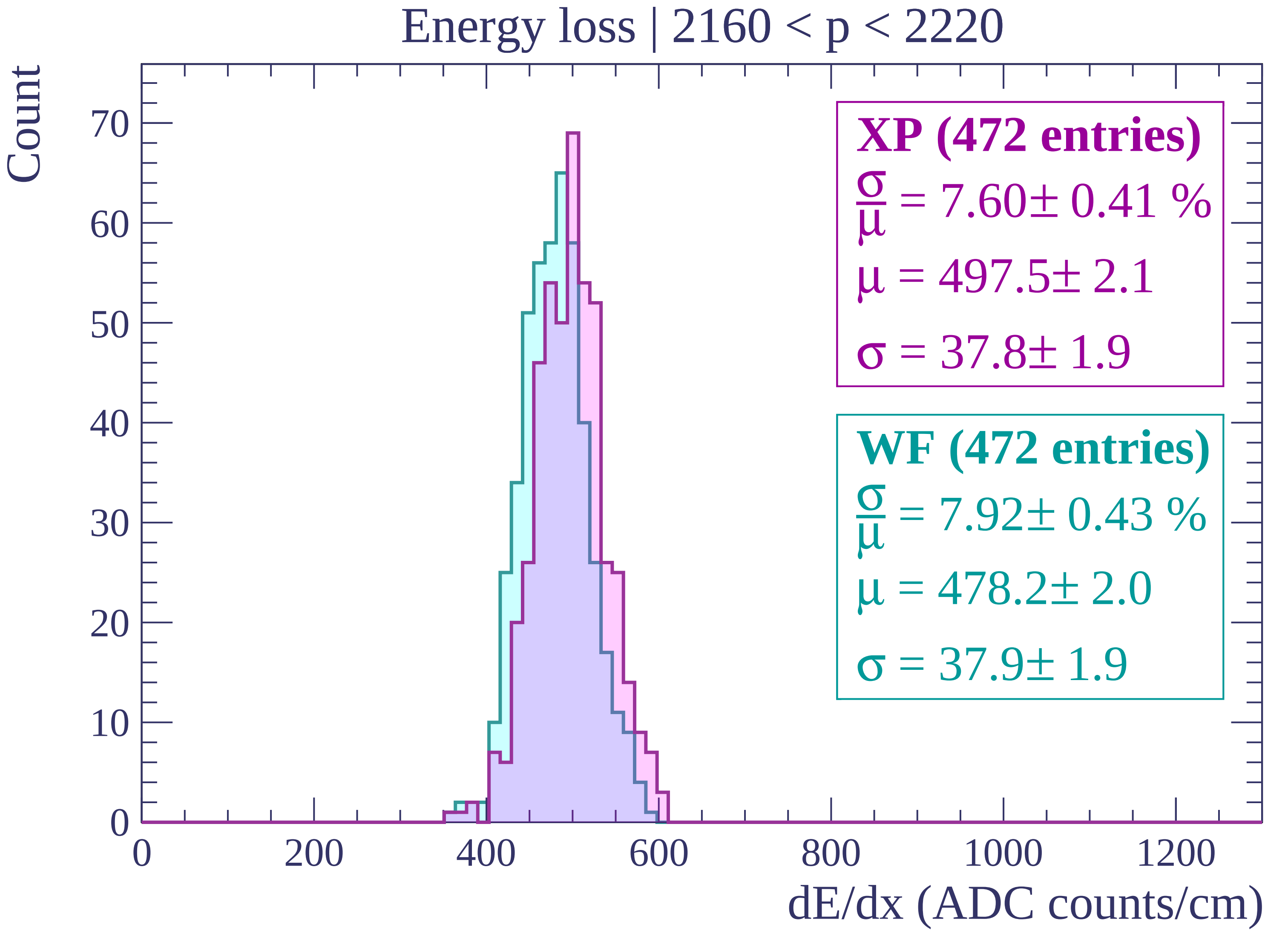


Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (539 entries)
σ/μ = 7.83± 0.40 %
μ = 492.6± 2.0
σ = 38.6± 1.8
WF (539 entries)
σ/μ = 7.65± 0.37 %
μ = 475.2± 1.8
σ = 36.4± 1.6

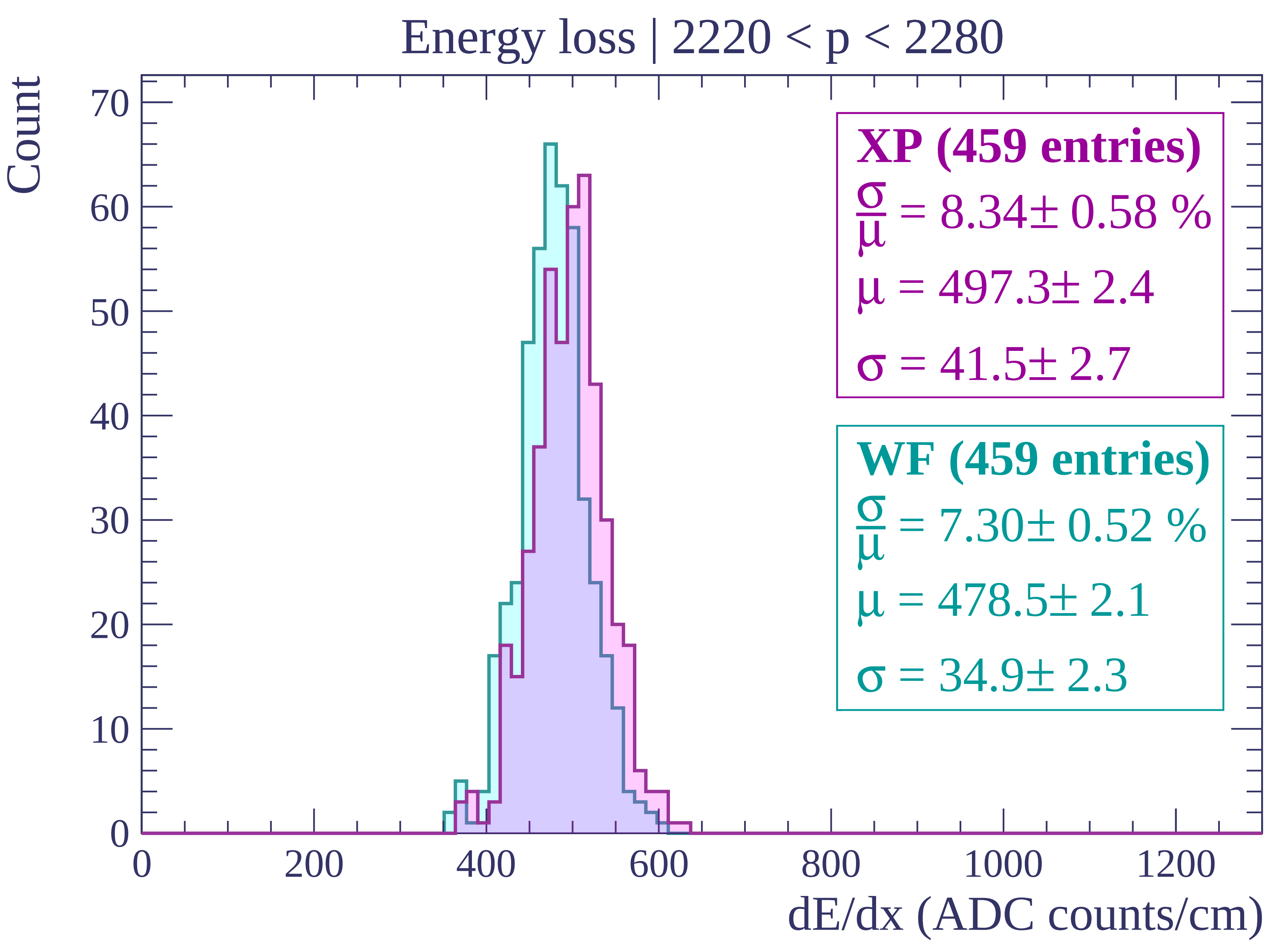


Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
XP (559 entries)
σ/μ = 7.60± 0.37 %
μ = 494.6± 1.9
σ = 37.6± 1.7
WF (559 entries)
σ/μ = 7.42± 0.38 %
μ = 475.7± 1.8
σ = 35.3± 1.7

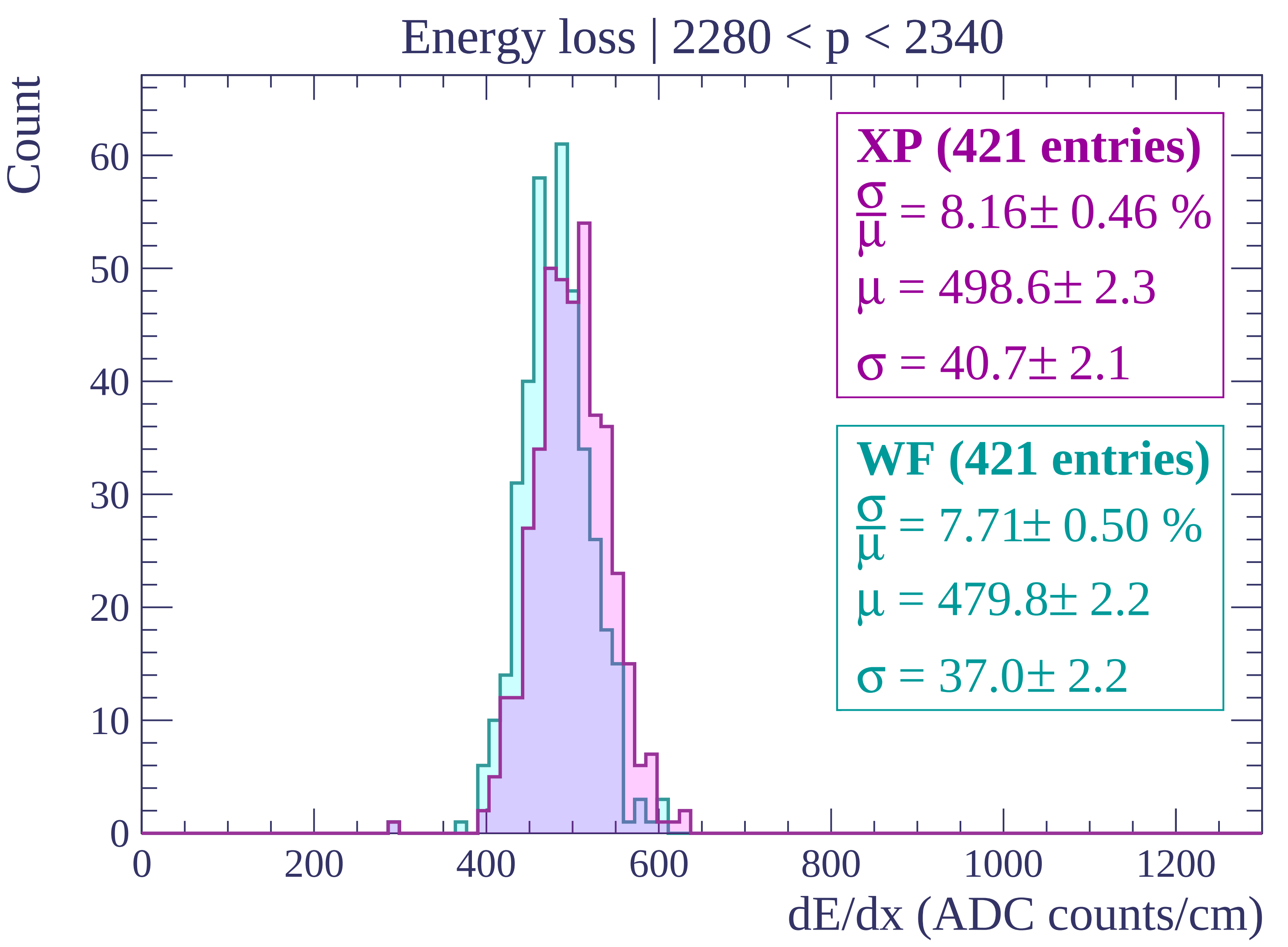

Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP (500 entries)
$\frac{\sigma}{\mu}$ = 7.64± 0.52 %
μ = 495.9± 2.1
σ = 37.9± 2.4
WF (500 entries)
$\frac{\sigma}{\mu}$ = 7.35± 0.40 %
μ = 476.5± 1.8
σ = 35.0± 1.7



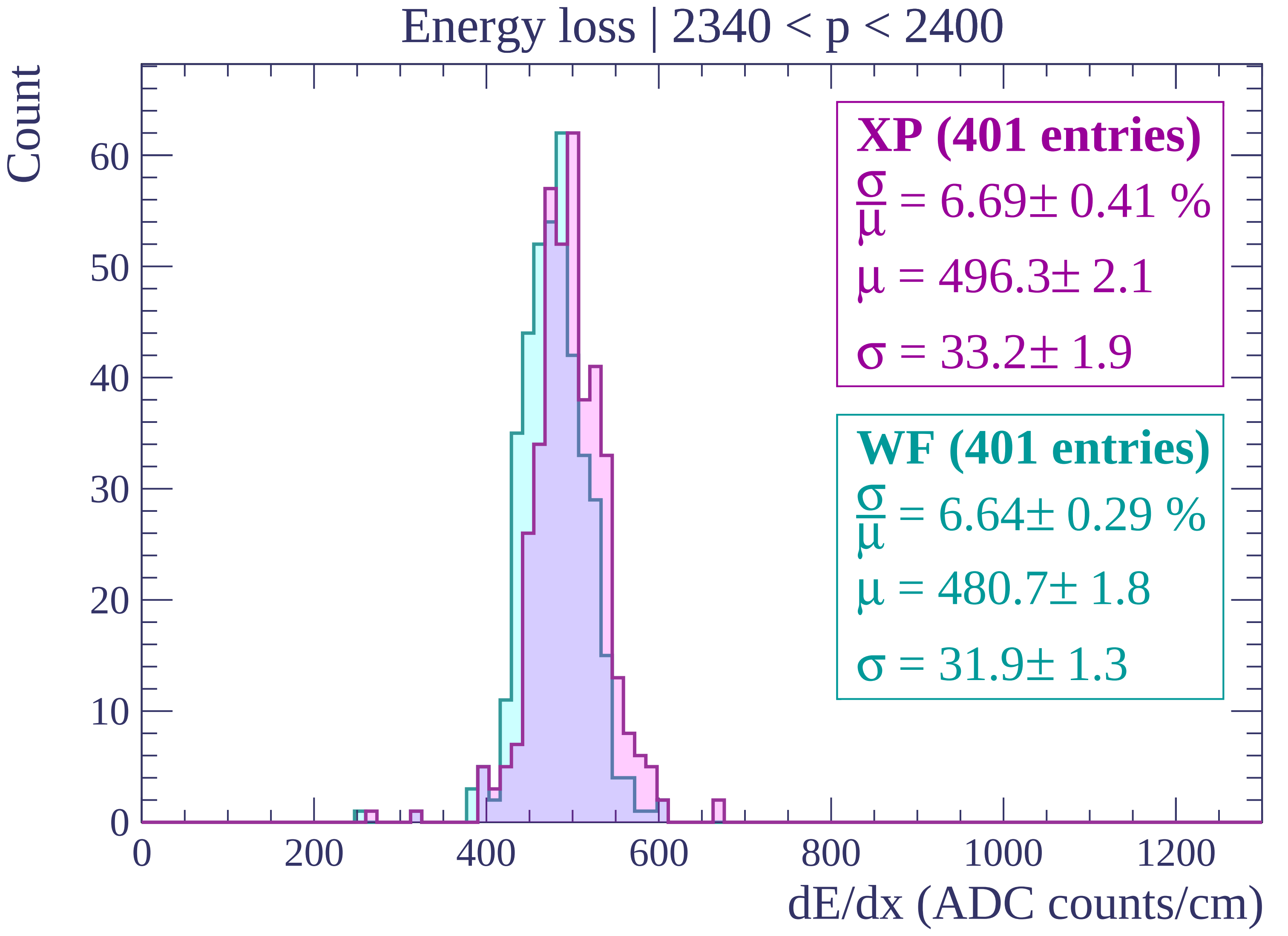
Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (472 entries)
σ/μ = 7.60± 0.41 %
μ = 497.5± 2.1
σ = 37.8± 1.9
WF (472 entries)
σ/μ = 7.92± 0.43 %
μ = 478.2± 2.0
σ = 37.9± 1.9



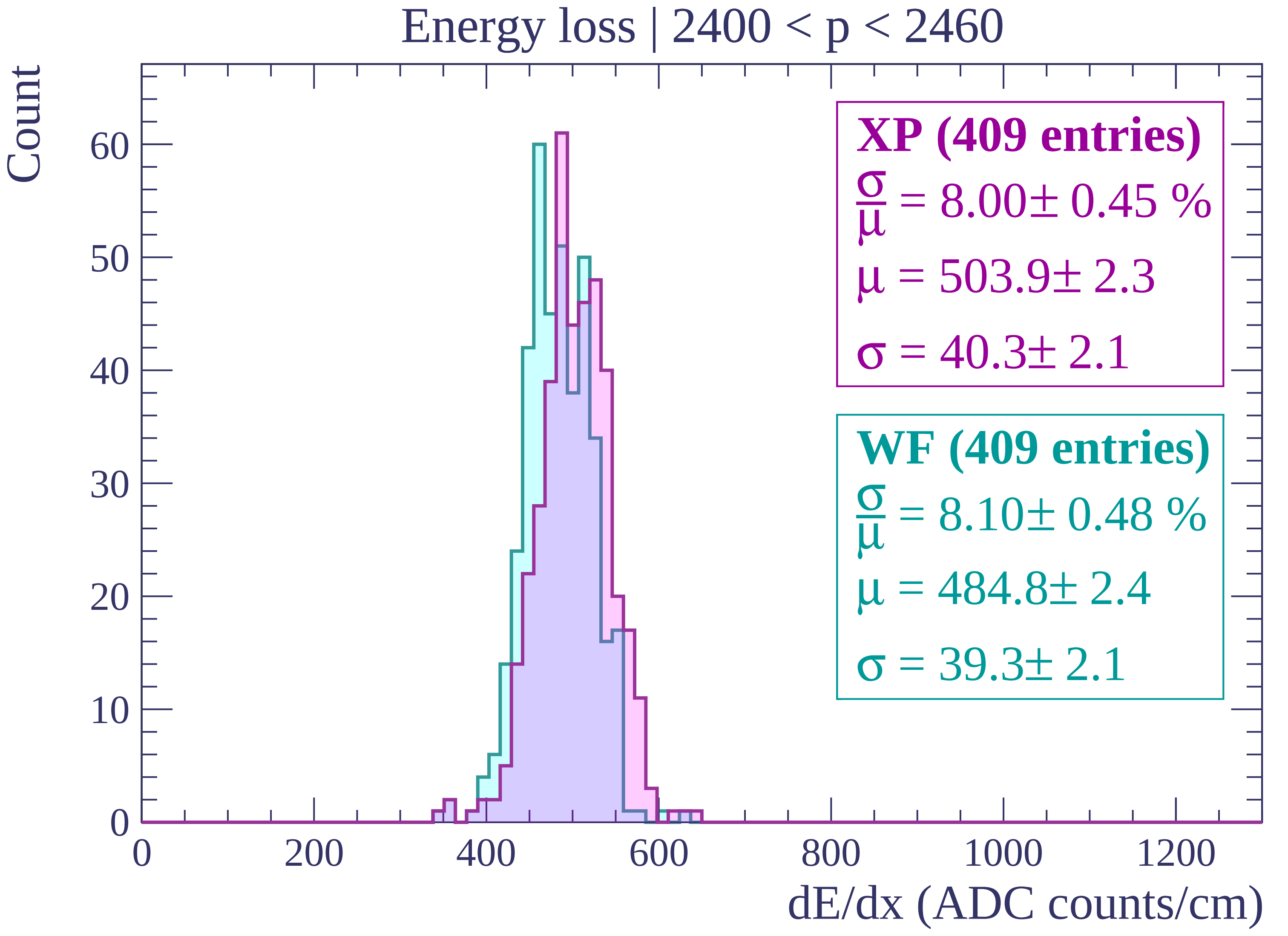
Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP (459 entries)
$\frac{\sigma}{\mu}$ = 8.34± 0.58 %
μ = 497.3± 2.4
σ = 41.5± 2.7
WF (459 entries)
$\frac{\sigma}{\mu}$ = 7.30± 0.52 %
μ = 478.5± 2.1
σ = 34.9± 2.3

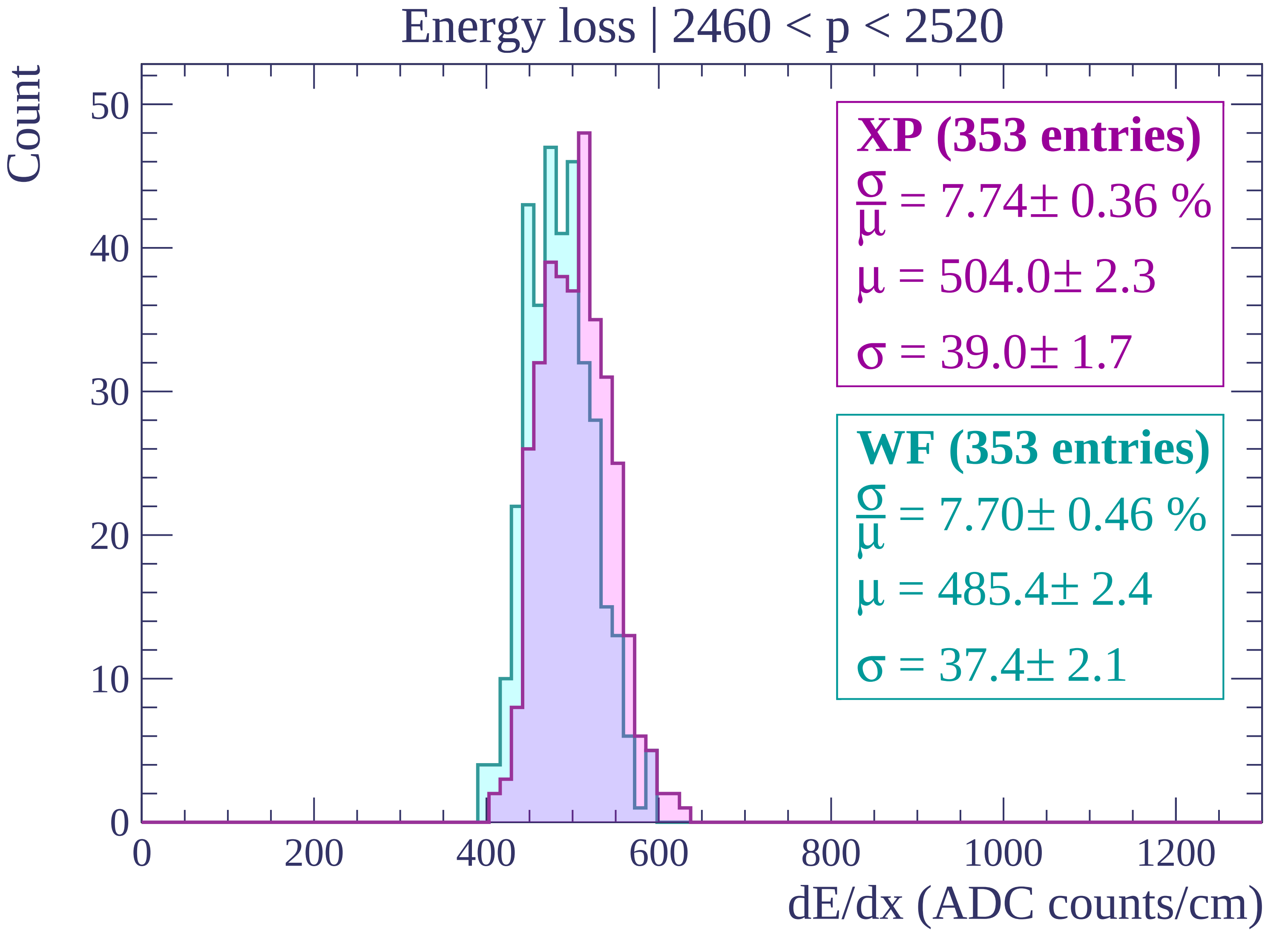


Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (421 entries)
σ/μ = 8.16± 0.46 %
μ = 498.6± 2.3
σ = 40.7± 2.1
WF (421 entries)
σ/μ = 7.71± 0.50 %
μ = 479.8± 2.2
σ = 37.0± 2.2

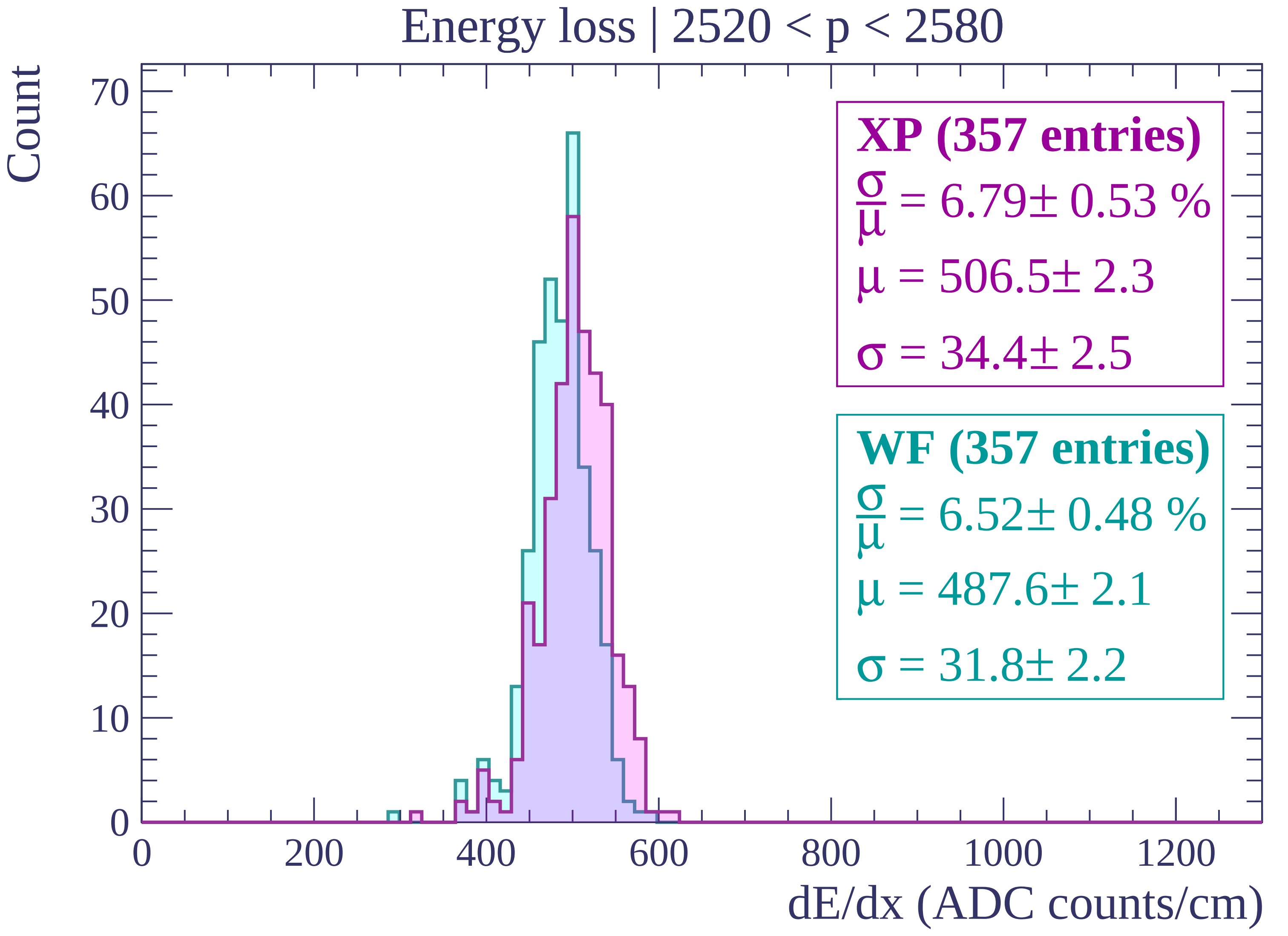


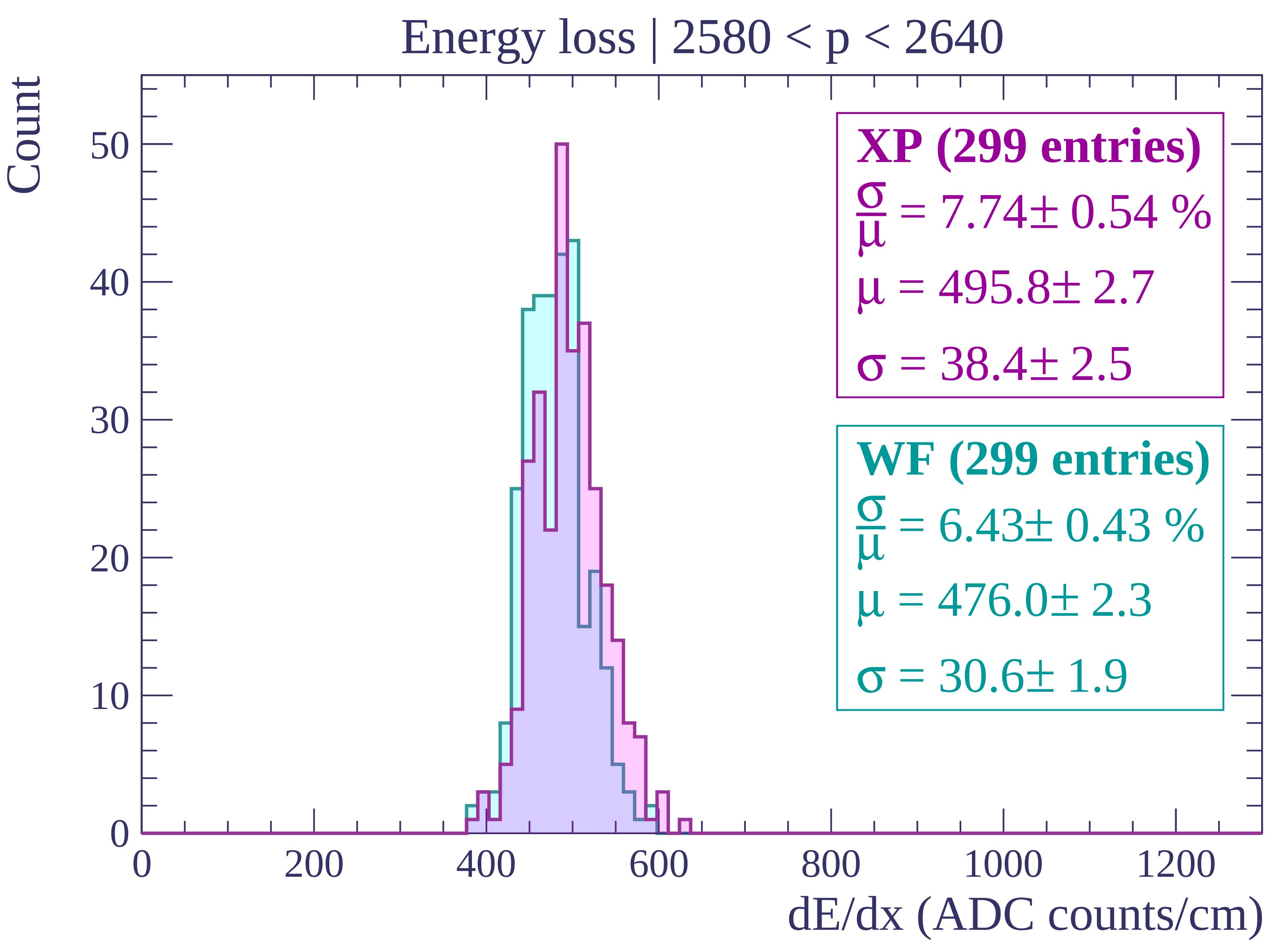
Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (401 entries)
σ/μ = 6.69± 0.41 %
μ = 496.3± 2.1
σ = 33.2± 1.9
WF (401 entries)
σ/μ = 6.64± 0.29 %
μ = 480.7± 1.8
σ = 31.9± 1.3


Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
XP (409 entries)
$\frac{\sigma}{\mu}$ = 8.00± 0.45 %
μ = 503.9± 2.3
σ = 40.3± 2.1
WF (409 entries)
$\frac{\sigma}{\mu}$ = 8.10± 0.48 %
μ = 484.8± 2.4
σ = 39.3± 2.1

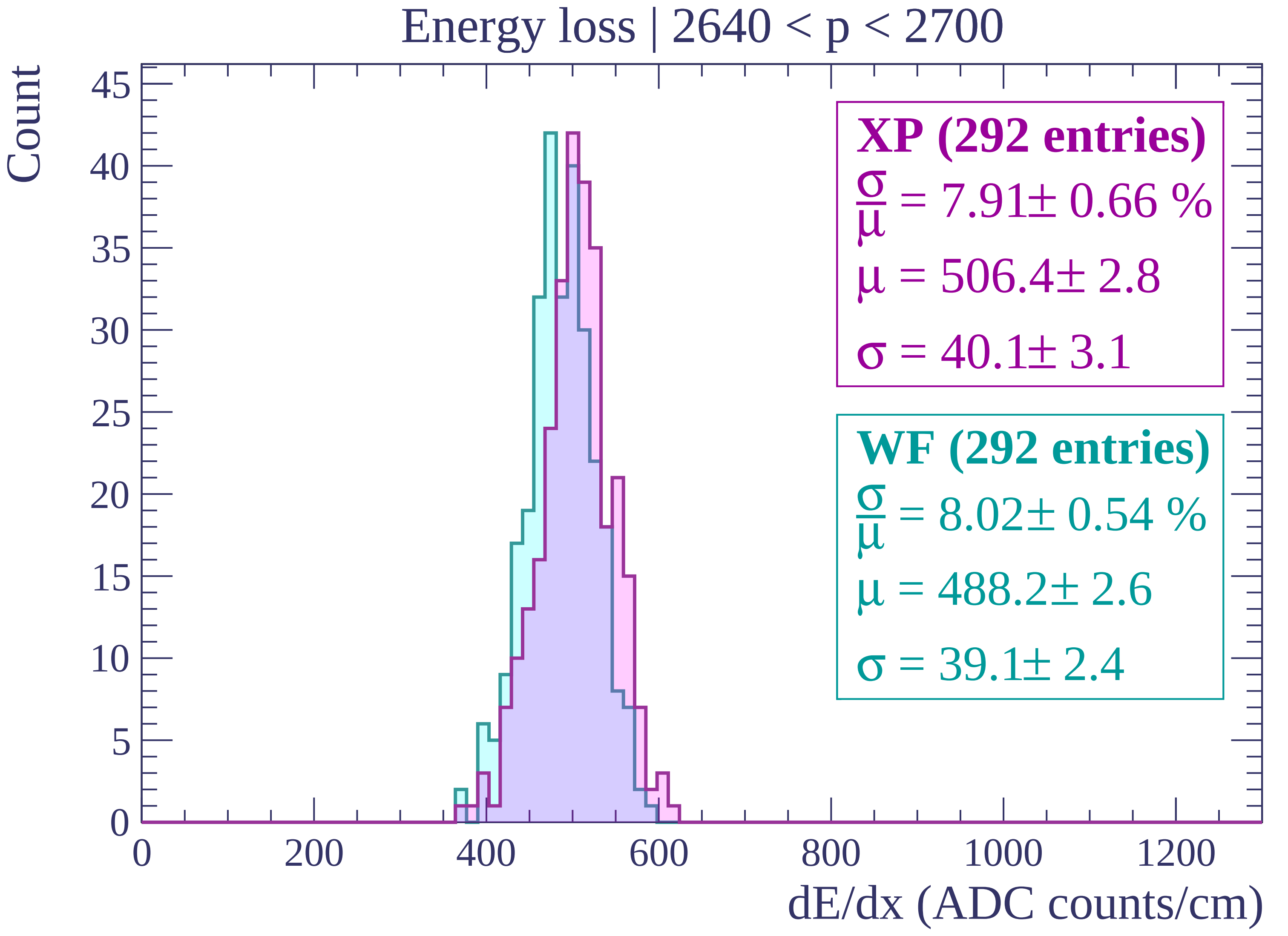


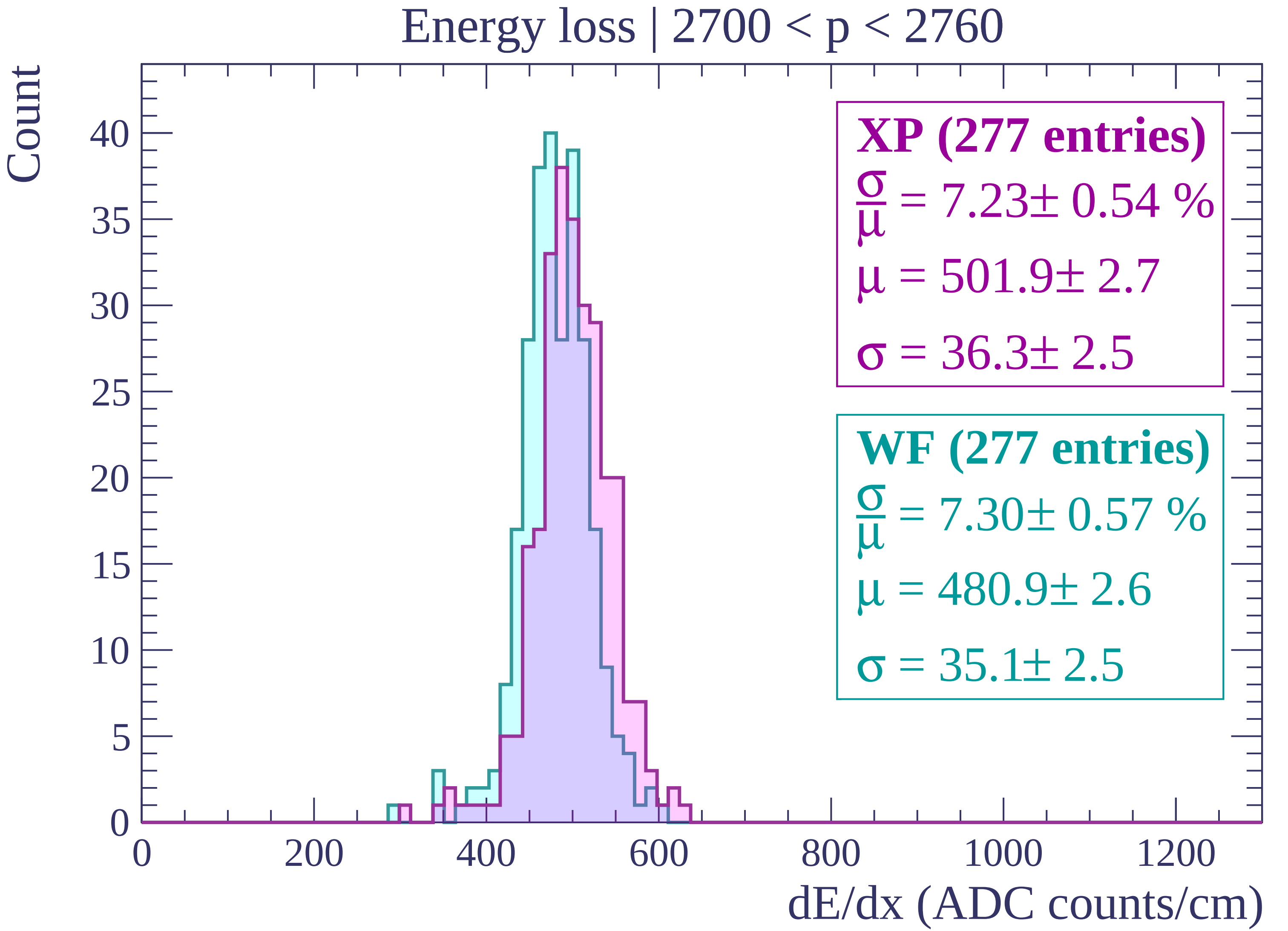
Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (353 entries)
σ/μ = 7.74± 0.36 %
μ = 504.0± 2.3
σ = 39.0± 1.7
WF (353 entries)
σ/μ = 7.70± 0.46 %
μ = 485.4± 2.4
σ = 37.4± 2.1


Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
0
200
400
600
800
1000
1200
XP (357 entries)
σ/μ = 6.79± 0.53 %
μ = 506.5± 2.3
σ = 34.4± 2.5
WF (357 entries)
σ/μ = 6.52± 0.48 %
μ = 487.6± 2.1
σ = 31.8± 2.2

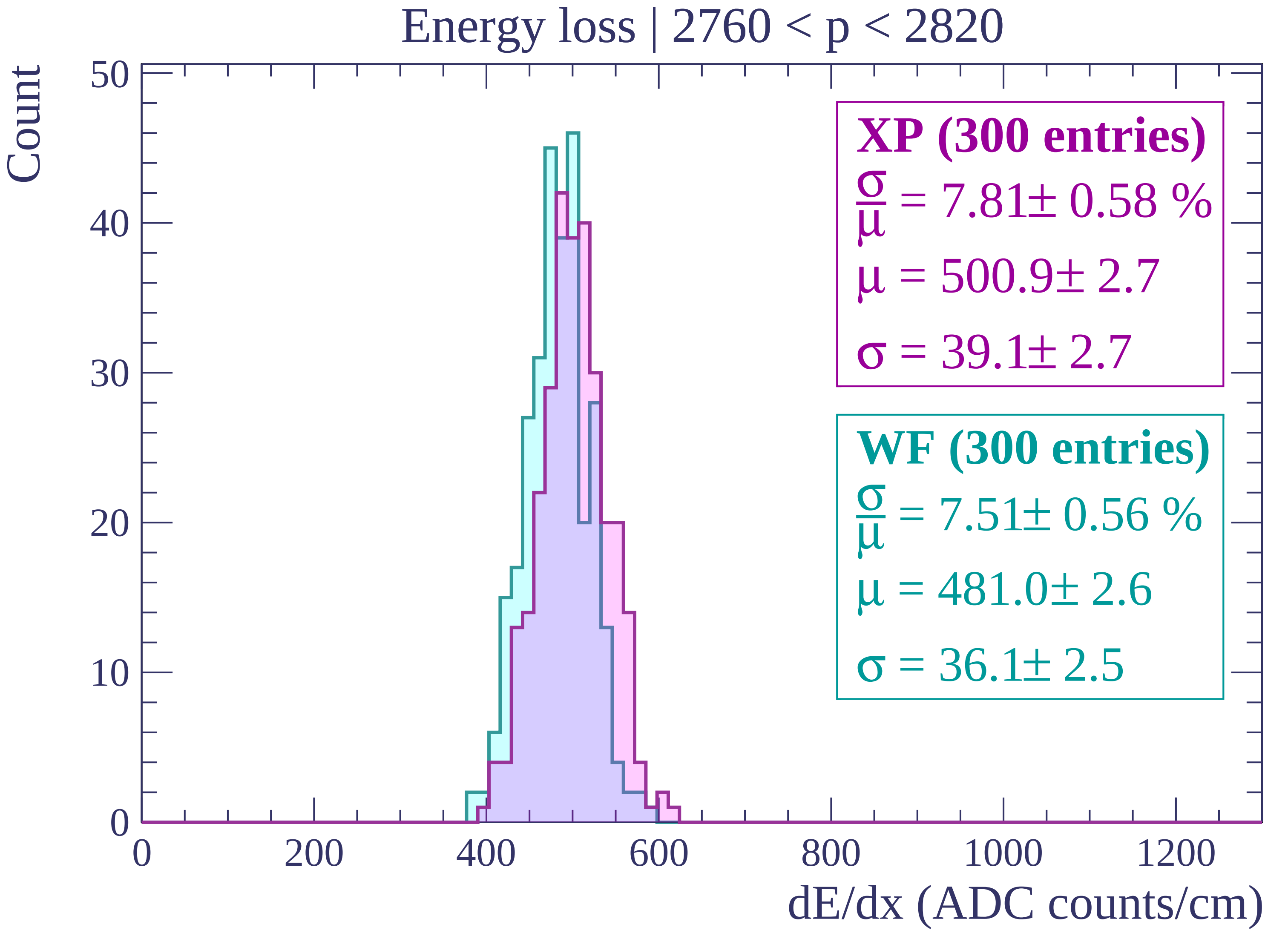


Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (299 entries)
σ/μ = 7.74± 0.54 %
μ = 495.8± 2.7
σ = 38.4± 2.5
WF (299 entries)
σ/μ = 6.43± 0.43 %
μ = 476.0± 2.3
σ = 30.6± 1.9

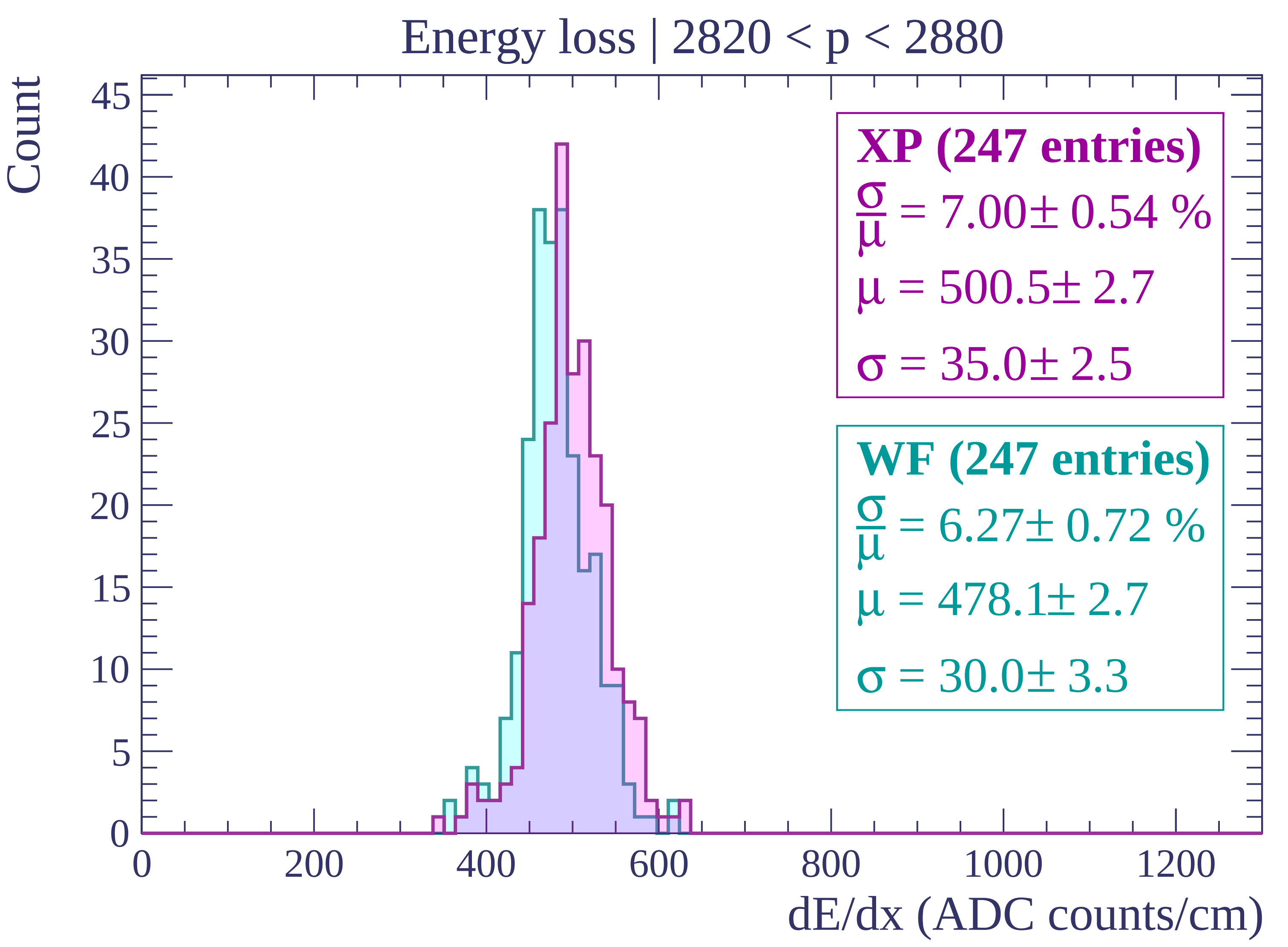

Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45
XP (292 entries)
$\frac{\sigma}{\mu}$ = 7.91± 0.66 %
μ = 506.4± 2.8
σ = 40.1± 3.1
WF (292 entries)
$\frac{\sigma}{\mu}$ = 8.02± 0.54 %
μ = 488.2± 2.6
σ = 39.1± 2.4

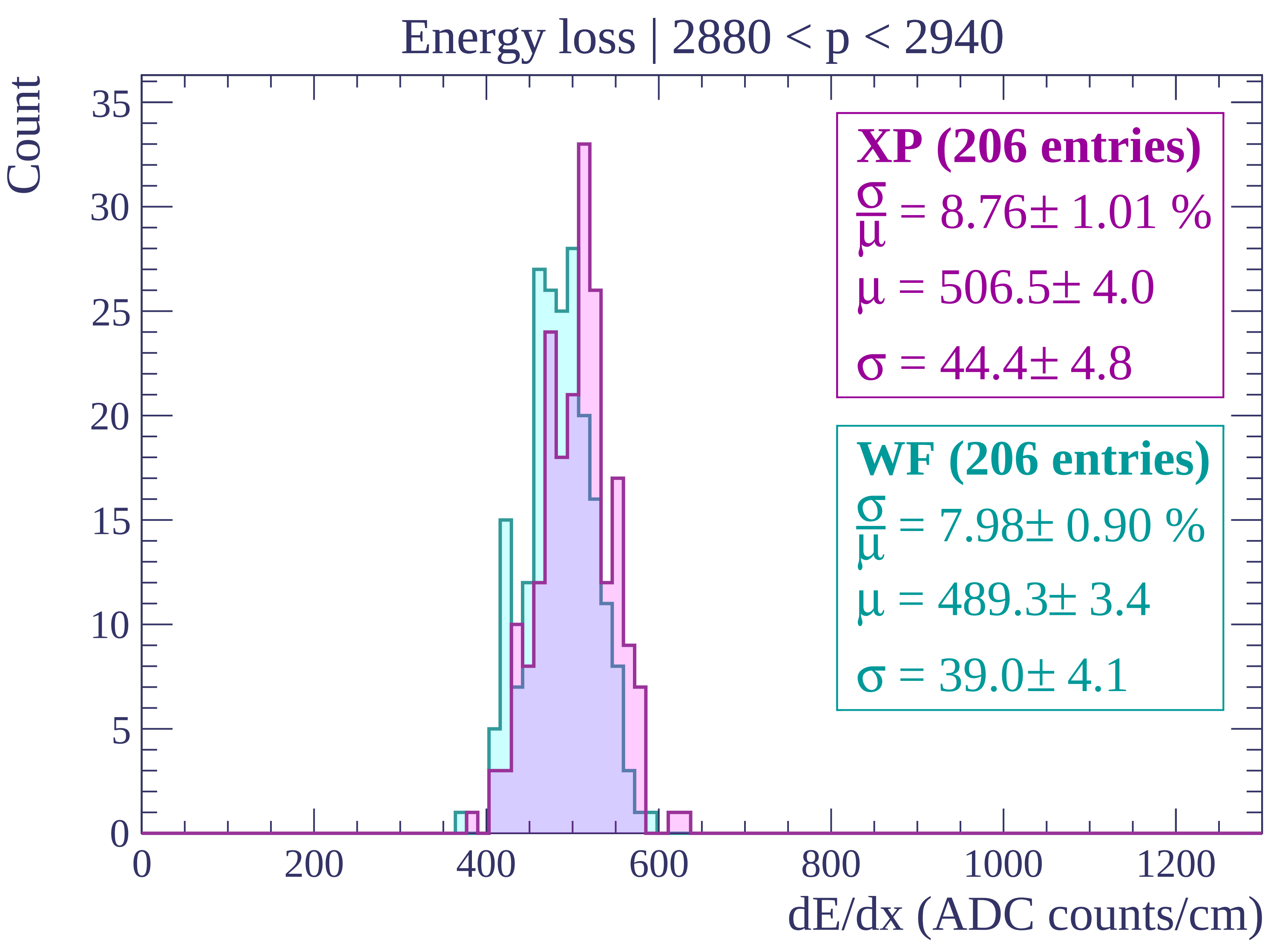


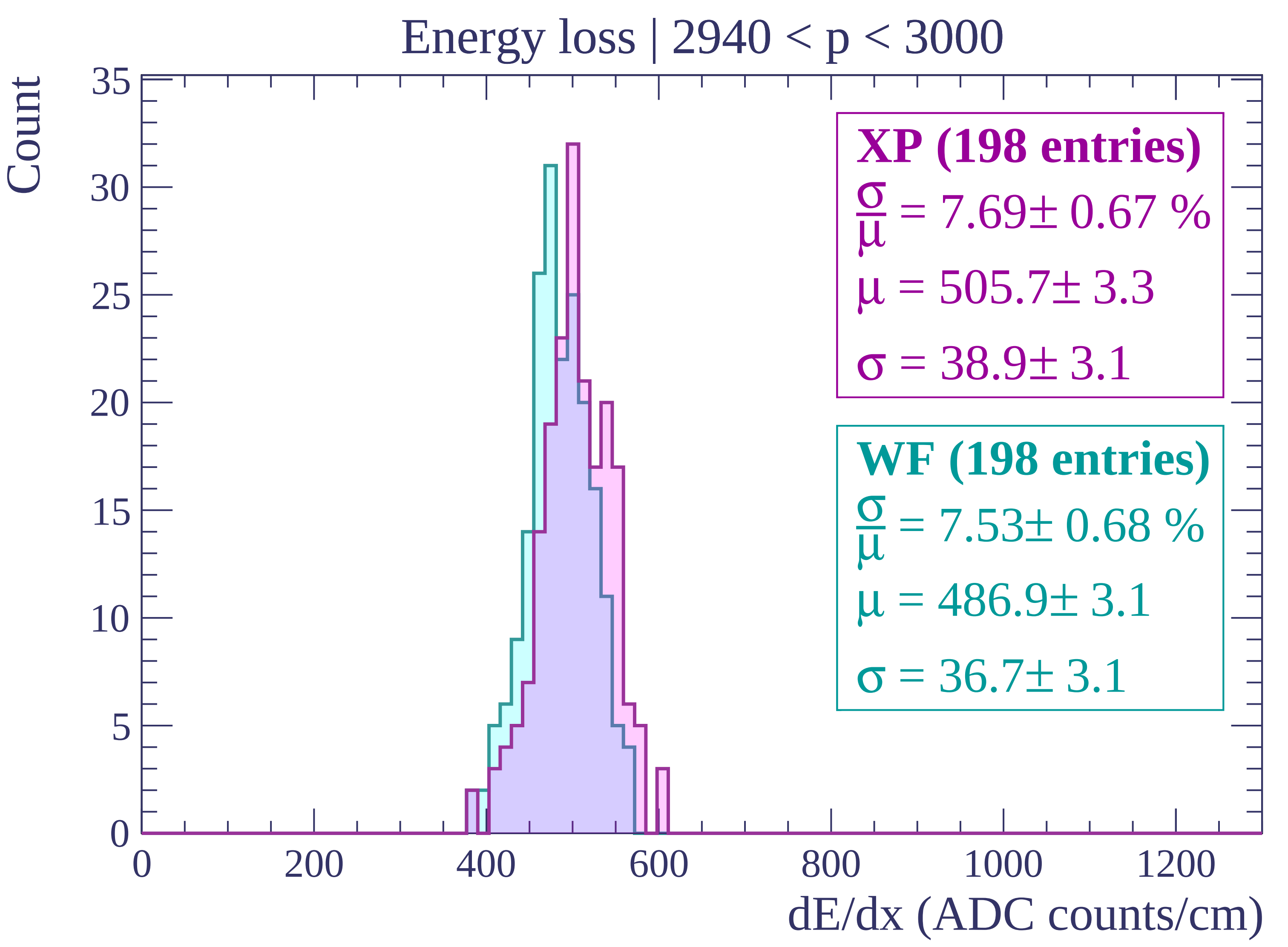
Energy loss | 2700 < p < 2760
Count
XP (277 entries)
σ/μ = 7.23± 0.54 %
μ = 501.9± 2.7
σ = 36.3± 2.5
WF (277 entries)
σ/μ = 7.30± 0.57 %
μ = 480.9± 2.6
σ = 35.1± 2.5
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200



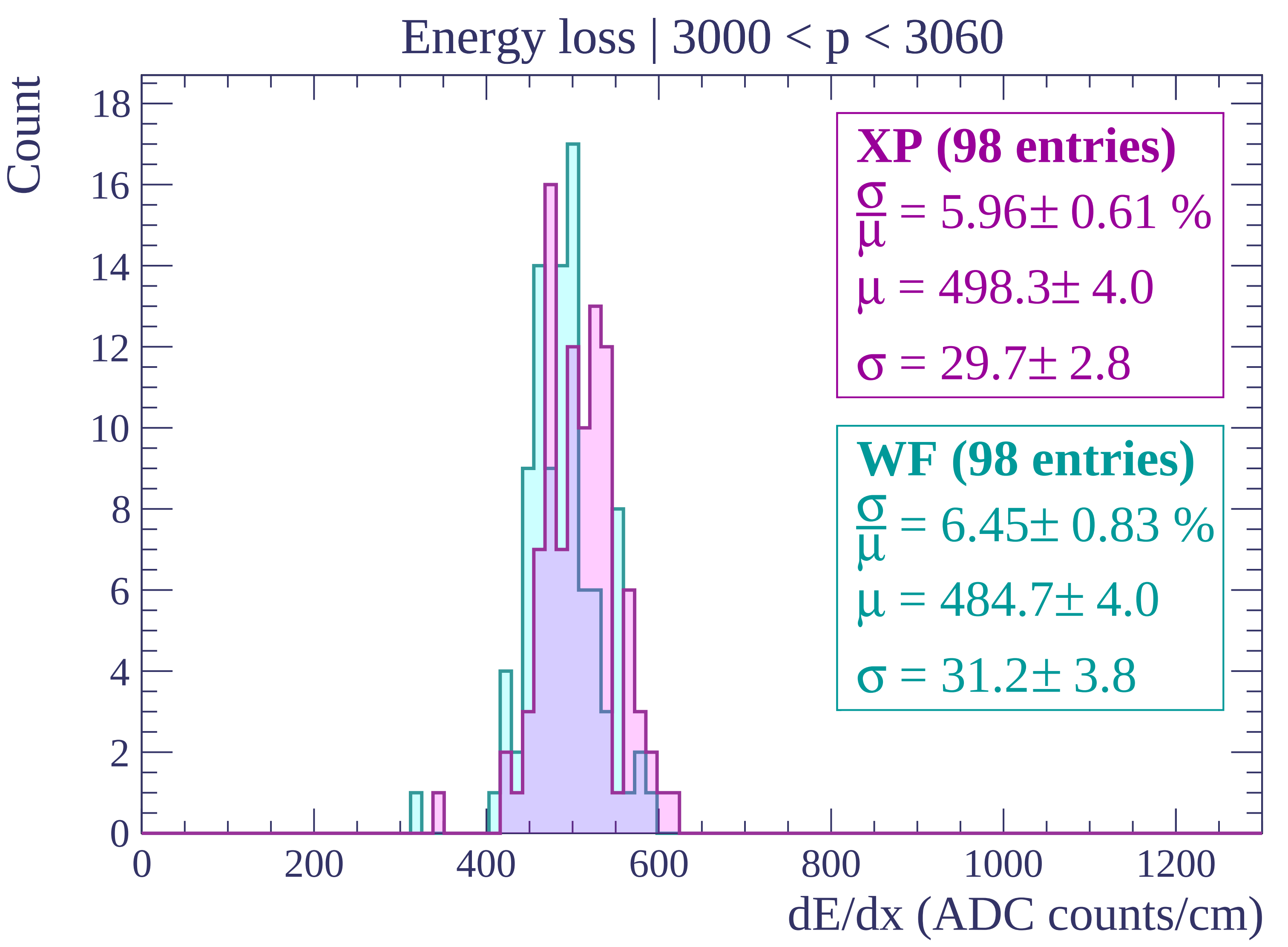
Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP (300 entries)
σ/μ = 7.81± 0.58 %
μ = 500.9± 2.7
σ = 39.1± 2.7
WF (300 entries)
σ/μ = 7.51± 0.56 %
μ = 481.0± 2.6
σ = 36.1± 2.5


Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
XP (247 entries)
$\frac{\sigma}{\mu}$ = 7.00± 0.54 %
μ = 500.5± 2.7
σ = 35.0± 2.5
WF (247 entries)
$\frac{\sigma}{\mu}$ = 6.27± 0.72 %
μ = 478.1± 2.7
σ = 30.0± 3.3
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45

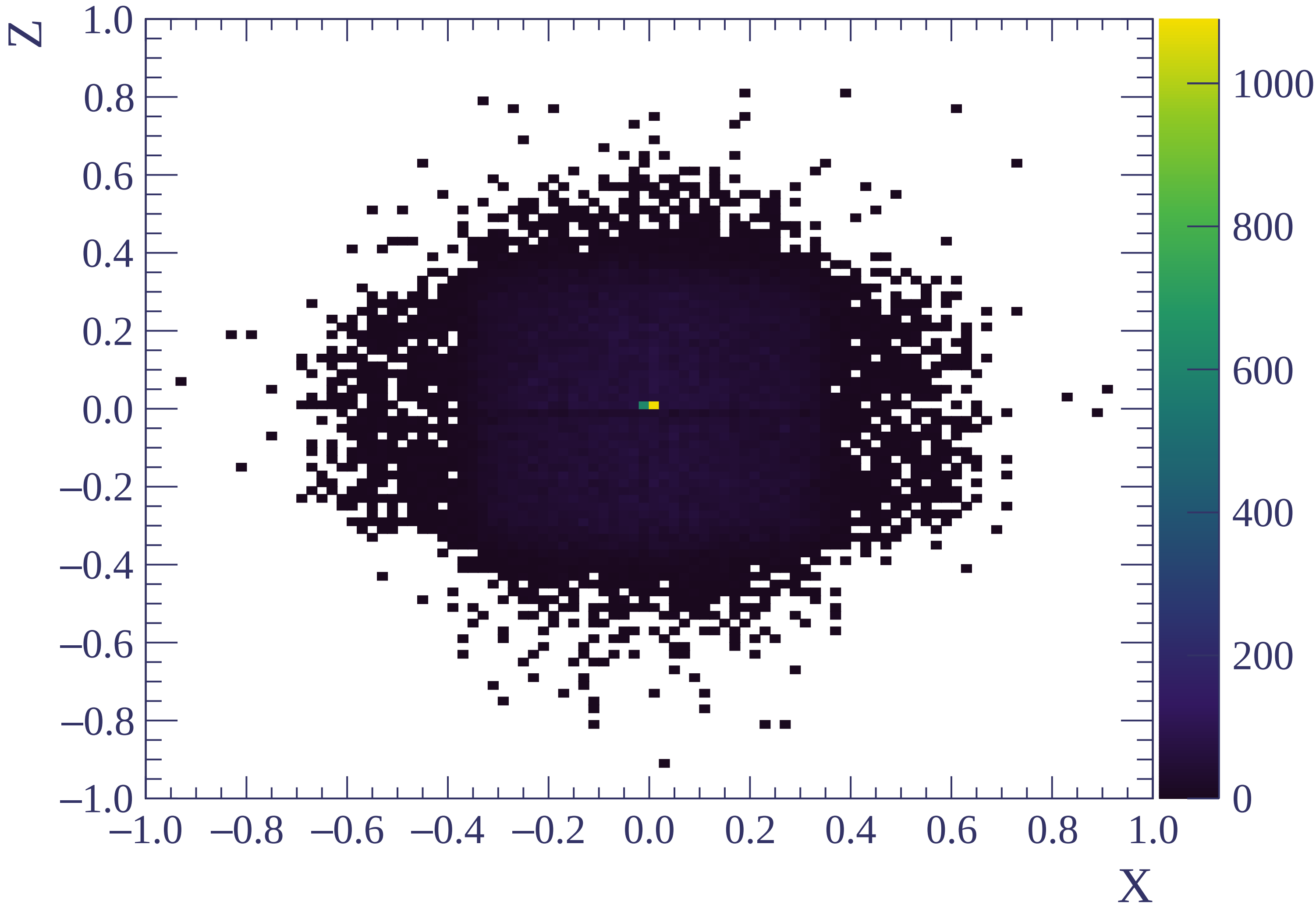

Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP (206 entries)
σ/μ = 8.76± 1.01 %
μ = 506.5± 4.0
σ = 44.4± 4.8
WF (206 entries)
σ/μ = 7.98± 0.90 %
μ = 489.3± 3.4
σ = 39.0± 4.1

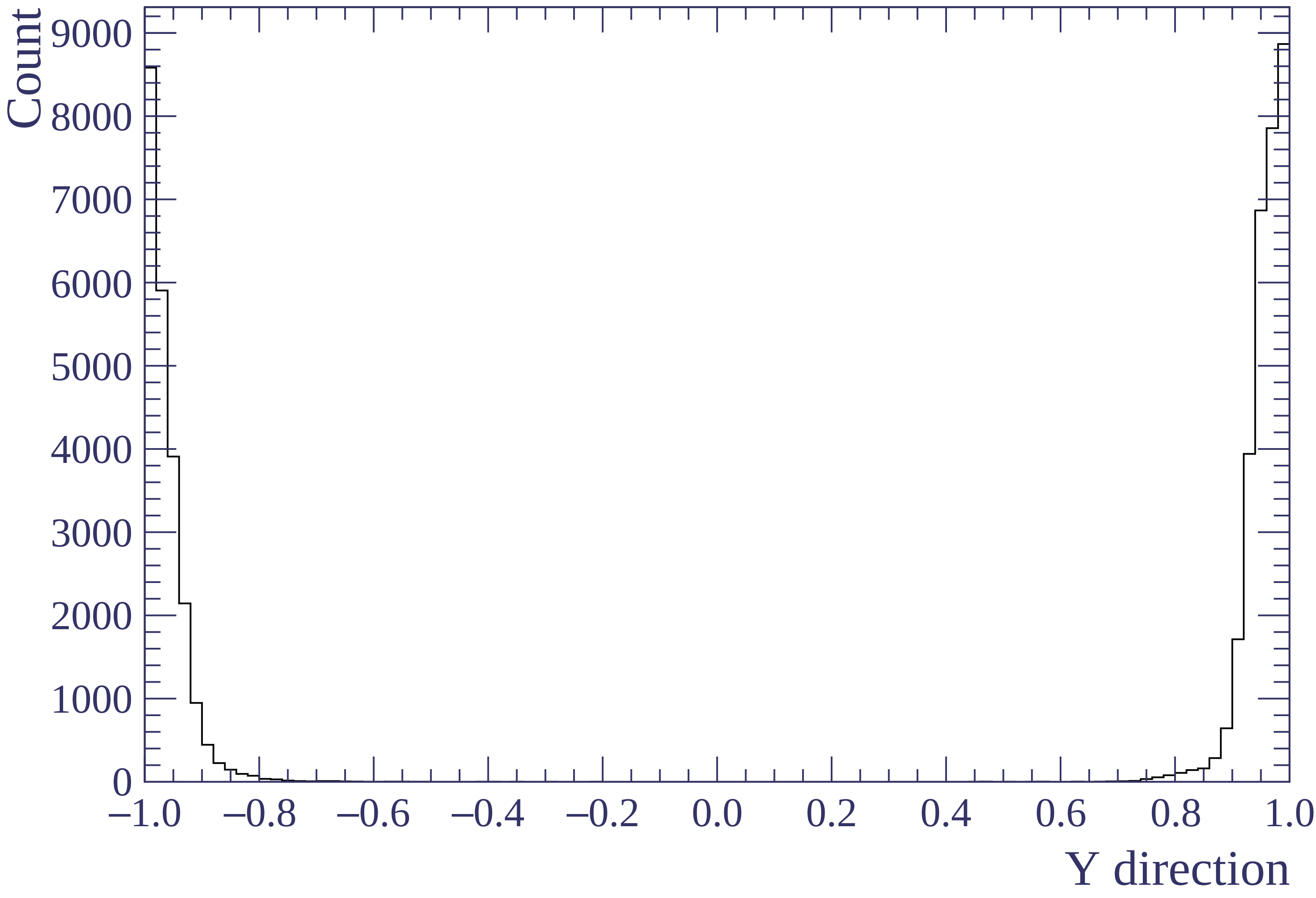


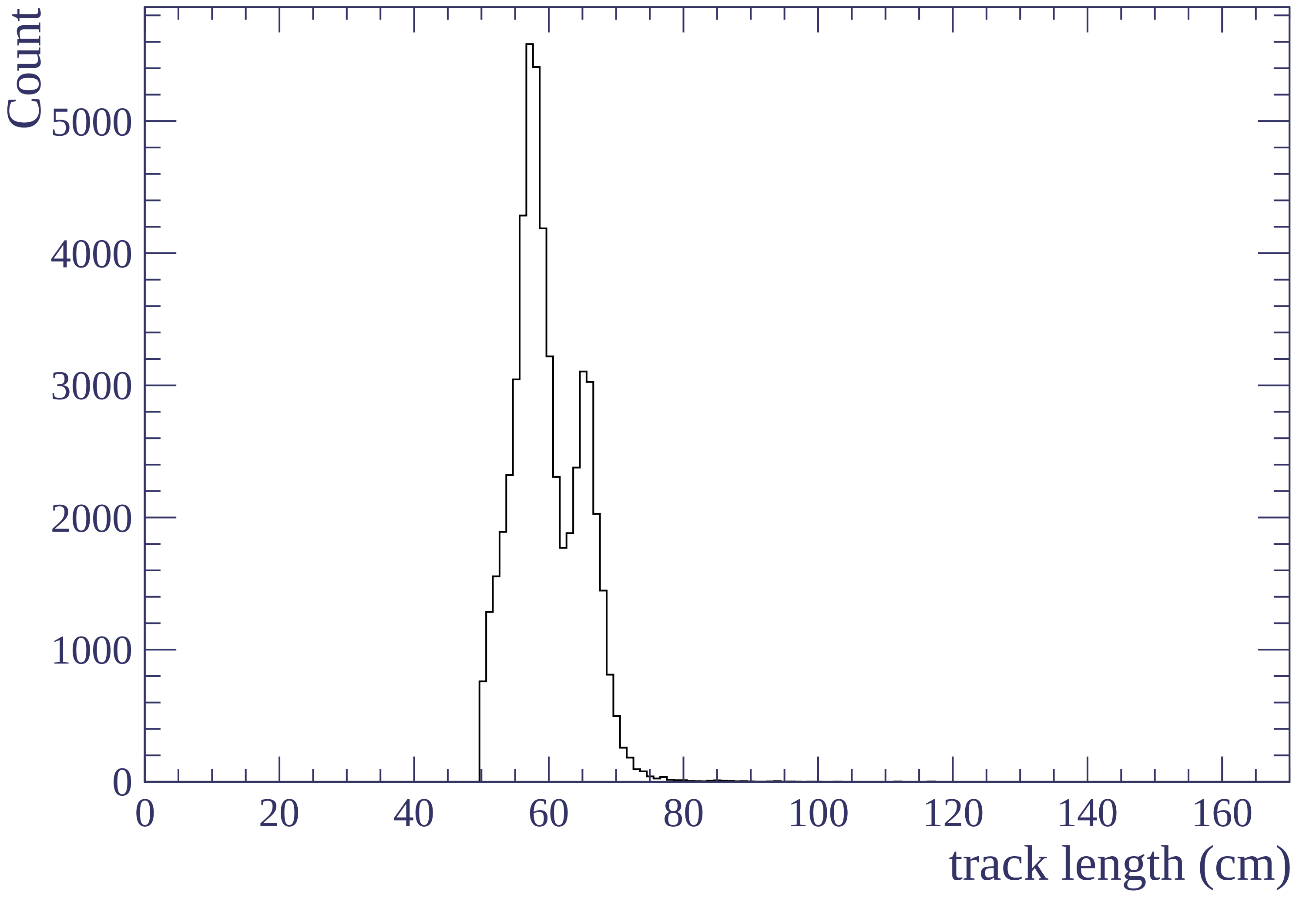
Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
XP (198 entries)
σ/μ = 7.69± 0.67 %
μ = 505.7± 3.3
σ = 38.9± 3.1
WF (198 entries)
σ/μ = 7.53± 0.68 %
μ = 486.9± 3.1
σ = 36.7± 3.1
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200



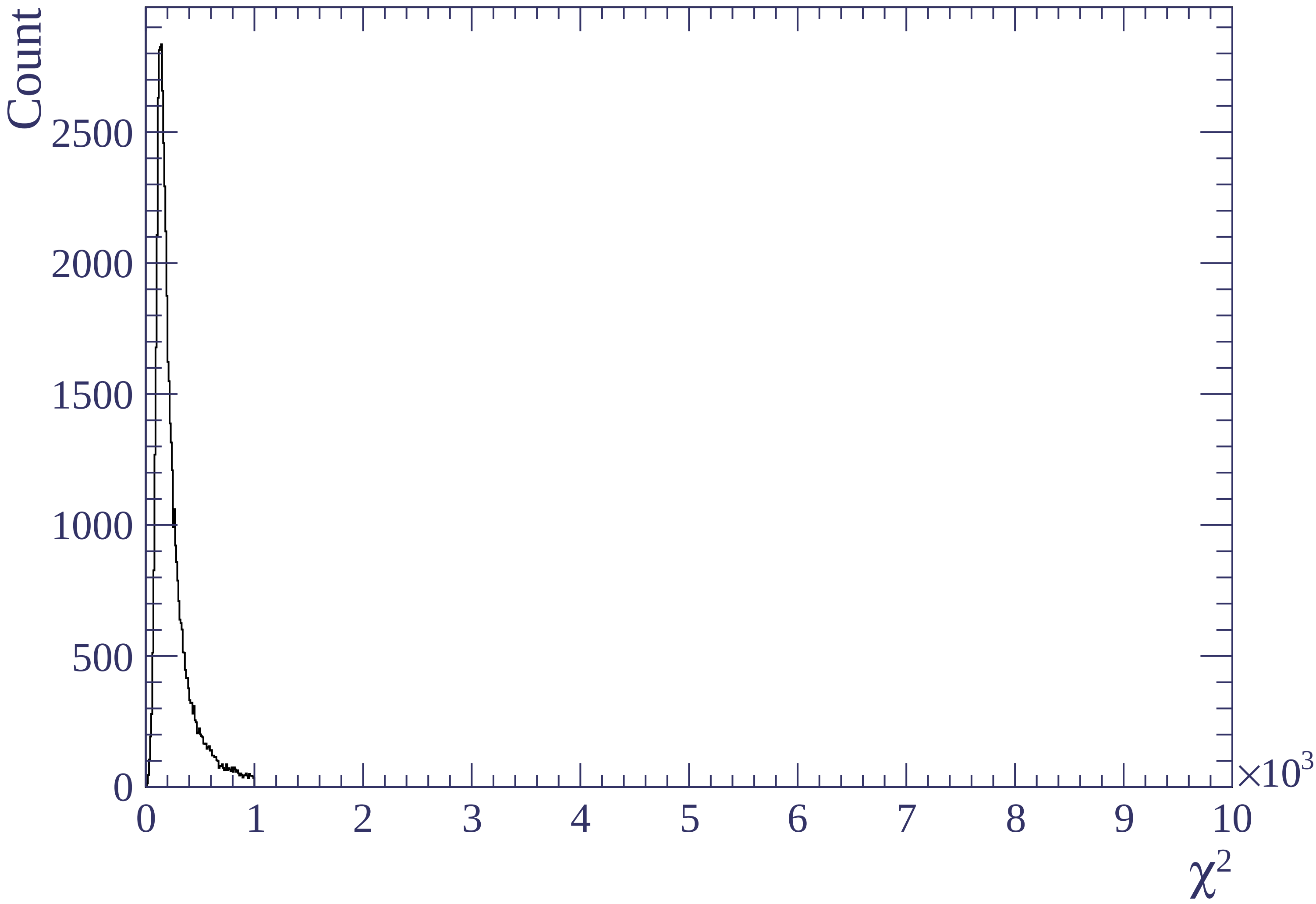
Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
XP (98 entries)
σ/μ = 5.96± 0.61 %
μ = 498.3± 4.0
σ = 29.7± 2.8
WF (98 entries)
σ/μ = 6.45± 0.83 %
μ = 484.7± 4.0
σ = 31.2± 3.8


Z
X
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
1000
800
600
400
200
0

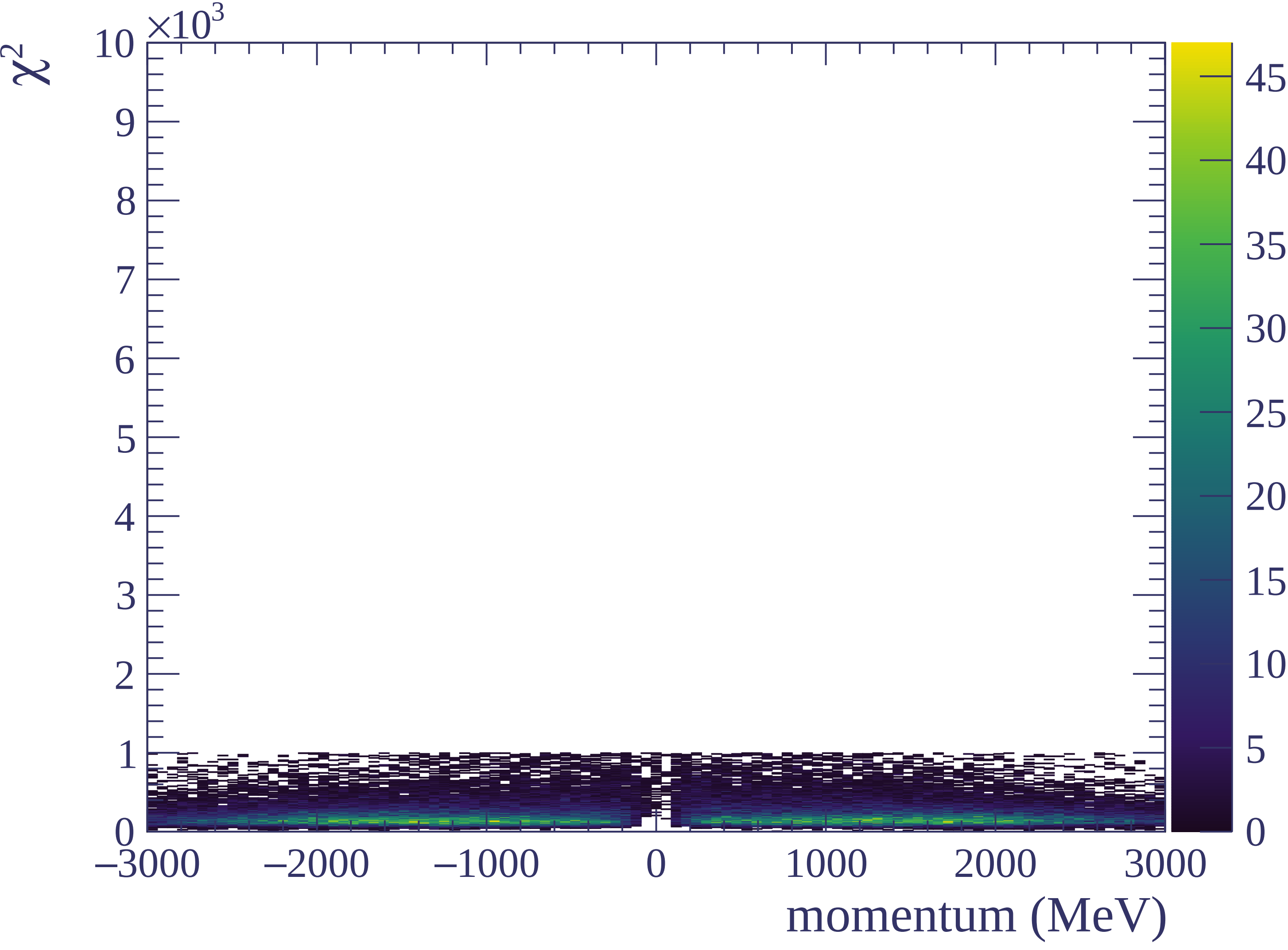

Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



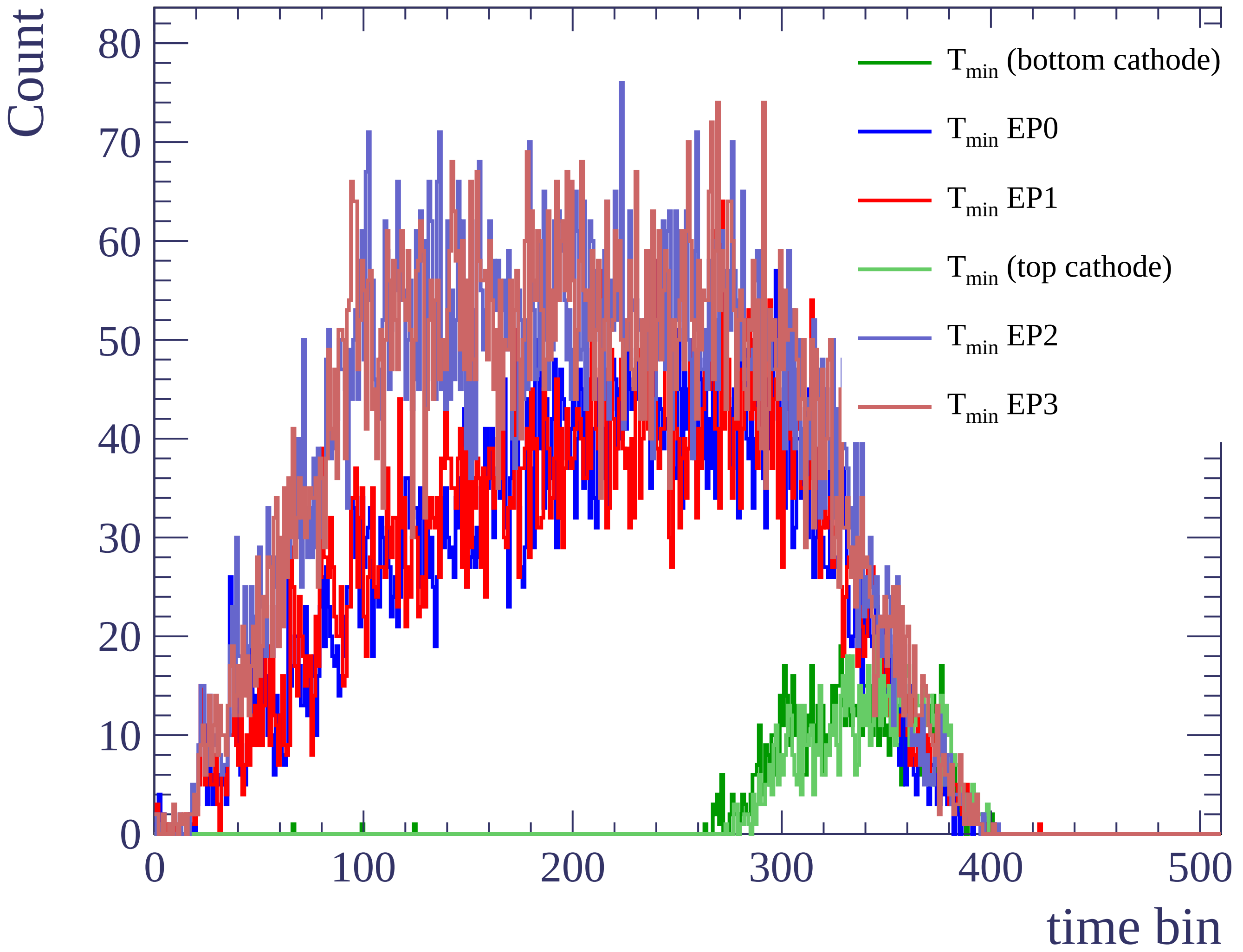
Count
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
160
track length (cm)



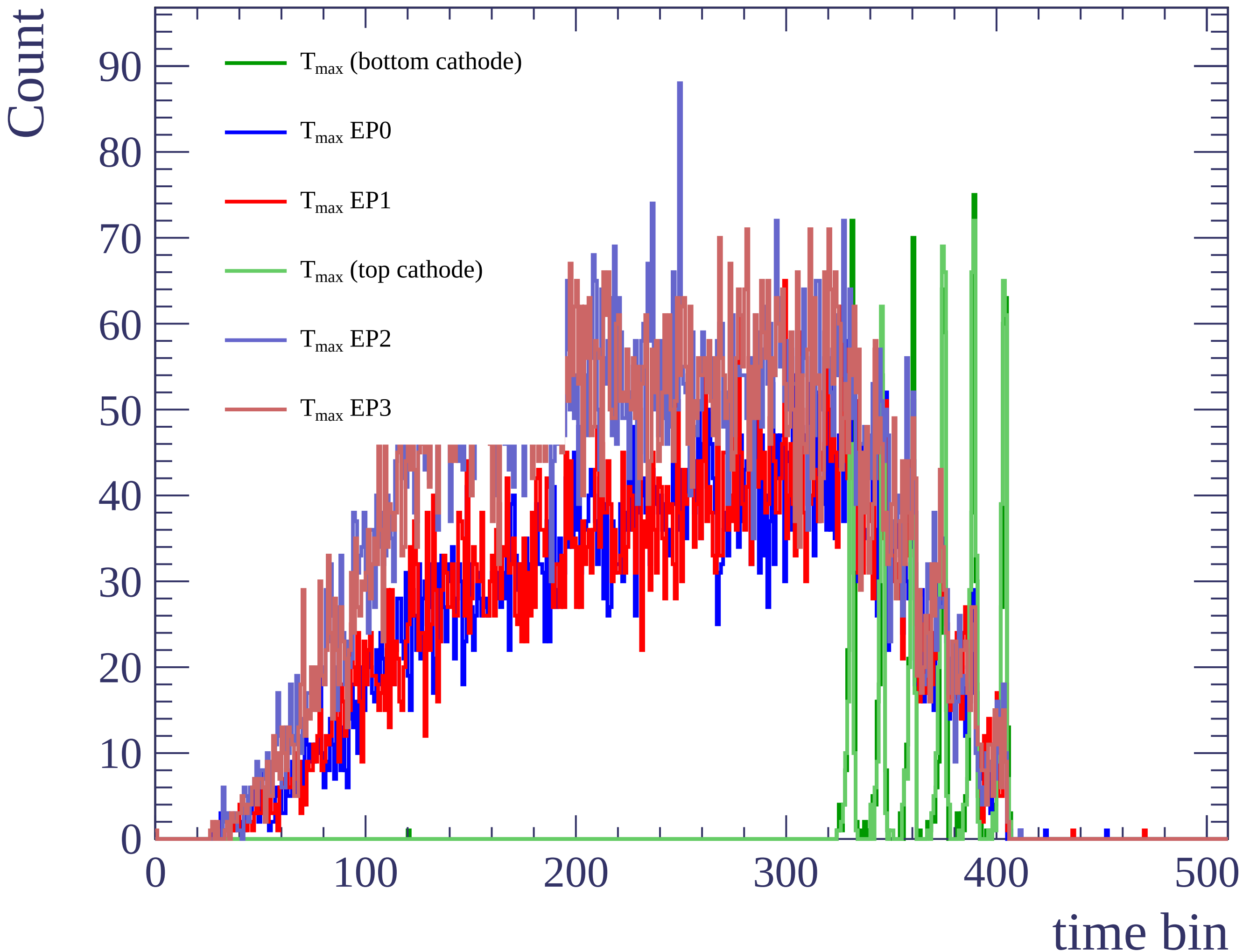
Count
2500
2000
1500
1000
500
0
0
1
2
3
4
5
6
7
8
9
10
$\times 10^3$
χ^2



×10^3
χ^2
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
45
40
35
30
25
20
15
10
5
0



Count
time bin
0
100
200
300
400
500
0
10
20
30
40
50
60
70
80
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
time bin
0
100
200
300
400
500
0
10
20
30
40
50
60
70
80
90
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3